

Total No. of Questions : 10]

SEAT No. :

P9418

[6182] - 701

[Total No. of Pages : 2

M.E. (Civil) (Construction Management)
APPLICATION OF STATISTICAL METHODS IN
CONSTRUCTION
(2017 Pattern) (Semester - I) (501021)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

- Q1)** a) Define probability and Explain the Baye's theorem. **[5]**
b) 10% of the bulbs produced in a factory are of red colour and 2% are red and defective. If one bulb is picked up at random, determine the probability of its being defective if it is red. **[5]**

OR

- Q2)** Four balls are to be drawn without replacement from a box containing 8 red and 4 white balls. If X denotes the number of red ball drawn, find the probability distribution of X. **[10]**

- Q3)** A car manufacturing factory has two plants, X and Y. Plant X manufactures 70% of cars and plant Y manufactures 30%. 80% of the cars at plant X and 90% of the cars at plant Y are rated of standard quality. A car is chosen at random and is found to be of standard quality. What is the probability that it has come from plant X? **[10]**

OR

- Q4)** Three machines E1, E2, E3 in a certain factory produce 50%, 25% and 25%, respectively, of the total daily output of electric tubes. It is known that 4% of the tubes produced one each of machines E1 and E2 are defective and that 5% of those produced on E3 are defective. If one tube is picked up at random from a day's production, calculate the probability that it is defective. **[10]**

P.T.O.

Q5) Random samples of size 225 are drawn from a population in which the proportion with the characteristic of interest is 0.25. Decide whether or not the sample size is large enough to assume that the sample proportion P is normally distributed. **[10]**

OR

Q6) A high-speed packing machine can be set to deliver between 11 and 13 ounces of a liquid. For any delivery setting in this range the amount delivered is normally distributed with mean some amount μ and with standard deviation 0.08 ounce. To calibrate the machine it is set to deliver a particular amount, many containers are filled and 25 containers are randomly selected and the amount they contain is measured. Find the probability that the sample mean will be within 0.05 ounce of the actual mean amount being delivered to all containers. **[10]**

Q7) A professor wants to know if her introductory statistics class has a good grasp of basic math. Six students are chosen at random from the class and given a math proficiency test. The professor wants the class to be able to score above 70 on the test. The six students get the following scores:

62, 92, 75, 68, 83, 95.

Can the professor have 90% confidence that the mean score for the class on the test would be above 70. **[10]**

OR

Q8) Explain in details 4 types of correlation with examples. **[10]**

Q9) Explain in detail. **[10]**

- a) Sensitivity analysis
- b) Role of mathematical model in construction management.

OR

Q10)a) Describe in brief ABC Analysis? **[5]**

b) Describe in brief EOQ in Civil Engineering? **[5]**



Total No. of Questions : 8]

SEAT No. :

P9419

[6182]-702

[Total No. of Pages :2

M.E. (Civil-Construction & Management)
MANAGEMENT & PROJECT PLANNING IN CONSTRUCTION
(2017 Credit Pattern) (Semester-I) (501022)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

- Q1)** a) Enlist and explain the various Functions of Management. [4]
b) Explain the role of Project Management Consultant in a building construction project. [5]

OR

- Q2)** a) Write a short note on “Line of Balance Technique.” [4]
b) Following table shows the activities of a construction project. [5]

Name of Activity	Immediate Predecessor Activity	Duration (Days)
A	--	4
B	--	3
C	A	6
D	A	5
E	B	4
F	C	3
G	C,D,E	7
H	F,G	5

- i) Draw a CPM Network. Show the critical path and find out the total project duration.
- ii) Calculate:
 - A) Total Float
 - B) Free Float
 - C) Independent Float
 - D) Interfering Float for each activity.

P.T.O.

- Q3)** a) What is mean by Network Crashing? Explain step by step procedure of crashing the network. [4]
b) Explain the factors to be considered while selecting a job for Work Study. [5]

OR

- Q4)** a) Explain in brief the application of MIS in construction. [5]
b) Write a short note on “Micro-motion Study”. [4]

- Q5)** a) Enlist and explain in detail various causes of trench collapse during excavation work. [8]
b) Write a short note on
i) Personal Protective Equipments [4]
ii) General causes of accidents on construction sites [4]

OR

- Q6)** a) What is HAZOP Technique? Explain HAZOP analysis process in detail. [8]
b) Write a detailed note on “Minimum Wages Act.” [8]

- Q7)** a) What are the different types of incentives? Explain the prerequisites for an effective incentive system. [8]
b) Write a brief note on:
i) Application of BIM in construction [4]
ii) Application of Artificial Neural Network in construction [4]

OR

- Q8)** a) Explain any two types of incentive schemes in detail. [8]
b) Write a brief note on:
i) Application of Genetic Algorithm in construction [4]
ii) Application of Fuzzy logic technique in construction [4]



Total No. of Questions : 8]

SEAT No. :

P9420

[6182]-703

[Total No. of Pages :2

M.E. (Civil Construction Management)
CONSTRUCTION TECHNOLOGY
(2017 Pattern) (Semester-I) (501023)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data if necessary.*
- 5) Use of cell phone is prohibited in the examination hall.*
- 6) Use of electronic pocket calculator is allowed.*

Q1) a) What are the methods of dewatering for shallow and deep excavations and explain vacuum dewatering with neat sketch. **[5]**

b) List out the types of grouting. Explain permeation grouting in details with soil suitability. **[4]**

OR

Q2) a) Enlist the methods of tunnel driving in soft soil and explain needle beam method in detail. **[5]**

b) Explain jet grouting technique and its application. **[4]**

Q3) a) Explain Table - formwork system, states its advantages over conventional shuttering system. **[4]**

b) What is Precast - Pre stressed pile? Explain a typical square pile section with diagram and its construction. **[5]**

OR

Q4) a) Discuss the merits and demerits of Drill and Blast Techniques for tunnel construction over other methods. **[5]**

b) With the help of neat labeled sketches enlist the sequence of construction of coffer dam constructed with MS sheet piles for a bridge pier construction. **[4]**

P.T.O.

- Q5)** a) Explain the sequential operations involved in construction of sheet pile cofferdam for construction of bridge pier. [8]
- b) Suggest the suitable type of cofferdam with reasons, explain its sequence of construction for following condition [8]
- Construction of pier in sea shore land
 - Excavation depth: 15m
 - Water level : 3m from bed
 - Soil condition : saturated

OR

- Q6)** a) Draw a neat & clean diagram of coffer dam showing load imposed on it. Explain any two loads in detail. [8]
- b) Explain with neat sketch the sequence of operations involved in construction of earth and rock fill cofferdam. [8]
- Q7)** a) List out types of caissons, explain floating caissons in detail. [4]
- b) Explain pneumatic caissons in details. [4]
- c) Focus on constructional sequence of caissons in details. [8]

OR

- Q8)** a) Enlist the types of caisson and explain the construction of pneumatic caissons with neat sketch. [8]
- b) Write a detail construction procedure of well foundation. [8]



Total No. of Questions : 8]

SEAT No. :

P9421

[Total No. of Pages : 2

[6182]-704

M.E.(Civil) (Construction & Management)

TOTAL QUALITY MANAGEMENT IN CONSTRUCTION

(2017 Pattern) (Semester - I) (501024)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q. 1 or Q. 2, Q. 3 or Q. 4, Q.5 or Q.6 and Q. 7 or Q. 8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

Q1) a) Explain Philip Cross by 14 Principles with respect to Quality. **[4]**

b) Explain Parato charts used for quality assessment. **[5]**

OR

Q2) a) Differentiate between Quality Assurance & Quality Control. **[4]**

b) How are dispersion methods applied in quality control of construction activities? **[5]**

Q3) a) What are the main objectives or goals of training programs focused on the preparation of checklist for RCC work? **[4]**

b) Explain the role of quality circles in an organization's quality management system. **[5]**

OR

Q4) a) How can a quality inspection team identify and address potential quality issues or non-conformities within a process or product? **[5]**

b) How do managerial competencies contribute to the effectiveness of human resources training and development? **[4]**

P.T.O.

- Q5) a)** Explain the certification process for ISO 9001? **[8]**
- b) How does ISO 9004 contribute to the overall quality management of an organization? **[8]**

OR

- Q6) a)** What are the key differences between ISO 9001 and ISO 9004? **[8]**
- b) Explain the eight principles of ISO 9001. **[8]**
- Q7) a)** Explain with neat sketch, Quality Functions Deployment (QFD). **[8]**
- b) Explain the role of Third Party Quality Audit in Infrastructural Projects. **[8]**

OR

- Q8) a)** Explain the different types of barriers that may affect TQM Implementation. **[8]**
- b) Elaborate the different principles used for TQM Implementation. **[8]**



Total No. of Questions : 10]

SEAT No. :

P9422

[6182]-706

[Total No. of Pages : 2

M.E. (Civil) (Construction Management)
CONSTRUCTION CONTRACTS ADMINISTRATION AND
MANAGEMENT
(2017 Pattern) (Semester - II) (501027)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

- Q1) a)** Explain with examples the valid excuses for non-performance permitted in the Indian Contract Act. **[5]**
- b) What is the role of project management services in construction project? **[5]**

OR

- Q2) a)** Explain in details standard forms of contracts & what is the importance of comparative statement after opening the Tenders? Give format for the same. **[5]**
- b) Explain Workmen's Compensation Act on construction projects with clauses. **[5]**
- Q3) a)** State the circumstances under which the lowest tender is rejected? **[5]**
- b) Circumstances under which the contractor shall be entitled to terminate the contract. **[5]**

OR

- Q4) a)** Discuss the advantages of DRB over dispute resolution mechanism. **[5]**
- b) Write in detail responsibilities and duties of the Employer for contracts under Red Book. **[5]**

P.T.O.

Q5) a) What is Construction Claims, Types of construction claims and explain settlement of claims. [5]

b) Write a brief meaning/ Provision for following under FIDIC conditions RED BOOK. [5]

i) Tests on completion & Tests after completion

ii) Employer's personnel & Contractor's personnel

OR

Q6) a) What are the different types of FIDIC books commonly used in projects and its uses? [5]

b) What is Extra items and causes of claims in construction industry. [5]

Q7) a) Describe the clauses of escalation of cost: What is an arbitral proceeding. Explain with example. [5]

i) Escalation formula for material cost

ii) Escalation formula for Labour cost

b) Explain in detail - Defect Liability Period. [5]

OR

Q8) a) Write a short note on-Conciliation related to contract. [5]

b) What is the importance of role of various stakeholders in prevention of disputes? [5]

Q9) a) Procedure of appeal against the awards. [5]

b) What are the various duties, power of arbitrators? [5]

OR

Q10) a) What is Conciliation and its provisions in the Act, also explain conduct of conciliation and arbitral proceedings? [5]

b) What is difference between Arbitration and Conciliation? [5]



Total No. of Questions : 8]

SEAT No. :

P-9699

[Total No. of Pages : 2

[6182]-707

M.E. (Civil-Construction & Management)
PROJECT ECONOMICS & FINANCIAL MANAGEMENT
(2017 Pattern) (CBCS) (Semester - II) (501028)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Use suitable data if necessary.

- Q1)** a) What will be the objectives of any business firm? [5]
b) Study of present economy is a very important factor for any business. Justify the statement. [4]

OR

- Q2)** a) What are the different types of risks that construction projects face? [5]
b) What do you understand by technical appraisal? Discuss with the help of example. [4]

- Q3)** a) What do you understand by concept of break even analysis? Explain with the help of figure. [5]
b) What are the different sources of long term finance? Explain in brief. [4]

OR

- Q4)** a) Discuss role of financial institutes in construction sector. [5]
b) Write down about SEBI and its regulations. [4]

- Q5)** a) Discuss in detail about Escrow accounts in PPP projects. [6]
b) The Financial details of “Swami infra” as on 2020 are as follows. Prepare a balance sheet of the company. [10]

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
Equity share capital	1,00,000	Creditors	5,000
Stock	7,500	Bill payable	3,750
Cash at bank	2,500	General reserve	12,500
P & L account	10,000	Debtors	5,000
Bill receivable	7,500	Provident funds	7,500
O/S expenses	1,250	Preliminary expenses	10,000
Land and building	50,000	Plant and machinery	57,500

P.T.O.

OR

Q6) a) Discuss method of reporting and recording site accounts between project and head office. [6]

b) A construction company undertakes large contracts. The following are the particulars related to contract carried during the year ended on 31st March 2021. Prepare a contract account and find out notional profit. How much profit is to be transferred to P & L account and in reserve? [10]

Particulars	Amt.(Rs.)	Particulars	Amt.(Rs.)
Work certified	1,65,000	Wages accrued on 31 st March	5,800
Work not certified	5,200	Direct expenses	6,300
Plant installed at site	12,500	Material on hand	3,500
Values of plant	18,500	Material returned to store	850
Material sent to site	76,400	Direct expenses accrued	900
Labor charges	70,000	Contract price	2,10,000
Establishment charges	5,900	Cash received from contract	1,50,000

Q7) a) Draft detailed project appraisal report for a construction of a Dam, considering technical, financial, social, environmental aspects. [8]

b) Discuss in detail any case study on Rapid bus transit system and its pros and cons. [8]

OR

Q8) a) Draft detailed project appraisal report for a construction of a closed conduit canal system to replace existing open canal, considering technical, financial, social, environmental aspects. [8]

b) Discuss in detail a case study of any govt. funded construction project with respect to project appraisal, raising of fund, risks involved, cost analysis etc. [8]



Total No. of Questions : 8]

SEAT No. :

P9423

[6182]-708

[Total No. of Pages : 4

M.E. (Civil-Constructions & Management)
OPERATIONS RESEARCH
(2017 Credit Pattern) (Semester - II) (501029)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
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- 6) *Use of electronic pocket calculator is allowed.*

Q1) a) Explain the necessity of operations research in Civil Engineering. **[4]**

b) Maximize $Z = 2x_1 + 4x_2 + 3x_3$ **[5]**

Subject to $3x_1 + 4x_2 + 2x_3 \leq 60$

$2x_1 + x_2 + 2x_3 \leq 40$

$x_1 + 3x_2 + 2x_3 \leq 80$

$x_1, x_2, x_3 \geq 0$

use simplex method.

OR

Q2) a) For each of the following function show whether it is convex, concave or neither. **[5]**

i) $F(x) = 15 - x^2$

ii) $F(x) = x^4 + 6x^2 + 10x$

b) Explain the advantages & limitations of Linear Programming models. **[4]**

P.T.O.

- Q3) a)** Explain the different applications of Assignment model. [4]
- b)** A company has proposals to invest Rs.600 million in three possible projects. The returns depends upon the level of investment as given below. Determine the amount can be invested each project so that the total returns are maximum. [5]

Investment in Rs.Million	Returns From Projects		
	A	B	C
0	0	0	0
100	70	80	120
200	170	160	180
300	320	220	300
400	360	330	350
500	420	440	410
600	520	510	500

OR

- Q4) a)** A deparatmental head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the task differ in their intrinsic difficulty. His estimates of the times each man would take to perform each task is given below in the matrix. Find total time for each tasks. [5]

		Suboridnates			
		I	II	III	IV
Tasks	A	16	52	34	22
	B	26	56	8	52
	C	76	38	36	30
	D	38	52	48	20

- b)** Write a note on - Golden Section method. [4]

- Q5) a)** Use the method of lagrangian multiplier [8]

$$\text{Maximize } Z = 6x_1 + 8x_2 - x_1^2 - x_2^2$$

$$\text{Subject to } 4x_1 + 3x_2 = 16$$

$$3x_1 + 5x_2 = 15$$

$$x_1 + x_2 \geq 0$$

- b)** Use the Newton's Modified method. [8]

$$\text{maximize } Z = 6x_1 - x_1^2 + 6x_2 - 3x_2^2 \text{ with initial value as } x^0 = (2, 2).$$

OR

Q6) a) Maximize $Z = 6x_1 - 2x_1^2 + 4x_2 - 2x_2^2 - 2x_1x_2$ with initial value (1, 1) using steepest gradient method. **[8]**

b) Maximize $f = x^3 (12 - x)$ in the range of (0, 12) within accuracy of 10%. Carry five stages iterations. Use Fibonacci method. **[8]**

Q7) a) A sample of 100 arrivals of automobiles at a toll booth is found to be according the following distribution: **[8]**

Time of Arrivals in min.	0.5	1.0	1.5	2	2.5	3	3.5	4	4.5	5
Frequency	2	6	10	24	20	15	10	7	4	2

A study of service time reveal the following distribution:

Service Time in min.	0.5	1	1.5	2	2.5
Frequency	13	22	37	20	8

Estimate the average waiting time, the percentage waiting time of vehicle, average idle time and percentage idle time of the server for 10 arrivals. Use following random No.

Arrival	16	77	23	02	77	28	06	24	25	93
ServiceTime	56	65	05	61	86	90	92	10	79	80

b) Four jobs 1, 2, 3 & 4 are to be processed on each of the five machines A, B, C, D, E in order ABCDE. Find total Minimum elapsed time if no passing of job is permitted. **[8]**

Machine job	A	B	C	D	E
1	7	5	2	3	9
2	6	6	4	5	10
3	5	4	5	6	8
4	8	3	3	2	6

OR

- Q8)** a) Define Queuing model. Write applications of Queuing models. Explain in detail Queuing model. **[8]**
- b) Reduce the following game by dominance and Find the game value. **[8]**

		Player - B			
		I	II	III	IV
Player- A	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8



Total No. of Questions : 10]

SEAT No. :

P9424

[6182]-710

[Total No. of Pages : 2

M.E. (Civil-Construction and Management)
ENVIRONMENT & ENERGY FOR SUSTAINABLE
CONSTRUCTION
(2017 Pattern) (Semester - III) (601033) (Credit System)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) Draw neat sketch wherever required.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data if necessary and clearly state.*
- 5) Use of cell phone is prohibited in examination hall.*
- 6) Use of electronic pocket calculator is allowed.*

- Q1) a)** Describe in detail social and economic impacts of expressways. **[5]**
- b) Explain in brief, why 3E's Environmental Economics, Ethics & Ecology of sustainable development concept is required to consider while developing mega projects? **[5]**

OR

- Q2) a)** Explain in brief basic steps in performing the socio economic assessment of Pune Metro Rail Project. **[5]**
- b) What are the Economic Environment considerations of multipurpose dam? Explain in detail. **[5]**

OR

- Q3) a)** Explain in detail provisions of Energy Conservation Act 2001 and its revisions. **[5]**
- b) Explain in brief revisions and present status of implementation of Electricity Act 2003. **[5]**
- Q4) a)** Explain in detail, how various laws and protocols of UN are relevant to curb a carbon change. **[5]**
- b) What is carbon credits and its trading. Explain its Benefits to developing countries. **[5]**

P.T.O.

Q5) a) Describe in detail role of government to promote energy conservation of projects. [5]

b) Explain how energy can be saved in lighting and ventilation? Explain in detail systems adopted for same now days. [5]

OR

Q6) a) Explain in detail efforts put by Government of India to cope up with climate change. [5]

b) Explain the procedures for CDM and its eligibility criteria. [5]

Q7) a) Explain in detail, how thermal comfort can be achieved in green building? [5]

b) What is ESCOs? Explain its role in development of energy efficient projects. [5]

OR

Q8) a) Explain how energy can be saved in air conditioning and ventilation system adopted now days? [5]

b) What is power factor? How it is improved? [5]

Q9) a) Explain energy efficient design of green buildings. Explain with suitable example. [5]

b) What is an active HVAC system used in energy efficient structures? State its significance. [5]

OR

Q10)a) Prepare detailed program for maintenance of energy. [5]

b) Explain in detail water heat recovery in steam plants and distribution systems. [5]



Total No. of Questions : 8]

SEAT No. :

P-9674

[Total No. of Pages : 2

[6182]-711

M.E. (Civil) (Construction & Management)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - III) (601034)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6. Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) List out Characteristics of a good research problem. [4]
b) Write objectives of literature survey and different sources of literature.[3]

OR

- Q2)** a) Explain formulation of research hypotheses. [3]
b) Write down need of literature survey. [4]

- Q3)** a) Explain the criteria of goodness of a measurement scale. [4]
b) Write notes on Validity and reliability. [4]

OR

- Q4)** a) What are qualitative methods of data collection. [4]
b) What are the basic concepts concerning testing of hypothesis? [4]

- Q5)** a) Write a note on the usage of ANOVA in linear regression analysis. Why do we use t-test in linear regression analysis? [8]
b) Explain cluster analysis & multidimensional scaling with example. [8]

OR

P.T.O.

Q6) a) What is the main difference between descriptive and inferential statistics. **[8]**

b) How many discriminant functions would you obtain in a 4 group discriminant analysis and why? What do you mean by misclassification in discriminant analysis? **[8]**

Q7) a) Write short notes on the techniques of writing report and characteristics of a good research report. **[6]**

b) Write a short note on Documentation in the context of a research report. Explain the significance of a research report. **[7]**

c) List out key elements of effective presentation. **[6]**

OR

Q8) a) Describe in brief, the structure of report. What do you mean by plagiarism? **[6]**

b) Write Notes on **[7]**

i) Publishing paper

ii) Patent procedure

c) What are need of effective documentation for report writing. **[6]**



Total No. of Questions : 10]

SEAT No. :

P9425

[Total No. of Pages : 2

[6182]-713

**M.E. (Civil Engg.)(ENVIRONMENTAL ENGINEERING)
ENVIRONMENTAL LEGISLATION AND MANAGEMENT SYSTEM
(2017 Pattern) (Semester-I) (501061)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right side indicate full marks.*
- 3) Assume Suitable data if necessary*
- 4) Question 09 and question 10 are compulsory.*

Q1) Explain in detail: Polluter Pays Principle. **[5]**

OR

Q2) Explain in detail; Concept of absolute liability. **[5]**

Q3) Explain in detail; Water Act with following points: **[5]**

- a) Objectives,
- b) Responsibilities of occupiers

OR

Q4) What are the powers of State Pollution Control Boards and Central Pollution Control Board in controlling air pollution? Support your answer in detail with suitable example. **[5]**

Q5) Write a detailed note on: Coastal Zone Regulations. Support your answer with suitable example. **[5]**

OR

Q6) Explain in detail: Public Interest Litigation. Support your answer with suitable example. **[5]**

P.T.O.

Q7) Explain in detail, with suitable example, the role of Judiciary and NGO:
International concern for environment. **[5]**

OR

Q8) Explain international and national efforts at environmental protection in detail.
Support your answer with suitable example. **[5]**

Q9) a) Explain in detail, the important powers and functions of the MPC board
under both the water and air acts. **[8]**

b) Explain the Role of SPCB in implementation of various notifications issued
by Central Govt. under Environment (Protection) Act, 1986, in detail. **[7]**

Q10) a) Explain in detail: Responsibilities of Pollution Control Boards under
Hazardous Waste rules and that of occupier. **[8]**

b) Explain in detail procedure of public hearing for obtaining Environmental
Clearance. **[7]**



Total No. of Questions : 8]

SEAT No. :

P9426

[Total No. of Pages : 1

[6182]-714

M.E. (Civil-Environmental Engg.)

ENVIRONMENTAL CHEMISTRY AND MICROBIOLOGY

(2017 Pattern) (Semester-I) (501062)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Figures to the right indicates full marks.*
- 3) *Neat diagram must be drawn wherever necessary.*
- 4) *Assume Suitable data, if necessary and clearly state.*
- 5) *Use of an electronic pocket calculator is allowed.*

Q1) Enlist the classes of colloidal dispersion. Explain the significance of colloids in environmental engineering. **[10]**

Q2) State the classification of pollutants with suitable examples of each. Explain in detail, the physical processes of formation of pollutants in the atmosphere. **[10]**

Q3) Enlist and explain in detail, the various health hazards due to Lead poisoning. **[10]**

Q4) Explain various pollution problems caused by pesticides. Discuss the problems due to DDT in detail wrt agricultural run off. **[10]**

Q5) Explain in detail structure and function of prokaryotic cell organelles. **[10]**

Q6) Explain in detail: the Biological process of wastewater treatment. Support your answer with suitable examples. **[10]**

Q7) Explain in detail the different stages of wastewater treatment wrt microbial aspects. **[10]**

Q8) What are biochemical cycles? Explain Atmospheric Cycle in detail. **[10]**



Total No. of Questions : 12]

SEAT No. :

P9427

[Total No. of Pages : 2

[6182]-715

M.E. Civil (Environmental Engineering)

PHYSICO-CHEMICAL PROCESS FOR WATER AND WASTE

WATER TREATMENT

(2017 Pattern) (Semester-I) (501063)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12,
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) Enlist the reactors that are used for wastewater treatment. Discuss any one.[5]

OR

Q2) It is required to prepare 1250 kg of a solution composed of 12 wt.% ethanol and 88 wt.% water. Two solutions are available, the first contains 5 wt. % ethanol, and the second contains 25 wt.% ethanol. How much of each solution are mixed to prepare the desired solution? [5]

Q3) What is the importance of Aeration in Wastewater Treatment? [5]

OR

Q4) Design a tube settler for increasing the capacity of an existing settling tank provided for a water treatment plant (WTP) from 0.1314-0.3285 m³/s. The existing tank dimensions are 9.14 m × 39.62 m × 3.66m . Raw water has a turbidity of 20-25 JTU and temperature of up to 4.4°C. Effluent turbidity desired is 1-5 JTU. [5]

Q5) Discuss the Energy of interaction between two colloidal particles in the form of attraction by van der Waal's forces and repulsion by electrostatic forces.[5]

OR

Q6) Water receiving an average degree of pretreatment and responding in average fashion to coagulation is to be filtered at a rate of 122 L/min/m² through a layer of sand grains 0.1 cm in diameter. Find the requisite minimal depth of sand that will prevent breakthrough of turbidity at a terminal loss of head of 2.44 m. Take value of the breakthrough index as 1×10^{-3} . [5]

P.T.O.

Q7) Draw and discuss the generalised biological process reaction in the Activated Sludge Process. [5]

OR

Q8) Discuss in details about Tube-Type Clarifier Package Plants. [5]

Q9) a) Discuss the influence of temperature and hydrogen ion concentration on anaerobic digestion. [7]

b) Discuss the four interacting microbial reactions in anaerobic digestion. [7]

OR

Q10)a) What is Nutrient Balance in Anaerobic Digesters? [7]

b) Discuss the microbial growth is a function of substrate concentration. [7]

Q11)a) Enlist and discuss the Oxidizing agents useful in Water and Wastewater Treatment [8]

b) Explain in detail the thermodynamics of redox reactions. [8]

OR

Q12)a) What are the types of RO membrane modules? Discuss any one in detail. [8]

b) Discuss the following. [8]

i) Concentration Factor

ii) Osmotic Gradient

iii) Osmotic pressure

iv) Tonicity



Total No. of Questions : 12]

SEAT No. :

P9428

[Total No. of Pages : 2

[6182]-716

M.E. (Civil) (Environment Engineering)

ENVIRONMENTAL SANITATION

(2017 Pattern) (Semester - I)(501064)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Figures to the right indicate full marks.*
- 2) *Your answer will be valued as a whole.*
- 3) *Assume suitable data, if necessary.*

Q1) Discuss the potential applications of epidemiology. **[5]**

OR

Q2) Why discharges of intestines communicated the diseases? **[5]**

Q3) Discuss the following- **[5]**

- i) Prevention of fly breeding
- ii) Insecticides

OR

Q4) Discuss any two disinfectants used in sanitation practices. **[5]**

Q5) What is industrial sanitation? Discuss the factors required for study of industrial sanitation? **[5]**

OR

Q6) What is industrial poisons? What are its effects? Give suitable example. **[5]**

Q7) Why Population habits and environmental conditions affect the rural sanitation. **[5]**

OR

Q8) Enlist and discuss any two rural sanitation improvement schemes. **[5]**

Q9) a) Discuss the One-Pipe system and Two pipe system in detail. **[7]**

- b) Enlist the factors to be considered for design of Water supply in buildings? Discuss any two. **[7]**

OR

P.T.O.

Q10)a) Discuss the Break pressure tank system and Hydro-pneumatic system in detail. [7]

b) Discuss the steps involved in Design of Water pipes. [7]

Q11)a) Discuss with suitable sketches about Gully Traps and Intercepting Traps. [8]

b) Discuss the working of Flusing Cistern, Wash Basin, Lavatory Basin and Sink. [8]

OR

Q12)a) Give detail note on Principles governing Design of Building Drainage.[8]

b) Discuss the Drainage design of high rise buildings. [8]



Total No. of Questions : 12]

SEAT No. :

P9430

[6182]-719

[Total No. of Pages : 2

M.E. (Civil) (Environmental Engg.)
INDUSTRIAL WASTE WATER MANAGEMENT
(2017 Pattern) (Semester - II) (501067)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4. Q5 or Q6. Q7 or Q8, Q9 or Q.10, Q.11 or Q12.
- 2) Figures to the right indicate full marks.
- 3) Your answers will be valued as a whole.
- 4) Assume suitable data, if necessary.
- 5) Use of electronic pocket calculator and steam table is allowed.

Q1) 'Differentiate between the followings- **[5]**

- a) Equalization and Neutralization
- b) Flotation and Precipitation

OR

Q2) Give detail note on Prevention and Control of Industrial Pollution. **[5]**

Q3) What is Wet Air Oxidation? Discuss in detail. **[5]**

OR

Q4) What is the application of Chemical oxidation process? Enlist the commonly use oxidants. **[5]**

Q5) Discuss the treatment & disposal method of effluent from food processing industry. **[5]**

OR

Q6) Discuss the manufacturing process and sources of effluents for paper and pulp industry. **[5]**

Q7) Differentiate between manufacturing process of Pharma industry and the Petroleum industry. **[5]**

OR

Q8) Discuss the treatment & disposal method of effluent from Atomic Energy Plant. **[5]**

P.T.O.

- Q9) a)** What is CETP? Discuss the common problems encountered in its management. **[7]**
- b)** Give note on reuse the waste water. Why it is important for industries? **[7]**

OR

- Q10) a)** Discuss the importance of Zero effluent discharge system. **[7]**
- b)** What are the points you will consider to plan the CETP? Discuss in details. **[7]**

- Q11)a)** Give note on polymer coagulation? How it is differ from other coagulation? **[8]**
- b)** Discuss major components of wastewater treatment plant for Pharmaceutical industry? **[8]**

OR

- Q12)a)** Discuss the Freundlich adsorption isotherm and Langmuir adsorption isotherm with their applications. **[8]**
- b)** What are the design components required for wastewater treatment plant for Petroleum industry? **[8]**



Total No. of Questions : 8]

SEAT No. :

P9431

[6182]-720

[Total No. of Pages : 2

**M.E. (Civil-Environmental Engineering)
AIR POLLUTION AND CONTROL
(2017 Pattern) (Semester - II) (501068)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume suitable data.

Q1) a) State the meteorological parameters and name its measuring instruments. **[4]**

b) A stack in an urban area is emitting 75 g/s of NO. It has an effective stack height of 110m. The wind speed at stack height is 5.60 m/s. It is a clear summer day with the sun nearly overhead (stability class B).

$\sigma_y = 290$ m, $\sigma_z = 220$ m. Estimate the ground level concentration at:**[5]**

- i) 2.5 km downwind on the centreline and
- ii) 2.5 km downwind, 0.15km off the centreline

OR

Q2) a) Enumerate and discuss the basic components and importance of air quality modelling. **[4]**

b) An industry utilizes 0.25 ML (million litres) of oil fuel per month. It has also been estimated that for every 1 ML fuel oil burnt in the factory, per year, the quantities of various pollutants emitted are given as : PM = 2.8 t/yr, SO₂ = 55 t/yr, NO_x = 5t/yr, HC = 0.35 t/yr, CO = 0.4 t/yr. Calculate height of chimney. required to be provided for safe dispersion of pollutants assuming 300 working days in a year. **[5]**

Q3) a) Explain the measures to be taken to control gaseous air pollutants. **[4]**

b) Determine the migration velocity for an existing ESP having collection plate area of 110 m², gas flow rate 2.5 m³/s and collection efficiency 99.5%. **[5]**

OR

P.T.O.

- Q4) a)** Describe the factors responsible for selection of particulate control equipment. **[4]**
- b) Find the collecting plate area and number of plates to be used in a horizontal flow single stage electrostatic precipitator handling an average gas flow of $2.5\text{m}^3/\text{s}$ from a pulverized coal fired boiler. Consider the plate of 4 m wide and 5.2 m high. The required collection efficiency of ESP is 98%. Take the drift velocity as 12 cm/s. **[5]**

- Q5) a)** List the devices and methods used for air pollutant sampling. **[8]**
- b) Explain with a neat sketch working of high volume sampler. **[8]**

OR

- Q6) a)** Compare national ambient air quality standards, 2009 and WHO air quality guidelines 2021. **[8]**
- b) State the objectives of ambient air monitoring and stack emission monitoring. **[8]**
- Q7) a)** Explain sick building syndrome and its solution. **[8]**
- b) Enumerate the odorous materials with respect to following industries. **[8]**
- i) Petroleum
 - ii) Chemical
 - iii) Tanneries
 - iv) Distillery

OR

- Q8) a)** Explain the radon removal techniques. **[8]**
- b) Describe the use of plants for control of indoor air pollution. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9432

[6182]-721

[Total No. of Pages : 2

M.E. (Civil-Environmental Engineering)
SOLID WASTE & HAZARDOUS WASTE MANAGEMENT
(2017 Pattern) (Semester - II) (501069)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt any 5 questions.*
- 2) Each question carries equal marks.*
- 3) Figures to the right indicate full marks.*
- 4) Your answers will be valued as a whole.*
- 5) Assume suitable data, if necessary.*
- 6) Use of electronic pocket calculator and steam table is allowed.*

Q1) Discuss the strategic aspects involved and their role in municipal solid waste management? **[10]**

Q2) The municipal solid waste contains organic and inorganic material. During the analysis by the combustion method, 0.20 gm of an organic substance yielded **[10]**

- a) 0.157 gm of carbon dioxide and 0.11 gm of water
- b) 74.6 ml of nitrogen gas at N.T.P. Find the empirical formula of the substance.

Q3) Discuss in detail about Microbiological Composting. Give its merits and demerits. **[10]**

Q4) Discuss the different stages involved in anaerobic digestion in details. **[10]**

Q5) What is recovery of resources? Draw a typical flow sheet for recovery of resources, conversion products, and energy from solid wastes. **[10]**

Q6) Discuss in details about Categorization of biomedical waste. **[10]**

P.T.O.

Q7) Why stabilization of hazardous waste important? Give note on most commonly used stabilization processes with their limitations. **[10]**

Q8) Give the detail note on **[10]**

- a) Protective aids
- b) Ignitability
- c) Autoclave
- d) Secured Landfill



Total No. of Questions : 12]

SEAT No. :

P9433

[6182]-723

[Total No. of Pages : 2

**M.E. (Civil-Environmental Engineering)
ENVIRONMENTAL IMPACT ASSESSMENT
(2017 Pattern) (Semester - III) (601073)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Figures to the right indicate full marks.*
- 3) *Your answers will be valued as a whole.*
- 4) *Assume suitable data if necessary.*
- 5) *Use of electronic pocket calculator and steam table is allowed.*

Q1) Discuss the significance of Environmental Impact Assessment (EIA). **[5]**

OR

Q2) Discuss the followings. **[5]**

- a) Environmental Impact Studies (EIS).
- b) Public Participation.

Q3) What is meant by baseline data collection? **[5]**

OR

Q4) What are the important requirements under Form 1 and Form 1 A? **[5]**

Q5) How the basic information of air quality can be collected during assessment?[5]

OR

Q6) Give note on:- **[5]**

- a) Prediction of noise levels.
- b) Assessment of impact, mitigations.

P.T.O.

Q7) What kind of Basic information of water quality is required for assessment?[5]

OR

Q8) Discuss the background information of soil environment required for assessment. [5]

Q9) a) Discuss collection of basic information on cultural resources. [7]

b) Give note on the followings. [7]

i) Resettlement and rehabilitation.

ii) Socioeconomic environment.

OR

Q10)a) Differentiate with an example between Cultural system and Cultural resources. [7]

b) Discuss about Public participation in environmental decision making?[7]

Q11)a) As an auditor, how you will categorize the industries seeking for environmental clearance? [8]

b) Discuss in details about Environmental management plan. [8]

OR

Q12)a) What is the Procedure for obtaining Environmental clearance for construction projects? Discuss. [8]

b) Discuss the Procedure for public hearing. [8]



Total No. of Questions : 8]

SEAT No. :

P9429

[6182]-724

[Total No. of Pages : 2

M.E. (Civil) (Environmental Engg.)

RESEARCH METHODOLOGY

(2017 Credit Pattern) (Semester - III) (601074)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicates full marks.
- 4) Assume suitable data, if necessary and clearly state.
- 5) Use of cell phone is prohibited in the examination hall.
- 6) Use of electronic pocket calculator is allowed.

Q1) a) Define Research. Explain any two methods of Research. [4]

b) What is the significance of Shodhganga in literature survey. [5]

OR

Q2) a) What are generic elements to be included in Research Proposal? [5]

b) Define literature and Literature survey [4]

Q3) a) What is meaning & significance of attitude measurement and scaling? [4]

b) What do you mean by testing of hypothesis? [5]

OR

Q4) a) What are the methods of data collection? Explain. [5]

b) Explain the criteria of goodness of a measurement scale. [4]

P.T.O.

- Q5) a)** What is correlation and regression analysis? What are their methods? State its significance with their values. **[8]**
- b)** How many discriminant functions would you obtain in a 4 group discriminant analysis and why? What do you mean by misclassification in discriminant analysis? **[8]**

OR

- Q6) a)** Discuss the important characteristics of factor analysis and discriminant analysis. **[8]**
- b)** What is Multi-dimensional Scaling? Explain metric & non- metric approach of MDS & also explain significance of MDS. **[8]**
- Q7) a)** Explain various steps involved in writing a research paper also explain how to search journal for paper publication. **[8]**
- b)** Explain the different steps involved in obtaining patent. Explain the difference between product patent and process patent. **[8]**

OR

- Q8) a)** Describe in brief, the structure of report. What do you mean by plagiarism? **[8]**
- b)** Write a research proposal for a suitable research problem (Any problem related to Civil Engineering can be considered) to a funding agency with reference to the following terms: Title, Introduction, origin of the problem, expected outcome, literature review, Significance of the study in the context of current status, objectives, methodology, year wise plan. **[8]**



Total No. of Questions : 12]

SEAT No. :

P9434

[Total No. of Pages : 2

[6182]-726

M.E. (Civil Engg.) (Geotechnical Engineering)

ADVANCE SOIL MECHANICS

(2017 Pattern) (Semester-I) (501121)

Time : 3Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q. 12.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and mention it clearly.*
- 5) *Use of non-programmable calculator is allowed.*

Q1) Derive an equation for vertical pressure at a depth considering an infinitely long line load of intensity 'q' per unit length acting on the surface of a semi infinite elastic medium. **[4]**

OR

Q2) Describe the influence of anisotropy in soil bodies. **[4]**

Q3) What is stress path? Explain stress path for the cases **[4]**

- i) Effective stress path
- ii) Total stress path

OR

Q4) Define soil failure. Briefly explain the various types of theories of soil failures related to strength. **[4]**

Q5) With a suitable sketch, explain in detail the Rebhann's graphical method for estimation of active earth pressure on retaining wall. **[4]**

OR

Q6) Describe stability of retaining wall against overturning, sliding and bearing capacity failure. **[4]**

P.T.O.

Q7) What is “smear zone” in sand drain? Explain the case of free - strain consolidation with no smear with suitable sketches. [4]

OR

Q8) Describe the concept of sand drain with neat sketch. [4]

Q9) a) Explain an elastic-soil-spring method of vibration for analysis of foundations. Also, state the assumptions made in the theory. [6]

b) Describe the general design criterias for the machine foundations. [4]

c) Write a note on “Dynamic loads due to vibrating machines”. [6]

OR

Q10)a) Explain elastic half space method. [6]

b) Write a note on free vibration in machine foundations. [6]

c) How do you determine mass spring constant and damping factor for a vibrating system. [4]

Q11)a) Explain two dimensional Amplitude method for the design of foundations of rotary machines. [6]

b) Write a short note on:

Construction details of machine foundations. [6]

c) Why vibration Isolation is required? Describe the methods to control vibrations why it is difficult to control low frequency vibrations. [6]

OR

Q12)a) Explain “use of isolators method” in vibration isolation with neat sketches. [6]

b) What are the properties of good vibrating isolation material? [6]

c) Write a note on: [6]

i) Importance of vibration isolation

ii) Rotary type machines



Total No. of Questions : 8]

SEAT No. :

P9435

[Total No. of Pages : 2

[6182]-727

M.E. (Civil-Geotechnical Engg.)

GROUND IMPROVEMENT TECHNIQUES

(2017 Pattern) (Semester-I) (501122)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagram must be drawn wherever necessary.*
- 4) *Assume Suitable data if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

Q1) a) Explain clay-water relations **[4]**

b) Discuss the principle and behaviour of chemically stabilised soils. **[5]**

OR

Q2) a) Describe identification of clay minerals. **[5]**

b) Which are the different methods of soil stabilization? **[4]**

Q3) a) How is the reaction of soil and lime? **[4]**

b) Explain vibro-flotation with its principle and sketches **[5]**

OR

Q4) a) Explain ash and slag stabilization techniques **[5]**

b) How is vacuum consolidation carried out? **[4]**

Q5) a) Write short notes on : **[8]**

i) Sand drains

ii) Granular trench stabilisation

b) Describe the procedure to determine the bearing capacity of lime group. **[8]**

OR

P.T.O.

- Q6)** a) Write notes on: [8]
i) Total and differential settlement
ii) Quality control methods for lime columns
b) What are lime columns and describe its applications? [8]

- Q7)** a) Explain compaction and jet grouting. [8]
b) How is grout injection measurement and monitoring carried out? [8]

OR

- Q8)** a) Explain the various grouting equipment with sketches. [8]
b) Describe soil grouting program. [8]



Total No. of Questions : 8]

SEAT No. :

P9436

[6182]-728

[Total No. of Pages :2

M.E. (Civil) (Geotechnical Engineering)
ROCK MECHANICS
(2017 Pattern) (Semester-I) (501123)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) State the field applications of rock mechanics. What is the significance of rock mechanics in tunnel Construction? [4]

b) Describe the classification of rock masses for engineering purpose. [5]

OR

Q2) a) List out the various types of rock blasting and explain the cushion blasting in detail. [4]

b) What are the objectives and limitations of Rock Grouting". [5]

Q3) a) Discuss the term "Over Coning". [4]

b) Explain hydraulic fracturing method of estimate the stress in rock. [5]

OR

Q4) a) Discuss the Griffth's crack theory. [5]

b) Describe the modes of rock failure. [4]

Q5) a) Explain structural discontinuities in rock masses. [8]

b) What are the rupture criteria for brittle rocks? [8]

OR

Q6) a) Discuss analysis of general slip surface. [8]

b) Explain water pressure and resistance of rock structures. [8]

P.T.O.

- Q7)** a) Describe the failure mechanisms of rock foundations. [8]
b) Explain the methods and techniques used for assessing the stress distribution and deflection in rock under a footing. [8]

OR

- Q8)** a) Explain the kinematic analysis of rock slopes. [8]
b) Explain the rock parameters required for the analysis of underground structures. [8]



Total No. of Questions : 8]

SEAT No. :

P9437

[Total No. of Pages : 2

[6182]-729

**M.E.(Civil) (Geotechnical Engg.)
GEO-ENVIRONMENTAL ENGINEERING
(2017 Pattern) (Semester - I) (501125)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q. 1 or Q. 2, Q. 3 or Q. 4, Q.5 or Q.6 and Q. 7 or Q. 8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

Q1) a) State the chemical and biological characteristics of solid waste. **[4]**

b) Explain the effects of pollutants on soil properties. **[5]**

OR

Q2) a) Discuss the “soil waste interaction”. **[4]**

b) Discuss the brief “Laboratory and field evolution of permeability”. **[5]**

Q3) a) State aspects are being considered for planning and design in waste disposal in land fill? **[5]**

b) Explain “Ash pond and Trailing ponds”. **[4]**

OR

Q4) a) Describe different types of barrier materials used in land fill. **[4]**

b) State the various disposal methods and explain any one. **[5]**

Q5) a) Describe in detail, “Geotechnical reuse of solid waste”. **[8]**

b) Explain in detail. “Demolition of waste”. **[8]**

OR

P.T.O.

- Q6) a)** State and explain the uses of waste in geotechnical engineering. **[8]**
b) Explain the different engineering properties of solid waste. **[8]**

- Q7) a)** Discuss the application of Geo-synthetics in waste disposal design. **[8]**
b) Describe the land fill construction. **[8]**

OR

- Q8) a)** Explain in detail single and double lined landfill. **[8]**
b) State and discuss different types and application of clay liners. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9438

[6182]-731

[Total No. of Pages : 2

M.E. (Civil-Geotechnical Engg.)
ADVANCE FOUNDATION ENGINEERING
(2017 Credit Pattern) (Semester - II) (501127)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2 , Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figure to the right indicates full marks.
- 4) Assume suitable data, if necessary and clearly state.
- 5) Use of cell phone is prohibited in the examination hall.
- 6) Use of electronic pocket calculator is allowed.

- Q1) a)** Explain in detail standard penetration test. **[5]**
- b)** A thin walled sampler was used in soft clay having **[4]**
- i) Inside and outside diameter 80mm & 72mm,
 - ii) Length of sampler is 550 mm,
 - iii) Sample recovered is 500 mm. Determine the recovery ratio and area ratio.

OR

- Q2) a)** Calculate the ultimate bearing capacity per unit area of a strip footing 1 m wide and 1 m [6] deep when unit weight of soil 18 kN/m³, cohesion is 20 kN/m², $\phi = 20^\circ$, $N_c = 17.5$, $N_q = 7.5$, $N_\gamma = 5$. **[5]**
- b)** State and explain types of samplers. **[4]**

- Q3) a)** Discuss 'Modulus of subgrade reaction'. **[4]**
- b)** Explain in short 'types of anchors'. **[5]**

OR

- Q4) a)** Determine the net ultimate bearing capacity of a mat foundation of 12 m $\times$ 10m on a saturated clay with $C_u = 95$ kN/m², $\phi = 0^\circ$, $D_f = 2$ m. **[5]**
- b)** Discuss 'design of anchors'. **[4]**

P.T.O.

Q5) a) A group of 9 piles arranged in square pattern with diameter and length of each pile as 30 cm and 9 m respectively, is used as a foundation in soft clay deposit. Tacking UCS of clay as 90 kN/ m² and pile spacing as 90 cm c/c. Find load capacity of group. Assume bearing capacity factor $N_c = 9$, Adhesion factor = 0.75, FOS = 2.5 [8]

b) Explain Negative skin friction. Discuss IRC 78 considerations for Piles socketing and termination depth. [8]

OR

Q6) a) Discuss 'Design of pile cap'. [8]

b) A group of nine piles arranged in a square pattern is to be proportioned in a deposit of soft clay to carry an incoming load of 550 kN. Calculate the settlement of the pile group by equivalent raft approach if the piles are square with side 300 mm and have a centre to centre spacing of 500 mm. The thickness of the stratum is 8 m and the piles are embedded to a length of 5 m in the clay. The properties of the clay are given as: $\gamma_{sat} = 19 \text{ kN/m}^3$, LL= 45% and $e = 1.5$. Ground water table is at the ground surface. [8]

Q7) a) Enlist the types of coffer dams and explain diaphragm type coffer dam. [8]

b) Explain well foundation with its different types. [8]

OR

Q8) a) Discuss different forces acting on well foundation. [8]

b) Explain Terzaghi's method to lateral stability of well foundation. [8]



Total No. of Questions : 8]

SEAT No. :

P9439

[6182]-732

[Total No. of Pages : 2

M.E. (Civil-Geotechnical Engineering)
CONSTRUCTION METHODS IN GEOTECHNICAL
(2017 Pattern) (Semester - II) (501128)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagram must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data if necessary and clearly state.*
- 5) Use of cell phone is prohibited in the examination hall.*
- 6) Use of electronic pocket calculator is allowed.*

Q1) a) Explain the construction of precast and cast-in-situ piles. **[4]**

b) Discuss pre-dredging and how it will effect on selection of cofferdams. **[5]**

OR

Q2) a) Explain pile raft foundation and its application. **[4]**

b) Explain the bracing systems for circular and rectangular cofferdams with neat sketches. **[5]**

OR

Q3) a) Discuss measures to resolve problems with tremic box. **[4]**

b) Describe the procedure of construction of pneumatic caisson with neat sketches. **[5]**

OR

Q4) a) Discuss, how Cofferdam will destroyed by surge. **[5]**

b) Describe the considerations for “Site and foundation preparation”for Caissons. **[4]**

P.T.O.

- Q5)** a) Explain in detail evaluation and planning required for rock excavation. **[8]**
b) Explain the following **[8]**
i) Trim blasting
ii) Over break control

OR

- Q6)** a) Describe the selection criteria of explosives for rock excavations. **[8]**
b) Discuss the process of loading and hauling equipment for rock excavation. **[8]**
- Q7)** a) Explain tunneling by TBM and its types. **[8]**
b) Explain how is tunneling in moderately firm rocks and ground carried out? **[8]**

OR

- Q8)** a) Describe sinking caisson method with its merits and demerits. **[8]**
b) Distinguish between Single stage and Multistage, stage method of tunneling. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9440

[6182]-733

[Total No. of Pages : 2

M.E. (Civil) (Geotechnical Engineering)
STABILITY OF SLOPES AND EARTH DAMS
(2017 Credit Pattern) (Semester - II) (501129)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Assume suitable data, if necessary.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

- Q1) a)** How is the factor of safety for finite and infinite slope determined? **[4]**
b) Compare the Bishop's and Swedish slip circle method for slope stability. **[5]**

OR

- Q2) a)** Calculate the factor of safety with respect to cohesion of a clay slope laid 1 in 2 to height 10m if the angle of internal friction $\phi = 10^\circ$, $C = 2.5 \text{ kN/m}^2$ and $\gamma = 19 \text{ kN/m}^3$. What will be the critical height of the slope in this soil? Assume $S_n = 0.064$. **[5]**
b) Under which conditions is long term stability of slope determined? **[4]**

- Q3)** Draw a neat sketch of earthen dam indicating all the components along with importance and design criteria for each component. **[9]**

OR

- Q4) a)** How is the stability of earth dam slopes ensured under different conditions? **[5]**
b) Explain the methods to reduce the construction pore pressure in earth dams. **[4]**

- Q5) a)** Explain the measures to control seepage through the cross section of dam and how can the quantity of seepage estimated? **[8]**
b) At the foot of a dam, the foundation soil has a void ratio of 0.72. The specific gravity of the soil solids is 2.65. To ensure safety against piping, the upward gradient must not exceed 30% of the critical gradient at which quick sand condition occurs. Estimate the maximum permissible upward gradient. **[8]**

OR

P.T.O.

- Q6)** a) What are the measures for preventing the phreatic line from intersecting the downstream slope? Explain with sketches. [8]
- b) Derive the two dimensional Laplace equation for flow through soil and discuss the characteristics of flownet. [8]

- Q7)** a) Explain the design considerations for earthen dam during sudden drawdown condition. [8]
- b) Specify the component of forces used in design of earthen dam. Explain significance of earthquake force. [8]

OR

- Q8)** a) Discuss the codal provisions for the stability of earthen dam under different conditions. [8]
- b) What measures are required to be taken to make the foundation strata suitable for safe earthen dam? [8]



Total No. of Questions : 8]

SEAT No. :

P9441

[6182]-735

[Total No. of Pages : 2

M.E. (Civil-Geotechnical Engineering)
GEOTECHNICAL EARTHQUAKE ENGINEERING
(2017 Pattern) (Semester - III) (601133)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

Q1) a) What is soil-structure interaction? Why is it important in geotechnical engineering? **[6]**

b) Explain ground motion parameters and its estimation. **[7]**

OR

Q2) a) Distinguish between 2-d and 3-d ground response analysis methods. **[6]**

b) Explain the term “spatial variability of ground motions”. **[7]**

Q3) a) Describe Seismic slope stability analysis. **[6]**

b) Discuss the different design parameters used for recording ground motion. **[6]**

OR

Q4) a) Discuss the “Effects of liquefaction”. **[6]**

b) Discuss the ‘Effect of local site conditions’ on ground motion. **[6]**

Q5) a) Explain Goodman and Seed approach. **[6]**

b) Explain the concept of responses analysis for embankment dams during earthquake. **[6]**

OR

P.T.O.

- Q6)** a) Describe the different types of damages occurs in earthen dam during earthquake. [6]
b) Explain Evaluation of liquefaction hazards. [6]

- Q7)** a) Discuss the method for mitigation of earthquake effects. [7]
b) Explain the term “Grouting of soil. [6]

OR

- Q8)** a) Explain the Bearing capacity of foundations and its importance during Seismic design of Retaining wall. [6]
b) Explain Seismic pressure on retaining wall with neat sketch. [7]



Total No. of Questions : 8]

SEAT No. :

P9442

[6182]-736

[Total No. of Pages : 2

M.E. (Civil)(Geotechnical Engineering)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - III) (601134)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figure to the right indicates full marks.*
- 4) Assume suitable data, if necessary and mention it clearly.*

Q1) a) Explain the technique in defining a research problem. **[5]**

b) Explain secondary sources of data and explain its limitations. **[4]**

OR

Q2) a) Explain the different online tools available for literature survey. **[5]**

b) Explain the research problem in a flowchart. **[4]**

Q3) a) Short note on Rating scales, Sampling techniques. Give examples for both. **[5]**

b) What is a hypothesis? Explain basic concepts concerning testing of hypothesis. **[4]**

OR

Q4) a) Discuss the condition in which analysis of variance techniques is used in research with a suitable example. **[5]**

b) What are the merits and demerits of collection of data through questionnaires? **[4]**

P.T.O.

Q5) a) Explain “Descriptive Statistics : Measures of Central Tendency, Measures of dispersion, Measures of Skewness and Measures of Relationship”. **[8]**

b) Explain the centroid method of factor Analysis. **[8]**

OR

Q6) a) Write a note on discriminant and cluster analysis. **[8]**

b) Explain multidimensional measurement and factor analysis. **[8]**

Q7) a) Explain the different steps involved in obtaining patent. Explain the difference between product patent and process patent. **[8]**

b) Write a note on elements of effective presentation and impact of presentation. **[8]**

OR

Q8) a) Discuss the important points to be considered during presenting a research idea. Also write a note on how to do patenting of a research idea. **[8]**

b) Explain in detail Mechanics of Writing Research Report with suitable example. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9443

[6182]-738

[Total No. of Pages :2

M.E. (Civil) (Hydraulics Engg.)

FLUID MECHANICS

(2017 Pattern) (Semester-I) (501041)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicates full marks.*
- 3) *Assume suitable data if necessary.*

- Q1)** a) Explain the term acceleration in fluid flow and indicate its components. Under what circumstances is the flow non-accelarating. [5]
- b) A stream function for two dimensional flow is given by $\psi = 3x^2 - y^2$. Determine equation of stream line. Find the components of velocity at (2,2). [5]

OR

- Q2)** a) Distinguish between source and sink. [4]
- b) Derive from first principle, the condition for irrotational flow. Prove that, for potential flow, both the stream function and velocity potential function, satisfy the Laplace equation. [6]
- Q3)** Derive an expression for the velocity distribution for viscous flow through a circular pipe. Also sketch the velocity distribution and shear stress distribution across a section of pipe. [8]

OR

- Q4)** Water is flowing between two large parallel plates which are 2mm apart. Determine
- a) Maximum velocity
 - b) The pressure drop per unit length
 - c) The shear stress at walls of the plate, if the average velocity is 0.4m/s. Take viscosity of water as 0.01 poise. [8]

P.T.O.

- Q5)** a) Explain Nikuradse's experiments on artificially roughened pipes and the results following these experiments. [8]
b) Describe with a neat sketch the construction of Moody's diagram. Explain its use. [8]

OR

- Q6)** a) i) What is eddy viscosity. How does it differ from the dynamic viscosity. [4]
ii) Find the relation between average velocity and maximum velocity in a circular pipe for a turbulent flow. [4]
b) A pipe of 150mm diameter and length 100m carries water at 20°C. If the pressure loss in pipe line is 26.613 kN/m², find whether the pipe will act as a smooth or rough pipe. [8]

Also determine:

- i) Maximum Velocity
ii) Average velocity
iii) Discharge. Take $k=0.01\text{mm}$
- Q7)** a) Obtain an expression for velocity of the sound wave in a compressible fluid in terms of change of pressure and change of density. [8]
b) i) State Bernoulli's theorem for compressible flow. Also state the continuity equation for one dimensional compressible flow. [4]
ii) Define Mach number. What is its significance in compressible fluid flows. [4]

OR

- Q8)** a) Derive the continuity equation for one dimensional compressible flow in differential form [8]
b) i) Find the sonic velocity for the crude oil of sp. gravity, 0.8 and bulk modulus 153036 N/cm². [4]
ii) Define the terms: Mach angle and Mach cone. [4]



Total No. of Questions : 8]

SEAT No. :

P9444

[6182]-739

[Total No. of Pages : 2

M.E. (Civil) (Hydraulics)
IRRIGATION & DRAINAGE ENGINEERING
(2017 Credit Pattern) (Semester - I) (501042)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

- Q1)** a) What are the factors that affect the choice of irrigation methods? [6]
b) What are the purposes of multi-crop irrigation scheduling? [6]

OR

- Q2)** a) Classify & the irrigation methods. [6]
b) Briefly explain conjunctive use of surface and groundwater. [6]

- Q3)** a) In case of lift irrigation system explain the concept of cost economics.[6]
b) What do you mean by sprinkler irrigation? Describe the components of sprinkler irrigation systems. [8]

OR

- Q4)** a) Explain operations and maintenance of drip assembly. [6]
b) Explain sprinkler system considering following points [8]
i) Criteria for system layout
ii) Selection of sprinkler and its spacing
iii) Discharge capacity of the pump
iv) Hydraulic design of sprinkler head
v) Main and
vi) Lateral pipe size

- Q5)** a) Explain the classification of salts affecting the soils and their characteristics. [6]
b) What are the similarities and differences between water erosion and wind erosion? [6]

OR

P.T.O.

- Q6)** a) Explain leaching and salinity control in soil. [6]
b) Explain how the soil conservation achieved by following methods [6]
i) Afforestation
ii) Checking Overgrazing
iii) Constructing Dams
iv) Changing Agricultural Practices.

- Q7)** a) What do you mean by water logging? What are its preventive & protective measures? [6]
b) Explain the concept of drainage by wells. [6]

OR

- Q8)** a) Explain the transient designs of surface drainage systems. [6]
b) What do you mean by soil erosion? Explain any two methods to avoid soil erosion. [6]



Total No. of Questions : 8]

SEAT No. :

P9445

[6182]-743

[Total No. of Pages : 4

M.E. (Civil-Hydraulics)

OPEN CHANNEL HYDRAULICS

(2017 Pattern) (Semester - II) (501047)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) A rectangular channel 3 m wide carries 3 m³/s of water at a depth of 1.0 m. If the width is to be reduced to 2.0 m and bed raised by 10 cm what would be the depth of flow in the contracted section? Neglect the loss of energy in transition. What maximum rise in the bed level of the contracted section is possible without affecting the depth of flow upstream of the channel? **[5]**

b) An overflow spillway has its crest at elevation 125.40 m and horizontal apron at an elevation of 95.00 m on the downstream side. Find the tailwater elevation required to form a hydraulic jump when the elevation of the energy line is 127.90 m. The C_d for the flow can be assumed as 0.735. The energy loss for the flow over the spillway face can be neglected. **[4]**

OR

Q2) a) A rectangular channel is 2.5 m wide and conveys a discharge of 2.75 m³/s at a depth of 0.9 m. A contraction of width is proposed at section in this canal. Calculate the water surface elevations in the contracted section as well as in an upstream 2.5 m wide section when the width of proposed contraction is 2.0 m. Neglect energy losses in the transition. **[4]**

b) A rectangular channel carrying a supercritical stream is to be provided with a hydraulic jump type of energy dissipator. If it is desired to have an energy loss of 5 m in the jump when the inlet Froude number is 8.5, determine the sequent depths. **[5]**

P.T.O.

- Q3) a)** Explain characteristics of M1, M2 and M3 profiles. **[5]**
- b)** A rectangular channel 3.0 m wide can carry a discharge of 3.6 m³/s at a normal depth of 1.2 m. Design a side weir so that it will pass all the flow in the canal when the discharge is 2.00 m³/s and will divert 0.6 m³/s when the canal discharge is 3.6 m³/s. **[4]**

OR

- Q4) a)** A river 50 m wide has an average bed slope of 1 in 10,000. Compute the approximate length of the backwater curve produced by a weir which raises the water surface immediately behind it by 3.0 m. What will be the rise in water level at a point 9.0 km upstream of the weir? The discharge in the river can be taken as 62.5 m³/s. Assume Chezy's constant $C = 45$. Use direct step method and take 2 steps only. **[5]**
- b)** Derive differential equation for spatially varied flow. **[4]**
- Q5) a)** A rectangular channel carries a flow with a velocity of 0.65 m/s and depth of 1.4 m. If the discharge is abruptly increased threefold by a sudden lifting of a gate on the upstream, estimate the velocity and the height of the resulting surge. **[8]**
- b)** Show that the continuity equation for a GVUF in a non-prismatic channel with no lateral in flow is. **[8]**

$$A \frac{\partial V}{\partial x} + VT \frac{\partial y}{\partial x} + \epsilon yV \frac{\partial T}{\partial x} + T \frac{\partial y}{\partial t} = 0$$

where $\epsilon =$ a coefficient which depends on the nature of the non-prismaticity of the channel with $\epsilon = 0.5$ and 1.0 for triangular and rectangular channels.

OR

- Q6) a)** A 2.0 m wide rectangular channel, 2 km long carries a steady flow of 4.6 m³/s at a depth of 1.15 m. The sides of the channel are 2.0 m high. If the flow is suddenly stopped by the closure of a gate at the downstream end, will the water spill over the sides of the channel? If there is no spillage, what minimum time interval must elapse before the arrival of the surge at the upstream end? **[8]**
- b)** Show that by using the Chezy's formula with $C = \text{constant}$, the ration $(V_w)_m / V_n$ of a monoclinal wave is 1.50 and 1.25 for wide rectangular and triangular channels respectively. **[8]**

- Q7) a)** The storage, elevation and outflow data of a reservoir are given below: **[10]**

Elevation (m)	Storage 10^6 m^3	Outflow discharge (m^3/s)
299.50	4.8	0
300.20	5.5	0
300.70	6.0	15
301.20	6.6	40
301.70	7.2	75
302.20	7.9	115
302.70	8.8	160

The spillway crest is at elevation 300.20 m. the following flood flow is expected into the reservoir.

Time (h)	0	3	6	9	12	15	18	21	24	27
Discharge (m^3/s)	10	20	52	60	53	43	32	22	16	10

If the reservoir surface is at elevation 300.00 m at the commencement of the inflow, route the flood to obtain

- i) The outflow hydrograph and
 - ii) The reservoir elevation vs time curve.
- b) Describe the Muskingum method of routing an inflow hydrograph through a channel reach. Assume the values of the coefficients K and x for the reach are known. **[6]**

OR

Q8) a) Define the problem of level pool routing. Describe a commonly used method of reservoir routing. **[7]**

b) The storage in the reach of a stream has been studied. The values of x and K in Muskingum equation haven been identified as 0.28 and 1.6 days. If the inflow hydrograph to the reach is as given below, compute the outflow hydrograph. Assume the outflow from the reach at $t = 0$ as $3.5 \text{ m}^3/\text{s}$. **[9]**

Time (h)	0	6	12	18	24	30
Inflow (m^3/s)	35	55	92	130	160	140



Total No. of Questions : 8]

SEAT No. :

P9446

[6182]-744

[Total No. of Pages : 2

M.E. (Civil) (Hydraulics)

SEDIMENT TRANSPORT & RIVER MECHANICS

(2017 Pattern) (Semester - II) (501048)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five Questions from eight Questions.*
- 2) *All questions carry equal marks.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of scientific calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) a) Differentiate the variation of form resistance and grain resistance in different flow regimes. **[5]**

b) Discuss in brief Origin and formation of sediments. **[5]**

Q2) a) Write short note on Bed Formations. **[5]**

b) What are the various approaches of establishment of incipient motion? **[5]**

Q3) a) What do you mean by suspended load, total load and wash load? **[3]**

b) Using Meyer-Peter and Muller method, estimate the bed load transport of sediment in an alluvial channel, 160 m wide and 4 m deep carrying a discharge of 300 m³/s. The channel bed slope is 1 in 3000 and the mean size of the sediment is 0.3 mm. **[7]**

Q4) a) Compare Kennedy method and Lacey method for the design of an irrigation channel Also Design an irrigation channel in alluvial soil according to Lacey's method for following data **[5]**

i) Full supply discharge : 40 m³/sec

ii) Lacey's silt factor - 0.9

iii) Channel side slope 0.5H : 1V

b) Explain the Tractive force approach method of channel design. **[5]**

P.T.O.

- Q5)** a) Explain [5]
i) Agradation
ii) Degradation
b) What is river gauging? Explain the methods of river gauging in detail.[5]
- Q6)** a) What are the objectives of river training and bank protection works? [5]
b) Write a short note on “Silting of reservoir” [5]
- Q7)** a) What are the preventive & protective measures of sediment control. [5]
b) Discuss the river training works in consideration with navigation and sediment control. [5]
- Q8)** a) How the Alluvial river models development will help in river training works? [5]
b) Explain Sediment transport through pipes, considering the following points [5]
i) Heterogeneous suspension
ii) Sliding bed Regime



Total No. of Questions : 8]

SEAT No. :

P9447

[6182]-747

[Total No. of Pages : 3

M.E. (Civil - Hydraulics)

OPTIMIZATION TECHNIQUES

(2017 Credit Pattern) (Semester - III) (601051)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary and clearly state.
- 5) Use of cell phone is prohibited in the examination hall.
- 6) Use of Non programmable electronic pocket calculator is allowed.

- Q1) a)** Food X contains 6 units of vitamin A per gram and 7 units of vitamin B per gram and costs 12 paise per gram. Food Y contains 8 units of vitamin A per gram and 12 units of vitamin B per gram and costs 20 paise per gram. The daily minimum requirement of vitamin A and vitamin B is 100 units and 120 units respectively. Find the minimum cost of product mix by the Simplex method (Big - M method). **[12]**
- b) Explain in brief “sensitivity analysis”. **[2]**

OR

- Q2) a)** An air-line that operates 7 days a week has the time table shown below (Table 2 a). Crew must have minimum layover of 5 hours between flights. Obtain the pairing of flights that minimizes layover time away from home assuming that the crew can be based at either of the two cities. The crew will be based at the city results in smaller layover. **[14]**

Table 2 a

Delhi-Jaipur			Jaipur-Delhi		
Flight	Depart	Arrive	Flight	Depart	Arrive
1	7:00 am	80:00 am	101	8:00 am	9:15 am
2	8:00 am	9:00 am	102	8:30 am	9:45 am
3	1:30 pm	2:30 pm	103	12 Noon	1:15 pm
4	6:30 pm	7:30 pm	104	5:30 pm	6:45 pm

OR

P.T.O.

- Q3) a)** Solve the following non-linear programming problem with use of Lagrangian multiplier technique: **[10]**

$$\text{Maximize } Z = 4x_1 - x_1^2 + 8x_2 - x_2^2$$

$$\text{Subject to } x_1 + x_2 = 2$$

$$x_1, x_2 \geq 0$$

- b) Differentiate between Golden section method and Fibonacci method. **[2]**

OR

- Q4) a)** Solve the following non-linear programming problem using Lagrangian multiplier technique: **[10]**

$$\text{Optimize } Z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2,$$

$$\text{Subject to } x_1 + x_2 + x_3 = 15,$$

$$2x_1 - x_2 + 2x_3 = 20,$$

$$x_1, x_2, x_3 \geq 0.$$

- b) Explain in brief the limitations of Fibonacci search method. **[2]**

- Q5) a)** A firm has divided its marketing area into three zones. The amount of sales depends upon the number of salesmen in each zone. The firm has been collecting the data regarding sales and salesmen in each area over a number of past years. The information is summarized in the Table 5 a. For the next year firm has only 9 salesmen and the problem is to allocate these salesmen to three different zones so that the total sales are maximum. **[10]**

Table 5 a (Profit in Thousands of rupees)

No of salesmen	Zone I	Zone II	Zone III
0	30	35	42
1	45	45	54
2	60	52	60
3	70	64	70
4	79	72	82
5	90	82	95
6	98	93	102
7	105	98	110
8	100	100	110
9	90	100	110

- b) Explain in brief the “Recursive equations”. **[2]**

OR

- Q6) a)** Explain in detail the distinguishing characteristics of Dynamic Programming. [6]
- b) Explain the procedure adopted in the analysis of Dynamic programming problems. [6]

- Q7) a)** A bakery keeps stock of a popular brand of cake. Daily demand based on past experience is given below: [10]

Daily Demand	0	15	25	35	45	50
Probability	0.01	0.15	0.20	0.50	0.12	0.02

Consider the following sequence of random numbers;

48, 78, 09, 51, 56, 77, 15, 14, 68 and 09.

- Using the sequence simulate the demand for the next 10 days.
 - Find the stock situation if the owner of the bakery decides to make 35 cakes every day
 - Also estimate the daily average demand for the cakes on the basis of the simulated data.
- b) Explain in brief “Queuing theory”. [2]

OR

- Q8) a)** Reduce the following game (Table 8 a) by dominance and find the game value. [10]

Player A	Player B				
		I	II	III	IV
	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8

- b) Explain in brief “Sequencing model”. [2]



Total No. of Questions : 8]

SEAT No. :

P9448

[6182]-748

[Total No. of Pages : 2

M.E. (Civil - Hydraulics)

RESEARCH METHODOLOGY

(2017 Credit Pattern) (Semester - III) (601052)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6 and Q. 7 or Q. 8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of Non programmable electronic pocket calculator is allowed.*

Q1) a) Explain in detail the following terms: **[6]**

- i) Characteristics of a good research problem and
- ii) Errors in selecting a research problem.

b) Discuss in details “Developing a research proposal with a suitable example”. **[6]**

OR

Q2) a) Explain in details various types of research. **[6]**

b) Explain in brief the process of research. **[6]**

Q3) a) Explain “elements and objectives of literature survey”. **[6]**

b) Explain styles of literature survey, and strategies of literature survey. **[6]**

OR

Q4) a) Explain : **[6]**

- i) Need of literature survey and
- ii) Sources of literature.

b) Explain in detail “Strategies of literature survey”. **[6]**

Q5) a) Enlist the various qualitative methods of data collection and explain any one in detail. **[6]**

b) Explain the various points considered by the researcher while determining the size of sample. **[6]**

OR

P.T.O.

- Q6) a)** Explain the following criteria for good measurement: **[6]**
- i) Reliability
 - ii) Validity
 - iii) Sensitivity
- b)** Explain the following terms with respect to classification of scale: **[6]**
- i) Single item versus multiple item scale and
 - ii) Comparative versus non-comparative scales.
- Q7) a)** Explain in detail the procedure for hypothesis testing. **[8]**
- b)** Explain the following related to testing of hypothesis: **[6]**
- i) Null hypothesis and
 - ii) Alternative hypothesis.

OR

- Q8) a)** Watching television reduces the amount of physical exercise causing weight gain. A sample of fifteen 10-year old children was taken. The number of kilograms each child was overweight was recorded (a negative number indicates the child is underweight). Additionally, the number of hours of television viewing was also recorded. These data are listed here.

TV (hours)	42	34	25	35	37	38	31	33	19	29	38	28	29	36	18
Over Weight (Kg)	18	6	0	-1	13	14	7	7	-9	8	8	5	3	14	-7

Fit the regression line and describe what the coefficients tell you about the relationship between the two variables. **[8]**

- b)** What do you mean by Plagiarism? Explain its significance related with writing and submitting a research paper for possible publication in a journal. **[6]**



Total No. of Questions : 8]

SEAT No. :

P9449

[6182]-750

[Total No. of Pages : 2

M.E. (Civil-Structures)

**THEORY OF ELASTICITY AND PLASTICITY
(2017 Pattern) (Semester - I) (501001)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary and clearly state.

Q1) a) Find the maximum shearing stress under a condition of plane stress for an element with state of stress at a point. **[5]**

b) Determine Airy polynomial for a cantilever having uniformly distributed load. **[4]**

OR

Q2) a) Write short notes on **[5]**

- i) Plane stress and Plane strain
- ii) Spherical coordinates

b) Derive the equilibrium equation for three dimensional problem considering body forces as F_x , F_y and F_z along X, Y and Z direction respectively. **[4]**

Q3) a) In case of Rotating Disks show that maximum tangential stress doubles when a small circular hole is made at the center of it. **[5]**

b) Determine the equation for angle of twist and stress induced in thin walled tube due to torsion. **[4]**

OR

Q4) a) What is Biharmonic equation? Solve for Biharmonic equation for the case of symmetrical stress distribution. **[5]**

b) For torsion of a bar of variable diameter find out the equation to determine magnitude of moment. **[4]**

P.T.O.

Q5) a) Derive the plane stress yield locus formula as per Tresca's yield criterion. **[8]**

b) Explain Mohr's theory of yielding and Tresca's associated flow rule. **[8]**

OR

Q6) a) Write short notes on isotropic hardening as per Von Mises criterion and Anisotropic hardening as per Bauschinger effect. **[8]**

b) Write short notes on Plastic bending and Torsion in elasto-plastic materials. **[8]**

Q7) a) Derive the expression for thermal stresses in a thick spherical shell considering elastic deformation and plastic yielding. **[8]**

b) Write a short note on ideally plastic solids of ductile and brittle materials with an suitable diagram. **[8]**

OR

Q8) a) Derive the expression of bending moment per unit width at the end of the first elastic/plastic phase of pure bending curved bar. **[8]**

b) Derive the equation of radial and tangential stress in plastic zones for thick walled cylinders under internal pressures considering elasto-plastic behavior. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9450

[6182]-751

[Total No. of Pages : 3

M.E. (Civil-Structures)

STRUCTURAL DYNAMICS

(2017 Pattern) (Semester - I) (501002)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of electronic pocket calculator is allowed.

Q1) A mass m is placed at the mid-span of a simply supported beam. To reduce the deflection of the beam, a spring is attached to the beam as shown in Fig. 1. Determine the value of k needed to reduce the deflection of the beam by 50 % of its original value. [9]

Consider $E = 206 \times 10^9 \text{ N/m}^2$, cross-section of the beam $(b \times d) = (150 \times 250)\text{mm}$, $L = 3.65 \text{ m}$ and $m = 450 \text{ kg}$.

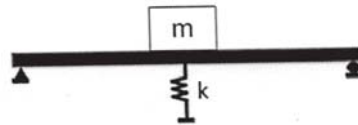


Fig. 1

OR

Q2) A 10 kg mass is attached to the lower end of a spring whose upper end is fixed. It vibrates with a natural period of 0.35 seconds. Determine the natural period when a 3 kg mass is attached to the midpoint of the same spring with the upper and lower ends fixed. [9]

Q3) A spring mass dashpot system has the following properties: $m = 150 \text{ kg}$, $k = 45.5 \times 10^3 \text{ N/m}$, and $c = 850 \text{ N.s/m}$. The system is initially at rest and in equilibrium. If the system is excited by a harmonic force of $50 \text{ N @ } 3.5 \text{ Hz}$, calculate the steady-state displacement amplitude. [9]

OR

P.T.O.

- Q4)** A building is modelled as a single degree of freedom system and is subjected to loading as shown in Fig. 2. Determine the response of the building. [9]

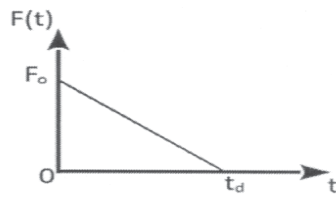


Fig. 2

- Q5) a)** Calculate the natural Frequencies and plot the mode shapes for the system shown in Fig. 3. Consider $m_1 = 100$ kg, $m_2 = 35$ kg, $k_1 = 5$ kN/m and $k_2 = 7.5$ kN/m. [8]

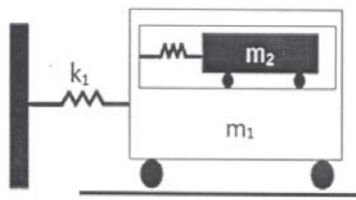


Fig. 3

- b)** Determine the Eigen values and Eigen vectors for the system shown in Fig. 4. [8]

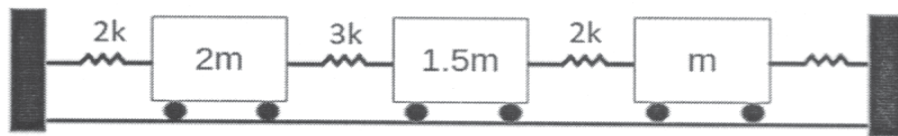


Fig. 4

OR

- Q6) a)** The typical elevation of an industrial building is in Fig. 5. The building consists of frames spaced at 4 m c/c. The design load on the roof is 5.5 kN/m² and on the intermediate floor is 7 kN/m². Determine the natural frequency and plot the mode shapes. Take $E = 21$ GPa. [12]

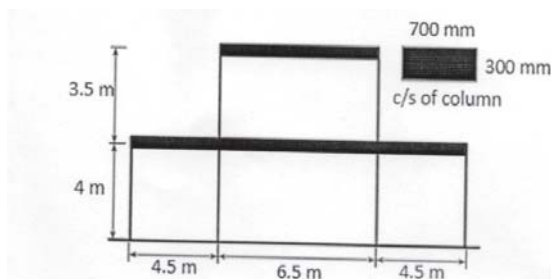


Fig. 5

- b)** Explain the significance of mode shapes with a suitable example. [4]

- Q7)** a) From the fundamental principle, derive the expression for transverse vibration of a simply supported beam carrying a uniformly distributed load w . [8]
- b) Derive the expression for transverse vibration of a beam fixed at one end and free at the other end. [8]

OR

- Q8)** a) List the methods available for the analysis of the transverse vibration of beams and explain any one method. [8]
- b) Derive the expression for transverse vibration of a beam fixed at both ends. [8]



Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages :2

P9451

[6182]-752

M.E. (Civil Structures)

ADVANCED DESIGN OF STEEL STRUCTURES

(2017 Credit Pattern) (Semester-I) (501003)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data if necessary and clearly state.*
- 5) Use of cell phone is prohibited in the examination hall.*
- 6) Use of electronic pocket calculator and relevant IS code are allowed.*

Q1) A castellated beam of span 20 m is to carry an imposed load 5 kN/m and dead load 4 kN/m, design the cross section. Also Check for moment and shear.[9]

OR

Q2) Design the column of 3.5 m length using tubular round section subjected to an axial load of 400 kN as per IS 806 and 1161. Both ends of the column are hinged. [9]

Q3) Two channel sections without bent lips 180mm × 50 mm are connected with webs to act as a beam. The thickness of channel section is 2.0mm. The effective span of simply supported beam is 5.0m. Determine the maximum uniformly distributed load excluding self weight, which may be supported by the beam. Adopts properties of the channel section from IS: 811-1961. [9]

OR

Q4) Find the geometry and draw elevation of the tower to be erected for transmission line at 0.90 power factor with the following data: Power conductor: 32 mm dia. aluminium cable steel reinforced conductors consist of 55 strands of 3 mm dia of aluminium and 8 strands of 3 mm dia of steel to be used, Unit weight of conductor: 17.00 N/m, Permissible axial tension: 36.00 kN, Young's modulus of elasticity: 0.842×10^5 N/mm², Coefficient of expansion: 0.00001992/°c, Shape factor: 0.67, Ground wire: 12mm diameter galvanised steel wire with permissible axial tension = 25.0kN. [9]

P.T.O.

Q5) A self supporting steel chimney is 80m high and its diameter at the top is 3.0 m. Design the plates for the stacks at 40m from the top and check the stresses. Adopt the wind force as per IS 875. [16]

OR

Q6) a) Explain in details different type of steel chimney with sketches. [8]
b) explain requirement of breech opening and ladder for steel chimney with suitable sketches. [8]

Q7) a) State and explain design steps of anchor bolt for steel chimney. [8]
b) State and explain in stability check for steel chimneys. [8]

OR

Q8) Design the base plate for the chimney subjected to wind load moment 11590 kNm and dead load 950 kN rest on M15 grade of concrete. [16]



Total No. of Questions : 8]

SEAT No. :

P9452

[6182]-753

[Total No. of Pages : 3

M.E. (Civil - Structures)

NUMERICAL METHODS IN STRUCTURAL ENGINEERING
(2017 Pattern) (Semester - I) (501004)

Time : 3 Hours]

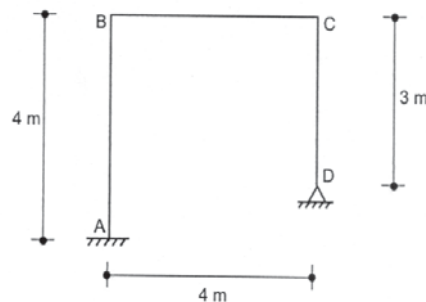
[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) If necessary, assume suitable data and indicate clearly.
- 4) Use of electronic pocket calculator is allowed.

- Q1) a)** Determine the member stiffness matrix $[S_m]_G$ in global coordinates for element BC fixed at end B and the transformation matrix $[R]$ of element AB fixed at end A of the frame shown in Figure 1. [5]

Take $E = 200 \text{ GPa}$, $A = 0.03 \text{ m}^2$, $I = 0.0003 \text{ m}^4$



- b)** Consider the following system of linear equations: [4]

$$3x + 2y - z = 8$$

$$2x - 6y + 2z = -4$$

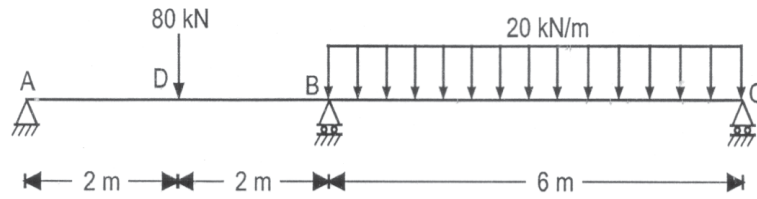
$$x + y + 5z = 9$$

Apply the Gauss Elimination method or Gauss Jordan method to solve the above system of linear equations. Show all the intermediate steps and the final solutions for the variables x , y , and z .

OR

P.T.O.

- Q2) a)** A beam AC with AD = 2m, DB = 2m and BC = 6m is loaded and supported as shown in figure. Using member stiffness approach of matrix method of analysis, determine the rotations at A, B and C. [5]



- b) Describe the Cholesky factorization method to find the solution to the problem. [4]

- Q3) a)** Use the Power Method to find the dominant eigenvalue and its corresponding eigenvector for the following matrix: [5]

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$$

Apply the Power Method with an initial guess for the eigenvector as [1, 1]. Perform two iterations only.

- b) For an integral $\int_a^b f(x)dx$, derive the one-point and two point Gauss quadrature rule. Use Gauss Quadrature formula to approximate the integral of the function $f(x) = x^3$ from $x = 2$ to $x = 5$. [4]

OR

- Q4) a)** Consider the following system of first-order ordinary differential equations: [5]

$$\frac{dy}{dx} = x + yz; \quad \frac{dz}{dx} = x^3 - y^3$$

Given initial conditions: $y(0) = 1$ and $z(0) = 2$.

Use the fourth-order Runge-Kutta method to approximate the values of y and z at $x = 0.1$. Perform two iterations to calculate $y(0.1)$ and $z(0.1)$.

- b) Given the function $f(x) = 3x^2 + 2x + 1$, calculate the area under the curve between $x = 0$ and $x = 4$ using the Trapezoidal rule, Simpsons rule, and Gauss Quadrature. [4]

- Q5) a)** A fixed beam is subjected to uniformly varying load with maximum intensity of 'w' at the mid span and zero intensity at the supports. Using Finite difference method, estimate the maximum deflection and moments at mid span of the beam. Divide the beam in four subintervals. [8]
- b)** Estimate the deflection at the nodal points of the grid, using grid interval of $h = a/2$ for a simply supported uniform plate of length $2a$ and width a . The plate supports a uniformly distributed load 'w' over the entire plate along with a central point load P . Use finite difference method. [8]

OR

- Q6) a)** Derive the expression for maximum axial buckling load for a column pinned at both ends using finite difference method. [8]
- b)** Derive the bi Harmonic Finite difference net for a plate simply supported at its edges. Explain how the boundary conditions affect the solution of the problem. [8]

- Q7) a)** Find the least square linear fit for the given data. [8]

x	1.5	1	2.8	0.4	1.3	2.0
y	4.0	3.3	6	2.4	3.3	5

- b)** Use the cubic spline interpolation technique to interpolate the function $f(x) = x^4$ on the interval $-1 \leq x \leq 1$ by the cubic spline $g(x)$ corresponding to the given nodes $x_0 = -1, x_1 = 0, x_2 = 1$. Ensure that the spline satisfies the clamped conditions $g'(-1) = f'(-1)$ and $g'(1) = f'(1)$. [8]

OR

- Q8) a)** Fit a Lagrangian polynomial to satisfy the points (1,1), (2,8), and (3,27). [8]
- b)** A structural engineering firm is analyzing the relationship between the height of a building (in meters) and the corresponding wind load on the building (in kN/m^2). They have collected data from various buildings to perform regression analysis for wind load prediction. The dataset consists of 6 buildings, each with their height (x) and corresponding wind load (y). The data is as follows: [8]

Building	1	2	3	4	5	6
Height (x) (m)	40	50	30	45	55	35
Wind Load (y) (kN/m^2)	1.5	2.0	1.3	1.8	1.6	2.8

Perform polynomial regression of order 2 (quadratic) to fit a curve to predict the wind load based on building height.



Total No. of Questions : 8]

SEAT No. :

P9453

[6182]-755

[Total No. of Pages : 2

M.E. (Civil-Structures)

FINITE ELEMENT METHOD

(2017 Pattern) (Semester - II) (501007)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Write short note on [6]

- i) Galerkin Method,
- ii) Collocation Method,
- iii) Least Square Method and
- iv) Rayleigh- Ritz Method

b) Write short note on step by step procedure of finite element analysis. [4]

OR

Q2) a) What are the criteria for convergence requirement of displacement function with suitable example. [5]

b) Draw 2D and 3D Pascal's triangle and hence write the displacement polynomial for 2D CST element and 3D tetrahedron element. [5]

Q3) a) What is aspect ratio of an element. How it affects the accuracy of FEM solution? Explain with suitable example. [5]

b) Derive stiffness matrix of two noded beam element using polynomial in cartesian coordinate system. Hence write stiffness matrix of beam element. [8]

OR

P.T.O.

Q4) a) Using the Lagrange interpolation function, derive shape functions of four noded rectangular element in natural coordinate system. [5]

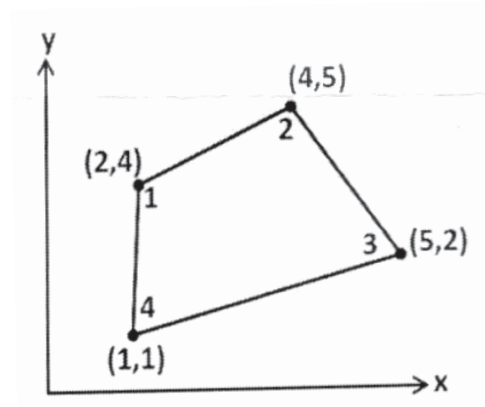
b) What is serendipity elements? Explain with suitable example. Hence, derive shape functions of eight noded serendipity elements in natural coordinate system. [8]

Q5) a) Explain conforming, non-conforming, C^0 , C^1 and C^2 elements with reference to plate bending element. [8]

b) Explain shear locking phenomenon. [5]

OR

Q6) Derive strain-displacement matrix for the quadrilateral element as shown in figure using isoparametric formulation. [13]



Q7) Derive stiffness matrix for the rectangular BFS plate bending element with 16 DOF. [14]

OR

Q8) a) Explain briefly the various factors to be considered in the development of curved shell 8 elements. [8]

b) Explain with suitable example, singly-curved and doubly-curved shells. [6]



Total No. of Questions : 8]

SEAT No. :

P9454

[6182]-756

[Total No. of Pages : 2

M.E. (Civil-Structures)

THEORY OF PLATES AND SHELLS

(2017 Pattern) (Semester - II) (501008)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary and clearly state.

- Q1)** a) Discuss the extension of Levy's method for the analysis of fixed plate subjected to uniformly distributed load ' q_0 '. [5]
- b) Distinguish between thin plate with small deflection and thin plate with large deflection. [4]

OR

- Q2)** a) Derive the moment curvature in the case of pure bending of plates. [5]
- b) Explain the stepwise procedure in Navier's method for bending analysis of a thin rectangular plate simply supported at all four edges. [4]
- Q3)** a) Obtain the expression for deflection in case of uniformly loaded circular plates with clamped edges. [5]
- b) Explain the different classification of shells with neat sketches. [4]

OR

- Q4)** a) A circular plate of radius ' a ' is clamped at edges. The plate carries a load of intensity ' q ' uniformly distributed over the entire surface of the plate. The thickness of the plate is ' h '. Analyze the plate from the basic principles and obtain the expression for bending moments at the center of the plate. [5]
- b) Explain about the advantages and disadvantages of the shells. [4]

P.T.O.

- Q5)** a) Find the equations of equilibrium in case of shells in the form of a surface of revolution and loaded symmetrically with respect to their axis. [8]
- b) Find the membrane stresses in a circular cylindrical shell subjected to snow load. [8]

OR

- Q6)** a) Find the membrane forces in a circular cylindrical shell subjected to a sinusoidal load. [8]
- b) State and explain the membrane theory cylindrical shells. [8]
- Q7)** a) Explain the principles of Lundgren's beam theory along with its application to cylindrical roof shells. [8]
- b) State the need of beam theory and also write the strain components in bending theory. [8]

OR

- Q8)** a) Derive the 8th order governing differential equations for cylindrical shell subjected to bending, according to the Finsterwalder theory. [8]
- b) Explain in details arch analysis for cylindrical shells. [8]



Total No. of Questions : 8]

SEAT No. :

P9455

[6182]-757

[Total No. of Pages : 2

M.E. (Civil) (Structures Engineering)
ADVANCED DESIGN OF CONCRETE STRUCTURES
(2017 Pattern) (Semester - II) (501009)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use IS 456, IS 1343, IS1893, IS 3370 and Non-programmable calculators.
- 5) Assume suitable data, if necessary.

- Q1) a)** Write a short note with sketches on characteristics of Yield lines. **[4]**
- b) Design a grid slab for a floor of hall 12m×15m having a square grid of 1.5m. Use M25 and Fe 500 steel. Take live load 6kN/m² and floor finish load 1.4kN/m². **[5]**

OR

- Q2) a)** Draw yield line pattern for: A rectangular slab with fixed support on three sides with shorter simply supported. **[4]**
- b) Design a grid slab for a floor of community hall 15m×18m having square grid of 1.5m. Use M25 and Fe 415. **[5]**

OR

- Q3) a)** Explain different types of flat slab and list the advantages of flat slab. **[4]**
- b) An open square tank 6m×6m×3m deep and supported 6m above the ground level on beams and columns. Design the beam and column of a tank. Use M20 and Fe 500 steel. **[5]**
- Q4) a)** Explain in detail wind load analysis of column for a water tank supported on four identical columns with their lower end fixed to the base. **[4]**
- b) Design an interior panel of flat slab for flexure with panel size 5.75m × 5.75m in dimensions. Take live load of 5 kN/m² and floor finish load of 1.3kN/m². Use M20 and Fe 500. **[5]**

P.T.O.

- Q5) a)** Design a circular bunker to store 60kN of coal take density of coal as 8kN/m^3 , angle of repose is 28° . Use M25 and Fe 500 steel. Draw the details of reinforcement in side wall and hopper. **[8]**
- b)** A concrete silo of diameter 11m, with height of cylindrical portion of 30m and contains coal weighing 8.0kN/m^3 . The coefficient of friction between grain and grain is 0.7, the coefficient of friction between grain and concrete is 0.7 and angle of repose is 35° . Determine the lateral pressure at 5m interval on the walls of silo. **[8]**

OR

- Q6) a)** Design a square bunker to store 100 tons of cement consider Unit weight of cement as 30kN/m^3 . Angle of repose is 29° . Use M25 and Fe 500. Draw reinforcement details for side walls and hopper bottom. **[8]**
- b)** A concrete silo of diameter 12m, with height of cylindrical portion of 20m and contains and contains Coke weighing 4.50kN/m^3 . The coefficient of friction between grain and grain is 0.83, the coefficient of friction between grain and concrete is 0.85 and angle of repose is 40° . Determine the lateral pressure at 5m interval on the walls of silo. **[8]**
- Q7) a)** A group of 30 piles are arranged in 5 rows and 6 in each row. The piles are 350mm in diameter and spaced 1.2m center to center. Each pile can carry 250kN load if it can act independently. Determine the carrying capacity of the pile group. **[8]**
- b)** Design a formwork for a slab of $3.5\text{m} \times 3.5\text{m}$ having thickness of 150mm. It is proposed to deposit concrete in one stage. **[8]**

OR

- Q8) a)** In a group of 25 piles of diameter 400mm and center to center spacing of pile being 1.2m. The piles are arranged in square area. Each pile is 13m long and taking $m = 0.7$ and $c = 50\text{kN/m}^2$. Ascertain whether the failure will occur with the piles acting individually or as a group. Also specify the failure load. **[8]**
- b)** Design a raft foundation for the following center to center distance of column in both directions is 2.4m. Taking column size as $300\text{mm} \times 300\text{mm}$, working load on each column is 600kN. The depth of strata is 1.5m. Use M25 and Fe500, $\text{SBC} 100\text{kN/m}^2$. Draw reinforcement details. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9456

[6182]-759

[Total No. of Pages : 2

M.E. (Civil) (Structural Engineering)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - III) (601013)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Write a note on “Types and Importance of Research Design. [4]
b) Write short notes on: [5]
i) Motivation in research.
ii) Objectives of research.

OR

- Q2)** a) What are the criteria or characteristics of good research problem? [4]
b) Enlist the various funding agencies for engineering research grants. Explain any one in detail. [5]
- Q3)** a) Distinguish between primary and secondary data in research work. [4]
b) Explain the purpose of literature survey. Explain with example the purpose and use of three different literature sources. [5]

OR

- Q4)** a) Write a short note on reviewing literature and writing a literature survey. [4]
b) How to write literature review to develop conceptual framework for research? [5]

P.T.O.

Q5) a) Critically examine the qualitative and quantitative research methods. Elaborate your answer by giving suitable example. **[8]**

b) Explain sampling concept with different sampling design in brief. **[8]**

OR

Q6) a) Explain the characteristics of good sample design. **[8]**

b) Explain different types of research hypothesis with suitable example. **[8]**

Q7) a) Write notes on: **[8]**

i) Intellectual Property Rights (IPR)

ii) What are the different tools used for making oral presentation effective?

b) Write a research proposal for a suitable research problem (Any problem related to Structural Engineering can be considered) to a funding agency with reference to the following terms: Title, Introduction, origin of the problem, expected outcome, literature review, Significance of the study in the context of current status, objectives, methodology, year wise plan. **[8]**

OR

Q8) a) Describe in brief, the structure of report. What do you mean by plagiarism? **[8]**

b) Write a research proposal for a suitable research problem (Any problem related to Civil Engineering can be considered) to a funding agency with reference to the following terms: Title, Introduction, origin of the problem, expected outcome, literature review, Significance of the study in the context of current status, objectives, methodology, year wise plan. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9457

[6182]-760

[Total No. of Pages : 2

M.E. (Civil) (Structure Engineering)
ANALYSIS AND DESIGN OF EARTHQUAKE RESISTANT
STRUCTURES
(2017 Pattern) (Semester - III) (601014)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*
- 7) *Is 1893 is allowed.*

Q1) a) Explain concept of stiffness, flexibility and ductility in earthquake design. **[5]**

b) What is the Modified Mercalli Intensity Scale? **[4]**

OR

Q2) a) What is liquefaction? State its effect on structures. **[5]**

b) Define response spectra and give the uses of a response spectrum **[4]**

Q3) a) Explain P – Δ effect. **[5]**

b) Explain effects of irregularity for earthquake resistant design. **[4]**

OR

Q4) a) Explain effects of irregularity and architectural planning for earthquake resistant design. **[5]**

b) Describe with neat figure divergent boundaries. **[4]**

P.T.O.

Q5) a) A RCC beam of rectangular section has to carry a distributed live load of 21 kN/m in addition to its own weight and a dead load of 25 kN/m. The maximum bending moment and shear force due to earthquake are 60 kN-m and 40 kN-m respectively. Centre to centre distance between supports is 6 m. Design the beam using M20 grade concrete and Fe 415 steel. **[12]**

b) Explain detailing of beams as per IS code provisions. **[4]**

OR

Q6) a) Design and detailing of shear wall the shear wall of length 13.4 m and thickness 200 mm. it is subjected to the following forces: **[12]**

Factored axial force $P_u = 8550$ kN

Factored Bending Moment = 47392.9 kN-m

Factored Shear force = 2063.2 kN

Material $F_{ck} = 25$ N/mm², $F_e = 415$ N/mm²

b) Explain detailing of beams as per IS code provisions **[4]**

Q7) a) Discuss the concept of flanged shear wall and state its advantages and dis-advantages. **[8]**

b) State IS 3370 code provisions for water tanks. **[8]**

OR

Q8) a) Ductile detailing consideration as per 13920:1993 for lap splices in beam, beam web reinforcement, column and joint detailing, travers reinforcement in columns. **[8]**

b) Write design procedure for water tank. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9458

[6182]-762

[Total No. of Pages : 2

M.E. (Civil)

WATER RESOURCE AND ENVIRONMENTAL ENGG.)

Planning And Management of Water Resources

(2017 Pattern) (Semester - I) (501081)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *All questions are compulsory.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right side indicate full marks.*
- 5) *Use of Calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

Q1) a) Explain environmental aspects of planning and management of water Resources. **[5]**

b) Comment on spatial and temporal characteristics of water resources. **[4]**

OR

Q2) a) Explain constraints like reversibility for development of water resources. **[5]**

b) Explain objective use of water resources planning. **[4]**

Q3) a) Explain how conjunctive use of water play important role in water management. **[4]**

b) Explain latest technique to measured sedimentation in a reservoir. **[5]**

OR

Q4) a) How groundwater content can be improved. **[4]**

b) Explain role of water doctrines to resolve disputes. **[5]**

Q5) a) Comment on causes of floods, effects of flood and methods to control floods. **[8]**

b) Discuss central water policy with regards to water allocation. **[8]**

OR

Q6) a) Explain essential data necessary for water power studies for storage project. **[8]**

b) Explain repayment of cost and allocation of cost for multipurpose project. **[8]**

P.T.O.

- Q7)** a) Explain financial performance analysis to check economic viability of long term sustainable projects. **[8]**
- b) Explain in detail basin planning (principles & procedures). **[8]**

OR

- Q8)** a) Explain with an example Inter Basin Water Transfer. **[8]**
- b) Explain discounting techniques for financial analysis of large dams. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9459

[6182]-767

[Total No. of Pages : 2

M.E. (Civil)(Water Resources and Environmental Engg.)

HYDROLOGY

(2017 Pattern) (Semester - II) (501087)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer all questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) a) Explain hydrologic cycle. State various rain-gauges Explain in detail modern method. **[5]**

b) Explain Stanford Watershed Model. **[4]**

OR

Q2) a) Draw a flow chart for mathematical methods in hydrology and explain stochastic hydrology applications. **[6]**

b) Explain log- normal distribution (statistical). **[3]**

Q3) a) What is design flood? How it is finalized while designing hydraulic structure. **[4]**

b) Explain Pearson Type - III method. **[5]**

OR

Q4) a) Write note on method of images for ground water management. **[5]**

b) Explain the Goodrich method of flood routing. **[4]**

P.T.O.

Q5) a) Explain step by step design of tube well & also explain types and construction methods for tube wells. **[8]**

b) What are causes of ground water pollution? State effects of ground water pollution. What are the methods to treat polluted ground water. **[8]**

OR

Q6) a) What affects the quality of ground water in India. **[8]**

b) Explain digital computer models. **[8]**

Q7) a) Explain any one widely used method of ground water recharge. **[8]**

b) State various methods to conserve ground water and explain any two and also explain how it helps in management of water resources. **[8]**

OR

Q8) a) Explain electric analog model for ground water modeling with neat sketches. **[8]**

b) Explain sand tank model and transparent model for ground water modeling with neat sketches. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9460

[6182]-769

[Total No. of Pages : 2

M.E. (Civil-Water Resources and Environmental Engineering)

ADVANCE WASTE WATER TREATMENT

(2017 Pattern) (Semester - II) (501089)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat figures wherever necessary.

Q1) a) What is the significance of providing grit chamber? Explain the working principle of grit chamber. **[4]**

b) Explain kinetics of biological growth and substrate limited growth. **[5]**

OR

Q2) a) Write working principle and mechanism of membrane filtration. **[4]**

b) What are the types of membrane filtration systems? Explain any one in detail. **[5]**

Q3) a) Write working principle and mechanism of micro filtration and nano filtration. **[4]**

b) Distinguish clearly between plug flow reactor and continuous stirred tank reactor with respect to design and operation point of view. **[5]**

OR

Q4) a) Explain activated sludge process. **[4]**

b) Write a short note on cross flow model. **[5]**

Q5) a) Write working mechanism and design parameters and application of phytoremediation technology for wastewater treatment. **[8]**

b) Write the working principle and mechanism of root zone cleaning system. **[8]**

OR

P.T.O.

- Q6)** a) Write effluent standards for disposal of the sewage. [8]
b) What are the fundamental considerations in land treatment systems? Write design objectives and site selection criteria of irrigation systems. [8]

- Q7)** a) Write principle, advantages and disadvantages of UASBR. [8]
b) What are the various stages of digestion? Explain the characteristics of biogas and its application. [8]

OR

- Q8)** a) Write methods of removal of nitrate and phosphate from the sludge and explain any one method. [8]
b) Write the various methods of sludge treatment and disposal. How can dry sludge be used for other purposes? [8]



Total No. of Questions : 8]

SEAT No. :

P9461

[6182]-772

[Total No. of Pages : 2

M.E. (Civil-Water Resources and Environmental Engineering)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - III) (601094)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Figures to the right indicate full marks.*
- 2) *Your answers will be valued as a whole.*
- 3) *Assume suitable data, if necessary.*

Q1) a) Discuss the various strategies of literature survey for Quantitative Research. **[5]**

b) Discuss the types of research hypotheses. **[4]**

OR

Q2) a) Enlist various sources for Literature review and discuss the strategy to be employed for Quantative research. **[5]**

b) Discuss the factors to be considered in selecting a research problem. **[4]**

Q3) a) Discuss the conditions in which Type I and Type II errors will be committed. **[5]**

b) Enlist methods of data collection. Discuss how secondary data will be collected. **[4]**

OR

Q4) a) Discuss the conditions in which t-test and z-test is used for Hypothesis testing. **[5]**

b) Explain conditions in which chi-square test is used for hypothesis testing are used. **[4]**

P.T.O.

- Q5)** a) Distinguish between discriminant analysis, cluster analysis. [8]
b) Explain Principal Component Analysis. Discuss the centroid method. [8]

OR

- Q6)** a) Discuss the important characteristics of One way ANOVA. [8]
b) Discuss Correlation and regression analysis. [8]

- Q7)** a) What are characteristics of good Abstract for a journal paper. Discuss as to why abstract is necessary. [8]
b) Discuss the importance of publishing a research work in a conference. Enlist the steps involved in publish a research paper in journal. [8]

OR

- Q8)** a) Describe, in brief, the layout of a research report, covering all relevant points. [8]
b) Discuss the points to be considered while giving a formal presentation. [8]



Total No. of Questions : 8]

SEAT No. :

P9462

[6182]-774

[Total No. of Pages : 3

M.E. (Mechanical) (Design/CADM&E/Automobile)

ADVANCED MATHEMATICS

(2017 Pattern) (Semester - I) (507201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of non-programmable electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Find the orthonormal basis for the Euclidean space $\mathbb{R}^3$, by applying Gram-Schmidt's method to the following vectors: (1, 1, 1), (1, 2, 2), (1, 1, 0) **[5]**

b) If $w = \phi + i\psi$ represents the complex potential for an electric field and

$$\psi = x^2 - y^2 + \frac{x}{x^2 + y^2}, \text{ determine the function } \phi. \quad \text{[5]}$$

Q2) a) Evaluate $\oint_C \frac{z+1}{z^3(z+4)} dz$, where C is the circle $|z| = 1$. **[5]**

b) Find the mechanical system which is governed by the differential equation and with the given initial condition $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = te^{-t}$, $y(0) = 1$, $y'(0) = -2$. **[5]**

Q3) a) Solve the initial value problem:

$$\frac{d^2y}{dx^2} + y = u(t-3), y(0) = 0, y'(0) = 1$$

Where u is the unit step function. **[5]**

b) Solve the following differential equation by series method.

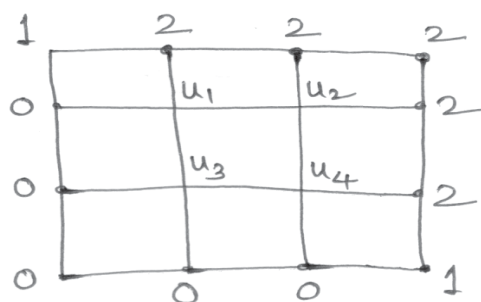
$$\frac{d^2y}{dx^2} + 4y = 0. \quad \text{[5]}$$

P.T.O.

- Q4) a)** Calculate by power method the largest eigen value and the corresponding eigen vector of the matrix:

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \text{ with } X_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [5]$$

- b) Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for the square mesh with boundary values as given below. [5]



- Q5) a)** Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ using schmidt Bendre's method, subject

to the conditions $u(0, t) = u(4, t) = 0$ for $t \geq 0$ and $u(x, 0) = \frac{x}{3}(16 - x^2)$ for $0 < x < 4$. By taking $h = 1$. [5]

- b) Using Rayleigh Ritz method, solve the boundary value problem $y'' + y = x^2$, ($0 < x < 1$) with condition $y(0) = y(1) = 0$. [5]

- Q6) a)** Find the extremal of the functional $\int [y^2 + (y')^2 + 2ye^x] dx$. [5]

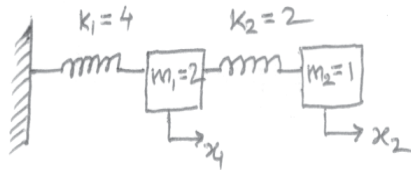
- b) Solve the linear equation system using least square method $x + 2y = 1$, $x + 5y = 2$, $x + 7y = 3$, $x + 8y = 3$. [5]

- Q7) a)** Find the map of the circle $|z - i| = 1$ under the transformation $w = \frac{1}{z}$ in to w - plane. [5]

- b) Establish the following relation [5]

$$e^{-2x} - e^{-3x} = \frac{10}{\pi} \int_0^\infty \frac{\lambda \sin \lambda x}{(\lambda^2 + 9)(\lambda^2 + 4)} d\lambda, x > 0.$$

- Q8) a)** For the following mass spring system there is no friction. Find the natural frequencies of the system and corresponding normal modes of vibration using matrix method. [5]



- b)** Solve $16 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$, given that [5]

i) $u(0, t) = 0$

ii) $u(5, t) = 0$

iii) $u(x, 0) = x^2(x - 5)$ and

iv) $\left(\frac{\partial u}{\partial t} \right)_{t=0} = 0$ by taking $h = 1$ & $t = 0.25$.



Total No. of Questions : 7]

SEAT No. :

P9463

[6182]-775

[Total No. of Pages : 2

M.E. (Mechanical) (Design Engineering)
MATERIAL SCIENCE AND MECHANICAL BEHAVIOR OF
MATERIALS
(2017 Pattern) (Semester - I) (502202)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of programmable calculator is not allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1) a)** Explain the applications of smart materials by considering functional attributes. **[5]**
- b)** Explain the constituents of composite materials in detail. **[5]**

- Q2)** State of stress at point A in a body is given by $\begin{bmatrix} 10 & 0 & 3 \\ 0 & 3 & 0 \\ 3 & 0 & 2 \end{bmatrix}$ and at point B **[10]**

it is given by $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -5 \end{bmatrix}$.

Determine which state is more critical to yielding if the following criteria of yielding are used:

- a) Octahedral normal stress
 - b) Octahedral shear stress
 - c) Maximum shear stress.
- Q3) a)** Explain Tresca's Criteria of Yielding and draw its Yield Locus. **[5]**
- b)** Describe the phenomenon of Bauschinger's effect in mild steel. **[5]**

P.T.O.

- Q4)** a) A 13 mm-diameter tensile specimen has a 50 mm gage length. The load corresponding to the 0.2 percent offset is 6800 kg and the maximum load is 8400 kg. Fracture occurs at 7300 kg. The diameter after fracture is 8 mm and the gage length at fracture is 65 mm. Calculate the standard properties of the material from the tension test. [6]
- b) Explain with neat sketch the Biaxial Tension (Hydraulic Bulge) Test. [4]
- Q5)** a) Two points on a true stress-strain curve for aluminum alloy are $\sigma = 222$ MPa at $\epsilon = 0.05$ and $\sigma = 303$ MPa at $\epsilon = 0.15$. Find the values of K and n using Power Law Approximation that best fit the data. Then using these values of K and n, predict the true stress at a strain of $\epsilon = 0.30$. [5]
- b) Explain isotropic and kinematic hardening models with neat sketch. [5]
- Q6)** An element of J_2 - material (deformation theory of plasticity) is subjected to a proportional loading path with stress ratio $\frac{\sigma}{\tau} = 2$. The material obeys Hook's law up to elastic limit. Post-yield behavior during simple tension of the material is given by $\frac{\sigma}{e} + \frac{\sigma - \sigma_y}{m}$. With Young's modulus $E = 207$ GPa, yield stress $\sigma_y = 207$ MPa, constant $m = 25$ GPa and Poisson's ratio = 0.3. Find all the components of normal and shear strains for stress state of $\sigma = 180$ MPa and $\tau = 90$ MPa. [10]
- Q7)** What is viscoelasticity? Explain Maxwell Model, Voigt Model and Maxwell- Voigt Model for rheological properties of viscoelastic material with neat sketch. [10]



[6182]-776

M.E. (Mechanical) (Design Engineering)
ADVANCED STRESS ANALYSIS
(2017 Credit Pattern) (Semester-I) (502203)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions out of 7.
- 2) All the questions should be solved in one answer book and attach extra supplements if required.
- 3) Neat diagram must be drawn wherever necessary.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary, but state the assumptions clearly.

Q1) a) Investigate function, $\phi_3 = \frac{a_3}{6}x^3 + \frac{b_3}{2}x^2y + \frac{c_3}{2}xy^2 + \frac{d_3}{6}y^3$

Is it permissible Airy's stress function?

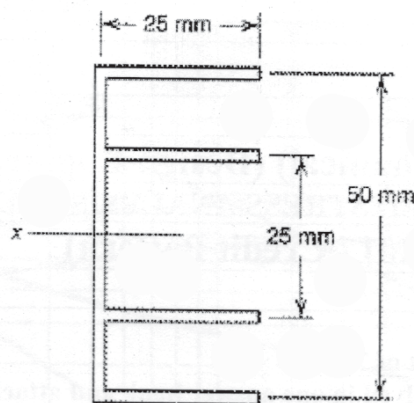
Under what conditions the expression above will represent pure bending condition? [7]

- b) Explain Plain Strain condition with suitable example. [3]

Q2) a) Derive Compatibility equation in the Cartesian coordinate system in terms of stress components. Explain the necessity of compatibility equation in stress analysis. [7]

- b) Sketch the situation under which conditions the beam is subjected to unsymmetrical bending. [3]

Q3) Derive an expression and locate the shear center for the beam cross-section shown in fig. [10]

The wall of the cross-section has a constant thickness $t = 2 \text{ mm}$ 

- Q4)** A solid bar of elliptical cross-section is subjected to pure torsion. Starting from the equation of an ellipse, determine for the bar maximum shearing stress and the angle of twist per unit length. Sketch the stress distribution along the boundary of an ellipse. [10]

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

- Q5) a)** A strain rosette gauge is mounted at a critical location on a free surface of the steel structural member. The strain gauge readings for a particular state of loading of the structure are- [6]
 $\epsilon_A = 400 \mu\text{m/m}$ (at 0° angle), $\epsilon_B = 375 \mu\text{m/m}$ (at 60° angle),
 $\epsilon_C = 200 \mu\text{m/m}$ (at 120° angle), Let $E = 200 \text{ GPa}$ and $\nu = 0.25$.
Determine the maximum principal strain, principal stress and maximum shear stress.
- b)** Explain Prandtl's membrane analogy, with a suitable example. [4]

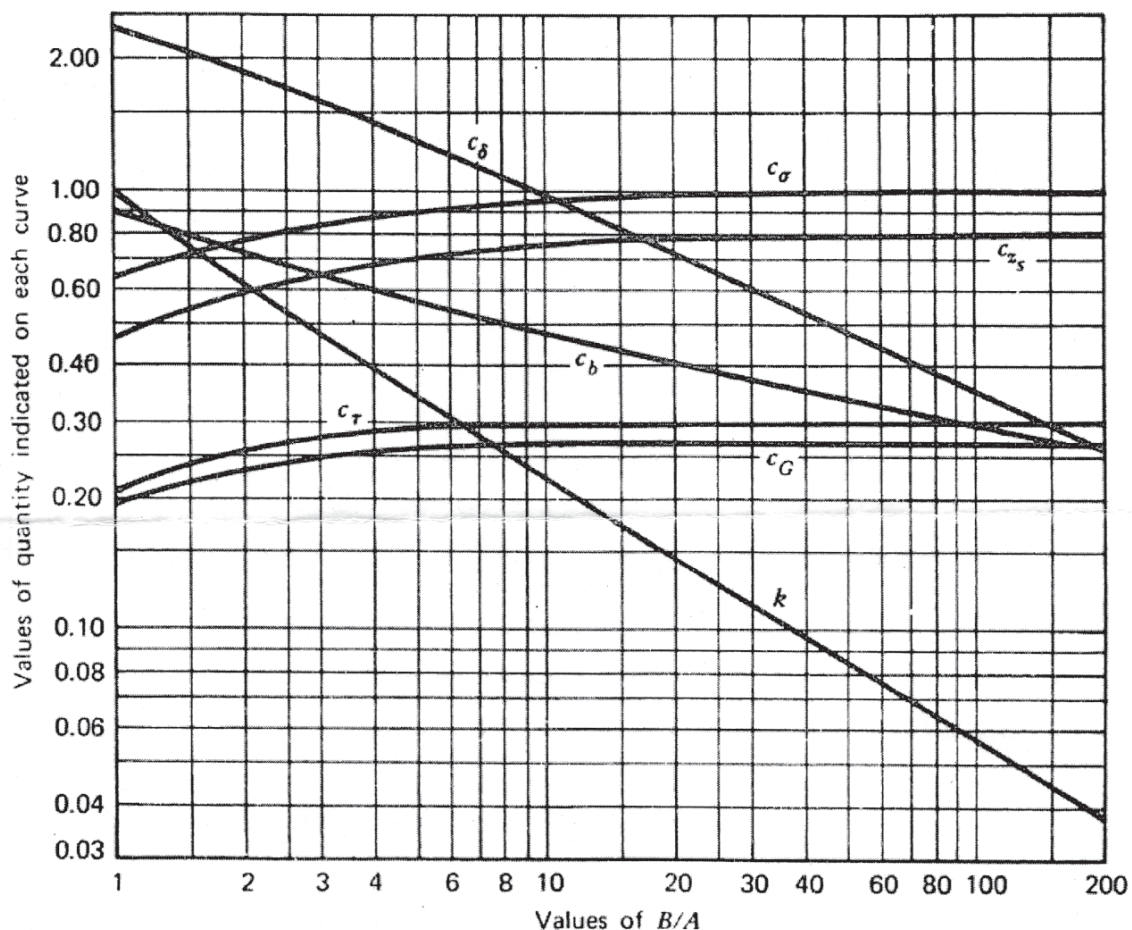
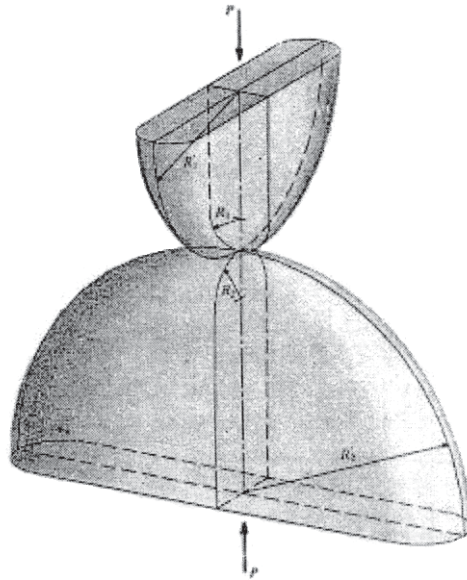


Fig: Stress and Deflection coefficient of two bodies in contact of a point

- Q6)** Two semicircular disks are shown in figure below made of steel ($E_1=E_2=200$ GPa and $\nu_1 = \nu_2 = 0.29$). The Radii of curvature of the two surfaces at the point of contact are $R_1=60$ mm, $R_1'=130$ mm, $R_2=80$ mm, and $R_2'=200$ mm. The angle α between the planes of minimum curvature is $\pi/3$ rad. If the load $P=5$ kN, Determine the maximum principal stress, maximum shear stress, and maximum octahedral shear stress in the disks and state at the location of the point where each of these stresses occurs. Determine the approach δ for the two disks because of load P . **[10]**



- Q7)** a) What are strain rosettes? Explain rosettes with suitable example? **[5]**
 b) Explain the effect of a stressed model in plane polariscopic with the suitable example. **[5]**



Total No. of Questions : 8]

SEAT No. :

P9465

[6182]-778

[Total No. of Pages : 2

**M.E. (Mechanical) (Design Engineering) (Heat Power Engg.)
(Common to All Mechanical Branches)**

RESEARCH METHODOLOGY

(2017 Credit Pattern) (Semester - I) (502104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Figures to the right indicate full marks.*
- 3) Neat diagrams should be drawn wherever necessary.*
- 4) Assume suitable data, if necessary.*
- 5) Use of logarithmic tables slide rule in Electronic pocket calculator is allowed.*

Q1) a) Discuss Research Design in case of Descriptive and Diagnostic Research studies. **[4]**

b) Explain Research process with neat block Diagram. **[5]**

OR

Q2) a) Explain the techniques of Defining a Research Problem. **[4]**

b) Explain Significance of Research. **[5]**

Q3) a) Write a note on setting up a Computer Model to Predict the Performance of an Experiment system. **[4]**

b) Explain the various types of Research. **[5]**

OR

Q4) a) Explain Meaning of following context of Research Design. **[4]**

i) Extraneous variables.

ii) Control

iii) Confounded Relationship

iv) Research Hypothesis

b) Explain Multi Scale Modeling with Neat Sketch. **[5]**

P.T.O.

- Q5)** a) Explain the role of DSP in data collection in noisy environment. [8]
b) Explain Regression analysis, Curve fitting and Developing Correlation.[8]

OR

- Q6)** a) Describe the outline and structure of research report. [8]
b) Explain dissemination of research findings and different steps and precaution while writing research report. [8]

- Q7)** a) Describe error analysis and methods to reduce errors in research process. [8]
b) Explain importance of literature survey during research works. [8]

OR

- Q8)** a) Explain the term Plagiarism. How can we avoid plagiarism in paper writing and thesis writing? [8]
b) Explain found of a research proposal clearly stating the sections to be included while writing a good research proposal. [8]



M.E. (Mechanical) (Design Engg.)
ANALYSIS AND SYNTHESIS OF MECHANISMS
(2017 Pattern) (Semester - II) (502207)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary*

Q1) a) Determine the degree of freedom of the mechanism shown in Figure 1.

[5]

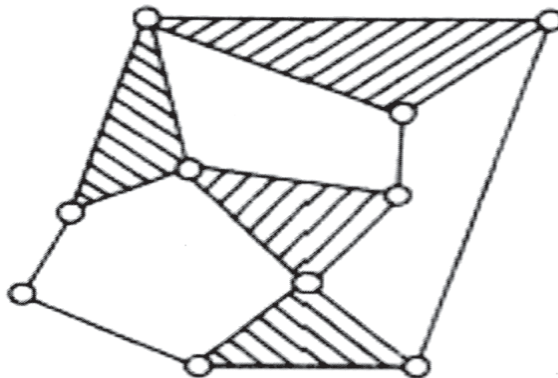


Figure 1

b) Discuss the procedure to determine velocity of a connecting rod in a slider crank mechanism when the crank is at 90° to IDC. **[5]**

Q2) In the mechanism shown in Figure 2, $\omega_2 = 10$ rad/sec (CW) (constant). Determine the velocity of link 6 using the auxiliary point method. **[10]**

Given:

$O_2A = 7.5$ cm, $AB = 5$ cm, $BC = 7.5$ cm, $O_5C = 6.25$ cm (and is vertical),
 $CD = 10$ cm, $BD = 5$ cm, $O_6D = 5$ cm,

P.T.O.

Angle $O_2AB = 110^\circ$, Angle $ABC = 115^\circ$, Angle $O_6DB = 117^\circ$, angle $O_5CB = 120^\circ$

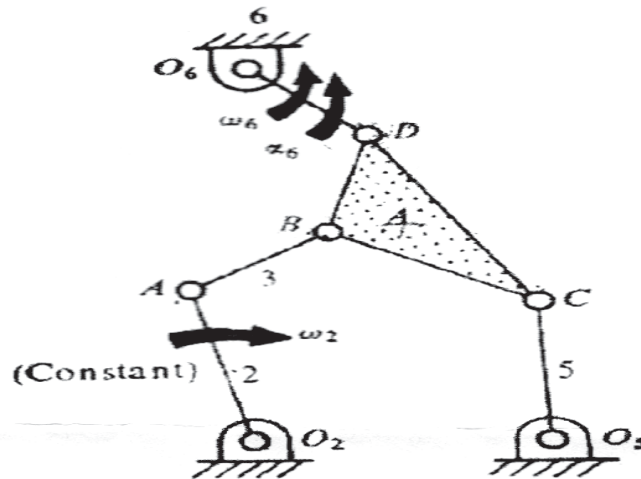


Figure 2

- Q3)** a) Discuss the step involved in determining the Inflection Circle using Bobilier Construction. [5]
 b) Explain the concept of Ball's point and give the application in dwell mechanisms. [5]
- Q4)** a) Write a note on 'Centre Point and Circle Point curves.' [5]
 b) Discuss the procedure to design a Four-link mechanism using the Inversion Method for the Function Generation task when Four positions of Input and Output link are known. [5]
- Q5)** a) Discuss the 'Complex Number Method for the Mechanism Synthesis'. [5]
 b) Indicate and prove the Robert-Chebyshev theorem. [5]
- Q6)** Discuss with adequate illustration, the Denavit - Hartenberg (D-H) parameters for the investigation of any spatial mechanism. [10]
- Q7)** a) Describe the term 'Kinematically Complex' mechanisms with suitable examples. [5]
 b) Explain Grashoff's condition and Grubler's criteria for the analysis of mechanisms. [5]



Total No. of Questions : 7]

SEAT No. :

P9467

[6182]-780

[Total No. of Pages : 2

M.E. (Mechanical) (Design Engineering)
ADVANCED MECHANICAL VIBRATIONS
(2017 Pattern) (Semester - II) (502208)

Time : 3 Hours]

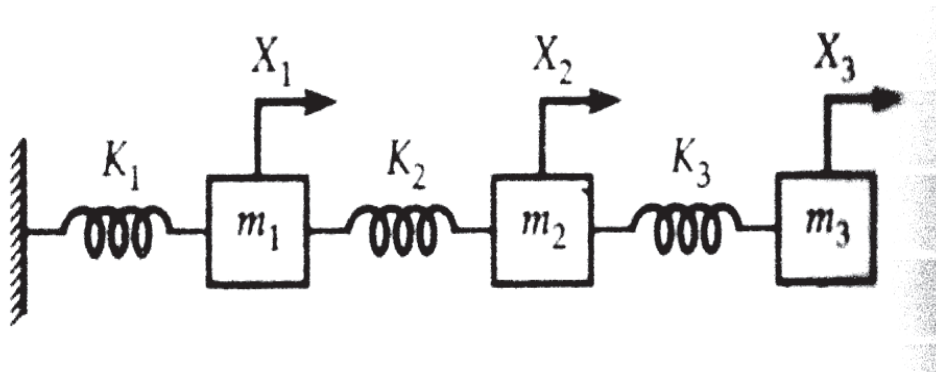
[Max. Marks : 50

Instructions to the candidates:

- 1) Solve any five questions.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data where ever necessary.
- 5) Figures to the right indicate full marks.

Q1) Find the flexibility influence coefficients for the following system.

[10]

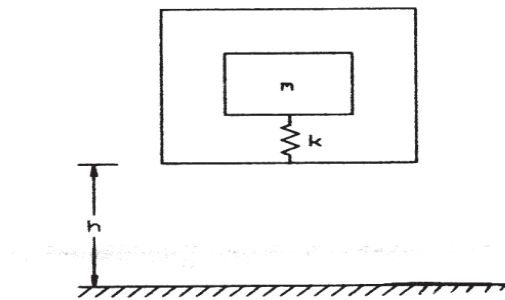


Q2) Determine the natural frequencies and mode shapes of torsional oscillations of a uniform shaft of length L , mass density ρ and the cross sectional polar moment of inertia J . The shaft is fixed at one end free at the other end. **[10]**

- Q3) a)** Explain FFT analyzer with schematic diagram of vibration measurement set up. **[5]**
- b)** Explain balancing of masses in single plane by vibration measurement technique. **[5]**

P.T.O.

- Q4)** A container having an apparatus suitably packaged inside it, is schematically represented in following fig. When the package is dropped on a hard surface during loading or unloading, analyse the system for its response. **[10]**



- Q5)** a) Explain how time domain and frequency domain techniques are used for condition monitoring and fault diagnosis. **[5]**
b) Explain Duhamel's Integral. **[5]**
- Q6)** A random signal has a spectral density that is a constant $S(f) = 0.004 \text{ cm}^2/\text{cps}$ between 20 and 1200 cps and that is zero outside this frequency range. Its mean value is 2.0 cm. Determine its rms value and its standard deviation. **[10]**
- Q7)** Write short notes on any three. **[10]**
- a) Auto correlation function.
 - b) Rayleigh Ritz method.
 - c) Analysis of narrow band system.
 - d) Experimental modal analysis.
 - e) Fault Diagnosis.



Total No. of Questions : 7]

SEAT No. :

P9468

[6182]-781

[Total No. of Pages : 2

M.E. (Mechanical - Design Engg.)
FINITE ELEMENT METHOD
(2017 Pattern) (Semester - II) (502209)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

Q1) a) Explain procedural steps in FEM.? Explain advantages and disadvantages of FEM over other methods. **[6]**

b) Short note on Rayleigh-Ritz method. **[4]**

Q2) a) Importance of boundary condition and explain

i) Essential boundary condition and

ii) Natural boundary condition **[5]**

b) Explain Timoshenko Beam Theory. Discuss its advantages and disadvantages. **[5]**

Q3) a) Explain the terms. **[6]**

i) Constant strain triangle (CST)

ii) Linear strain triangle (LST)

b) Evaluate the Integral using Gauss Quadrature Formula **[4]**

$$I = \int_{-1}^1 (2 + x + x^2) dx$$

Q4) a) Explain significance of mesh preparation in FEM with suitable example. **[4]**

b) Explain hexahedral and tetrahedral elements used in 3D problems in stress analysis. **[6]**

P.T.O.

- Q5) a)** Explain the following terms **[6]**
- i) Shear locking phenomenon
 - ii) Hour glass phenomenon in context with FEM
- b) Differentiate between Kirchhoff's and Mindlin's theory for thin plate formulations. **[4]**
-
- Q6) a)** Explain different source of non-linearity encountered in structural analysis. **[4]**
- b) Differentiate between Newton Raphson and modified Newton Raphson procedure for solving non-linear problems. **[6]**
-
- Q7) a)** Differentiate between 'h' & 'p' refinement techniques with suitable example. **[4]**
- b) Explain the different types and significance of symmetries used in FEM? **[6]**



Total No. of Questions : 7]

SEAT No. :

P9469

[6182]-783

[Total No. of Pages : 2

M.E. (Mechanical) (Design Engg.)

OPTIMISATION TECHNIQUES

(2017 Pattern) (Semester - III) (602213)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) What do you mean by optimisation? State different applications of engineering optimisation? **[5]**

b) Explain the characteristics mathematical models? **[5]**

Q2) Minimize the function $f(x) = 0.65 - [0.75/(1 + x^2)] - 0.65x \tan^{-1}(1/x)$ using Quasi-Newton method with starting point $x_1 = 0.1$ and step size $\Delta x = 0.01$. Use $|f'(x_{i+1})| \leq \varepsilon$ for checking convergence. Take $\varepsilon = 0.01$. **[10]**

Q3) Minimize the function using the golden section method with $n = 6$ $f(x) = 0.65 - [0.75 / (1 + x^2)] - 0.65x \tan^{-1}(1/x)$ within the range $[0,3]$. **[10]**

Q4) Maximize the following function with simple simplex method. **[10]**

$$\text{Maximize } Z = 60 X_1 + 50 X_2$$

$$\text{Subject to, } X_1 + 2X_2 \leq 40$$

$$3X_1 + 2X_2 \leq 60$$

$$X_1, X_2 \geq 0$$

P.T.O.

Q5) Explain in brief following methods of optimization. (any 2) **[10]**

- a) Simulated Annealing
- b) Genetic Algorithm
- c) Artificial neural network
- d) Teaching Learning Based Optimization

Q6) Write a short note on. **[10]**

- a) Evolutionary structural optimization for displacement or stiffness.
- b) Evolutionary structural optimization based on stress level.

Q7) a) What is topology optimization? Explain with suitable example. **[5]**

b) Write a short note Bi-Directional evolutionary optimization? **[5]**



Total No. of Questions : 10]

SEAT No. :

P9470

[6182]-784

[Total No. of Pages : 3

M.E. (Mechanical) (Design Engineering)
MECHANICAL MEASUREMENTS AND CONTROLS
(2017 Pattern) (Semester - III) (602214)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q. 3 or Q.4, Q.5 or Q.6, Q.7or Q.8, Q.9. or Q.10.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicates full marks.*
- 4) Use of electronic scientific pocket calculator is allowed.*
- 5) Assume suitable data, if necessary.*

Q1) a) Write a note on Working of LVDT with neat Sketch. **[5]**

b) Explain with neat sketch Resistance Thermometer. **[5]**

OR

Q2) a) Explain how FFT analyzer can be used for vibration measurement. **[5]**

b) Explain Laser Doppler Vibrometer in details. **[5]**

Q3) a) Explain Analog and Digital Signals with examples. **[5]**

b) Explain Bit Width. What is the difference between 32-bit and 64-bit system? **[5]**

OR

Q4) a) Explain ADC and List various types of ADC. **[5]**

b) What is Band Pass filter? Draw and explain the frequency response of a band pass filter. **[5]**

P.T.O.

- Q5) a)** Write a note on importance of controllability in control system. [5]
b) Reduce the block diagram Shown in Fig. Q5b. [5]

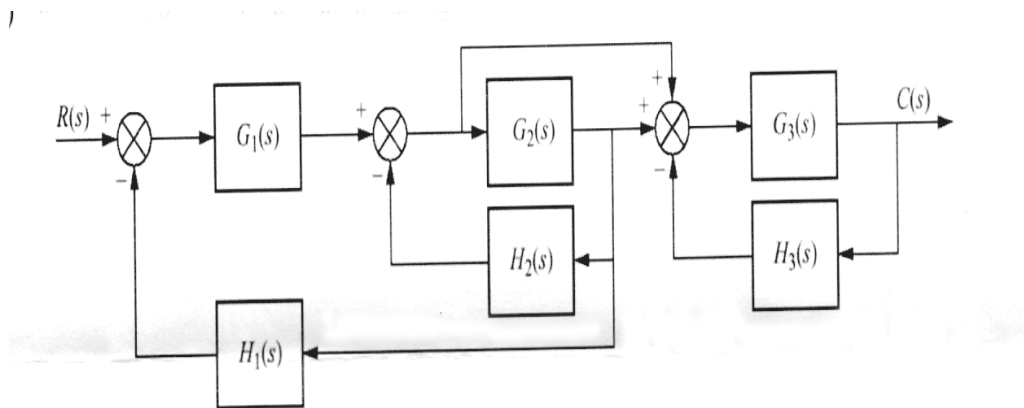


Fig. Q5b

OR

- Q6) a)** State and explain Routh Hurwitz criterion to determine Poles & Zeros of system. [5]
b) Explain the importance of Transient response specifications w.r.t. performance of control system. [5]
- Q7) a)** Write a note on proportional + integral + derivative controller. [4]
b) For the system shown in Fig. Q7b below, assume $m = 1$ kg, $k = \text{stiffness} = 2\text{N/m}$ and $d = \text{damping} = 0.5 \text{Ns/m}$. also $F = \text{force input}$ in N and $y = \text{displacement output}$ in M. [6]

For this system

- Determine the transfer function $y(s) / F(s)$
- Identify the location of poles and zeros
- Comment on the stability of system

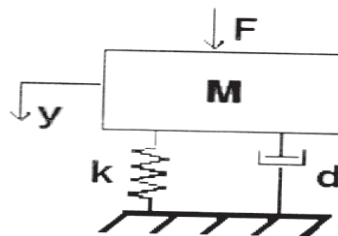


Fig. Q7b

OR

- Q8)** a) Explain two steps of Routh Hurwitz criterion in detail. [5]
b) Find the stability of feedback control system $2s^4+s^3+3s^2+5s+10=0$. [5]

- Q9)** a) Explain Phase margin and gain margin. [5]
b) Explain frequency domain analysis using Bode Plot. [5]

OR

- Q10)a)** Explain with example. [6]
i) Stable System
ii) Unstable System
iii) Critically or Marginally Stable System
b) Explain poles and Zeroes of control system. Explain Lyapunov's criterion for stability of system. [4]



Total No. of Questions : 8]

SEAT No. :

P9471

[Total No. of Pages : 3

[6182]-786

M.E. (Mechanical Engg.) (Energy/Heat Power Engg.)
ADVANCED MATHEMATICS AND NUMERICAL METHODS
(2017 Pattern) (Semester - I) (507101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any Five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Using Newton-Raphson method, find a root of the equation $x^3 - 3x + 1 = 0$, correct to 3 decimal places. [5]

b) Fit a straight line to the following data: [5]

x	6	7	7	8	8	8	9	9	10
y	5	5	4	5	4	3	4	3	3

Q2) a) From the given table find $F(22)$ using Newton's forward interpolation formula. [5]

x	20	25	30	35	40	45
F(x)	354	332	291	260	231	204

b) Solve the system of the equations by using LU decomposition method. [5]

$$3x + 2y + 7z = 4$$

$$2x + 3y + z = 5$$

$$3x + 4y + z = 7$$

Q3) a) By Cubic Spline, find $f(1.4)$ for the data. [5]

x	1	2	3	4
y	0	1	2	4

b) Use Romberg's method to compute $\int_0^1 \frac{dx}{1+2x}$. [5]

P.T.O.

Q4) a) Using power method to find the dominant eigen value of a matrix. [5]

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 10 \end{bmatrix}$$

b) Using three point Gaussian quadrature formula to evaluate $\int_0^1 \frac{dx}{x+1}$. [5]

Q5) a) Using House-Holders methods reduce the following matrix A to the tridiagonal form. [5]

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 4 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

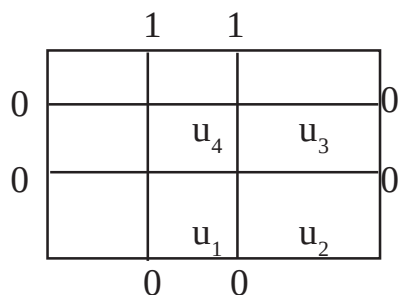
b) Compute $y(0.3)$, given that $\frac{dy}{dx} + y + xy^2 = 0$, $y(0.1) = 0.9006$ taking $h = 0.1$, using Runge-kutta method of fourth order. [5]

Q6) a) Find y at $x = 2$ using Adam's-Bashforth method given that. [5]

$$\frac{dy}{dx} = \frac{x+y}{2}, y(0) = 2, y(0.5) = 2.636$$

$$y(1) = 3.595, y(1.5) = 4.968$$

b) Use Jacob's iteration method to solve the partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ represented by mechanical system. (carry out five iterations). [5]



Q7) a) Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 2$, $u(0, t) = u(2, t) = 0$, $t > 0$ and

$u(x, 0) = \sin\left(\frac{\pi x}{2}\right)$, $0 \leq x \leq 2$ using $\Delta x = 0.5$ and $\Delta t = 0.25$ by crank-Nicholson method. [5]

b) Apply Gauss-seidel method to solve the following equations: [5]

$$20x + 4y - z = 32$$

$$2x + 17y + 4z = 35$$

$$x + 3y + 10z = 24$$

Q8) a) Use Henu's method to solve the initial value problem $\frac{dy}{dx} + xy^2 = 0$, $y(0) = 2$ to get $y(0.2)$ using step size of 0.2 [5]

b) Solve the wave equation for a vibrating string $u_{tt}(x, t) = 4u_{xx}(x, t)$ for $0 \leq x \leq 4$ and $0 \leq t \leq 2$. With the boundary conditions $u(0, t) = 0$, $u(4, t) = 0$, $t > 0$ and the initial conditions [5]

$$\left. \begin{array}{l} u(x, 0) = x(4 - x) \\ \frac{\partial u}{\partial t}(x, 0) = 0 \end{array} \right\} 0 \leq x \leq 4$$

taking $h = 1$, $k = \frac{1}{2}$ (solve for one period)



Total No. of Questions : 7]

SEAT No. :

P9472

[6182]-787

[Total No. of Pages : 2

M.E. (Mechanical) (Heat Power Engineering)
ADVANCED THERMODYNAMICS AND COMBUSTION
TECHNOLOGY
(2017 Pattern) (Semester - I) (502102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five questions.*
- 2) *Draw neat diagrams whenever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Use of steam tables, Mollier charts, scientific calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) State and explain law of corresponding states. Explain compressibility chart with suitable sketch. **[6]**
b) Explain Gibb's phase rule with example. **[4]**
- Q2)** a) Explain P-T diagram for a pure substance. **[4]**
b) One kg of water at 0°C is brought into contact with a heat reservoir at 90°C. When the water has reached 90°C find. **[6]**
i) Entropy change of water
ii) Entropy change of the heat reservoir
iii) Entropy change of the universe
- Q3)** a) State & Explain the third law of thermodynamics. **[4]**
b) A 50cm long, 800W electric resistance heating element whose diameter is 0.5 cm is immersed in 40kg of water initially at 20°C. Assuming the water container is well insulated, determine how long it will take for this heater to raise the water temperature to 80°C. Also, determine the minimum work input required and exergy destruction for this process in kJ Take $T_0 = 20^\circ\text{C}$ and $C_p = 4.18\text{kJ/kg}^\circ\text{C}$. **[6]**
- Q4)** a) Derive clapeyron - Clausius equation. **[5]**
b) Derive the relation **[5]**

$$(C_p - C_v) = -T \left(\frac{\partial V}{\partial T} \right)_P^2 \left(\frac{\partial P}{\partial V} \right)_T$$

P.T.O.

Q5) a) What is enthalpy of formation? How does it differ from the enthalpy of combustion? [4]

b) Liquid propane (C_3H_8) enters a combustion chamber at $25^\circ C$ at a rate of 1.2 kg/min where it is mixed and burned with 150% excess air that enters the combustion chamber at $12^\circ C$. If the combustion is complete and the exit temperature of the combustion gases is 1200K. determine

[6]

i) the mass flow rate of air and

ii) the rate of heat transfer from the combustion chamber take

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{285k}$ kJ/kmol	$\bar{h}_{298k}$ kJ/kmol	$\bar{h}_{1200k}$ kJ/kmol
$C_3H_8(e)$	-118910	-	-	-
O_2	0	8296.5	8682	38447
N_2	0	8286.5	8669	36777
$H_2O(g)$	-241820	-	9904	44380
CO_2	-393520	-	9364	53848

Q6) a) Define fugacity and activity. [4]

b) State and explain Dalton's law of partial pressure and Amagat's law. [6]

Q7) a) Explain thermodynamics of Aging and Death. [5]

b) If the average meal consumed by an adult consists of about 45.0% carbohydrate, 15.0% protein and 40.0% fat, determine [5]

i) the specific energy content of an average meal with natural state foods and

ii) the total mass of an average meal needed to provide 10.5 MJ per day.



Total No. of Questions : 7]

SEAT No. :

P9473

[Total No. of Pages : 2

[6182]-788

M.E. (Mechanical-Heat Power)
ADVANCED FLUID MECHANICS
(2017 Pattern) (Semester-I) (502103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

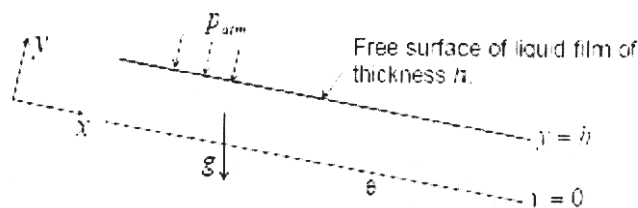
- 1) Answer any five questions.
- 2) Draw diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data wherever necessary.

Q1) a) Derive an expression for conservation of mass and momentum using differential analysis. [7]

- b) Water flows in a circular pipe. At one section, the diameter is 0.3m, the static pressure is 260 kPa, the velocity is 3m/s and the elevation is 10 m above ground level. The elevation at a section downstream is 0 m, and the pipe diameter is 0.15 m. Find the gauge pressure at the downstream section. Frictional effects may be neglected. Assume density of water to be 999 kg/m³. [3]

Q2) Derive generalized expression of Navier Stokes equations. [10]

Q3) a) Derive an expression for velocity and shear stress profile for fully developed flow down an inclined plane as shown in fig. Also estimate maximum shear stress for $\theta = 30^\circ$, $h=10$ mm, dynamic viscosity of liquid = 0.4 kg/m.s. $\rho = 920$ kg/m³ [5]



P.T.O.

- b) Explain the term; [5]
 i) Vorticity
 ii) Stream function
 iii) Velocity potential
- Q4)** a) Explain different types of free turbulent flows. What is a need of turbulence modeling. [5]
 b) Determine the radial pressure gradient at radius $r = 1.5$ m for a two-dimensional fluid motion defined by the stream function. [5]
 $\psi = \log_e r$
- Q5)** a) Obtain momentum-integral equation for the boundary layer. [5]
 b) A general sinusoidal velocity profile for laminar boundary-layer flow on a flat plate can be taken as
 $u = A \sin (By) + C$
 State the three boundary conditions applicable to the velocity profile, and thus evaluate the constants A, B, C. [5]
- Q6)** a) Derive an expression for lift on rotating cylinder. (Kutta-Joukowski theorem) [5]
 b) Water is flowing through a rough pipe, 400 mm in dia., at the rate of 2.5 m³/s. Calculate the power loss in overcoming friction per km length of pipe, if the roughness element height (ϵ) = 0.25 mm. Also calculate the maximum velocity. for laminar flow $f = \frac{64}{Re}$ [5]
 for turbulent flow $\frac{1}{\sqrt{f}} = 2 \log_{10} \left(\frac{R}{\epsilon} \right) + 1.74, u_c = \sqrt{\frac{\tau}{\rho}}$
 $\frac{u_{\max}}{u_c} = 8.5 + 2.5 \log_e \left(\frac{R}{\epsilon} \right)$
- Q7)** a) Develop area-velocity relationship in terms of Mach number and discuss effect of variation of area for subsonic, sonic and supersonic flows. [5]
 b) A normal shock wave in air Mach number is 1.8. The static pressure and temperature of the air upstream of the shock wave are 100 kN/m² and 15°C respectively, determine the Mach number, pressure, and temperature downstream of the shock. [5]



Total No. of Questions : 8]

SEAT No. :

P9474

[6182]-791

[Total No. of Pages : 3

M.E. (Mechanical-Heat Power Engineering)

ADVANCED HEAT TRANSFER

(2017 Pattern) (Semester - II) (502107)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary*

Q1) a) Write an expression for three dimensional heat diffusion equation in cylindrical Coordinates for unsteady heat conduction and deduce it into Fourier, Poisson and Laplace equation. **[5]**

b) Explain different initial and boundary conditions. **[5]**

Q2) a) What are Heisler charts? Explain their significance in solving transient conduction problems? **[4]**

b) A 60 mm thick large steel plate ($k = 36 \text{ W/mK}$, $\alpha = 0.040 \text{ m}^2/\text{h}$) initially at 400°C is suddenly exposed to an environment with $h = 220 \text{ W/m}^2\text{K}$ and temperature 35°C . Determine the centerline temperature and temperature inside the plate 15 mm from the midplane after 240 seconds. **[6]**

Q3) a) Explain the significance of fully developed flow and entrance length in forced convection problems. **[5]**

b) Write a note on Natural convection from finned surfaces and PCBs. **[5]**

P.T.O.

Q4) In a wind tunnel 15°C air at 300 m/min flows over a square flat plate of 1m side. The plate temperature is 35°C . Estimate. **[10]**

- a) Rate of heat transfer from the plate from one side
- b) Initial rate of cooling/hour of the plate if mass of the plate is 5 kg, C_p of the plate is 877 J/kg K.
- c) If the flow is turned off, compute the heat flow rate from the upper surface of the plate in still air at 15°C .
- d) Rate of percentage change in heat flow.

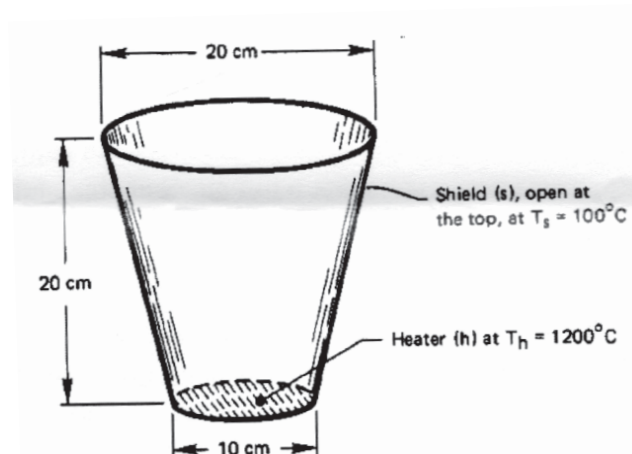
Use $Nu = 0.664 Re^{0.5} Pr^{0.33}$ and $Nu = 0.15 Ra^{0.33}$

Take properties of air as $k = 0.0281 \text{ W/mK}$, $\mu = 1.96 \times 10^{-5} \text{ kg/ms}$, $\rho = 1.087 \text{ kg/m}^3$ $C_p = 1.007 \text{ kJ/kg K}$.

Q5) a) How does the presence of a non-condensable gas in a vapor influence the condensation heat transfer? **[5]**

- b) With the help of neat figure explain the flow regimes for forced convection boiling in a tube. Show the variation in heat transfer coefficient over the length. **[5]**

Q6) a) A heater (h) shown in Figure, radiates to the partially conical shield, (s) that surrounds it. If the heater and surround are black, calculate the net heat transfer between heater to the shield. **[5]**



- b) Write a note on Radiation exchange between Emitting and Absorbing gases. **[5]**

Q7) Write a difference between.

[10]

- a) Nusselt and Biot number
- b) Velocity boundary layer in external and internal flow
- c) Solar and Thermal radiation
- d) Thermal resistance and contact resistance

Q8) a) Does the orientation of a heat pipe affect its performance? Does it matter if the evaporator end of the heat pipe is up or down? Explain. **[5]**

- b) Define characteristics of black body with the help of temperature, wave length and direction. **[5]**



Total No. of Questions : 7]

SEAT No. :

P9475

[6182]-792

[Total No. of Pages : 2

**M.E. (Mechanical) (Heat Power Engg.)
AIR CONDITIONING TECHNOLOGY
(2017 Pattern) (Semester - II) (502108)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any 5 questions out of 7.
- 2) Draw neat labeled diagrams wherever necessary.
- 3) Assume suitable data if necessary and mention it clearly.
- 4) Use of steam table, p-h chart, Psychrometric chart and scientific calculator is allowed.
- 5) Figures to the right side indicate full marks.

Q1) a) Explain the significance of Dry bulb temperature, Relative humidity, Bypass factor and Air washer. **[4]**

b) What is Air refrigeration system? Explain with neat sketch of Vapor Compression Cycle. **[6]**

Q2) a) What are the sources of indoor air pollution. **[4]**

b) Moist air enters an insulated duct at the rate of 10 kga/s at 20°C, 50 relative humidity and standard atmospheric pressure. It is heated by a heater of 25.0 kW capacity and steam at the rate of 0.1 kg/s is injected at 100°C. Find the outlet state, and the sensible and latent heat transfer rates. **[6]**

Q3) An air conditioning system is to take in outdoor air at 10°C and 30% relative humidity at a steady rate of 45m³/min and to condition it to 25°C and 60% relative humidity. The outdoor air is first heated to 22°C in the heating section and then humidified by the injection of hot steam in the humidifying section. Assuming the entire process takes place at a pressure of 1 bar. Determine. without using the psychrometric chart: **[10]**

a) The rate of heat supply in the heating section; and

b) The mass flow rate of steam required in the humidifying section.

P.T.O.

Q4) A laboratory having an unusually large latent heat gain is required to be air conditioned. The design conditions and loads are as follows: **[10]**

Summer design conditions: 40°C DBT, 27°C WBT

Inside design conditions: 25°C DBT, 50% RH

Room sensible heat: 34.9 kW

Room latent heat: 18.6 kW

The ventilation air requirement is 85 cmm. Determine the following:

- a) Ventilation load.
- b) Room and effective sensible heat factors.
- c) Apparatus dew point and amount of reheat for economical design.
- d) Supply air quantity.
- e) Condition of air entering and leaving coil and supply air temperature.
- f) Grand total heat.

Assume a suitable bypass factor.

Q5) a) Explain the types of Duct systems. Describe the procedure for the Duct design. **[5]**

b) With the help of example, explain equal friction method of Duct design. **[5]**

Q6) a) Explain the chilled water systems and their types. **[4]**

b) Short note on : **[6]**

i) Two stage Evaporative cooling

ii) Types of ventilation system.

Q7) a) Explain the significance of chiller ceiling and chiller beams. **[4]**

b) Write suitable Air conditioning equipment for the commercial building. **[6]**



Total No. of Questions : 8]

SEAT No. :

P9476

[6182]-793

[Total No. of Pages : 2

M.E. (Mechanical) (Heat Power Engineering)
MEASUREMENTS AND CONTROLS
(2017 Pattern) (Semester - II) (502109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables, slide rule, Mollier charts, electronic pocket.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Explain Allen salt velocity method and ultrasonic flow meter. [5]
b) Explain sources of systematic errors. [5]
- Q2)** a) Differentiate between null and deflection type instruments, Comment also on “Null method is often preferred over deflection method.” [5]
b) Enumerate the types of errors in measurements. Explain standard and probable error. [5]
- Q3)** a) Explain the following terms: [6]
i) Static error
ii) Dynamic error
iii) Probable error
iv) Mean Standard Deviation
b) Explain regression analysis. [4]
- Q4)** a) Calculate correlation coefficient from following data: [6]
 $N = 10, \Sigma X = 350, \Sigma Y = 310, \Sigma (X - 35)^2 = 162,$
 $\Sigma (Y - 31)^2 = 222, \Sigma (X - 35)(Y - 31) = 92.$ Also find regression line Y on X.
b) Explain the properties of regression and correlation coefficient. [4]

P.T.O.

- Q5)** a) What are the method of measuring humidity and liquid level? [5]
b) Which parameters are measured during vibration measurements? Explain piezoelectric accelerometer with neat sketch. [5]
- Q6)** a) Explain the construction and working of Foil type heat flux gauge. [5]
b) How do you measure frequency and phase angle. [5]
- Q7)** a) Explain any one instrument used for noise measurement. [5]
b) Explain construction and working of the torque measurement devices.[5]
- Q8)** a) Explain in detail working of electronic proportional and derivative controller. [5]
b) Design a control system for temperature control in a metal-melting furnace. [5]



Total No. of Questions : 7]

SEAT No. :

P9477

[6182]-795

[Total No. of Pages : 2

M.E. (Mechanical) (Heat Power)
COMPUTATIONAL FLUID DYNAMICS
(2017 Pattern) (Semester - III) (602113)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of electronic pocket calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Find the divergence of $\phi = V(u, v, w) = (4x, 2xy, 3z)$. **[5]**
- b) Explain the significance of the mass conservation equation in fluid mechanics and derive an expression for the same. **[5]**
- Q2)** a) Explain in detail Dirichlet and Robins boundary conditions. Draw neat sketch. **[4]**
- b) Explain the difference between explicit and implicit methods with its significant. **[6]**
- Q3)** a) Describe the physical meaning of Substantial and Local Derivative. **[4]**
- b) Derive the discretized form of the steady, one-dimensional, heat conduction equation. **[6]**
- Q4)** a) Write two dimensional heat convection-diffusion equation and discretize it with finite volume method. **[7]**
- b) Define Peclet number and state its importance? **[3]**

P.T.O.

Q5) a) Derive a solution of Navier-Stoke's equation for incompressible flow using SIMPLE algorithm. **[6]**

b) Explain the significance of 1-D transient convection-diffusion system. **[4]**

Q6) What are the difficulties in solving Navier-stokes Equations numerically? Discuss the remedy over the difficulties. Draw and discuss required control volumes. **[10]**

Q7) a) Explain the **[6]**

i) $k-\varepsilon$ turbulence model and

ii) $k-\omega$ turbulence model

b) Explain Reynolds average Navier stoke (RANS) in details. **[4]**



Total No. of Questions : 7]

SEAT No. :

P9478

[Total No. of Pages : 2

[6182]-796

**M.E. (Mechanical) (Heat Power Engineering)
DESING OF HEAT TRANSFER EQUIPMENTS
(2017 Pattern) (Semester - III) (602114)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right side indicates full marks.*
- 4) *Use of steam tables, scientific calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Explain in brief various applications of Heat exchangers. **[5]**

b) Explain heat exchanger design methodology with flowchart. **[5]**

Q2) a) Explain the Bell Delaware method for heat exchanger design. **[5]**

b) A double pipe counter flow heat exchanger is to cool ethylene glycol ($C_p = 2560 \text{ J/kg}^\circ\text{C}$) flowing at a rate of 3.5 kg/s from 80°C to 40°C by water ($C_p = 4180 \text{ J/kg}^\circ\text{C}$) that enters at 20°C and leaves 55°C . The overall heat transfer coefficient based on the inner surface area of the tube is $250 \text{ W/m}^2\text{C}$. Determine **[5]**

- i) The rate of heat transfer
- ii) Mass flow rate of water and
- iii) The heat transfer surface area on the inner side of the tube.

Take $C_{pw} = 4.18$ & C_p of ethylene glycol = $2.5 \text{ kJ/kg}^\circ\text{C}$.

Q3) a) State the assumptions made for pressure drop analysis of heat exchanger and give importance of pressure drop in heat exchanger analysis. **[5]**

b) What is chain rule methodology? How it is applied in heat exchange design. **[5]**

P.T.O.

Q4) In an oil to water heat exchanger, the oil enters the exchanger at 100°C with a heat capacity rate of 3700 W/K . Water is available at 15°C and 0.6 kg/s . Determine the exit temperature in **[10]**

- a) Counter flow and
- b) Parallel flow arrangements for $U = 500 \text{ w/m}^2\text{k}$ and surface area of 10m^2 . Consider $C_p = 1.88$ and 4.19 J/g.k for oil and water respectively. If the ratio of convection thermal resistances of oil to water is 1.2 and the wall and fouling resistance are negligible, calculate the wall temperature at each end of the counterflow and parallel flow exchanger

Q5) a) What are the different parameters contribution the pressure losses in shell and tube heat exchanger. **[5]**

- b) How cooling towers are classified? Explain the various factors affecting the performance of cooling tower. **[5]**

Q6) a) What are the different parameters cobsidered for furnaces design. **[5]**

- b) With neat sketch, explain the construction and working of Heat piple.**[5]**

Q7) Write short notes on (any two) **[5×2=10]**

- a) Heat transfer augmentation techniques.
- b) Insulation materials of furnaces.
- c) Kern's Method.



Total No. of Questions : 12]

SEAT No. :

P9479

[6182]-798

[Total No. of Pages : 2

M.E. (Computer Engg.)/(Data Science)/(Artificial Intelligence and Data Science)

**MATHEMATICAL FOUNDATION FOR DATA SCIENCE
(2017 Pattern) (Semester - I) (510301)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Assume Suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right side indicate full marks.

Q1) Write Difference between B tree & B+tree. Explain B + tree with example.[8]

OR

Q2) Represent given graph using adjacency list. Explain any shortest path algorithm and using same find the shortest path from vertex S to Vertex T using shortest path algorithm. [8]



- Q3) a) Compare Poisson distribution and Binomial distribution. [5]**
b) Explain two central tendency and dispersion measures of numerical data with example. [4]

OR

Q4) Bag A contains 10 marbles of which 2 are red and 8 are black. Bag B contains 12 marbles of which 4 are red and 8 are black. A ball is drawn at random from each bag. Find the probability that: [9]

- a) both are red
- b) both are black
- c) one black and one red
- d) at least one red

P.T.O.

Q5) Explain Chi-Square Tests and t-test with example. [9]

OR

Q6) Explain ANOVA Coefficient with example. [9]

Q7) Discuss different methods of to calculate Karl Pearson's Coefficient of Correlation. [8]

OR

Q8) a) Use these methods to normalize the following group of data: [4]

200, 300, 400, 600, 1000

i) z-score normalization

ii) z-score normalization using the mean absolute deviation instead of standard deviation.

b) Find covariance for following data set $x = \{2, 5, 6, 8, 9\}$
 $y = \{4, 3, 7, 5, 6\}$ Comment on the movement of X and Y values with respect to covariance. [4]

Q9) What is Jacobian? Let $x(u, v) = u^2 - v^2$, $y(u, v) = 2uv$. [8]

Find the Jacobian $J(u, v)$.

OR

Q10) Solve the system of equation $x_1 + x_2 + x_3 = 1$, $3x_1 + x_2 - 3x_3 = 5$ and $x_1 - 2x_2 - 5x_3 = 10$ by LU decomposition method. [8]

Q11) Suppose we have the following dataset with one response variable y and two predictor variables X1 and X2. Fit a multiple linear regression model to this dataset. [8]

X1	60	62	67	70	71	72	75	78
X2	22	25	24	20	15	14	14	11
Y	140	155	159	179	192	200	212	215

OR

Q12) Explain any one probabilistic model with example. [8]



Total No. of Questions : 12]

SEAT No. :

P9480

[6182]-799

[Total No. of Pages : 2

M.E. (Computer Engineering)/(Artificial Intelligence and Data Science)/(Master of Data Science)

**BASICS OF DATA SCIENCE
(2017 Pattern) (Semester - I) (510302)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume suitable data, if necessary.*

Q1) a) What is data science? Explain following steps involved in data science.[5]

- i) Data Exploration
- ii) Data Modelling

b) What is the role of a Data Scientist in the industry? [4]

OR

Q2) a) Explain the various types of data. [4]

b) What is big data? Explain characteristics of big data. [5]

Q3) While analyzing data, it is important to prepare the data in a format appropriate for analysis of the samples. State and explain steps involved in data preparation with example. [8]

OR

Q4) Explain following continuous distribution function in detail. [8]

- i) Gaussian distribution
- ii) Exponential distribution

Q5) What is naïve bayes classifier? Why is it called as naïve? Write how naïve bayes classifier works and explain with example. [9]

OR

Q6) Write kNN algorithm and explain with example. Also, state the cons of the kNN algorithm. [9]

P.T.O.

Q7) What is data visualization? Explain where it is used for? **[8]**

OR

Q8) What are the different types of data visualization? Explain any four of them. **[8]**

Q9) What is a recommender system? Explain long tail phenomenon in recommendation system. Also, state the applications of recommendation system. **[8]**

OR

Q10) How does content based recommendation system work? State pros and cons of content based recommendation system. **[8]**

Q11) Explain Social Networks as Graphs and following varieties of social networks. **[8]**

- i) Collaboration Networks
- ii) Email Networks

OR

Q12) Write Girvan-Newman algorithm and explain with example. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9481

[Total No. of Pages : 1

[6182]-800

M.E. Computer Engg. (Artificial Intelligence and Data Science)

ARTIFICIAL INTELLIGENCE

(2017 Pattern) (Semester-I) (510501)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) What is artificial Intelligence? Give the different definitions with the concept of Rationality. **[9]**

OR

Q2) Explain different utility functions in making simple decisions. **[9]**

Q3) Give the heuristic functions for 8 puzzle problem and explain with example. **[9]**

OR

Q4) Explain reinforcement learning with an example. **[9]**

Q5) a) Give Horn clauses and Definite clauses with example. **[8]**

b) Explain forward chaining algorithm with an example. **[8]**

OR

Q6) a) Explain Recommender system architecture with functions. **[8]**

b) How AI can be used in Sentiment analysis? **[8]**

Q7) a) How planning can be solved as forward state space search? Explain with example. **[8]**

b) How is the knowledge represented using First order logic? **[8]**

OR

Q8) a) Explain Hill climbing search with example. **[8]**

b) How genetic algorithm can be used to solve optimization problems? **[8]**



Total No. of Questions : 6]

SEAT No. :

P9482

[6182]-801

[Total No. of Pages : 2

**M.E. (Computer Engineering) (Artificial Intelligence and Data
Science) (Cyber Security)**

**RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (510101)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat diagrams wherever necessary.*

Q1) a) What is importance of Research? What are Research paradigms? Elaborate the terms like research objectives and research outcomes in detail? **[9]**

OR

b) What is the significance of the code of ethics in research? Which are the 'ethical aspects' affecting and influencing engineering research? **[9]**

Q2) a) What is a research plan? What are elements getting generically included in Research Proposal? **[8]**

OR

- b) Elaborate the significance of the following in research methodology **[8]**
- i) h-index
 - ii) Patents as a type of research publication
 - iii) Research Gate in literature survey
 - iv) Paraphrasing in literature survey

Q3) a) What is the hypothesis and the Null hypothesis? Differentiate between One- dimensional statistics and Two-Dimensional Statistics? **[8]**

OR

- b) State the use of the following tools **[8]**
- i) ANOVA
 - ii) SOFA
 - iii) t-test
 - iv) CAT (Qualitative analysis tool)

P.T.O.

Q4) a) State the importance of optimization in engineering research? With the help of an example, explain the monte Carlo Optimization Method? **[8]**

OR

b) State the simplex optimization steps? What are constraints and cost function? State the similarity and differences in simplex and gradient methods of optimization? **[8]**

Q5) a) What are the guidelines for conducting surveys? What is role of statistical analysis in surveys? What are guidelines for survey reporting? **[8]**

OR

b) What is significance of the research proposal? What is an outline of a research proposal? What is the criteria for evaluating the research proposal? **[8]**

Q6) a) What is research Thesis? What are generic contents of the organization of thesis? Why the 'appendix' part in the thesis is important? **[9]**

OR

b) Elaborate following research outcomes and when which publication is to be attempted. **[9]**

- i) InPASS (Indian Patents Advanced Search System)
- ii) Copyright
- iii) Research Paper



Total No. of Questions : 8]

SEAT No. :

P9483

[6182]-803

[Total No. of Pages : 2

M.E. (Artificial Intelligence and Data Sciences)

DATA WAREHOUSING AND MINING

(2017 Pattern) (Semester - II) (510308)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) In context of “failures of past decision-support systems”, what are the factors relating to the inability to provide strategic information? **[4]**
- b) List different types of data warehouse Architectures and Explain any one. **[4]**
- c) What is data Mart? **[2]**
- Q2)** a) How does a snowflake schema differ from a STAR schema? **[4]**
- b) Name five types of the major data transformation tasks. Give an example for each. **[4]**
- c) What is fact table? **[2]**
- Q3)** a) Give three reasons why you think ETL functions are most challenging in a data warehouse environment. **[4]**
- b) Draw and explain Major steps ETL. **[4]**
- c) What is Dimension table? **[2]**
- Q4)** a) Discuss two reasons why feeding data into the OLAP system directly from the source operational systems is not recommended. **[4]**
- b) Explain Rollup and Drill down OLAP Operations with example. **[4]**
- c) What is meant by slice-and-dice? **[2]**

P.T.O.

- Q5)** a) Describe data classification with Decision tree classifier. [4]
b) Draw and explain phases in Knowledge discovery in brief. [4]
c) What is Data Mining? [2]
- Q6)** a) What are the major deployment activities in data warehouse deployment? [4]
b) Explain how data partition helps in data warehouse as a performance enhancement technique? [4]
c) Why parallel processing is required in data warehouse? [2]
- Q7)** a) What is market basket analysis? Give two examples of this application in business. [5]
b) Give three broad reasons why you think data mining is being used in today's businesses. [5]
- Q8)** a) Explain in brief applications of the data mining. [5]
b) Explain various OLAP models in brief [5]



Total No. of Questions : 12]

SEAT No. :

[Total No. of Pages : 2

P9484

[6182]-804

ME (AI & DS)

MACHINE LEARNING

(2017 Pattern) (Semester - II) (510309)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) Explain supervised learning and Unsupervised learning with example? [9]

OR

Q2) Write a short note on (Any 3) [9]

- a) Geometric model
- b) Probabilistic Models
- c) Logical Models
- d) Parametric and non-parametric models

Q3) What is principal component Analysis (PCA), when it is used? Explain in Detail. [9]

OR

Q4) Write a short note on [9]

- a) Linear discriminant analysis
- b) Bias/Variance tradeoff
- c) Probably Approximately Correct learning.

Q5) a) What do you mean by a linear regression? Which applications are best modeled by linear regression? [9]

OR

P.T.O.

Q6) What do you mean by support vector machine? Explain with example. **[9]**

Q7) Explain Decision tree algorithm ID3, Minimum Description length decision trees and regression trees. **[9]**

OR

Q8) Write a short note on Ensemble methods Bagging and Boosting. **[9]**

Q9) Define Bayes Theorem. Elaborate Naive Bayes Classifier working with example. **[9]**

OR

Q10) Write a short note on **[9]**

- a) Hidden Markov model
- b) Gaussian Mixtures and compression based models

Q11) Explain Text mining tasks like semantic analysis, author profiling and author identification. **[5]**

OR

Q12) Write a short note on Diagnosis of crop disease? **[5]**



Total No. of Questions : 12]

SEAT No. :

P9585

[6182]-805

[Total No. of Pages : 2

**M.E. (Artificial Intelligence and Data Science)
VIRTUAL REALITY AUGMENTED REALITY
(2017 Pattern) (Semester - II) (510504)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data if necessary.*

Q1) Why geometric modeling need to study? How it rotations works. **[9]**

OR

Q2) What do understand taxonomy of mixed reality. **[9]**

Q3) Explain visual rendering with suitable example. **[9]**

OR

Q4) Justify the difference between ray tracing and shading models. **[9]**

Q5) Write a short note on augmented reality application. **[9]**

OR

Q6) Explain various component which are used augmented reality. **[9]**

Q7) Explain marker less approaches? How it work. **[9]**

OR

Q8) What are marker types? Why they are important. **[9]**

P.T.O.

Q9) Explain the overview of game development in Unity IDE. **[9]**

OR

Q10) Why high level concept of content creation are important? **[9]**

Q11) How setup unity with virtual reality. **[5]**

OR

Q12) Explain vuforia environment for application development. **[5]**



Total No. of Questions : 8]

SEAT No. :

P9486

[6182]-807

[Total No. of Pages : 2

M.E. (Artificial Intelligence and Data Sciences)

SOFT COMPUTING AND DEEP LEARNING

(2017 Pattern) (Semester - III) (610501)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What are various fuzzy set operations? Explain 3 operations with examples. [4]
b) Write two applications each for Neural Networks and Genetic Algorithms. [4]
c) What are features of soft computing? [2]
- Q2)** a) Explain any three fuzzy membership functions. [4]
b) Write and explain algorithm for Genetic algorithm. [4]
c) Explain concept of Fitness function with example. [2]
- Q3)** a) Consider some suitable problems and compare necessity of Shallow neural networks and Deep neural networks for solving such a problems. [4]
b) Draw and explain architecture of Feed forward Neural networks. [4]
c) What is Perceptron? [2]
- Q4)** a) Explain the vanishing gradient problem, and ways to mitigate it. [4]
b) Explain in brief the heuristic to avoid bad Local Minima and the heuristics for faster in Neural Network? [4]
c) What's the main drawback of the ReLU activation function? [2]

P.T.O.

- Q5)** a) Explain working of Convolutional neural networks with help of Convolutional layer, Pooling layer and Fully-connected (FC) layer. [4]
- b) Explain deep art and deep dream in deep learning? [4]
- c) What is fooling the Convolutional network? [2]
- Q6)** a) Explain how Recurrent Neural Network is useful in natural language processing. [4]
- b) Explain LSTM architecture with different types of gates used in it. [4]
- c) Explain how GRU is different than LSTM? [2]
- Q7)** a) Design a Fuzzy Logic Controller for Air conditioning systems. [5]
- b) Explain how CNN can be applied to solve problem of Image classification for identifying cat vs dogs using CNN. [5]
- Q8)** a) Explain any one Hybrid soft computing and with suitable example show that how hybridization is required in solving this problem. [5]
- b) Design solution to resolve traveling salesman problem by applying Genetic algorithm. [5]



Total No. of Questions : 12]

SEAT No. :

P9487

[Total No. of Pages : 2

[6182]-809

M.E. (Computer Data Science) (Artificial Intelligence)

SCALABLE DATA SCIENCE

(2017 Pattern) (Semester - III) (610502)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10, Q11 or Q12.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) Elaborate various challenges associated with big data analysis and discuss the use of bloom filter with suitable example. **[9]**

OR

Q2) Explain Misra-Gries sketch algorithm with suitable example. **[9]**

Q3) Compare and contrast RDDs versus Data Frames. What are the benefits of using Data frames over RDDs? **[9]**

OR

Q4) Explain the use of RDDs transformations and actions WRT Apache Spark SQL. Compare and contrast RDDs versus Data Frames. **[9]**

Q5) Explain statistical calculations skewness, Kurtosis, Covariance, Covariance matrices using suitable examples. **[9]**

OR

Q6) List various correlation methods to achieve parallelism in Apache spark. **[9]**

Q7) Enlist various API types that are compatible with Spark? How is feature selection used in pipeline? **[9]**

OR

P.T.O.

Q8) Explain default level of parallelism in Apache Spark and give suitable example of K-Means clustering in Spark. **[9]**

Q9) Elaborate different kinds of graphs used in Spark? What are the structural operators provided in the GraphX library? **[9]**

OR

Q10) Can Apache Spark handle graph processing? Justify your answer. **[9]**

Q11) Elaborate set operations using Spark R data frames. Explain advantages and limitations of using R environment. **[5]**

OR

Q12) Explain how Spark is good at low latency work loads like graph processing and machine learning? **[5]**



Total No. of Questions : 8]

SEAT No. :

P9488

[Total No. of Pages : 2

[6182]-811

M.E. (Chemical Engineering)

MATHEMATICAL AND STATISTICAL METHODS

(2017 Pattern) (Semester - I) (509101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) a) Solve the differential equation by power series method $y' - y = 0$. **[5]**

b) Expand the following function in terms of Legendre polynomials in the interval $-1 < x < 1$ $f(x) = x^2 + 2x^2 - x - 3$. **[5]**

Q2) a) What does convergence of a power series mean? Why it is important. **[4]**

b) Explain Laplace equation in cylindrical form. **[6]**

Q3) a) Find the Laurent series expansion of $(z-1)/z^2$ for $|z - 1| > 1$. **[5]**

b) Obtain the Laplace's equation in two dimensions for heat flow. **[5]**

Q4) a) Evaluate $\int_0^{1+i} (x^2 + iy) dz$ along the paths $y = x$ and $y = x^2$. **[6]**

b) Find the Laurent's expansion of $f(z) = (7z - 2)/(z + 1) z(z - 2)$ in the region $1 < z < 3$. **[4]**

P.T.O.

- Q5) a)** Solve the following system of ODE using Laplace transform method using initial conditions as $x(0) = 1, y(0) = 0$. **[4]**

$$x'' - y = e^t$$

$$y + z = \sin t$$

- b)** Genetic theory state that children having one parent of blood type A and the other blood type B will always be one of three types. A, AB, B and that the proportion of three types will on an average be as 1:2:1. A report states that out of 300 children having one A parent and B parent, 30% were found to be types A, 45% type AB and remainder type B. Test the hypothesis by χ^2 test. (P^* = probability under H_0 that $\chi^2 > \chi^2$ at 5%). **[6]**

Degree freedom	1	2	3	4	5
P^*	3.841	5.991	7.815	9.488	11.070

- Q6) a)** The mean of two single large samples of 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from same population of standard deviation 2.5 inches (Test at 5% level of significance). **[5]**

- b)** Explain residual error in regression model. **[5]**

- Q7) a)** Discuss the procedure to calculate the mean from ungrouped data? **[6]**

- b)** Discuss brief about test of goodness of fit. **[4]**

- Q8) a)** Fit a straight-line $y = a + bx$ to the following data by method of least squares. **[5]**

$x:$ 0 1 3 6 8

$y:$ 1 3 2 5 4

- b)** Comment on factorial experimentation with suitable example. **[5]**

x x x

Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

P9489

[6182]-812

M.E. (Chemical)

PROCESS OPTIMIZATION

(2017 Pattern) (Semester - I) (509102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Discuss Optimization. Explain Nature and Scope of Optimization. [5]
b) Write short note on unimodal and multimodal function. [5]

- Q2)** a) Explain in detail Optimality Conditions for One Dimensional and Multi Dimensional Optimization. [5]
b) Write short note on convex and concave function. [5]

- Q3)** a) Using pattern search method, perform one iteration to minimize the function $f(x) = X_1 - X_2 + 2X_1^2 + 2X_1X_2 + X_2^2$ Assume initial point as (0, 0). [5]
b) Explain steepest descent method for multi-variable optimization. [5]

- Q4)** Write short notes on:
a) Secant method. [5]
b) Evolutionary optimization method. [5]

- Q5)** Use Fibonnaci method to maximize $Z = 24X - 2X^3$ in the range 0 to 10 to an accuracy of 1% carry out 4 iteration. [10]

P.T.O.

Q6) a) What is fitness function in Genetic Algorithm? [5]

b) What is 'Mutation' in Genetic Algorithm? [5]

Q7) Minimize using Linear Programming. [10]

$$F = 4X + 5Y$$

Subjected to

$$10X + Y > 10$$

$$5X + 4Y > 20$$

$$3X + 7Y > 21$$

$$X + 12Y > 12$$

$$X > 0 \text{ and } Y > 0$$

Q8) a) Priori approach for Multi-objective optimization. [5]

b) What is fitness function in Genetic Algorithm? [5]



Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

P9490

[6182]-813

M.E. (Chemical)

ADVANCED SEPARATION PROCESSES

(2017 Pattern) (Semester-I) (509103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*

Q1) a) Distinguish between conventional and advanced separation processes in chemical Engineering. **[5]**

b) Explain in brief surface based solid-liquid separations involving Sirofloc filter. **[5]**

Q2) a) Describe in brief spiral wound and hollow fibre membrane. **[5]**

b) Explain : **[5]**

i) Reverse Osmosis

ii) Dialysis

Q3) a) Explain in brief : Design aspects of Nanofiltration. **[5]**

b) State the principles of prevaporation and explain its mechanism in brief. **[5]**

Q4) a) Explain in brief different types of adsorption techniques & laws of adsorption isotherms. **[7]**

b) State advantages of Reactive Adsorption. **[3]**

Q5) a) What do you mean by chromatography? Explain its importance in chemical Industries. **[5]**

b) Explain in brief Ion exchange chromatography. **[5]**

P.T.O.

Q6) a) Describe azeotropic distillation and reactive distillation. **[6]**

b) Write a short note on supercritical fluid extraction. **[4]**

Q7) Explain in brief Industrial effluent treatment by using modern techniques. **[10]**

Q8) a) Describe in brief : HPLC for analytical separation. **[5]**

b) Describe equipment used for electrophoresis. **[5]**



Total No. of Questions : 8]

SEAT No. :

P9491

[6182]-814

[Total No. of Pages : 2

M.E. (Chemical Engineering)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (509104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) Explain Significance of Research. [5]
b) Write a note on manpower needs of a research project. [5]

- Q2)** a) Compute the coefficient of correlation between X and Y using following data. [5]

X	1	3	5	7	8	10
Y	8	12	15	17	18	20

- b) Discuss fundamentals of time series analysis and spectral analysis along with one application in research work. [5]
- Q3)** a) Explain ANOVA with suitable example. [5]
b) Write an explanatory note on time series analysis of probability distribution with a suitable example. [5]
- Q4)** a) For effective research what are the different methods used for collection of data? Explain the advantages and disadvantages of each method. [6]
b) What is design of Design of Experiments? Explain with suitable example. [4]

P.T.O.

- Q5)** a) Discuss the steps involved in publishing a research idea in a journal. [6]
b) A sample of 10 is drawn randomly from a certain population. The sum of the squared deviations from the mean of the given sample is 50. Test the hypothesis that the variance of the population is 5 at 5 per cent level of significance. [4]
- Q6)** a) Explain the sources which can and/or should be used for selecting a research problem. [5]
b) Write an explanatory note on citation methods for project writing. [5]
- Q7)** a) Write a note on the layout and contents of research report. [5]
b) What do you understand by term “Intellectual Property Rights (IPR)”?
What is importance of IPR in research work? What benefits researcher can receive through generation of IPR? [5]
- Q8)** a) Write an explanatory note on “interdisciplinary research”. [5]
b) Write an explanatory note on research hypothesis. [5]



Total No. of Questions : 8]

SEAT No. :

P9492

[6182]-816

[Total No. of Pages : 2

M.E. (Chemical Engineering)
ADVANCED TRANSPORT PHENOMENA
(2017 Pattern) (Semester - II) (509107)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions*
- 2) *Assume suitable data, if necessary*
- 3) *Neat diagrams must be drawn wherever necessary*
- 4) *Use of electronic pocket calculator is allowed*

Q1) Derive the momentum flux relation in a Couette viscometer. which is a rotational viscometer in which the viscosity of a liquid is determined by measuring the torque required to turn a solid object in contact with the liquid. Apply the equations of continuity and motion for constant ρ and μ to the tangential flow in the annular region around the bob. **[10]**

$$T_z = (-\tau_{r\theta})|_{r=\kappa R} \cdot 2\pi\kappa RL \cdot \kappa R = 4\pi\mu\Omega_0 R^2 L \left(\frac{\kappa^2}{1-\kappa^2} \right)$$

Where, Ω_0 = constant angular velocity, R =radius of the rotating outer cylinder, κR = radius of the stationary inner cylinder, L = length of cylinders.

Q2) An incompressible fluid flows through a tube of circular cross section, for which the tube radius changes linearly from R_0 , at the tube entrance to a slightly smaller value R_L at the tube exit. Assume that the Hagen-Poiseuille equation is approximately valid over a differential length, dz . of the tube so that the mass flow rate is

$$W = \frac{\pi[R(z)]^4}{8\mu} \rho \left(-\frac{dp}{dz} \right)$$

Solve the equation to get the mass flowrate 'w' relation for the problem conditions. **[10]**

P.T.O.

- Q3)** A sphere of radius R and thermal conductivity k_1 , is embedded in an infinite solid of thermal conductivity k_0 . The center of the sphere is located at the origin of coordinates. and there is a constant temperature gradient A in the positive z direction far from the sphere. The temperature at the center of the sphere is T^0 . Derive the steady-state temperature distributions in the sphere T_1 , and in the surrounding medium T_0 to show that [10]

$$T_1(r, \theta) - T^0 = \left[\frac{3k_0}{k_1 + 2k_0} \right] Ar \cos \theta$$

- Q4)** Discuss the boundary layer theory for non-isothermal flow around a submerged object. [10]
- Q5)** Gas A is absorbed by a stationary liquid solvent S, the latter containing solute B. Species A reacts with B in an instantaneous irreversible reaction according to the equation $aA + bB \rightarrow \text{Products}$. It may be assumed that Fick's second law adequately describes the diffusion processes, since A, B, and the reaction products are present in 'S' in low concentrations. Obtain expressions for the concentration profiles. [10]
- Q6)** Discuss the time-smoothing of the equations of continuity and motion for component 'A' disappearing by an n^{th} order chemical reaction in rectangular coordinates. [10]
- Q7)** A liquid 'B' is flowing very slowly around a spherical bubble of gas 'A' of radius 'R' (creeping flow). Develop the relation for the determination of the rate of mass transfer of 'A' into the surrounding fluid ' N_{A0} ', if the solubility of gas 'A' in liquid 'B' is C_{A0} . State the assumptions. [10]
- Q8)** An electrical heating coil of diameter D is being designed to keep a large tank of liquid above its freezing point. It is desired to predict the temperature that will be reached at the coil surface as a function of the heating rate Q and the tank surface temperature T^0 . This prediction is to be made on the basis of experiments with a smaller, geometrically similar apparatus filled with the same liquid. Discuss a suitable experimental procedure for making the desired prediction based on dimensional analysis. Temperature dependence of the physical properties, other than the density, may be neglected. The entire heating coil surface may be assumed to be at a uniform temperature T_1 . [10]



Total No. of Questions : 8]

SEAT No. :

P9493

[6182]-817

[Total No. of Pages : 2

M.E. (Chemical)

**ADVANCED PROCESS CONTROL
(2017 Pattern) (Semester - II) (509108)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) The impulse response readings of a system strongly suspected to be first order is given in the table below. From the supplied data, find the steadystate gain and the time constant for this system. Assume $\Delta t = 0.5$ min. How good is the postulate that the system is first order? **[10]**

Time t_k (min)	Impulse Response $g(t_k)$
0	4.22
0.5	3.03
1	1.84
1.5	1.12
2	0.68
2.5	0.41
3	0.25
3.5	0.15
4	0.09
4.5	0.06
5	0.03
5.5	0.02
6	0.01
6.5	0.008
7	0.005

P.T.O.

Q2) Explain the cascade control system of a process furnace with neat process instrumentation and control diagram. **[10]**

Q3) A 2×2 MIMO process has the following steady-state gain matrix, **[10]**

$$K = \begin{bmatrix} 1 & 0 \\ 10 & 1 \end{bmatrix}$$

Calculate the singular values and the condition number of the system and comment on the conditioning of the system.

Q4) Explain the standard input excitation functions with equations and graphs. What is PRBS? How it is more helpful in process identification? **[10]**

Q5) a) Explain the concepts of Internal Model Control with respect to controlling a first order process. Draw block diagram and give equations of same. **[5]**

b) Explain the concepts of Dynamic Matrix Control (DMC) with equations and the response graph. **[5]**

Q6) Explain the top-down analysis of a typical plant wide control design problem along with all the steps involved. **[10]**

Q7) Explain the stability analysis of a typical first order discrete-time systems in the complex plane. **[10]**

Q8) Discuss the propagation and recirculation of disturbances for suitable process examples. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9494

[6182]-818

[Total No. of Pages : 2

M.E. (Chemical Engineering)
ADVANCED REACTION ENGINEERING
(2017 Pattern) (Semester - II) (509109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) Discuss the various factors responsible for determination of rate controlling step in the fluid-particle reaction. **[10]**

Q2) Explain modeling diffusion without reaction. **[10]**

Q3) Explain in brief Langmuir-Hinshelwood model. **[10]**

Q4) Explain Rideal Eiley Model and two phase fluidised bed model. **[10]**

Q5) Explain various types of heterogeneous reactions. **[10]**

Q6) The first order reaction $A \rightarrow B$ was carried out over two different-sized pellets. The pellets were contained in a spinning basket reactor that was operated sufficiently high rotation speeds that external mass transfer resistance was negligible. The results of two experimental runs made under identical conditions are given in table Estimate the Thiele modulus and effectiveness factor for each pellet. How small should the pellets be made to virtually eliminate all internal diffusion resistance? **[10]**

RunNo.	Measured Rate (mol/g.cat.s) $\times 10^5$	Pellet Radius (m)
Run 1	3.0	0.01
Run 2	15.0	0.001

P.T.O.

Q7) a) Determine the amount of catalyst required in packed bed reactor for 80% conversion of 1000 mol/min if $C_{A0} = 8 \text{ mol/m}^3$ of feed. **[6]**

b) Write note on isothermal and adiabatic fixed bed reactors with suitable illustration. **[4]**

Q8) Show that $X_{T-L-S} = X_{mm}$ for a first order reaction $A \xrightarrow{k} B$. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9495

[Total No. of Pages : 1

[6182]-820

M.E. (Chemical)

PROCESS MODELING AND SIMULATION

(2017 Pattern) (Semester - III) (509113)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any 5 questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*
- 4) *Use of logarithmic tables slide rule, Mollier charts, and electronic pocket calculator and steam table is allowed.*

Q1) Explain with suitable example modeling. **[10]**

Q2) Explain it in detail, a steady state model with suitable examples. **[10]**

Q3) Explain different methods of Design of Experiments. **[10]**

Q4) Develop a model for Plug flow Reactor. **[10]**

Q5) Write an assumption for a model of Packed Extractor & develop a model. **[10]**

Q6) Give the simulation scheme for CSTR. Explain it with a case study. **[10]**

Q7) A patient just had surgery and is required to have at least 81 units of drug D1 and 120 units of drug D2 each day. Assume that an over dosage of either drug is harmless. Each gram of substance M contains 10 units of D1 and 8 units of drug D2 and each gram of substance N contains 2 units of D1 and 4 units of D2. Now suppose that both M and N contain an undesirable drug D3, 3 units per gram in M and 1 unit per gram in N. Find how many grams of substances M and N should be taken in order to meet the requirements and minimize the intake of D3 at the same time. **[10]**

Q8) Explain numerical methods pertaining to Chemical engineering. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9496

[Total No. of Pages : 2

[6182]-821
M.E. (Chemical Engg.)
ADVANCED THERMODYNAMICS
(2017 Pattern) (Semester - III) (509114)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*
- 4) *Use of electronic pocket calculator is allowed.*

Q1) The excess enthalpy for a liquid mixture of species 1 and 2 at fixed T and P are represented by the equation:

$$H^E = x_1 x_2 (40 x_1 + 20 x_2)$$

Where H^E is in J/mol Determine the expressions for H_1^E and H_2^E as functions of x_1 . **[10]**

Q2) Calculate the values of fugacity f_i and the fugacity coefficient Φ_i for water at 300 °C and 100 bar. For the integration following data is given: $\Phi_i^{\text{sat}} = 0.784$, $P_i^{\text{sat}} = 8592.7$ kPa, Molar vol. of saturated liquid water at 300 °C = 25.28 cm³/mol.

[10]

Q3) Estimate the adiabatic flame temperature when ethane C₂H₆ (g) is burned with 150% theoretical air. The fuel and air enter the burner at 1 bar and 298 K and the combustion products leave the burner at 1 bar. The molar heat capacities of CO₂, H₂O, O₂ and N₂ are constant and are given by 53.1 J/mol K, 41 J/mol K, 34.3 J/mol K and 32.2 J/mol K respectively. Heat of reaction at standard temperature is - 1429.37 kJ/mol. **[10]**

Q4) A isomerisation reaction $A \rightarrow B$ occurs in a liquid phase, where A and B are miscible liquids for which: $GE / RT = 0.1 x_A x_B$. If $\Delta G_{298}^\circ = -1$ kJ/mol, What is the equilibrium composition of the mixture at 25°C? How much error is introduced if one assumes that A and B form an ideal mixture? **[10]**

P.T.O.

- Q5)** Suppose a certain quantity of energy Q is transferred from a body at constant temperature T_1 to another body at constant temperature T_2 ($T_1 > T_2$). Is there any loss in the available energy due to the process of energy transfer, if the ambient temperature is T_0 ? If so derive the loss in the available energy. **[10]**
- Q6)** Explain the concept of Partition Function in statistical thermodynamics in detail with example. **[10]**
- Q7)** When an ordinary rubber band is hung from a clamp and stretched with constant downward force F by a weight attached to the bottom end, gentle heating is observed to cause the rubber band to contract in length. To keep the length L of the rubber band constant during heating, F must be increased. The stretching work is given by $dw = FdL$. From this information, find the sign of the partial derivative $(\partial T / \partial L)_{S,P}$. Then predict whether stretching of the rubber band will cause a heating or a cooling effect. **[10]**
- Q8)** Discuss the effect of gravity on composition of a binary mixture by deriving the appropriate thermodynamic relations. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9497

[Total No. of Pages : 2

[6182]-823

M.E. (Chemical) (Environmental Engineering)

APPLIED STATISTICS FOR ENVIRONMENTAL ENGINEERING

(2017 Pattern) (Semester - I) (509131)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Growth of bacteria (N) in a culture after 1 hr is given in the following table

t	0	1	2	3	4	5	6
N	32	47	65	92	132	190	275

Fit a curve of the form $N = a b^t$ and estimate N when $t = 7$ using least square method. **[5]**

b) Explain test of skewness. **[5]**

Q2) An analysis of monthly wages paid to the workers of two firms A and B belonging to same industry gives the following results: **[10]**

	Firm A	Firm B
Number of workers	500	600
Average daily wage, Rs./-	186	175
Variance of distribution of wages	81	100

- a) Which firm, A or B, has a largest wage bill
- b) In which firm, A or B, is there greater variability in individual wages.
- c) Calculate
 - i) the average daily wage and
 - ii) the variance of distribution of wages of all the workers in the firms A and B taken together.

P.T.O.

- Q3)** a) Explain Convexity of frequency related to kurtosis. [5]
 b) Discuss relation between moments about mean in terms of moments about any point. [5]
- Q4)** a) Explain the method of moments generating function for the CHI-square distribution. [5]
 b) Explain the tests of significance for attributes. [5]
- Q5)** a) Define the terms [3]
 i) Precision
 ii) Parameters
 iii) Factorial experiment
 b) State the advantages of completely randomized experimental design. [7]
- Q6)** a) What is the analysis of variance in two way classification model. [5]
 b) Explain the stepping stone method. [5]
- Q7)** a) Explain Transportation algorithm. [5]
 b) Explain graphical method for sequential sampling. [5]
- Q8)** a) What are the unbound solutions in simplex method. [5]
 b) Find the optimal assignment of four jobs and four machines when the cost of assignment is given by following table : [5]

	J1	J2	J3	J4
M1	10	9	8	7
M2	3	4	5	6
M3	2	1	1	2
M4	4	3	5	6

x x x

Total No. of Questions : 8]

SEAT No. :

P9498

[6182]-824

[Total No. of Pages : 2

**M.E. Chemical (Environmental Engg.)
ENVIRONMENTAL MANAGEMENT
(2017 Pattern) (Semester - I) (509132)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Explain the concept Green funding and taxes. [5]
b) Discuss the various National standards for Environmental Management.[5]
- Q2)** a) Discuss trade and environmental management. [5]
b) Explain EIA evaluation in India. [5]
- Q3)** a) What are the impact of Social sub-systems of environment on Business? [5]
b) What are the features of the directive principles of state policy? [5]
- Q4)** Discuss the various Environmental Policies of the Government of India for Industrial Location with respect to Ecology. [10]
- Q5)** a) Explain fundamental rights and duties of Indian citizens. [5]
b) Write short note on Safety Related rules in Chemical industry. [5]
- Q6)** a) Explain the Effectiveness of international environmental regimes. [5]
b) Give the difference between regulations and laws. [5]

P.T.O.

- Q7)** a) Explain the role of Central Pollution Control Board for sustainable growth. [5]
b) Write short note on Kyoto protocol. [5]
- Q8)** a) Explain the Environmental Policies of the Government of India for conservation forest. [5]
b) Discuss general features of annual report on Ministry of Environment for current year. [5]



Total No. of Questions : 8]

SEAT No. :

P9499

[6182]-825

[Total No. of Pages :1

M.E. Chemical (Environmental Engineering)

ENVIRONMENTAL CHEMISTRY

(2017 Pattern) (Semester-I) (509133)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any 5 questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Use of logarithmic tables slide rule, Mollier charts, and electronic pocket calculator and steam table is allowed.*
- 5) Assume suitable data if necessary.*

Q1) What are Colloids? Explain the use of colloids in Environmental Engineering with suitable example. **[10]**

Q2) Discuss greenhouse gases and global warming. **[10]**

Q3) Explain the Fate of chemicals in aquatic environment. **[10]**

Q4) What are the ill effects of particulate matter on human health? **[10]**

Q5) What is CO₂ capture mechanism? **[10]**

Q6) What are the sources of particulate matter in Thermal power plant? **[10]**

Q7) What are different effects of Air pollutants on materials? **[10]**

Q8) What is Anion exchange capacity? How to determine it. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9500

[6182]-826

[Total No. of Pages : 1

M.E. (Chemical) (Environmental Engineering)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - I) (509134)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any 5 questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data if necessary.*

Q1) a) Explain in detail about the objectives of research. [5]

b) What is types of research problem? [5]

Q2) What is the necessity of defining a research problem? Explain. [10]

Q3) What are the Guidelines for design of experiments in research? [10]

Q4) What are the different types of graphs? Explain the applications with suitable examples. [10]

Q5) How does the case study method differ from the survey method? Analyze the merits and limitations of case study method in sociological research. [10]

Q6) Describe the precautions that the researcher should take while interpreting his findings. [10]

Q7) "Report writing is more an art that hinges upon practice and experience". Discuss with suitable case study. [10]

Q8) How are computers used as a tool in research? Explain giving examples.[10]



Total No. of Questions : 8]

SEAT No. :

P9501

[6182]-828

[Total No. of Pages : 2

M.E. Chemical (Environmental Engg.)
WASTE WATER TREATMENT & DESIGN
(2017 Pattern) (Semester - II) (509137)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any 5 questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam table is allowed.
- 5) Assume suitable data, if necessary.

Q1) Explain configurations for ideal and nonideal reactors. **[10]**

Q2) A treated wastewater with a flow rate of 1000 L/min is to be treated with PAC to reduce the concentration of residual organics measured as TOC from 5 to 1 mg/L. The Freundlich adsorption isotherm parameters were developed. Assuming the following data apply, determine the PAC requirements to treat the wastewater flow. If the PAC costs Rs. 0.50/ kg, estimate the annual cost for treatment, assuming the PAC will not be regenerated. Compound - mixed organics. **[10]**

Initial Concentration $C_0=5$ mg/L

Final Concentration $C_g = 1$ mg/L

GAC density = 450 g/L

Freundlich capacity factor, $k_f = 150$ (mg/g) (L/mg) $1/n$

Freundlich intensity parameter, $1/n = 0.5$

Q3) a) Explain PSA in the process of adsorption. **[5]**

b) Explain the design of GAC contractors. **[5]**

P.T.O.

- Q4)** Determine the theoretical amount of air required at 20 °C to reduce the ammonia concentration from 40 to 1 mg/L in a treated wastewater with a flow rate of 4000 m³/d. Assume the pH of the wastewater has been increase to a value of 11, the Henry's constant for ammonia at 20°C is 0.75 atm., and the air entering the bottom of the tower does not contain any ammonia. **[10]**
- Q5)** a) How activated carbons are prepared and explain carbon regeneration and reactivation. **[5]**
b) Explain the concept of uptake capacity. **[5]**
- Q6)** a) Discuss process design consideration for Rotating Biological Contactors (RBC). **[5]**
b) What are the biological properties of waste. **[5]**
- Q7)** Determine the capacity of a chlorinator for a treatment plant with an average wastewater flow of 1000 m³/d. The peak daily factor for the treatment plant is 3.0 and the maximum required chlorine dosage is to be 20 mg/L. **[10]**
- Q8)** a) Explain Principle and working of Trickling filter. **[5]**
b) Discuss the general design consideration for anaerobic treatment process. **[5]**



Total No. of Questions : 8]

SEAT No. :

P9502

[Total No. of Pages : 1

[6182]-829

M.E. (Chemical Engineering) (Environmental)

SOLID WASTE MANAGEMENT

(2017 Pattern) (Semester - II) (509138)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Assume suitable data, if necessary.*

Q1) Give the chemical and biological properties of the solid waste. **[10]**

Q2) Illustrate the compositions of solid waste. **[10]**

Q3) Give the various types of solid waste collection system and explain the general consideration for waste storage at source. **[10]**

Q4) Analyze the economics of solid waste generation rate. **[10]**

Q5) Give the advantages and disadvantages of various technological options of composting of solid waste. **[10]**

Q6) Illustrate the parameters affecting on the recovery of energy from the solid waste. **[10]**

Q7) Explain the elements of functional management plan for solid waste system. **[10]**

Q8) Develop and explain the essential components required and site selection for the land filling methods. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9503

[Total No. of Pages : 1

[6182]-830

M.E. (Chemical) (Environmental Engineering)

INDUSTRIAL WASTE TREATMENT

(2017 Pattern) (Semester - II) (509139)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.*
- 2) Assume suitable data, if necessary.*
- 3) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is permitted.*

Q1) Explain the treatment techniques for removal of the following pollutants from Industrial waste: **[10]**

- a) Phosphates
- b) Oil & Grease.

Q2) Write short notes Plate and frame Filters. **[10]**

Q3) Explain Primary treatments for industrial Wastewater. **[10]**

Q4) What are clean up and cleaner technologies? Explain with proper examples.**[10]**

Q5) Explain the treatment techniques for the removal of Heavy Metals from industrial wastewater with neat flow diagram. **[10]**

Q6) Define COD. Explain the procedure to determine COD. What are the limitations of COD test? **[10]**

Q7) Explain the concept. objective, design and cost-benefit analysis of common ETP plant. **[10]**

Q8) Write notes on: Flow chart for sludge treatment & disposal. **[10]**



Total No. of Questions : 8]

SEAT No. :

P9504

[6182]-832

[Total No. of Pages : 1

M.E. (Chemical - Environmental Engineering)
REMOTE SENSING AND GIS APPLICATION IN
ENVIRONMENTAL ENGINEERING
(2017 Pattern) (Semester - III) (509143)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any 5 questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Use of logarithmic tables slide rule, Mollier charts, and electronic pocket calculator and steam table is allowed.*
- 5) Assume suitable data, if necessary.*

Q1) Explain in brief about the various satellites in orbit and their sensors. **[10]**

Q2) What is remote sensing? Explain its application? Discuss Passive Remote Sensing system. **[10]**

Q3) a) What is FCC and explain its significance in identification of surface object? **[5]**
b) Explain Digital system of remote sensing. **[5]**

Q4) Explain the principle of classifying Imaging sensor. Explain thermal sensing system. **[10]**

Q5) Explain concept of G.I.S. and state any four components of G.I.S. **[10]**

Q6) What are the limitations of GIS? Explain it with case study. **[10]**

Q7) Explain how GIS is useful in monitoring water pollution studies? **[10]**

Q8) Explain different GIS software's and explain one of them with suitable example **[10]**



Total No. of Questions : 8]

SEAT No. :

P9505

[Total No. of Pages : 2

[6182]-833

M.E. Chemical (Environmental Engg.)

**INDUSTRIAL POLLUTION PREVENTION & CLEANER
PRODUCTION
(2017 Pattern) (Semester - III) (509144)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Use of calculator is allowed.*
- 5) Assume suitable data, if necessary.*

Q1) Discuss about Environmental Policies and Regulations to encourage Pollution Prevention and Cleaner Production. **[10]**

Q2) a) Discuss the various Sustainability Strategies for pollution control and cleaner Production. **[5]**

b) Discuss the various key indicators for the sustainability. **[5]**

Q3) Explain the role of government of industry in pollution prevention and cleaner Production. **[10]**

Q4) Discuss the role of Process and Equipment Optimization in pollution prevention and control. **[10]**

Q5) Explain the applications of Internet Information pollution prevention and cleaner **[10]**

a) Production.

b) Raw material substitution

Q6) Discuss the technical and Environmental Feasibility analysis of pollution prevention and cleaner production program. **[10]**

P.T.O.

- Q7)** a) Explain the role of Cost Analysis for pollution prevention and control. **[5]**
b) Write a note on Environmental Audit. **[5]**

Q8) Write short notes on.

- a) International Environmental Standards - ISO 14001. **[5]**
b) Elements of life cycle costing. **[5]**



Total No. of Questions : 6]

SEAT No. :

P9506

[6182]-835

[Total No. of Pages :2

M.E. (Computer Engineering)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester-I) (510101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat diagrams wherever necessary.*

Q1) a) Describe research process steps in detail? What are the techniques involved in defining a research problem and research objectives? [9]

OR

b) What is the code of ethics in engineering research? State the IEEE and ACM Code of ethics in engineering research? [9]

Q2) a) What is Literature Survey? How to enumerate research gap, research objectives from literature survey? [8]

OR

b) What is the significance of following in research methodology [8]

- i) h-index
- ii) Google Scholar
- iii) Impact Factor
- iv) Paraphrasing

Q3) a) What is the hypothesis and the Null hypothesis? How statistical analysis helps in testing of hypothesis? [8]

OR

b) Explain the methods used for collecting the data for research? Also, elaborate the advantages and disadvantages of each of these data collection methods? [8]

P.T.O.

Q4) a) State the importance of optimization in engineering research? With the help of an example, explain what gradient optimization is? [8]

OR

b) State the simplex optimization steps? What are constraints and cost function? State the similarity and differences in simplex and gradient methods of optimization? [8]

Q5) a) What are the guidelines for conducting surveys? How are respondents identified? What are human factors associated with surveys conducted for research? [8]

OR

b) What do you understand by the research proposal? What is an outline of a research proposal? What is the criteria for evaluating the research proposal? [8]

Q6) a) What are the contents of Thesis in general? What is significance of References in Thesis? What are characteristics of a good thesis and What are cautions in Thesis writing? [9]

OR

b) When to use the respective research outcome of publishing research, explain with example [9]

i) Copyright

ii) Patent

iii) Research Paper

iv) Survey paper



Total No. of Questions : 7]

SEAT No. :

P9507

[6182]-836

[Total No. of Pages : 1

M.E. (Computer Engineering)
BIO-INSPIRED OPTIMIZATION ALGORITHMS
(2017 Pattern) (Semester - I) (510102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Q.No.7 is compulsory, solve any 5 from Q.No.1 to Q.No. 6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Explain the philosophy of natural computing. [4]
b) Illustrate the process of problem solving as a search track. [4]
- Q2)** a) Explain the term standard Evolution algorithm and its advantages. [4]
b) Explain the term Evolutionary biology. [4]
- Q3)** a) Give basic principle of Swarm Intelligence system. [4]
b) Interpret the biological terminology into Ant Colony Optimization and Ant Clustering Algorithm. [4]
- Q4)** a) Explain CUCKOO search algorithm. [4]
b) Explain Bat algorithm and discuss rules of Bat algorithm. [4]
- Q5)** a) Summarize aiNet learning algorithm. [4]
b) Discuss procedure to implement monitoring phase of Negative Selection Algorithm. [4]
- Q6)** a) Discuss architecture of framStick. [4]
b) Illustrate boid flocking. [4]
- Q7)** a) Explain any one application of Genetic algorithm. [6]
b) What is artificial life & what are the goal of artificial life. [4]



Total No. of Questions : 8]

SEAT No. :

P9508

[6182]-837

[Total No. of Pages :1

M.E. (Computer Engineering)
SOFTWARE DEVELOPMENT & VERSION CONTROL
(2017 Pattern) (Semester-I) (510103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data if necessary.*

Q1) What is a quality attribute? Explain any four quality attributes of software. **[9]**
OR

Q2) Explain the behaviour models in UML. **[9]**

Q3) Explain the data centered architecture with advantages and Limitations. **[9]**
OR

Q4) Give the features of product line architecture. **[9]**

Q5) a) What is the importance of documenting software architecture? Give the template for documenting software architecture. **[8]**

b) How the architecture design is carried out in agile projects. **[8]**

OR

Q6) a) How do you evaluate and select build tools? **[8]**

b) Explain any four source code management core concepts. **[8]**

Q7) a) Explain the centralized version control with an example. **[8]**

b) Give the features of version control tools. **[8]**

OR

Q8) a) Write short note on: **[8]**

i) Distributed version control

ii) Branching

b) Give any four version control tools. **[8]**



Total No. of Questions : 12]

SEAT No. :

P9509

[6182]-838

[Total No. of Pages : 2

M.E. (Computer Engineering)
EMBEDDED AND REAL-TIME OPERATING SYSTEM
(2017 Pattern) (Semester - I) (510104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What are the characteristics of embedded systems? [5]
b) What is the use of watchdog timer in embedded system? [3]

OR

- Q2)** a) What are the challenges in the design of embedded system? [3]
b) With a diagram explain the different hardware units of embedded system. [5]

- Q3)** a) Give examples of Large scale embedded systems. Which processors are used in development of Large scale systems. [5]
b) Explain the features of ARM processor. [4]

OR

- Q4)** a) What are the common structural units of the processors? [4]
b) With neat diagram explain System on Chip (SOC) [5]

- Q5)** a) Briefly explain RS-232C, RS-485 communication buses. [4]
b) What is the use of PCI and PCI-X buses? [4]

OR

- Q6)** a) Give the brief description of SPI and SCI. [4]
b) Explain the working of CAN Bus and its applications. [4]

P.T.O.

- Q7)** a) Which parameters are used to categorize the tasks? [4]
b) Explain the use of precedence graph with example. [4]

OR

Q8) Explain the Earliest Deadline First EDF algorithm with example. [8]

Q9) What are the ways to enforce mutual exclusion? Explain with example. [9]

OR

Q10) Explain priority inversion with an example. [9]

Q11) Explain Embedded software development tools and Debugging techniques. [8]

OR

Q12) Write short notes on any two. [8]

- a) RTLinux
- b) Embedded software development tools.
- c) MicroC/OS-II



Total No. of Questions : 10]

SEAT No. :

P9510

[6182]-840

[Total No. of Pages : 3

M.E. Computer (Computer Networks)

OPERATION RESEARCH

(2017 Pattern) (Semester - II) (510108)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt all questions*
- 2) *Assume Suitable Data, if necessary.*
- 3) *Neat Diagram must be drawn wherever necessary.*
- 4) *Figures to the right indicate full marks.*

Q1) A plant manufactures 2 products X and Y. The profit contribution of each product has been estimated as Rs. 50 for product A and Rs. 60 for product B. Each product passes through 4 departments of the plant. The time required for each product and total time available in each department are as follows:

[4]

Department	Hours Required		Available hours during the month
	Product X	Product Y	
1	2	3	800
2	3	2	1000
3	1	1	700
4	3	4	500

The company has a contract to supply at least 380 units of product B per month. Formulate the problem as L.P.P.

Q2) Solve the following LPP by using Simplex Method or graphical method. **[5]**

Maximize $z = 3x_1 + 2x_2$

Subject to constraint

$$x_1 + x_2 \leq 40$$

$$x_1 - x_2 \leq 20$$

$$x_1, x_2 \geq 0$$

P.T.O.

- Q3)** The utility data for a network are given below. Determine the total float and identify the critical path. [5]

Activity	1-2	1-4	1-7	2-3	3-6	4-5	4-8	5-6	6-9	7-8	8-9	9-10
Duration	3	2	2	5	2	6	8	5	4	4	6	3

- Q4)** A firm makes two products P1 & P2 and has production capacity of 18 tons per day. P1 & P2 require the same production capacity. The firm must supply at least 4 tons of P1 and 6 tons of P2 per day. Each ton of P1 & P2 requires 60 hours of machine work each. Maximum machine hours available are 720. Profit per ton for P1 is Rs. 160 & P2 is Rs. 240. Find optimal solution by graphical method. [5]

- Q5) a)** Obtain the initial feasible solution for following transportation problem by Vogel's Approximation method. [5]

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	34

- Q6)** Solve using Graphical Method. [5]

Maximize $Z = 3X_1 + 4X_2$ subject to the constraints

$$3X_1 + 5X_2 \leq 30,$$

$$2X_1 + 2X_2 \leq 20 \text{ and } X_1, X_2 \geq 0$$

- Q7)** Tasks A to I constitute a project. The precedence relationships are:

A<D; A<E; B<F; D<F; C<G; C<H; F<I; G<I.

Draw a network to represent the project and find the minimum time to complete the project when time, in days, of each task is as follows: [5]

Task:	A	B	C	D	E	F	G	H	I
Time :	8	10	17	10	16	8	18	14	9

Also identify the critical Path

Q8) Determine the initial basic feasible solution to the following transportation problem using least cost method. [5]

		Destination				Availability
		D1	D2	D3	D4	
Origin	O1	1	2	1	4	20
	O2	3	3	2	1	40
	O3	4	2	5	9	20
	O4	5	3	6	10	20
Requirement		20	40	30	10	

Q9) Solve by principle of dominance : [5]

	A1	A2	A3	A4
B1	6	8	3	13
B2	4	1	5	3
B3	8	10	4	12
B4	3	6	7	12

Q10) Write short note on application of Game Theory. Write any four limitations of Game Theory. [5]



Total No. of Questions : 12]

SEAT No. :

P9511

[Total No. of Pages : 2

[6182]-841

**M.E. (Computer Engineering)
SYSTEM SIMULATION AND MODELING
(2017 Pattern) (Semester - II)(510109)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Describe the different systems properties. [3]
b) What are different components of system explain with an example. [5]

OR

- Q2)** a) What are the different ways to study the system? Explain with a diagram.[5]
b) Describe the system approach of analysis, design and control. [3]

- Q3)** a) What are the Statistical properties of U[0,1] generators explain with example. [5]
b) Describe the random walk. [4]

OR

- Q4)** a) Determine the sequence of numbers generated by the LCG with $a=5$, $c=3$, $m=16$, & $Z_0=7$. [5]
b) Describe the white noise? [4]

- Q5)** Consider the sinusoidal signal $x(t)=\sin(t)$ over the interval $[0,10]$ with sampling frequency of $f_1=5$ samples per second and de sampling frequency of $f_2=10$ samples per second. Show the results of sampling the continuous signal and then trying to invert the sampling process. [8]

OR

- Q6)** Consider the following boundary value problem defining a signal $x(t)$ on the time interval $[1,3]$ $t^2x'' - 2tx' + 2x + 4 = 0$ $x(1) = 0$ $x(3) = 0$. Find the unique analytical solution $x(t)$ that satisfies both the differential equation and boundary conditions. [8]

P.T.O.

Q7) Consider a single-stage distributed delay with constant delay Δ . Assuming a zero initial state at $t=0$ and unit step input, derive an explicit solution for the output $y(t)$. [8]

OR

Q8) With an illustrative Example explain the simulation of queuing System. [8]

Q9) a) Describe the type of simulations with regards to output analysis.
Give example for each. [6]

b) Explain the problem of initial transient. [3]

OR

Q10) Why the replication/deletion approach for estimation of mean is preferred over other methods justify your answer. [9]

Q11) Explain the importance of modeling machine down time correctly with the help of suitable example. [8]

OR

Q12) Write short notes on [8]

- a) Simulation Software
- b) Application of Simulation in Manufacturing.



Total No. of Questions : 12]

SEAT No. :

P9518

[Total No. of Pages : 2

[6182]-841A
M.E. (Computer Networks)
NETWORK DESIGN AND ANALYSIS
(2017 Pattern) (Semester - II)(510209)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*

Q1) Define and explain User, Application, Device, Network and others requirements. **[8]**

OR

Q2) What are different phases of network design. Explain in details. **[8]**

Q3) What is need of NAT? Explain NAT working using suitable example. **[8]**

OR

Q4) What is IP Packet fragmentation? Explain with diagram. **[8]**

Q5) What are queue performance parameters? Explain M/M/1 model. **[8]**

OR

Q6) Explain use of queuing system in network design. Analyze M/M/n model using transition diagram. **[8]**

Q7) Explain Dijkstra's Algorithm with suitable example. **[9]**

OR

Q8) Explain Bellman's algorithm with suitable example. **[9]**

Q9) Explain various QoS mechanism to ensure in Qos network. **[8]**

OR

Q10) Explain in detail Queue management algorithm. **[8]**

Q11) What are next generation networks? Explain with examples. **[9]**

OR

Q12) Write a note on "Wireshark". **[9]**



Total No. of Questions : 8]

SEAT No. :

P9512

[Total No. of Pages : 2

[6182]-842

M.E.(Computer Engineering)

MACHINE LEARNING

(2017 Pattern) (Semester - II) (510110)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Q.7 and Q.8 are compulsory to attempt.*
- 3) *Figures to the right side indicates full marks.*
- 4) *Neat diagrams must be drawn wherever necessary.*
- 5) *Use of calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

Q1) a) Differentiate predictive and descriptive learning with suitable example.[5]

b) Explain supervised machine learning with suitable examples. [5]

OR

Q2) a) What is classification? Explain Binary and multi-class classification with example. [5]

b) Differentiate between Generative and Discriminative Machine Learning Model. [5]

Q3) a) Discuss PAC learning in detail. [5]

b) What is Linear Discriminant Analysis (LDA)? Explain LDA with suitable example. [5]

OR

Q4) a) What is PCA? When do you use it. [5]

b) Explain bias and variance. What is the relationship between bias and variance. [5]

Q5) a) How does logistic regression differ from linear regression? [5]

b) Draw diagram for multilayer perceptron and explain in detail. [5]

OR

P.T.O.

Q6)

Customer	Age	Loan	Default
John	25	40000	N
Smith	35	60000	N
Alex	45	80000	N
Jade	20	20000	N
Kate	35	120000	N
Mark	52	18000	N
Anil	23	95000	Y
Pat	40	62000	Y
George	60	100000	Y
Jim	48	220000	Y
Jack	33	150000	Y
Andrew	48	142000	?

[10]

Write K-Nearest Neighbors algorithm. Suppose you have given the following data where Customer, Age and Loan are input variables and default is the dependent/class variable.

Apply KNN algorithm to predict the class of new data point Age = 48 kg and Loan = 142000 using Euclidean distance. Assume $k = 5$.

Q7) a) Explain Entropy and Information Gain in Decision Trees? **[5]**

b) What is Expectation-Maximization (EM) algorithm? **[5]**

Q8) Analyze the limitations and potential improvements of 'Topic models of the Underground Internet Economy'. **[10]**



Total No. of Questions : 12]

SEAT No. :

P9519

[Total No. of Pages : 2

[6182]-842 A

M.E. (Computer Networks)

DATA ALGORITHMS

(2017 Pattern) (Semester - II) (510210)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) Explain Secondary sort in Hadoop and how it works? **[8]**

OR

Q2) What is POJO moving average? Write solution using queue. **[8]**

Q3) What is left outer join operations in database? What is significance of left outer join in real world situation? Give suitable SQL for the same? With the help of SQL explain what is mean by Left in left outer join. **[8]**

OR

Q4) Explain Left outer join data flow in phase 1 and 2 with the help of classes in Hadoop. **[8]**

Q5) What is moving average? And how do we calculate it? Give the suitable example. **[9]**

OR

Q6) “One can sort the input data and then easily select the K largest records from the sorted file. This is often the most efficient method for very large K.” to implement this sort give the construct of Map and Reduce functions. **[9]**

P.T.O.

Q7) What is market basket analysis? What are the association rules? What is the role of association rules in market basket analysis? In the following example calculate support, confidence and lift **[8]**

* Assume there are 100 customers

* 10 of them bought milk, 8 bought butter and 6 bought both of the

OR

Q8) What is POJO? How does POJO handle the common friends solutions? Give the pseudo code snippet? How the same common friends solutions resolve using MAP-REDUCE? **[8]**

Q9) Explain in brief five method template for implementing scatter search. **[9]**

OR

Q10) What is the scatter search algorithm? What are its characteristics? And Give the algorithmic procedure. **[9]**

Q11) Explain Bellman's equation and acyclic graph? What is significance of bellman's equation in shortest path finding? **[8]**

OR

Q12) Explain network simplex algorithm and minimum cost flow problem. **[8]**



Total No. of Questions : 12]

SEAT No. :

P9513

[6182]-844

[Total No. of Pages : 2

M.E. (Computer Engg.)

FAULT TOLERANT SYSTEMS

(2017 Pattern) (Semester - III) (610101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer six questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right side of question indicate full marks.*

Q1) Compare Information redundancy and Software redundancy? **[9]**

OR

Q2) Explain various Empirical Models in detail with examples? **[9]**

Q3) Explain in detail Parallel fault Simulation Algorithm with its application? **[9]**

OR

Q4) Explain in detail about Concurrent Fault Simulation algorithm? **[9]**

Q5) What is Faulty Mesh Networks? Explain in detail Faulty Mesh Networks with suitable example? **[8]**

OR

Q6) Write short notes on Node Labeling Technique with suitable case study? **[8]**

Q7) Write short notes on

- a) BSN Fault Diameter
- b) BSN Reliability **[8]**

OR

Q8) Explain in detail HINs and Hierarchical Cubic Network (HCN)? **[8]**

P.T.O.

Q9) Explain Classification of Fault-Tolerant Switching Architectures? [8]

OR

Q10) Explain various Classifications of Fault-Tolerant Switching Architectures? [8]

Q11) Explain various checkpoint schemes for wireless networks? [8]

OR

Q12) Explain : [8]

- a) Faults
- b) Errors
- c) Failures
- d) Process resilience in terms of Definition and example.



Total No. of Questions : 12]

SEAT No. :

P9520

[Total No. of Pages : 2

[6182]-844 A

M.E. (Computer Networks)

FAULT TOLERANT SYSTEM

(2017 Pattern) (Semester - III) (610201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer six questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right side of question indicate full marks.*

Q1) Explain Software Redundancy, and Information Redundancy, in detail with examples? **[9]**

OR

Q2) Explain Combinatorial (Continuous) Model in detail with example? **[9]**

Q3) Differentiate Serial Fault Simulation Algorithm versus Parallel Fault Simulation? **[9]**

OR

Q4) What is fault modeling? Explain fault modeling levels in detail? **[9]**

Q5) Explain Depth-First Search routing algorithm for fault tolerant in Hypercube?**[8]**

OR

Q6) Explain Node Labeling Techniques in detail? **[8]**

Q7) What is Hierarchical Cubic Network (HCN)? Explain HCN (2, 2) network in detail with diagram? **[8]**

OR

P.T.O.

Q8) Explain Edges Groups, Construction, Degree and Diameter in Block-Shift Network (BSN) in detail? **[8]**

Q9) Explain The Hyper-Torus Network (HTN) and Hierarchical Cubic Network (HCN). **[8]**

OR

Q10) Explain BSN Connectivity and BSN Reliability. **[8]**

Q11) Compare issues in various checkpoint schemes for wireless networks? **[8]**

OR

Q12) Explain the Architecture-Dependent Fault Tolerance in detail **[8]**



Total No. of Questions : 12]

SEAT No. :

P9514

[Total No. of Pages : 3

[6182]-845

M.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2017 Pattern) (Semester - III) (610102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figure to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of Scientific calculator is permitted.

Q1) a) Compute the edit distance between Paris and Alice using Levenshtein edit distance method. **[5]**

b) Explain information retrieval architecture. **[4]**

OR

Q2) a) Construct k-gram indexes for query retrieval (wild card query). **[5]**

b) What is need of spelling correction? Explain any one context sensitive spelling correction method. **[4]**

Q3) You are asked to design a modern information management system for a large multi-volume encyclopedia like the Encyclopaedia Britannica. **[9]**

Describe in detail the ways in which you would organize and index the encyclopedia text, and the facilities you would offer the system user.

OR

Q4) a) Explain the statistical properties of terms in information retrieval. Explain with example. **[5]**

b) What role does stemming play in indexing? **[4]**

Q5) a) Why are query terms and document terms weighted in modern information retrieval systems? Describe a method for weighting terms. **[4]**

b) Define following terms in detail. **[4]**

i) Field

ii) IDF

iii) Zone

iv) Term-document matrix

OR

P.T.O.

Q6) a) Explain vector space model for ranking the documents. [4]

b) Consider the following training example. [4]

Example	DocID	Query	ST	SB	Judgement
1	34	Linux	1	1	R
2	32	Penguin	0	1	R
3	35	System	0	1	R
4	36	Mining	0	0	N
5	87	Training	1	1	R
6	54	Database	0	1	R
7	7	Linux	1	0	N

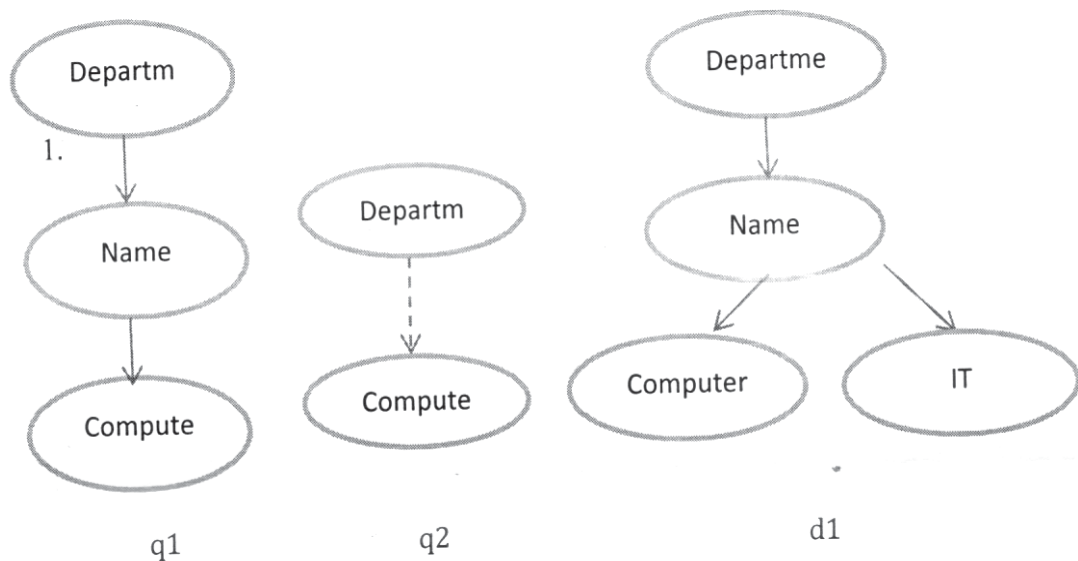
Computer weighted zone score for each (query, document) example.

Q7) a) Explain the need of component coverage and topical relevance in evaluation of XML document. [4]

b) Explain the challenges in XML retrieval. [4]

OR

Q8) a) Compute the similarity of quest q1, q2 with document d1 using context resemblance functions. [4]



b) Compare text-centric and data centric XML retrieval. [4]

- Q9) a)** Suppose we have collection that consists of the four documents given in the table below. **[4]**

Doc id	Document Text
1	Click go the shears boys click click click
2	Click click
3	Metal here
4	Metal shears click here

Build a query likelihood language model (unigram model) for this document collection for the query ‘click shear’s and rank the documents.

- b) Explain the types of language models with suitable example. **[4]**

OR

- Q10)a)** Explain the query likelihood model with suitable example. **[4]**

- b) Estimating the query generation probability in detail with example. **[4]**

- Q11)a)** Explain the issues in the classification of text document. **[4]**

- b) Explain Single-link and complete-link clustering with suitable example. **[4]**

OR

- Q12)a)** Explain in detail latent semantic indexing. **[4]**

- b) Explain the clustering method performance metrics. **[4]**



Total No. of Questions : 8]

SEAT No. :

P9521

[Total No. of Pages : 2

[6182]-845A

**M.E. Computer (Computer Networks)
MOBILE AD HOC NETWORKS
(2017 Pattern) (Semester - III) (610202)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

- Q1)** a) What is multipath propagation? Explain how it affects signal quality. [6]
b) What are the advantages of reservation-based MAC protocols over contention based MAC protocols? [6]

OR

- Q2)** a) Differentiate between cellular networks and Ad Hoc wireless networks.[6]
b) Discuss the differences in topology reorganization in DSDV and CGSR routing protocols. [6]

- Q3)** a) Comment on the scaling properties of source-initiated and receiver initiated multicast protocols with respect to the number of sources and receivers in the group. [6]

- b) What are the limitations of the IEEE 802.11 MAC protocol that prevent it from supporting QoS traffic? [6]

OR

- Q4)** a) Explain Multicast routing protocol based on zone routing in detail. [6]
b) Compare and contrast the IEEE 802.11e MAC protocol with the DBASE protocol. [6]

- Q5)** a) What are the major differences between ad hoc wireless networks and sensor networks? [7]

- b) What are the advantages of a clustered architecture over a layered architecture in a sensor network? [6]

OR

P.T.O.

- Q6)** a) How does the hybrid usage of TDMA and FDMA minimize energy consumption? [6]
- b) Describe the Sybil attack and the sinkhole attack on sensor networks.[7]

- Q7)** a) Compare and contrast UWB communication with conventional wideband communication techniques based on spread spectrum techniques. [6]
- b) Give two advantages and disadvantages in using laser diodes as light sources for optical wireless networks. [7]

OR

- Q8)** Write Short Notes on (Any Two) [13]
- a) Software defined Radio-Based Multimode Systems.
- b) Optical Wireless WDM.
- c) The service provider model for Wi-Fi systems.



Total No. of Questions : 12]

SEAT No. :

[Total No. of Pages :2

P9515

[6182]-847

**M.E. (Computer Networks)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester-I) (510201)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8. Q.9 or Q.10, Q.11 or Q.12.*
- 2) Assume suitable data if necessary.*
- 3) Figures to the right indicates full marks.*

Q1) Write short note on:

- a) Objectives of research [5]
- b) Motivation in research [5]

OR

Q2) Explain in detail the flow chart of Research Process? [10]

Q3) Explain suitable research proposal in terms of title, design, details and budget for outcomes. [8]

OR

Q4) Explain Various Funding resources and Research grants and for innovative research. [10]

Q5) Explain the various sources of error and uncertainty in measurement. [8]

OR

Q6) Explain One-way (or single factor) ANOVA technique? [8]

Q7) Explain Monte Carlo optimization technique in detail? [8]

OR

Q8) Write short notes on Google Optimization Tool Open MDAO. [8]

P.T.O.

Q9) Explain why a good literature survey is important to narrow the research problem and the technique that might be used. [8]

OR

Q10) Explain importance of Statistical analysis and Reporting in research? [8]

Q11) Define and Explain Copyright and Patents in modern research? [8]

OR

Q12) Explain technique of interpretation? What are precautions needed in interpretation? [8]



Total No. of Questions : 12]

SEAT No. :

P9516

[6182]-848

[Total No. of Pages : 2

M.E. (Computer Engineering) (Computer Networks)

NETWORK SECURITY

(2017 Pattern) (Semester - I) (510202)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Black figures to the right indicate full marks.*
- 4) *Neat diagrams must be drawn wherever necessary.*
- 5) *Assume Suitable data if necessary.*

Q1) How to detect and prevent black hole attack in mobile adhoc network? Explain with an example. **[9]**

OR

Q2) Elaborate security threats to WSN and its countermeasures. **[9]**

Q3) Explain the process of scanning with Nmap tool. Scan a single port, Scan a range of ports, Scan 100 most common ports (fast) and Scan all 65535 ports. **[8]**

OR

Q4) Explain Nmap port scan types: Scan using TCP SYN scan (default), Scan using TCP connect in detail. **[8]**

Q5) What are the preventive measure against Cross Site Scripting (XSS) attacks? How to reduce your chances of being attacked? **[8]**

OR

Q6) Write short on: **[8]**

- i) BeEF with Better Cap,
- ii) BeEF with SET.

P.T.O.

Q7) What are Cryptographic Hash functions? How do hackers obtain the hash?
Which tools are used to get the hash? **[8]**

OR

Q8) What is Open Web Application Security Project (OWASP) ZAP? Which types of vulnerability are tested on it for weak points of targeted website?**[8]**

Q9) Elaborate the web app vulnerabilities and its security tools. **[8]**

OR

Q10)How VLAN hopping attack is executed? State and explain the Consequences of VLAN Hopping. **[8]**

Q11)Write short note on **[9]**

- i) Spoofing
- ii) DNS Spoofing
- iii) Address Spoofing.

OR

Q12)What is Evil Twin attack? Is it on wired network or wireless network? Justify.**[9]**



Total No. of Questions : 8]

SEAT No. :

P9517

[Total No. of Pages : 2

[6182]-849

M.E. Computer (Computer Networks)

WIRELESS SENSOR NETWORKS

(2017 Pattern) (Semester-I) (510203)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Assume Suitable data if necessary.*

Q1) a) Discuss process of aggregate data from a sensor network with Syntax of an SQL query **[6]**

b) List some example prototype sensor node architectures. Explain IMote Node XYZ Node and Hogthrob Node Architecture in detail. **[6]**

OR

Q2) a) Draw the Architecture of a wireless sensor node and explain sensing subsystem in detail. **[6]**

b) Explain Application-Specific Integrated Circuit (ASIC) with its basic building blocks, also discuss Advantages, Disadvantages and Applications of ASICs. **[6]**

Q3) a) How does the S-MAC protocol reduce the duty cycles of sensor nodes? How does it address the hidden-terminal problem? **[6]**

b) Describe LEACH operations a setup phase and a steady-state phase with figure. **[6]**

OR

Q4) a) Explain how the Z-MAC protocol allows nodes to determine their own local time frames instead of using a single global time fram. What are the disadvantages of Z-MAC. **[6]**

b) Explain Multihop range estimation using Euclidean distance estimation in the absence of direct connectivity examples, also comment on “DV-Hop” and “DV-Distance” methods. **[6]**

P.T.O.

- Q5)** a) Explain Single packet delivery using multiple paths by providing alternative routes, ReInForM, HHB and HHBA [7]
b) Explain Single packet delivery using a single path, is it efficient to use only end-to-end acknowledgments? [6]

OR

- Q6)** a) RMST is designed toward guaranteed delivery of large blocks of data from sensors to sinks, while design of RMST what repair mechanisms required on different layers. [6]
b) Explain Mechanisms for congestion detection and congestion handling in WSN. [7]

- Q7)** a) Discuss various Challenges of providing Security in Wireless Sensor Networks. [6]
b) In order to defend against the many possible attacks in a WSN, explain various security protocols and other defense mechanisms [7]

OR

- Q8)** Write short notes (any two) [13]
a) Security attacks on Routing and on Transport Layer
b) TinySec and Localized Encryption and Authentication Protocol (LEAP) protocols
c) Physical Layer and Link layer DoS attacks



Total No. of Questions : 12]

SEAT No. :

P9522

[6182]-849A

[Total No. of Pages :2

M.E. (Computer Engineering) (Data Science)

BIG DATA ANALYTICS

(2017 Pattern) (Semester-I) (510303)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Assume suitable data if necessary.*
- 5) *Use of Scientific Calculator is permitted.*

Q1) Explain Data Analytics Cycle with suitable diagram. **[9]**

OR

Q2) Define Big data? What are the characteristics of Big Data? Explain any 2 Real-World Problems that Big Data Can Solve. **[9]**

Q3) What is Apache HBase? List out the name of the key components of HBase. **[8]**

OR

Q4) Explain the important features and Core Components of Hadoop. **[8]**

Q5) What is a Zookeeper? Describe the benefits of using a zookeeper? **[8]**

OR

Q6) Explain the Apache Flume in Hadoop? Explain the features of Apache Sqoop. **[8]**

Q7) What are the important components of the Spark ecosystem? What are the different cluster managers available in Apache Spark? **[8]**

OR

P.T.O.

Q8) Explain the functions of Spark SQL? How can you connect Hive to Spark SQL? [8]

Q9) What are the main characteristics of PySpark? What do you think PySpark is important in Data Science? [9]

OR

Q10) What is Text Mining in Data analytics? What is stemming and Lemmatization? [9]

Q11) Define Big Data Visualization? Explain the Benefits of good data visualization? [8]

OR

Q12) What are the different Data Visualization Techniques used in data science? Explain in brief uses of Tableau for data visualization? [8]



Total No. of Questions : 12]

SEAT No. :

P-9685

[Total No. of Pages : 2

[6182]-850

M.E. (Computer) (Computer Networks)
HIGH PERFORMANCE NETWORKS
(2017 Course) (Semester - I) (510204)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Distinguish between ISO-OSI and TCP/IP model with a neat diagram. **[4]**
b) Write short notes on : **[4]**
i) X.25
ii) Frame relay

OR

- Q2)** a) Distinguish between Fast Ethernet and Gigabit Ethernet. **[4]**
b) Explain the design consideration of high-performance networking. **[4]**
- Q3)** a) Draw Ethernet Frame format. **[4]**
b) List some of the applications of Gigabit Ethernet. **[4]**

OR

- Q4)** a) Differentiate between Ethernet 10/100/1000 speeds. **[4]**
b) Explain automatic configuration of Gigabit Ethernet. **[4]**
- Q5)** a) Define the features of a router. **[4]**
b) Describe the process of Packet Scheduling in router. **[5]**

OR

P.T.O.

- Q6)** a) Draw the architecture of a router. [5]
b) Write short notes on : [4]
i) Resource Allocation in routers.
ii) Multicasting packets in routers.

- Q7)** a) Define MPLS, also explain the purpose of IP over MPLS. [5]
b) Explain any one of the MPLS signaling protocol. [4]

OR

- Q8)** a) Explain in detail MPLS forwarding operations. [5]
b) Explain any two MPLS encapsulation standards. [4]

- Q9)** a) Explain about Large-Scale Optical Switches. [4]
b) What is Optical Cross Connects (OXC). [4]

OR

- Q10)** a) Write a note on routing in All-optical Switch. [4]
b) Explain wavelength allocation in networks. [4]

- Q11)** a) Distinguish between Borderless Network Architecture and Borderless Campus Network. [4]
b) Write short note on Multi-Tier Borderless campus design model. [4]

OR

- Q12)** a) Explain the design principles of Campus Network Design. [4]
b) Define distribution layer network design model. [4]



Total No. of Questions : 12]

SEAT No. :

P9523

[6182]-852

[Total No. of Pages : 2

M.E. (Computer Engineering) (Data Science)

DATA WAREHOUSING AND MINING

(2017 Pattern) (Semester - II) (510308)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) What is Data Warehouse? List and explain the major critical business subjects for the following companies? **[9]**

- a) A domestic hotel chain.
- b) An international manufacturing company.

OR

Q2) How are the top-down and bottom-up approaches for building a data warehouse different? Discuss the merits and disadvantages of each approach. **[9]**

Q3) Explain the architectural components of Data warehousing in three major areas with suitable diagram. **[9]**

OR

Q4) What are the platform options for the staging area of the Data Warehouse? Compare the options and mention the advantages and disadvantages. **[9]**

Q5) Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit. Design a star schema diagram for the above Data Warehouse. **[8]**

OR

P.T.O.

Q6) Explain the essential differences between the MOLAP and ROLAP models?
Also list a few similarities. **[8]**

Q7) What do you understand about the information delivery framework in the data warehouse environment? Explain in detail. **[8]**

OR

Q8) Describe briefly two major features of the Web-enabled data warehouse. Explain How do the Internet, intranet, and extranet apply to the data warehouse? **[8]**

Q9) List the major phases of a data mining operation. Out of these phases, pick two and describe the types of activities in these two phases. **[8]**

OR

Q10) Suppose that the data mining task is to cluster points (with (x, y) representing location) into three clusters, where the points are

A1(2,10), A2 (2,5), A3 (8,4), B1 (5,8), B2 (7,5), B3 (6,4), C1 (1,2), C2 (4,9).

The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show only the three cluster centers after the first round of execution. **[8]**

Q11) What are the components that make up the physical model? How are these related to components of the logical model? **[8]**

OR

Q12) List the Post deployment administration areas for data warehousing and explain any two in details. **[8]**



Total No. of Questions : 12]

SEAT No. :

P9524

[Total No. of Pages : 3

[6182]-853

M.E. (Computer Engineering) (Master of Data Science)

MACHINE LEARNING

(2017 Pattern) (Semester - II)(510309)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks
- 4) Assume suitable data, if necessary.

Q1) a) Define Multi-class classification. Explain One vs One and One vs All methods of multi-class classification in detail with example. **[5]**

b) Explain supervised and unsupervised machine learning with example. **[4]**

OR

Q2) a) Enlist and explain any two Machine learning models. **[5]**

b) Differentiate between Parametric and non-parametric models. **[4]**

Q3) Write and explain steps used in Principal Component Analysis with example. **[8]**

OR

Q4) What is concept learning? Explain hypothesis space in detail. **[8]**

Q5) What is logistic Regression? Explain how logistic regression is used for classification. **[8]**

OR

P.T.O.

- Q6) a)** Given the following data where X and Y are the 2 input variables and Class is the dependent variable. [4]

X	Y	class
2	3	P
2	2	N
1	2	P
1	0	P
1	-1	N
0	2	N
0	1	P
-1	1	N

Predict the class of New data point $x=1$ and $y=1$ using kNN where value of k is 3. To which class the new point belongs to?

- b) Explain simple neural network in detail. [4]

- Q7)** What is Bagging and Boosting ensemble methods? Explain the difference between Bagging and Boosting ensemble methods. [9]

OR

- Q8)** What is decision tree? Write a decision tree algorithm and explain with an example. [9]

- Q9)** Explain Expectation-Maximisation method in detail. [8]

OR

- Q10)** Explain normal distribution and its geometric interpretations in detail. [8]

Q11) Write a short note on

[8]

- a) Credit card fraud detection
- b) Detecting malicious websites in adversarial classification.

OR

Q12) Write a short note on

[8]

- a) Learning to rate vulnerabilities and predict exploits
- b) Language identification.



Total No. of Questions : 12]

SEAT No. :

P9525

[Total No. of Pages : 2

[6182]-854

M.E. (Computer Engineering Data Science)

SOFT COMPUTING

(2017 Pattern) (Semester - II)(510310)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Compare soft and hard computing with their methods. **[5]**

b) Give the applications of genetic algorithms. **[4]**

OR

Q2) a) Explain union, intersection, complement and bounded difference operators in fuzzy set with an example. **[5]**

b) What is binary Fuzzy relation? Explain with an example. **[4]**

Q3) a) Explain any three fuzzy membership functions. **[5]**

b) What is fuzzification process? **[4]**

OR

Q4) a) Explain the rule based form for fuzzy rules. **[5]**

b) Explain any two methods of defuzzification. **[4]**

Q5) Explain any four operators of genetic algorithm. [8]

OR

Q6) Explain in brief the following terms on Genetic algorithms. [8]

a) Individual

b) Population

c) Gene

d) Fitness function

Q7) Explain the concept of McCulloch-Pitts neuron. [8]

OR

Q8) Explain the perceptron training algorithm for single output class. [8]

P.T.O.

Q9) How associative memory can store patterns? [8]

OR

Q10) Explain the bidirectional associative memory network. [8]

Q11) Explain the features of neuro fuzzy hybrid system. [8]

OR

Q12) Explain the genetic fuzzy rule based system. [8]



Total No. of Questions : 12]

SEAT No. :

P9526

[Total No. of Pages : 2

[6182]-856
M.E. (Computer) (Data Science)
DEEP LEARNING
(2017 Pattern) (Semester - III) (610301)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10, Q11 or Q12.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*
- 5) *Use of cell phone is prohibited in the examination hall.*
- 6) *Use of electronic pocket calculator is allowed.*

Q1) Differentiate between shallow and deep neural networks. **[9]**

OR

Q2) What is backpropagation? Explain the algorithm in detail. State the advantages and disadvantages. **[9]**

Q3) What is gradient decent. Explain following three variants of Gradient Descent: Batch, Stochastic and Mini batch. **[9]**

OR

Q4) What is PCA (Principle Component Analysis)? Explain the assumptions and limitations of PCA? Enlist the applications of PCA. **[9]**

Q5) What is Bias Variance trade - off? Explain with suitable examples. **[8]**

OR

Q6) What is the need of Ensemble methods? Discuss in brief about Ensemble Methods. **[8]**

P.T.O.

Q7) How to apply guided backpropagation to enhance the interpretability of CNN. Justify with suitable example. **[8]**

OR

Q8) Explain the different layers in CNN. Why do we use a Pooling Layer in a CNN? **[8]**

Q9) In what situations would you prefer to use LSTMs over simple Neural Nets? Explain the architecture of an LSTM network. **[8]**

OR

Q10) How does the attention mechanism solve the encoder-decoder bottleneck problem? Elaborate on the working of attention mechanism for images. **[8]**

Q11) Explain Generative Adversarial Network and Compare Variational Autoencoders and Generative Adversarial Network? **[8]**

OR

Q12) How to apply autoencoders for dimensionality reduction? Discuss the types of Autoencoders. **[8]**



Total No. of Questions : 12]

SEAT No. :

P9527

[Total No. of Pages : 2

[6182]-857

M.E. (Computer Engineering) (Master of Data Science)

DATA MODELING AND VISUALIZATION

(2017 Pattern) (Semester - III) (610302)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Define Data Visualization and terms of definition. **[4]**

b) What are the four steps in working with data? Explain. **[4]**

OR

Q2) How Fitt's law is helpful in visualization? Justify your answer with example. **[8]**

Q3) a) How the histogram and scatter plots are useful in visualization of Numerical data? **[4]**

b) How the streamlines are used in visualization of the Flow? **[4]**

OR

Q4) a) Explain the box plot used in understanding the data dispersion with the help of diagram. **[4]**

b) Which parameters can be used to Measure the Dispersion of Data? Give examples for any two. **[4]**

Q5) Explain the node link diagram and shade maps with example. **[8]**

OR

Q6) How Tree maps and Heat maps are used for visualization of non numeric data? Explain with example. **[8]**

P.T.O.

Q7) Explain the steps of Principle component Analysis (PCA) with example. **[9]**

OR

Q8) Explain PROCLUS, a top-down subspace approach. **[9]**

Q9) Explain the Gestalt laws of Proximity and similarity with Example. **[9]**

OR

Q10) Describe the process of pattern perception of figure and ground with examples. **[9]**

Q11) How to evaluate a good visualization. Justify your answer. **[8]**

OR

Q12) What are the evaluation methods based on psychophysics? **[8]**



Total No. of Questions : 12]

SEAT No. :

P9528

[Total No. of Pages : 2

[6182]-859

M.E. (Computer Engg.) (Cyber Security)

MATHEMATICAL FOUNDATIONS FOR CYBER SECURITY

(2017 Pattern) (Semester-I) (510401)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data if necessary.*
- 4) *Neat diagram must be drawn wherever necessary.*

- Q1)** a) Distinguish between $\mathbb{Z}$ and $\mathbb{Z}_n$. Which set can have negative integers? How can we map an integer in $\mathbb{Z}$ to an integer in $\mathbb{Z}_n$. **[4]**
- b) What is linear Diophantine equation of two variables? How many Solutions can such an equation have? How can the solutions be found? **[4]**

OR

- Q2)** a) Define a residue class and least residue? **[4]**
- b) Define Linear Congruence? What algorithm can be used to solve an equation of type $ax \equiv b \pmod{N}$? How can we solve a set of linear equations? **[4]**
- Q3)** a) Define an algebraic structure? Discuss three algebraic structures. **[4]**
- b) Define a ring. Distinguish with cumulative ring. **[4]**

OR

- Q4)** a) Show how a polynomial can represent an n-bit word? **[4]**
- b) For the Group $G = \langle \mathbb{Z}_6^*, x \rangle$ **[4]**
- i) Prove that it is an abelian group.
 - ii) Show the result of $5*1$ and $1+5$.
 - iii) Show that why we should not worry about division by zeros in this group.

P.T.O.

- Q5) a)** Define the following functions and their applications [4]
 i) $\pi(n)$ Function
 ii) Euler's totient Function
- b)** Using the approximation find. [5]
 i) The number of primes between 1,00,000 and 2,00,000
 ii) The number of composite integers between 1,00,000 and 2,00,000
- OR
- Q6) a)** Find some primes in the form $5k+1$, $5k+2$, $5k+3$, $5k+4$ where k is positive integer. [4]
b) Show that $2^{24}-1$ and $2^{16}-1$ are composites. [5]
- Q7) a)** Determine whether the following Mersenne numbers are primes: [4]
 M_{23} , M_{29} , M_{31}
b) Determine how many of the following integers pass the primality test? [4]
 100, 110, 130, 150, 200, 250, 271, 341, 561. Use base 2.
- OR
- Q8) a)** Use $a = 2$, $x = 3$, and a few primes to show that if p is prime, the following congruence $(x - a)^p \equiv (x^p - a) \pmod{p}$ holds. [4]
b) Find the value of x for the following sets of congruence using Chinese remainder theorem- [4]
 i) $X \equiv 2 \pmod{7}$, and $x \equiv 3$ and -9
 ii) $X \equiv 4 \pmod{5}$, and $x \equiv 10 \pmod{11}$
- Q9) a)** What is pseudorandom number generator? Discuss in detail. [4]
b) Write a short note on stochastic Process Markov Chain. [4]
- OR
- Q10) a)** Discuss the Monte Carlo algorithm for probability. [4]
b) Discuss Baye's theorem and its application in probability theory. [4]
- Q11) a)** What is Hadamard Code? Discuss in detail. [4]
b) Write short note on Goppa Codes. [5]
- OR
- Q12) a)** What is the concept of Minimum distance? Explain with example. [4]
b) Discuss Code families and Asymptotically good codes. [5]



Total No. of Questions : 6]

SEAT No. :

P9529

[6182]-860

[Total No. of Pages : 1

M.E. (Cyber Security)

MODERN CRYPTOGRAPHY

(2017 Pattern) (Semester - I) (510402)

Time :3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve any five questions from Q.No. 1 to Q. No. 6.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) State and Explain the second principle of Modern Cryptography: Reliance on Precise Assumptions [5]
b) Provide a formal definition of the Gen, Enc and Dec algorithms for mono-alphabetic substitution. [5]
- Q2)** a) Present a construction of a variable output-length pseudorandom generator from any pseudorandom function. Prove your construction secure. [5]
b) Let G be a pseudorandom generator and define $G'(s)$ to be the output of G truncated to n bits (where s is of length n). Prove that the function $F_k(x) = G'(k) \oplus x$ is not pseudorandom. [5]
- Q3)** a) In RSA algorithm, if $p=7$, $q=11$ and $e=13$, what will be the value of d ? [5]
b) Explain the Diffie-Hellman key exchange algorithm. [5]
- Q4)** a) What is the difference between a Digital Signature and a Digital Certificate? [5]
b) What is steganography? Present the types of steganography. [5]
- Q5)** a) Describe the Cyber Governance Issues. [5]
b) Explain the cyber security management cycle. [5]
- Q6)** a) Discuss the cyber security metrics. [5]
b) What is security framework? [5]



Total No. of Questions : 8]

SEAT No. :

P9530

[6182]-861

[Total No. of Pages :2

M.E. (Computer Engineering) (Cyber Security)
SECURE SOFTWARE DESIGN, CODING PRACTICES & ETHICS
(2017 Pattern) (Semester-I) (510403)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data if necessary.*

Q1) a) What is mean by system complexity? What is role of software assurance in system development? [4]

b) Discuss about benefits of early detection of defects in software development? [5]

OR

Q2) a) What are the threats and sources present in system design? Explain each in detail. [4]

b) Discuss about OWASP Security knowledge framework [5]

Q3) a) Discuss about SQUARE Model and output in detail. [4]

b) What are the architectural risk analysis in secure software? Explain in detail. [5]

OR

Q4) a) Discuss about requirement elicitation method in detail. [4]

b) Which are the different guidelines and attack patterns? Explain in detail. [5]

Q5) a) Write short note on: [8]

i) Security failures

ii) Maturity of practice

b) Discuss about Governance and security. [8]

OR

P.T.O.

- Q6)** a) Write short note on: [8]
i) System complexity drivers
ii) Deep technical problem
b) Discuss about Software penetration testing. [8]

- Q7)** a) What do you mean by Agile development process? Explain with suitable example. [8]
b) Describe agile vulnerability management with example [8]

OR

- Q8)** a) Write short note on any two following: [8]
i) Getting security into requirements
ii) Code review for security
iii) Agile security testing
b) Discuss about Risk based Software testing [8]



Total No. of Questions : 6]

SEAT No. :

P-9531

[Total No. of Pages : 2

[6182]-862
M.E. (Cyber Security)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (510101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat diagrams where necessary.*

Q1) a) What is the significance of the code of ethics in research? Which are the 'ethical aspects' affecting and influencing engineering research? **[9]**

OR

b) What are Research paradigms? Why research is needed? Elaborate the terms like research objectives and research outcomes in detail? **[9]**

Q2) a) Elaborate the significance of the following in the research methodology. **[8]**

- i) i10-index
- ii) Copyright
- iii) Bibliometric Analysis in literature survey
- iv) Paraphrasing in a literature survey

OR

b) Differentiate between the Research Plan and Research proposal? What are elements getting generically included in the Research Proposal? **[8]**

Q3) a) Differentiate between One-dimensional statistics and Two-Dimensional Statistics with suitable examples? **[8]**

OR

b) State the use of the following tools **[8]**

- i) ANOVA
- ii) NOST-Dataplot
- iii) Paired t-test
- iv) CAT (Qualitative analysis tool)

P.T.O.

Q4) a) State the importance of optimization in engineering research. With the help of an example, explain the Gradient Optimization Method? **[8]**

OR

b) State the simplex optimization steps? What are constraints and cost function? State the similarities and differences in simplex and Monte Carlo methods of optimization? **[8]**

Q5) a) What is the significance of the research proposal? What is the criteria for evaluating the research proposal? What is a Research plan? **[8]**

OR

b) What are Bibliometric Surveys? What is role of statistical analysis in surveys? What are the guidelines for survey reporting? **[8]**

Q6) a) Elaborate following research outcomes and when which publication is to be attempted. **[9]**

i) InPASS (Indian Patents Advanced Search System)

ii) Patent

iii) Survey Paper

OR

b) What are the generic contents of the organization of the thesis? Why the 'appendix' part in the thesis is important? What is the meaning of citing references in the Thesis? **[9]**



Total No. of Questions : 10]

SEAT No. :

P9532

[Total No. of Pages : 2

[6182]-865

**M.E.(Computer Engineering) (Cyber Security)
DISASTER RECOVERY AND MANAGEMENT
(2017 Pattern) (Semester - II) (510409)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is Disaster? Compare Between Hazard And Disaster. [4]
b) What are the repercussions of natural disasters? [4]

OR

- Q2)** a) What is Natural Disaster? Give the causes of Natural Disaster. [4]
b) Explain the effects of nuclear accidents? [4]
- Q3)** a) What is a seismic zone also called? What are the 4 seismic zones of India? [4]
b) What are the applications of remote sensing in disaster risk management? [6]

OR

- Q4)** a) Which areas are prone to cyclonic and coastal hazards in India? Why are coastal areas of India prone to cyclones? [4]
b) What are the important measures to be taken in community level of disaster preparedness? Explain. [6]
- Q5)** a) What is the concept of disaster risk reduction? Explain national disaster risk. [4]
b) Explain Techniques Of Risk Assessment. [4]

OR

P.T.O.

- Q6)** a) What are Global and national disaster risk situation. [4]
b) What are the concepts and elements of disaster risk. [4]
- Q7)** a) Explain Global Co- Operation In Risk Assessment And Warning. [4]
b) How people's participation in Risk Assessment. [4]

OR

- Q8)** a) Explain the role of People's Participation in Strategies for Survival. [4]
b) Explain disaster risk in detail. [4]
- Q9)** a) What do you mean by mitigation? Explain Concept And Strategies Of Disaster Mitigation. [8]
b) Why is disaster mitigation important? Compare Structural Mitigation And Non-Structural Mitigation. [8]

OR

- Q10)** a) Explain Prevention and mitigation of Common Disasters in India. [8]
b) What is mitigation of disaster example? Explain Emerging Trends in Mitigation. [8]



Total No. of Questions : 12]

SEAT No. :

P9533

[Total No. of Pages : 2

[6182]-869

M.E. (Cyber Security)

CYBER SECURITY AND IT INFRASTRUCTURE PROTECTION

(2017 Pattern) (Semester - III) (610402)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10, Q11 or Q12.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*

Q1) Define cybercrime and discuss two examples of cybercrimes. Explain how cybercrimes can impact individuals and organizations. **[8]**

OR

Q2) Explain the concept of cyber warfare and provide one example of a significant cyber warfare event. **[8]**

Q3) Enumerate five basic cybersecurity steps that individuals or organizations can take to enhance their online security. Provide a brief explanation of each step. **[8]**

OR

Q4) Define risk management in the context of cybersecurity and explain why it is crucial for businesses. Provide two examples of cybersecurity risks and discuss how they can be managed effectively. **[8]**

Q5) Describe the basic framework for Infrastructure Management, highlighting its key components. **[8]**

OR

Q6) Highlight two major benefits of implementing an Infrastructure Management system and provide examples or scenarios illustrating their practical importance. **[8]**

P.T.O.

Q7) Discuss the significance of data storage in IT infrastructure and mention two common data storage technologies or methods. **[9]**

OR

Q8) What are types of Computer Software used for Infrastructure Management in Business? **[9]**

Q9) How operational excellence and customer intimacy is achieved? Explain in detail. **[8]**

OR

Q10) Name one ethical issue and one social issue related to the use of information systems in society. **[8]**

Q11) What are the challenges in Storage Infrastructure Management? Brief. **[9]**

OR

Q12) Define the risk triad in the context of storage security. Describe each element of the risk triad and explain how they are interconnected. **[9]**



Total No. of Questions : 8]

SEAT No. :

P9534

[Total No. of Pages : 2

[6182]-871

M.E. (Electrical) (Control Systems)

OPTIMIZATION TECHNIQUES IN CONTROL SYSTEM

(2017 Pattern) (Semester-I) (503101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagram must be drawn wherever necessary.
- 4) Use of algorithmic tables slide rule, Mollier charts, and electronic pocket calculator and steam table is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) Define engineering optimization? What are the engineering applications of optimization. **[4]**

b) Use analytical method to investigate for extreme points

$$F(X) = 2X_1^3 + X_2^3 + X_1^2 + 2X_2^2 + 4 \quad \text{[5]}$$

OR

Q2) Find the Maxima and Minima if any of the function $F(X) = 2x^3 - 18x^2 + 27x - 9$. **[9]**

Q3) Use a Lagrangian multiplier to minimize the function $f(x) = -2X_1^2 + 5X_1X_2 - 4X_1^2 + 18X_1$ subject to $g(x) = X_1 + X_2 = 7$ and $X_1, X_2 \geq 0$ **[9]**

OR

Q4) Explain what you understand by dual simplex method. **[9]**

Q5) Explain the method of conversion of non-serial system to serial system problem. **[16]**

OR

Q6) Explain final value and initial value problems dynamic programming in continuous time systems. **[16]**

P.T.O.

- Q7)** a) Conversion of a Zero-One Polynomial Programming Problem into a Zero-One LP Problem. **[8]**
- b) Explain the Gomory's cutting plane method. **[8]**

OR

- Q8)** Explain the basic concept of stochastic linear and nonlinear programming. **[16]**



Total No. of Questions : 6]

SEAT No. :

P-9535

[Total No. of Pages : 2

[6182]-873

**M.E. (Electrical) (Control Systems)
AUTOMATION IN MANUFACTURING
(2017 Pattern) (Semester - I) (503102)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer Questions from Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) Figures to the right indicate full marks.*
- 3) Neat diagrams must be drawn wherever necessary.*

- Q1)** a) 'Write a Short note on servomechanism. [5]
- b) Explain construction of Ratchet and Pawl Mechanism. [5]
- c) What is directional control Valve? Explain operation of single Solenoid valve. [4]
- d) What is cyber Physical System and explain its features. [4]

OR

- Q2)** a) Explain Hall Effect sensor. [4]
- b) Explain any one type of stepper motor. [5]
- c) Differentiate Pneumatic and Hydraulic actuating systems. [5]
- d) Explain architecture of Cyber Physical system. [4]

- Q3)** a) List out the benefits of Digital Manufacturing? [8]
- b) Explain the architecture of Digital Manufacturing. [8]

OR

P.T.O.

- Q4)** a) Explain different methods of Digital manufacturing science? Explain any one. [8]
b) Explain Operation Reference Mode of Digital Manufacturing. [8]

- Q5)** a) Explain characteristics of Industry 4.0. [8]
b) Explain the benefits of Industry 4.0. [8]

OR

- Q6)** a) Explain working of Industry 4.0.Explian in detail. [8]
b) List out the drivers of Industry 4.0. Explain in detail. [8]



Total No. of Questions : 5]

SEAT No. :

P-9536

[Total No. of Pages : 2

[6182]-874

M.E. (Electrical) (Control Systems)
NON - LINEAR CONTROL SYSTEM
(2017 Pattern) (Semester - I) (503103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Figures to the right indicate full marks.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Use of algorithmic tables slide rule, Mollier charts, and electronic pocket Calculator and steam table is allowed.*
- 4) *Assume suitable data if necessary.*
- 5) *Q.5 is compulsory.*

Q1) a) Explain following terms [6]

- i) Singular points
- ii) Critical Jump resonance
- iii) Limit Cycle

b) Draw the phase plane trajectory for the following equation using Isocline method. [6]

$$\ddot{x} + 2\varepsilon w \dot{x} + w^2 x = 0, \quad \text{given } \varepsilon = 0.15, w_n = 1 \text{ rad / sec}, c(0) = 1.5 \text{ and } \dot{c}(0) = 0.$$

c) An autonomous system is given by [6]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Examine the Stability using Liapunov Direct Method.

OR

Q2) a) Express Scalar function given below in Quadratic form & test it for definiteness [6]

$$V(x) = 8x_1^2 + 5x_2^2 + x_3^2 + 4x_1x_2 - 2x_1x_3 - 3x_2x_3$$

P.T.O.

- b) Explain delta method of constructing a phase plane trajectory for a given system. [6]
- c) Explain Popov criterion for absolute stability. [6]
- Q3)** a) Explain the terms : [6]
- i) Sliding Phase
 - ii) Reaching Phase
 - iii) Chattering as used in sliding control
- b) Write a short note on Input-Output Linearization. [5]
- c) Write a short note on Chattering as used in Sliding Control. [5]
- OR
- Q4)** a) Explain the following (any two) : [16]
- i) Concept of Variable Structure Control
 - ii) Properties of Sliding Mode Control
 - iii) Effect of disturbance on Sliding Mode Control
- Q5)** Explain any two of the following : [16]
- a) Explain any non-linear system design using sliding mode technique.
 - b) Effect of disturbance on Sliding Mode Control.
 - c) Matched & Unmatched uncertainty in case of Sliding Mode Control.



Total No. of Questions : 3]

SEAT No. :

P-9537

[Total No. of Pages : 2

[6182]-875

M.E. (Electrical) (Control Systems)

RESEARCH METHODOLOGY & LINEAR ALGEBRA

(2017 Pattern) (Semester - I) (503104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Figures to the right indicate full marks.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Use of algorithmic tables slide rule, Mollier charts, and electronic pocket Calculator and steam table is allowed.
- 4) Assume suitable data if necessary.

Q1) Solve any three.

- a) Explain about research proposal. Also elaborate about role of literature survey in writing research proposal. [6]
- b) Explain standard writing format for research scheme of AICTE & DST.[6]
- c) Elaborate types of technical paper with its peculiarities. [6]
- d) Compare the structure of survey, conference and Journal paper.
- e) $V = (3, 7, -4)$ is linear combination of vectors $U_1 = (1, 2, 3)$, $U_2 = (2, 3, 7)$, $U_3 = (3, 5, 6)$ find scalar x, y, z for the linear combination. [6]

Q2) Solve any two.

- a) Explain Homogeneous and non homogeneous equations. [8]
- b) By Gaussian elimination form solve the given equations. [8]
$$x - 3y - 2z = 6, \quad 2x - 4y - 3z = 8, \quad -3x + 6y + 8z = -5$$
- c) Explain i) Row canonical form ii) Rank of Matrix. [8]
- d) Write short note with illustration Echelon Matrix. [8]

P.T.O.

Q3) Solve any two.

- a) Explain eigen values and eigen vectors with illustration. [8]
- b) Prove diagonal matrix $B = P^{-1}AP$ where A is system matrix P is any non singular matrix. [8]
- c) Explain with illustration Diagonal form and Jordan form of Matrix representation. [8]
- d) Find diagonal matrix eigen values eigen vectors, characteristic equation

for system Matrix $A = \begin{bmatrix} 2 & 3 \\ 7 & 11 \end{bmatrix}$. [8]



Total No. of Questions : 6]

SEAT No. :

P9538

[6182]-877

[Total No. of Pages : 2

M.E. (Electrical Engineering) (Control System)
MULTIVARIABLE & OPTIMAL CONTROL SYSTEM
(2017 Pattern) (Semester - II) (503107)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right indicate full marks.*
- 3) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 4) Assume suitable data, if necessary.*

- Q1) a)** Explain the definition of controllability and observability of multivariable controlsystem. **[4]**
- b) Define observer. What are advantages of observer in a given system. **[5]**
- c) Explain Concept of state observer and disturbance observer, with it's advantages and limitations. **[5]**
- d) State the infinite time state regulator problem. **[4]**

OR

- Q2) a)** Explain the state space model theory. **[4]**
- b) Elaborate sliding mode observer and give it's robustness properties. **[5]**
- c) Discuss the nonlinear Extended state observer with it's advantages, applications and limitations. **[5]**
- d) Explain **[4]**
- i) Minimum fuel problem
 - ii) State regulator problem

P.T.O.

Q3) State the Pontryagin's minimum principle. Discuss step by step procedure of solving optimal control problem using Pontryagin's minimum principle. **[16]**

OR

Q4) a) Explain the optimal control theory. **[10]**

b) Define Hamiltonian. Derive state, Co-state and control equations. **[6]**

Q5) a) Discuss in detail the Bang - bang control strategy. **[8]**

b) Explain the optimal control law with example. **[8]**

OR

Q6) a) Explain the minimum time optimal control problem. **[8]**

b) Define and explain singular control solution. **[8]**



Total No. of Questions : 6]

SEAT No. :

P9539

[Total No. of Pages : 2

[6182]-878

**M.E. Electrical (Control Systems)
Control of Power Electronics Circuits
(2017 Pattern) (Semester - II) (503108)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Figures to the right indicate full marks.*
- 2) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Use of algorithmic tables slides rule, Mollier charts, and electronic pocket Calculator and steam tables is allowed.*
- 5) Assume suitable data, if necessary.*

- Q1)** a) Explain the roles & objectives of power electronic converters. [6]
- b) Explain different control families of power electronic converter. [6]
- c) Write a note on Sampled data type model. [6]

OR

- Q2)** a) Explain in brief Control principals of Power Electronic Conveters. [12]
- b) State Relations Between Modelling and Control. [6]
- Q3)** a) Draw the circuit diagram of Buck-Boost Converter & explain circuit topologies of it. [12]
- b) Write a note on switched type model. [4]

OR

- Q4)** a) Explain pole placement based on observer design for linear feedback control. [8]
- b) State equilibrium point and static transfer function for Buck and Boost converter. [8]

P.T.O.

- Q5)** a) Explain variable structure control of power electronic converters. [8]
- b) Explain in detail grid connected single phase DC-AC converter with examples. [8]

OR

- Q6)** Explain variable structure control design with one application as DC-DC power stage. [16]



Total No. of Questions : 8]

SEAT No. :

P9540

[Total No. of Pages : 2

[6182]-879

M.E.(Electrical-Control System)

DIGITAL SIGNAL PROCESSING AND ITS APPLICATIONS

(2017 Pattern) (Semester - II) (503109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q. 1 or Q. 2, Q. 3 or Q. 4, Q.5 or Q.6, Q.7 or Q. 8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Explain the linear convolution and its properties. **[5]**

b) Discuss the Z transform and inverse Z-transform along with its basic properties. **[4]**

OR

Q2) a) Discuss the DTFT Definition and its properties. **[5]**

b) Define DFT of discrete-time sequence. Explain the various properties of DFT. **[4]**

Q3) a) What is the difference between DFT and DTFT? Compare Radix-2 DIT and DIF FFT algorithms. **[4]**

b) Explain Fast Fourier Transform: Radix 2 DIT and DIF Algorithms. **[5]**

OR

Q4) a) Explain in detail low pass Butterworth filter approximation. Comment on the Passband and Stopband characteristics of Butterworth filter. **[5]**

b) Realize the IIR systems by Direct form I and Direct form II. **[4]**

Q5) a) Explain the process of windowing using illustrations. What is a Kaiser window function? Obtain its frequency-domain characteristics. **[8]**

b) Explain the symmetric and antisymmetric FIR filters with block diagram. **[8]**

OR

P.T.O.

Q6) a) Obtain Direct form and Cascade form realizations for the transfer function of an FIR system. [6]

b) Compare FIR and IIR filters. What is Hanning window function? Obtain its frequency-domain characteristics. [10]

Q7) a) Discuss design of Discrete PID controller using DSP. [8]

b) Discuss advanced PWM generation by DSP. [8]

OR

Q8) a) Explain various methods of measurement of frequency by DSP. [8]

b) Discuss the various methods of speed control of electrical machines by DSP. [8]



Total No. of Questions : 6]

SEAT No. :

P9541

[Total No. of Pages : 2

[6182]-881

M.E. (Electrical Engineering) (Control System)

ADVANCED DRIVES AND CONTROL

(2017 Credit Pattern) (Semester - III) (603101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figure to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable additional data, if necessary.*
- 5) *Use of non-programmable calculator is allowed.*

- Q1)** a) State and explain the factors affecting the time and energy during starting of electric drive System. [4]
- b) Compare the performance of converter fed DC motor with chopper fed DC motor. [5]
- c) Explain the effect of space harmonics and pulsating torques developed in the inverter fed three phase induction motor. [5]
- d) Write a note on PWM inverters. [4]

OR

- Q2)** a) Explain criteria for selection of drive components. [4]
- b) Explain static krammer control drive in induction motor. [5]
- c) Draw a neat diagram of the system model and derive the transfer function of the converter fed DC motor. [5]
- d) Explain sensorless vector control of induction motor. [4]

Q3) Explain the following in detail:

- a) Switched Reluctance Motor [8]
- b) Sinusoidal SPM drive [8]

OR

P.T.O.

Q4) With reference to synchronous motor drives explain following.

- a) D-q axis equivalent circuit. [8]
- b) Flux weakening operation-the control strategy. [8]

- Q5)**
- a) Explain micro computer control of electric drive [8]
 - b) Explain phase locked loop in closed loop control of electric drive. [8]

OR

- Q6)**
- a) Explain the industrial application of PID controller. [8]
 - b) Explain design procedure of current control in closed loop control of electric drive. [8]



Total No. of Questions : 7]

SEAT No. :

P9542

[Total No. of Pages : 2

[6182]-882

M.E. (Electrical) (Control Systems)
SYSTEM IDENTIFICATION AND ADAPTIVE CONTROL
(2017 Pattern) (Semester - III) (603102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Qu1 or 2, Qu3 or 4, Qu6 or Qu7, Qu5 is compulsory.*
- 2) Figures to the right side indicate full marks.*
- 3) Neat diagram must be drawn whenever necessary.*
- 4) Use of algorithmic tables slide rule, Mollier charts and electronic pocket calculator and steam table is allowed.*
- 5) Assume suitable data, if necessary.*

Q1) a) Explain the parametric and non parametric methods of system identification. **[4]**

b) Compute the QR factorization of matrix: $\begin{bmatrix} 12 & 27 \\ 4 & 2 \\ 6 & 10 \end{bmatrix}$ **[5]**

OR

Q2) a) Derive least square estimate of θ for model given by $y = \Phi\theta$. **[5]**

b) What is the importance of persistently exciting input signal in system identification? What order of p.e. is white noise? **[4]**

Q3) Write short notes on any three: **[9]**

- a) Instrumental variable method
- b) Recursive estimation
- c) Pattern recognition
- d) Bayesian learning
- e) Maximum likelihood method
- f) Model structure

OR

P.T.O.

Q4) a) Prove the matrix inversion lemma: [4]

$$[A + BCD]^{-1} = A^{-1} - A^{-1}B[C^{-1} + DA^{-1}B]^{-1}DA^{-1}$$

$$[A + BCD]^{-1} = [A + BCD] \{A^{-1} - A^{-1}B[C^{-1} + DA^{-1}B]^{-1}DA^{-1}\} I = I.$$

b) With the help of block diagram explain 'self tuning regulator'. [5]

Q5) a) Write short note on [8]

i) MIT rule

ii) Kalman filter as a state estimator

b) What are various adaptive schemes and how they are implemented? [8]

Q6) With the help of both MIT rule and Lyapunov theory, design an MRAS for

system described by $G(s) = \frac{b}{s+a}$ where a and b are unknown. The controller

is given by $u(t) = \theta_1 u_c(t) + \theta_2 y(t)$ and the desired closed loop model is

$\frac{dym}{dx} = -a_m y_m + b_m u_c$. Draw simulation diagram and compare two methods.

$$\text{Assume } V(\theta) = \frac{1}{2} \left[e^2 + \frac{1}{b\gamma} (b\theta_2 + a - a_m)^2 + \frac{1}{b\gamma} (b\theta_1 - b_m)^2 \right]. \quad [16]$$

OR

Q7) Consider a position servo described by $\frac{dv}{dt} = -av + bu$ and $\frac{dy}{dt} = v$; where

a and b are unknown. Assume the control law $u = \theta_1(u_c - y) - \theta_2 v$ is used and that it is desired to control the system in such a way that the transfer function

from command signal to process output is given by $G_m(s) = \frac{w^2}{s^2 + 2\xi ws + w^2}$

determine an adaptive control law that adjusts the parameters so that the desired objective is obtained. [16]



Total No. of Questions : 5]

SEAT No. :

P9543

[Total No. of Pages : 2

[6182]-884

M.E. Electrical (Power Electronics and Drives)

LINEAR SYSTEMS THEORY AND DESIGN

(2017 Pattern) (Semester-I) (503301)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Assume Suitable data. if necessary.*
- 4) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*

Q1) a) Use the vector product to show whether the following sets of vector are orthogonal or not. **[3]**

i) $a = i + 3j + 2k, b = i + 3j - 5k$

ii) $a = 5i - 5j, b = 3i$

iii) Prove your answers using the dot product.

b) If $a = 5i + 4j + 2k, b = 4i - 5j + 3k$ and $c = 2i - 1j - 2k$, compute **[4]**

i) the scalar product $a \cdot b$ and the angle between the vectors a and b

ii) the magnitude and direction cosines of the vector product ($a \times b$) and also the angle that the product vector makes with the vector c .

c) Find the scalar product of the vectors $a = 3i + 8j + k$ and $b = 6i + 8j + 2k$ **[2]**

Q2) a) Describe the span of the following set of vectors. **[4]**

i) $v_1 = (0, 1, 1)$ and $v_2 = (1, 1, 0)$

ii) $v_1 = 1, v_2 = x$ and $v_3 = x^3$

b) Explain, with examples, the following characteristics of square matrices: **[5]**

i) Symmetric

ii) Skew-symmetric

iii) Diagonal

iv) Unit

OR

P.T.O.

Q3) a) Solve the following sets of equations using the matrix method [5]

$$4x - 5y + 6z = 3$$

$$8x - 7y - 3z = 9$$

$$7x - 8y + 9z = 6$$

b) The height of 59 trees was recorded and data was grouped in a frequency distribution table as shown below: [4]

Height (m)	5-9	10-12	13-15	16-18	19-28
Number of trees	14	18	15	4	8

i) Draw a frequency histogram for this data

ii) Use the mid-interval value to calculate the mean height.

Q4) a) The heights of a certain group of boys follow a normal distribution with mean 150.3 cm and variance 25 cm. Find the probability that a boy picked at random has height: [8]

i) less than 153 cm

ii) More than 158 cm

iii) Between 150 cm and 158 cm

iv) More than 10 cm difference from the mean height.

b) A manager wants to see how the mac computers are distributed in his company branches according to region. Data collected were as follows: North East (12), South West (21), East (10) and Mid West (7). Total number of mac computers was 50. Calculate the chi-square statistic for this data. [8]

Q5) a) Explain any two methods of data representation and parameter estimation in detail. [8]

b) Fit the curve $y = a/x^2$ using method of least square to the following data: [8]

x	10	20	30	40	50	60
y	0.02245	0.00545	0.00295	0.00145	0.00095	0.00065



Total No. of Questions : 8]

SEAT No. :

P-9544

[Total No. of Pages : 2

[6182]-885

M.E. Electrical (Power Electronics & Drives)
MODELING AND ANALYSIS OF ELECTRICAL
MACHINES
(2017 Pattern) (Semester - I) (503302)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Figures to the right indicate full marks.*
- 2) *Assume suitable data, if necessary.*

- Q1)** a) Draw necessary diagrams to explain the common essential features of rotating electrical machine. [5]
- b) Explain what is meant by 'power invariance' in transformation theory applied to Electrical machines. [5]

OR

- Q2)** a) Derive the transformations for voltages between two-phase ($d-q$) rotating winding and two-phase (α, β) rotating winding. [6]
- b) Discuss Kron's primitive machine with necessary sketches, How are the various windings of a machine represented in the primitive machine. [4]
- Q3)** a) Discuss the transfer function model of DC shunt motor where speed is the response and armature voltage as ref.input. [5]
- b) Derive the transfer function model of DC series motor. [5]

OR

- Q4)** a) Discuss the Principle of field orientation and types of vector control methods. [4]
- b) Write and discuss two axis ($d-q$) Voltage equation matrix model for induction machine in various reference frames. [6]

P.T.O.

Q5) a) A 3-phase 50 Hz cylindrical-rotor synchronous machine has the following parameters:

Self-inductance for phase a = 3.15 mH

Armature leakage inductance = 0.35 mH

For this machine calculate the mutual inductance between armature phases and its Synchronous reactance. [7]

b) Derive expressions for armature mutual inductance of salient pole synchronous machine from a consideration of its basic parameters. [8]

OR

Q6) a) From the torque matrix of a 3-phase salient pole alternator and its phasor diagram, derive an expression for synchronous power in terms of the load angle. [8]

b) Starting from the impedance matrix of a 3-phase salient pole synchronous machine, obtain an expression for instantaneous electromagnetic torque. [7]

Q7) a) Discuss Performance prediction of - Induction machine with stator electric transients neglected in a linearised machine model. [7]

b) Explain Linearization of machine equations and displacement theory. [8]

OR

Q8) Write short notes on :

a) Eigen values of typical induction machine [8]

b) Eigen values of typical synchronous machine [7]



Total No. of Questions : 6]

SEAT No. :

P-9677

[Total No. of Pages : 2

[6182]-886A

M.E. (Electrical) (Power Electronics & Drives)

POWER CONVERTERS - I

(2017 Pattern) (Semester - I) (503303)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic calculator is allowed.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Explain the following ratings related to SCR [8]

- i) Surge current Rating
- ii) di/dt rating
- iii) Various Anode voltage ratings

b) Explain six step voltage source inverter (120 deg mode) with necessary waveforms and derive line-line output voltage equation using Fourier series. [10]

OR

Q2) a) A single phase full a wave ac voltage regulator feeds a load of $R = 20$ ohms with an input voltage of 230 V and 50 Hz. Firing angle of both the thyristors is 45 degrees. Calculate [10]

- i) rms value of output voltage
- ii) load power and input power factor
- iii) average and rms current of thyristors

b) Explain the Buck Converter with necessary wave forms. [8]

Q3) a) Explain single phase bi directional controller with R load. [8]

b) Explain integral control method and Phase angle control method in a single phase AC/AC voltage regulator. [8]

OR

P.T.O.

- Q4)** a) Explain the internal and external methods to control the output voltage of inverter. [8]
- b) A single phase voltage controller is employed for controlling the power flow from 230 V 50Hz into a load circuit containing $R = 4 \Omega$ and $\omega L = 3\Omega$. Calculate [8]
- i) The control range of firing angle
- ii) maximum value of RMS load current maximum power and power factor
- Q5)** a) With a neat diagram and relevant waveform explain the working of a single phase to single phase step down midpoint type cycloconverter supplying RL load for discontinuous load current. [8]
- b) A three phase cycloconverter feeds a single phase load of 190 V at 45 A at power factor of 0.7 lagging Determine [8]
- i) required supply voltage
- ii) thyristor rating power factor of supply current

OR

- Q6)** a) Draw the circuit diagram and explain the principle of operation of a 1 phase bridge type Cycloconverter. What is the relation between triggering angles of the thyristor of positive and negative converters. [8]
- b) With a neat diagram and relevant waveform explain the working of a single phase to single phase step down midpoint type cycloconverter supplying RL load for continuous load current. [8]



Total No. of Questions : 8]

SEAT No. :

P-9546

[Total No. of Pages : 2

[6182]-887

M.E. Electrical (Power Electronics & Drives)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - I) (503304)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable additional data, if necessary.*
- 5) *Use of non-programmable calculator is allowed.*

Q1) a) List out the characteristics of Good Research. **[4]**

b) Explain the necessity of defining the problem. **[5]**

OR

Q2) a) Explain the methods of data collection. **[4]**

b) What are the measures of central tendency? **[5]**

Q3) a) Discuss the important features of LaTeX for report writing. **[4]**

b) Explain how to make title page using LaTeX for a thesis. **[5]**

OR

Q4) a) What is plagiarism and how to avoid it? **[4]**

b) Explain the various types of IPR. **[5]**

Q5) a) What are the steps in research proposal writing? Explain all the steps. **[8]**

b) Explain the structure of a Survey paper in detail. **[8]**

OR

P.T.O.

- Q6)** a) Explain the various parts of the project report writing. [8]
b) Explain review paper, comparison paper, poster paper and conference paper. [8]

- Q7)** a) What is difference between references and bibliography? Explain the various formats of writing bibliography. [8]
b) Explain the significance of technical report writing. [8]

OR

- Q8)** a) Explain the importance of effective communication. [8]
b) How to make oral presentation? Comment on planning, preparation and delivery. [8]



Total No. of Questions : 6]

SEAT No. :

P9547

[6182]-889

[Total No. of Pages : 2

M.E. (Electrical) (Power Electronics and Drives)

AC AND DC DRIVES

(2017 Pattern) (Semester - II) (503307)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain different types of industrial loads. **[6]**
- b) Explain Steady state analysis of separately excited DC motor fed by chopper. **[6]**
- c) Explain with neat circuit diagram operation of Separately excited DC motor connected to 1 phase fully controlled converter. Derive expression showing speed torque relationship. Draw output voltage and current waveforms. **[6]**

OR

- Q2)** a) Explain four quadrant operation of Electric drive with suitable example. **[6]**
- b) Explain principle of operation of VSI driven Induction motor. **[6]**
- c) How motor duty, heating and cooling affects the temperature of motor? **[6]**
- Q3)** a) Compare DC motor control analogy with vector controlled induction motor. **[8]**
- b) Explain the principle of vector control and its implementation for rotor flux oriented control. Justify your explanation with necessary phasor diagram and block diagrams. **[8]**

OR

P.T.O.

- Q4)** a) Explain field oriented control of induction motor. [8]
- b) Giving implementation block diagram discuss direct vector control of induction motor. [8]
- Q5)** a) Describe the operation of PMDC motor drive and derive the expression for torque generated in it. [8]
- b) With the help of neat block diagram explain how the speed is controlled in BLDC motor. [8]

OR

- Q6)** a) Discuss the control circuit & performance characteristics of switched reluctance motor. [8]
- b) What is multi-stack stepper motor? Discuss performance characteristics of this motor. [8]



Total No. of Questions : 6]

SEAT No. :

P9548

[Total No. of Pages : 2

[6182]-890

M.E. Electrical (Power Electronics and Drives)

POWER CONVERTERS - II

(2017 Pattern) (Semester - II) (503308)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figure to the right indicate full marks.
- 3) Use of electronic calculator is allowed.
- 4) Assume suitable data, if necessary.

Q1) a) Explain about the following with neat circuit diagrams. **[10]**

- i) Sinusoidal Pulse Width Modulation,
- ii) Modified SPWM
- iii) Comparison of PWM techniques

b) Explain about the following with neat circuit diagrams. **[8]**

- i) Concept of Multilevel,
- ii) Diode Clamped inverter,
- iii) Cascaded multi-level inverter

OR

Q2) a) Explain about Concept of basic resonant circuit and their Classification with neat circuit diagrams. **[10]**

b) Explain the concept of ZVS and ZCS with neat diagram and waveforms. **[8]**

Q3) a) With neat diagram explain Thermal modeling of power switching devices. **[8]**

b) Explain about Electrical equivalent thermal model of heat sink and derived the necessary equations. **[8]**

OR

P.T.O.

Q4) a) Explain the design for heat sink for SCR and derive the required equations. **[8]**

b) Explain the Selection procedure for Heat Sink for IGBT. **[8]**

Q5) a) With a neat diagram and relevant waveform explain the magnetic design for Specific inductor for induction heating load applications and its procedure. **[8]**

b) Explain about the Need of Snubber circuit and Types of Snubber circuits. **[8]**

OR

Q6) a) With a neat diagram and relevant waveform explain the working of Turn-On snubber and its design Turn-Off snubber. **[8]**

b) With a neat diagram and relevant waveform explain the working of explain over voltage Snubber. **[8]**



[6182]-891

M.E. (Electrical) (Power Electronics & Drives)**ADVANCED CONTROL SYSTEMS****(2017 Pattern) (Semester - II) (503309)****Time : 3 Hours]****[Max. Marks : 50****Instructions to the candidates:**

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Assume suitable data if necessary,*
- 3) *Use of scientific calculator is allowed.*

- Q1) a)** Consider the system with PID controller shown in figure 1(A). Obtain K_p , T_i and T_d using Zeigler-Nichols rules. **[6]**

where, $G_e(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$, $G(s) = \frac{1}{s(s+1)(s+5)}$



- b) Write the short note on regular form of sliding mode control. **[3]**

OR

- Q2) a)** Explain in depth, the Zeigler-Nichol's methods for tuning of PID controller. **[6]**

- b) Explain in brief about the sliding surface design. **[3]**

- Q3) a)** Explain in brief the mathematical model of sliding mode observer. **[5]**

- b) Explain observer based control boost converter. **[4]**

OR

- Q4) a)** Write the short note on direct sliding control for Buck type DC/DC converter. **[4]**

- b) Write the short note on application of non-linear slide mode observer in real life system with advantages and disadvantages. **[5]**

Q5) a) Discuss the following terms pertaining in sense of Lyapunov's stability criterion : **[8]**

- i) Asymptotic stable system
- ii) Asymptotic stable in large
- iii) Limit cycles

b) Describe the variable gradient method for constructing the Lyapunov's function to determine the stability of non linear system. **[8]**

Q6) a) Explain the followings with an example : **[8]**

- i) Input-State Linearization
- ii) Feedback Linearization

b) Write the short note on normal form and zero dynamics in concern with the input output linearization. **[8]**



Total No. of Questions : 8]

SEAT No. :

P9550

[Total No. of Pages : 2

[6182]-893

**M.E. (Electrical Engineering) (Power Electronics & Drives)
SPECIAL APPLICATIONS OF POWER ELECTRONICS
(2017 Pattern) (Semester - III) (603301)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) Figures to the right indicate full marks.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Assume suitable additional data, if necessary.*
- 5) Use of non-programmable calculator is allowed.*

Q1) a) Explain series connected multi terminal HVDC link with suitable diagram. **[4]**

b) Explain with neat diagram modelling of thyristor switched capacitor (TSC). **[5]**

OR

Q2) a) Draw a neat diagram of thyristor switched reactor (TSR). **[4]**

b) Explain the concept of HVDC light technology and what are its advantages. **[5]**

Q3) a) Write application of Internet of things (IOT) for smart grid power applications. **[4]**

b) Write a short note on concept of cyber security. **[5]**

OR

Q4) a) Write a short note on Bluetooth. **[4]**

b) Explain the function of actuators used for signal acquisition and control in smart metering. **[5]**

P.T.O.

- Q5) a)** Write a short note on Wide Area Measurement System (WAMS). Draw a neat block diagram. **[8]**
- b)** Explain Distributed Energy Resources and their interconnection with a hosting grid. **[8]**

OR

- Q6) a)** With block diagram, explain the operation of Phase measurement Unit (PMU). **[8]**
- b)** With reference to Smart Grid. Explain. **[8]**
- i) Concept of Smart Grid
 - ii) Need of Smart Grid.

- Q7) a)** Explain the working principle of Electronic ballast with neat diagram. **[8]**
- b)** Describe following power electronics applications using various possible techniques : Induction heating. **[8]**

OR

- Q8) a)** Explain the concept of Hybrid vehicle system. List out advantages and weakness of Hybrid vehicle system. **[8]**
- b)** Identify the various voltage disturbances likely to occur in power system operation. How can it be control using uninterrupted power supplies? Explain its working with the help of block diagram. **[8]**



Total No. of Questions : 6]

SEAT No. :

P9551

[Total No. of Pages : 1

[6182]-894

**M.E. (Electrical) (Power Electronics & Drives)
ENERGY MANAGEMENT AND POWER QUALITY
(2017 Pattern) (Semester - III) (603302)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figure to the right indicate full marks.*
- 3) *Use of calculator is allowed.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Discuss Grounding aspects in power electronics circuits. [9]
b) Explain reasons for Grounding and various Grounding Problems. [9]

OR

- Q2)** a) What are the various effects of the power quality disturbances? [9]
b) Explain energy saving methods and optimal selection criteria of industrial Fans. [9]

- Q3)** a) What is power quality benchmarking? Explain in detail. [8]
b) Write the procedure to design a harmonic filter. [8]

OR

- Q4)** a) Discuss in detail various voltage regulation devices. [8]
b) Write detail note on Voltage sags and interruptions. [8]

- Q5)** a) Explain the application of intelligent system in power quality monitoring. [8]
b) List various power quality monitoring instruments and briefly explain their use. [8]

OR

- Q6)** a) Write detail note on Power quality monitoring standards. [8]
b) Explain objectives and consideration of power quality monitoring. [8]



Total No. of Questions : 3]

SEAT No. :

P-9552

[Total No. of Pages : 2

[6182]-896

M.E. (Electrical Power System)
COMPUTER APPLICATIONS IN POWER SYSTEM
(2017 Pattern) (Semester - I) (503201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) Solve any THREE :

[18]

- a) Draw surface constraint diagram showing behavior and side constraints. Also indicate bounded and unbounded, acceptable and unacceptable points on surface constraint.
- b) Explain Univariate optimization technique.
- c) Derive the model of converter for AC DC load flow
- d) Derive load flow equations used in decoupled load flow method with assumptions.

Q2) Solve any TWO.

[16]

- a) The incremental production costs of both the units are

$$\frac{dF_1}{dP_1} = 0.4P_1 + 40 \text{ Rs/MWh} \quad \text{and} \quad \frac{dF_2}{dP_2} = 0.5P_2 + 30 \text{ Rs/MWh}$$

Determine economical schedule for the sharing of 200 MW demand considering generation limit on each unit as 25MW and 150MW. Determine the saving obtained by economical loading over equal load sharing.

P.T.O.

- b) State economical dispatch problem with
 - i) Unconstrained without loss
 - ii) Constrained without loss
 - iii) Unconstrained with loss
 - iv) Constrained with loss
- c) Explain Newton's method for economic load dispatch with algorithm considering limits.

Q3) Solve any TWO.

[16]

- a) Derive B-coefficient formula used in economic dispatch.
- b) The transmission loss coefficients B_{mn} expressed in pu on the base of 100MVA of a power system network having three plants are given by

$$B = \begin{bmatrix} 0.01 & -0.001 & -0.002 \\ -0.001 & 0.02 & -0.003 \\ -0.002 & -0.003 & 0.03 \end{bmatrix}$$

Three plants supply power of 100MW, 200MW and 300MW respectively into the network. Calculate the transmission loss and incremental transmission losses of the plant in MW

- c) Show that transmission loss formula is a function of generation and load.



Total No. of Questions : 5]

SEAT No. :

P-9553

[Total No. of Pages : 2

[6182]-897

M.E. (Electrical Power Systems)

**POWER SECTOR ECONOMICS AND MANAGEMENT
(2017 Pattern) (Semester - I) (503202)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right indicate full marks.*
- 3) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 4) Assume suitable data, if necessary.*

Q1) Attempt any three from following :

[18]

- a) Explain the role Power Grid Corporation and Rural Electrification Corporation.
- b) Explain principles of tariff setting
- c) For a project initial investment of Rs. 500000 is required. The cash flows generated for five years are Rs. 100000, Rs. 200000, Rs. 50000, Rs. 150000 and Rs. 100000. Calculate net present value for a project with discounting factor as 8%.
- d) Explain various key performance parameters used, by regulator to assess the performance of various utilities.
- e) Explain Single Buyer and vertically integrated power network model.
- f) Explain rate of return and cost plus regulations.

Q2) a) Explain demand elasticity and price elasticity. Also discuss demand and price curves in power market. **[8]**

b) What are the ancillary services? Explain ancillary service market. **[8]**

OR

P.T.O.

- Q3)** a) Compare future and forward market. [8]
b) Explain the process of market settlement. Also discuss the factors affecting market settlement process. [8]

- Q4)** a) What are the firm transmission rights? What are the effect of these on system operation? [8]
b) Explain transmission cost allocation methods. [8]

OR

- Q5)** a) Explain transmission planning and its need in restructure power industry. [8]
b) What are the effects of congestion? How these can be managed? [8]



[6182]-898

M.E. (Electrical) (Power Systems)
POWER SYSTEM MODELING
(2017 Pattern) (Semester - I) (503203)

Time : 3 Hours]**[Max. Marks : 50****Instructions to the candidates:**

- 1) Solve Total Three questions. Answer 1 question from Q1 and Q2, Q3 or Q4, and Q5 or Q6 each.
- 2) Assume suitable data if necessary.
- 3) Write down all the assumptions made.
- 4) Given $F^{abc} = [P] F^{dq0}$, where Park's transformation

$$[P] = \begin{bmatrix} k_d \cos \theta & k_q \sin \theta & k_0 \\ k_d \cos \left(\theta - \frac{2\pi}{3} \right) & k_q \sin \left(\theta - \frac{2\pi}{3} \right) & k_0 \\ k_d \cos \left(\theta + \frac{2\pi}{3} \right) & k_q \sin \left(\theta + \frac{2\pi}{3} \right) & k_0 \end{bmatrix}$$

$$\left(\text{where, } k_d = k_q = \sqrt{2/3} \text{ and } k_0 = \sqrt{1/3} \right)$$

- Q1)** Derive the model of short and 'pi' circuit medium transmission line. Draw an equivalent circuit diagram. Also comment about the symmetry of the model. Write down assumptions involved if any. **[18]**

OR

- Q2)** A synchronous generator is represented by an EMF behind a synchronous reactance. For this machine governor controls the shaft input power and the excitation system controls the internally generated EMF E_g . A synchronous generator is operating at a lagging power factor with current I_1 , internally generated voltage E_{g1} , and terminal voltage V_t . Assume that the input power is held constant by governor. With these initial operating conditions, assume that the excitation is increased to new value E_{g2} . Assume that the bus voltage is held constant by other machine operating in parallel with this machine. Draw equivalent circuit diagram and with the help of phasor diagram, determine the new value of current I_2 , the new power factor $\cos \Phi_2$, and the new torque angle δ_2 . **[18]**

P.T.O.

Q3) Write down with the help of suitable diagram working of stationary diode excitation control scheme of alternator. Discuss difference between static and rotating diode type excitation control scheme. **[16]**

OR

Q4) Explain with the help of suitable diagram working of static and brushless excitation control scheme of alternator. **[16]**

Q5) State need of Clarke's transformation and Kron's transformation and write about their applications in power system. Also write down their relationship with Park's transformation. **[16]**

OR

Q6) With the help of appropriate transformation using α - β variables, develop the model of long transmission line. Discuss about the assumptions and approximations involved in it. **[16]**



Total No. of Questions : 6]

SEAT No. :

P-9555

[Total No. of Pages : 1

[6182]-899

M.E. (Electrical)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - I) (Power Systems) (503204)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any three questions from question 1 to 4.*
- 2) *Questions 5 and 6 are compulsory.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right indicate full marks.*
- 5) *Use of logarithmic table, slide rule and non-programmable calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

Q1) What is the importance of literature survey? Explain in details. **[6]**

Q2) What are the types of technical papers? Explain in detail. **[6]**

Q3) Describe the process of obtaining a patent along with the legal procedure. **[6]**

Q4) Explain the Elimination method of solving linear simultaneous equations with example. **[6]**

Q5) a) Write a detailed note on convex programming problem. **[8]**

b) Explain in detail - Finite difference approximations of partial derivatives with example. **[8]**

Q6) Write short notes on the following with applicable example in electrical engineering: **[16]**

- i) Particle Swarm Optimization
- ii) Genetic algorithm
- iii) Linear regression



Total No. of Questions : 5]

SEAT No. :

P9556

[6182]-901

[Total No. of Pages : 2

M.E. (Electrical) (Power Systems)
POWER SYSTEM DYNAMICS
(2017 Pattern) (Semester - II) (503207)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any three sub questions out of four in Q.1.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Write down all the assumptions made.

Q1) a) A generator is connected to an infinite bus through external impedance. The generator is represented by a voltage source in series with a reactance. If infinite bus voltage = 1.0, $Z_e = 0.1 + j0.2$, $X_g = 0.3$ (all in p.u.), for $P_b = 1.0$ p.u. Find the equilibrium values of power angle in the range of angle (-180, 180). Test their stability. (P_b is the received power at the infinite bus). Assume infinite bus angle as zero. **[6]**

b) What is Voltage Stability? Explain CIGRE task force definitions of voltage stability. **[6]**

c) What are various modes of power system oscillations? Explain block diagram of PSS. **[6]**

d) Obtain state space representation of synchronous machine model 1.0 for small signal stability analysis. **[6]**

Q2) Derive the stator equations and rotor equations for synchronous machine dynamics connected to infinite bus for model 1.1. Also state the modifications required for model 1.0. **[16]**

OR

P.T.O.

Q3) Explain computation of initial conditions for dynamics of synchronous machine connected to infinite bus. **[16]**

Q4) Explain synchronizing and damping torque analysis and its application for **[16]**

- a) System without AVR
- b) Fast excitation system.

OR

Q5) Draw overall block diagram of SMIB system with Haffron Philips constants. Apply Routh Hurwitz criterion to system with static exciters and obtain necessary stability conditions. **[16]**



Total No. of Questions : 6]

SEAT No. :

P9557

[Total No. of Pages : 2

[6182]-902

**M.E. Electrical (Power Systems)
Power System Planning & Reliability
(2017 Pattern) (Semester - II) (503208)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, and Q.No.5 or Q.No.6.*
- 2) Neat Diagram must be drawn wherever necessary.*
- 3) Figure to the right indicate full marks.*
- 4) Assume suitable data, if necessary.*
- 5) Use of non programmable scientific calculator is allowed.*

Q1) a) What are the objectives of load forecasting? Explain the method of reactive load forecasting. **[9]**

b) What is probability in context with reliability? Explain Regression analysis. **[9]**

OR

Q2) a) Elaborate the steepest descent method used for the mathematical model in generation planning. **[9]**

b) Explain the concept and evaluation technique of loss of load expectation (LOLE) index. **[9]**

Q3) a) Describe inter regional transmission systems of India. **[8]**

b) Write down the procedure of evaluating system indices in transmission network. **[8]**

OR

Q4) a) List the main causes of failure of transmission line, towers and substation equipments. **[9]**

b) Explain the steps of short term transmission planning. **[7]**

P.T.O.

- Q5) a)** Explain the monitoring of system through reconfiguration algorithm. **[8]**
- b) How the effect of weather is taken into consideration in distribution system planning **[8]**

OR

- Q6) a)** State the roles of disconnectors, circuit breakers and isolators in electrical distribution network.
[8]
- b) Elaborate on partial load transfer how it is useful in distribution planning.**[8]**



Total No. of Questions : 5]

SEAT No. :

P-9558

[Total No. of Pages : 2

[6182]-903

M.E. (Electrical) (Power System)

HVDC AND FLEXIBLE AC TRANSMISSION

(2017 Course) (Semester - II) (503209)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Assume suitable data, if necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculators is allowed.*

Q1) Attempt any three of the following : [18]

- a) Explain different configuration of multi terminal HVDC system.
- b) Explain 12-pulse converter operation in HVDC system.
- c) Explain HVDC protection schemes used against over - voltage and over current.
- d) Describe the difference between EHVAC and HVDC transmission.
- e) Describe the harmonics in HVDC systems.
- f) What is FACTS? Explain their controller types.

Q2) a) Describe the applications of SVC and STATCOM. [8]

- b) Explain the working principle of Static Synchronous Series Compensator (SSSC). [8]

OR

Q3) a) Write the comparison of different SVC's in terms of performance, efficiency & limitations. [8]

- b) Explain the need of FACTS controller in modern power system. List the advantages of different FACTS controllers. [8]

P.T.O.

- Q4)** a) Give the comparison of UPFC with other FACTS devices. [8]
b) Explain the block diagram of overall UPFC control structure. [8]

OR

- Q5)** a) Explain how a UPFC is different than a simple VSC. [8]
b) Explain the steady state UPFC model for Power flow studies. [8]



Total No. of Questions : 3]

SEAT No. :

P9559

[Total No. of Pages : 2

[6182]-905

M.E. (Electrical) (Power Systems)

ADVANCED POWER SYSTEM PROTECTION

(2017 Pattern) (Semester - III) (603201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*
- 4) *Assume suitable data wherever necessary.*
- 5) *Use of non-programmables scientific calculator is allowed.*

Q1) Attempt any three.

- a) Draw and explain in detail equivalent circuit of protection PT. **[6]**
- b) Explain protection of transformer using digital filtering scheme. **[6]**
- c) Draw and explain block diagram of numerical relay. State function of each block. **[6]**
- d) Elaborate various types of faults in power systems. **[6]**
- e) A 1200/5, C400 with secondary resistance 0.61Ω and a 2.0Ω is the relay burden. Based on the accuracy classification, what is the maximum symmetrical fault current that may be applied to this CT without exceeding a 10% ratio error? **[6]**

Q2) Attempt any two.

- a) What is meant by Over-reach of distance relay? How this problem is solved in three-stepped distance protection? **[8]**
- b) Describe digital protection scheme for transmission line based upon fundamental signals. **[8]**
- c) Explain amplitude comparison travelling wave relay scheme; how it is useful for protection of forward as well as reverse faults in transmission line. **[8]**

P.T.O.

Q3) Attempt any two.

- a) Explain how primary and backup relay settings are calculated to get proper relay coordination. **[8]**
- b) Describe under reach and over reach phenomenon in over current protection of transmission line. **[8]**
- c) Elaborate the applications of computer graphics in electrical protection system. **[8]**



Total No. of Questions : 5]

SEAT No. :

P9560

[Total No. of Pages : 2

[6182]-906

**S.Y.M.E. Electrical (Power System)
POWER QUALITY ASSESSMENT & MITIGATION
(2017 Pattern) (Semester - III) (603202)**

[Time : 3 Hours]

[Max. Marks : 50]

Instructions to the Candidate:

- 1) *Attempt Q.No.1 is compulsory.*
- 2) *Answer Q. No.2 or Q. No.3, Q.No.4 or Q. No.5.*
- 3) *Neat diagram must be drawn wherever Necessary.*
- 4) *Figures to the right indicate full marks.*
- 5) *Assume suitable data, If necessary.*
- 6) *Use of logarithmic tables slide rule, mollier charts and electronic pocket calculator and steam table is allowed.*

Q1) Attempt any three of the following:- [18]

- a) Explain term distortion power , displacement and distortion Power Factor.
- b) Explain source of voltage Flicker & mitigation techniques.
- c) How power quality events are classified.
- d) What is harmonic? Explain causes of harmonics.
- e) Explain purpose of grounding? How grounding is associated with power quality.

Q2) a) Explain Effect of system response under presence of harmonics. Also explain Principles of harmonic controls. [8]

b) Explain working operation of shunt active Filters. [8]

OR

Q3) a) Explain Conventional & advanced filter design criteria. [8]

b) Explain the role of computer tools in harmonic analysis. [8]

P.T.O.

- Q4)** a) Explain Power quality monitors. [8]
b) Explain typical Instrument used for power quality monitoring. [8]

OR

- Q5)** Explain any two of the following:- [16]
a) Why selection of transducers is Important in power quality assessment?
b) What is the role of custom power devices for power quality mitigation.
c) Discuss the process of flicker monitoring.

❧ ❧ ❧

Total No. of Questions : 8]

SEAT No. :

P-9561

[Total No. of Pages : 2

[6182]-908

M.E. (Electronics) (Digital System)
ANALOG & DIGITAL CMOS DESIGN
(2017 Pattern) (Semester - I) (504101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five Questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) What are MOSIS Scalable CMOS Design Rules? What are the advantages using it? [5]

b) Explain advantages of stick diagram and draw it for logic : $\overline{(A+B \cdot C)}$ [5]

Q2) a) What do you mean by photolithography and Ion Implantation? [5]

b) Explain need of layout Design rules and list them with proper sketches. [5]

Q3) a) What is RC-delay model? Illustrate it with example of any circuit. [5]

b) Explain the various components of Delay in Logic gates. [5]

Q4) a) What are the efforts to compensate delay in Logic gates? [5]

b) Differentiate Logical, Electrical and Parasitic delay. [5]

P.T.O.

- Q5)** a) With neat diagram explain Cascode Amplifiers. What are the advantages? [5]
- b) Explain the implementation of : [5]
- i) Diode and
- ii) Current Sinks and Voltage Source.
- Q6)** a) Write short notes on : [5]
- i) Current mirrors,
- ii) Voltage Reference
- b) Explain the operation of sense Amplifier in SRAM & DRAM. [5]
- Q7)** a) Draw and explain BiCMOS Circuits. [5]
- b) With schematic diagram, explain Sense Amplifier Circuits. [5]
- Q8)** a) Compare different logic circuit families. [5]
- b) What are the specific methodologies used for Low Power Logic Design? [5]



Total No. of Questions : 8]

SEAT No. :

P-9562

[Total No. of Pages : 2

[6182]-909

M.E. (Electronics) (Digital Systems)
MULTIRATE SIGNAL PROCESSING
(2017 Pattern) (Semester - I) (504102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any five questions.*
- 2) *Assume suitable data if necessary.*
- 3) *Figures to the right indicate marks.*

Q1) a) Design a 2-stage interpolator which is used to increase sampling rate from 10KHz to 200KHz. [5]

Pass band ripple : 0.01

Stop band ripple : 70 dB

Pass band : 0 to 90 Hz

b) Explain steepest descent algorithm of adaptive filter. What are the limitations of this algorithm. [5]

Q2) a) What are the characteristics of polyphase filters? How are the polyphase structures efficient? [5]

b) For a given signal $x(n)$ considering of single pulse of length N

$$x(n) = 1 \quad \text{for } n = 0, 1, 2, \dots, N - 1$$

$$= 0 \quad \text{otherwise}$$

Use Prony's method to model $x(n)$ with one pole & one zero. [5]

Q3) a) Explain the design of phase shifter in multirate system. [4]

b) What is the characteristics of QMF's frequency response? How to design these filters. [4]

c) Write a note on oversampling ADC/DAC. [2]

P.T.O.

Q4) a) Decompose the following sequence using Harr wavelet. $f = (2, 5, 8, 9, 7, 4, -1, + 1)$ [4]

b) Explain how to use wavelet transform for filtering. [4]

c) Determine the output $y(n)$ in terms of input $x(n)$ for the multirate system as shown in fig. [2]

$$x(n) \rightarrow \boxed{5\uparrow} \rightarrow \boxed{10\downarrow} \rightarrow \boxed{2\uparrow} \rightarrow y(n)$$

Q5) a) Explain use of adaptive filter for echo cancellation. [4]

b) How is the implementation of decimeter & interpolator made efficient? [4]

c) Explain the properties of Haar wavelet. [2]

Q6) a) Transfer function $H(z) = \frac{(1 + 0.85z^{-1})}{(1 - 0.65z^{-1})}$.

Perform polyphase decomposition of $H(z)$ into 4 section. [4]

b) What is the need of DFT filter banks? Explain synthesis filter bank. [4]

c) Draw & explain all pole structure. [2]

Q7) a) Explain signal compression using DWT. [4]

b) Explain 3D spectrogram. [4]

c) Calculate time bandwidth product for wavelet decomposition. [2]

OR

Q8) a) Write a note on sampling rate conversion by a arbitrary factor. Also comment on sequence of multirate. [4]

b) Derive the expression for output spectrum of decimeter in terms of input spectrum. [4]

c) Define continuous wavelet transform. [2]



Total No. of Questions : 8]

SEAT No. :

P-9563

[Total No. of Pages : 2

[6182]-910

**M.E. (Electronics) (Digital System)
EMBEDDED SYSTEM DESIGN
(2017 Pattern) (Semester - I) (504103)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data, if necessary.*

Q1) a) Explain details of ARM-CORTEX series. **[5]**

b) Describe features of Intel Galileo Gen2. **[5]**

Q2) a) Describe significance of CMSIS standards. **[5]**

b) Draw and explain architecture of LPC-1768. **[5]**

Q3) a) Explain Micro C/OS II. **[5]**

b) Draw structure of Android & explain with features. **[5]**

Q4) a) Draw kernel of RTOS & explain. **[5]**

b) Describe interrupt & synchronization in RTOS. **[5]**

Q5) a) Explain importance of embedded Linux. **[5]**

b) Write different steps to implement device driver for any hardware. **[5]**

P.T.O.

- Q6)** a) What are different hardwares & its drives? Explain any one. [5]
b) Explain Design aspects in implementation of drives. [5]
- Q7)** a) Explain any one design application. [5]
b) What are the important points of EMI certification. [5]
- Q8)** a) Write testing & failure analysis of EMD product. [5]
b) Draw & explain automatic metering system. [5]



Total No. of Questions : 8]

SEAT No. :

P-9564

[Total No. of Pages : 2

[6182]-911

**F.Y. M.E. (Electronics)/(Digital Systems)/(E & TC)
(Microwave)/(E & TC lot and Sensor System)/
(Signal Processing)/(VLSI & Embedded System)/
(Wireless Communication Technology)/
(Communication Network)**

**RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (504104)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions out of eight questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data if necessary.*
- 4) *Use of scientific calculator is allowed.*

- Q1)** a) What are Research Characteristics? [5]
b) Compare Research methods verses Research Methodology. [5]

- Q2)** a) Explain Objectives and types of research. [5]
b) Compare Quantitative Research verses Qualitative Research. [5]

- Q3)** a) What is significant role of literature review in defining a problem? [5]
b) Identify gap areas from literature review and Explain in detail. [5]

- Q4)** a) How to Define the research problem? [5]
b) Write short note on to publish a Research patent. [5]

- Q5)** a) What is need of research design? [5]
b) Explain steps in sample design and criteria for selecting sampling procedure. [5]

P.T.O.

- Q6)** a) Explain technique of developing measurement tool and important scaling technique. [5]
b) Explain methods of data collection in research and types of analysis. [5]
- Q7)** a) Write a different steps in the preparation of thesis writing. [5]
b) Explain any one tool for Documentation and presentation. [5]
- Q8)** a) Write short note on Intellectual property rights and patent law. [5]
b) Explain different types of technical papers. [5]



Total No. of Questions : 8]

SEAT No. :

P9565

[6182]-913

[Total No. of Pages : 2

M.E. (Electronics) (Digital Systems)
ADVANCED DSP PROCESSORS
(2017 Pattern) (Semester - II) (504107)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions out of eight questions.**
- 2) Figures to the right indicate full marks.**
- 3) Assume suitable data if necessary.**
- 4) Use of scientific calculator is allowed.**

- Q1) a)** Explain Very Long Instruction Word architecture. **[5]**
- b) Explain Parallelism and data address generators in digital signal processors. **[5]**
- Q2) a)** Explain pipelining and circular buffering used in Digital signal processors. **[5]**
- b) How many buses are there in TMS320C54XX processor. **[5]**
- Q3) a)** Explain DMA controller in Black fin processor. **[5]**
- b) Explain software development flow of Blackfin Processor. **[5]**
- Q4) a)** Explain following processes. **[5]**
- Sampling and quantization
 - Coder-decoder
- b) How to create and build a project in Code Composer Studio. **[5]**

P.T.O.

- Q5)** a) Explain Accumulators and Barrel Shifter of TMS320C54XX. [5]
b) Explain Multiplier/Adder unit of TMS320C54XX. [5]
- Q6)** a) What is the difference between fixed-point and floating-point representations? [5]
b) Explain memory map of Black fin BF533 processor. [5]
- Q7)** a) What is need of antialiasing filter prior to down sampling and anti-imaging filter after up sampling a signal? [5]
b) Draw & Explain architecture of TMS 320C54XX. [5]
- Q8)** a) Draw & Explain architecture of TMS 320C67XX. [5]
b) Explain barrel shifter with block schematic. [5]



Total No. of Questions : 8]

SEAT No. :

P9566

[Total No. of Pages : 2

[6182]-914

M.E.(Electronics - Digital Systems)

PLD's And ASIC Design

(2017 Pattern) (Semester - II) (504108)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions from the following.*
- 2) *Figure to the right indicates full marks.*
- 3) *Draw neat diagram wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Explain Top down approach to System Design. [5]
b) What is System on Chip? What are advantages and disadvantages of SOC? [5]
- Q2)** a) State the FSM issues and explain any one of them. [5]
b) Write a short note on Pipelining. [5]
- Q3)** a) Write Short notes on: [6]
i) Operator Interference in VHDL
ii) Library and Packages in VHDL
b) Differentiate between functions and procedures in VHDL. [4]
- Q4)** a) Draw and explain the architecture Complex Programmable Logic Device XC 9500. [5]
b) Explain the Data Flow and Behavioral modeling techniques in VHDL with proper examples. [5]
- Q5)** a) Explain the architecture and functional block of Cool Runner CPLD. [5]
b) What do you understand by FPGA? Explain with the applications. [5]
- Q6)** a) Write the short note on Xilinx Virtex II Pro. [5]
b) What are the Common In-Circuit Debug Approaches for FPGAs. Explain any one of these approaches. [5]

P.T.O.

- Q7) a)** What is the standard cell based ASIC design? What is the roll of cell libraries in ASIC design? In short, explain the steps in ASIC cell libraries design. **[5]**
- b) Write a short note on ASIC design EDA Tools. **[5]**
- Q8) a)** Write a short note on Physical Design for ASIC covering following points:
- i) Floor Planning,
 - ii) Partitioning,
 - iii) Power Planning,
 - iv) Placement,
 - v) Clock Tree Synthesis,
 - vi) Routing and,
 - vii) DFM. **[7]**
- b) What is the difference between a microprocessor and IP Core? Which processors are available as IP Cores? **[3]**



Total No. of Questions : 8]

SEAT No. :

P-9567

[Total No. of Pages : 2

[6182]-915

M.E. (Electronics) (Digital System)

IMAGE PROCESSING AND COMPUTER VISION

(2017 Pattern) (Semester - II) (504109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five of the following questions.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Draw neat diagram wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Write short note on Hough transform and its applications. [5]
b) Explain in brief different filtering methods for image enhancement. [5]

- Q2)** a) Explain following methods of 2D transformations with the help of appropriate matrix and equations: affine transform, projective transform and scaled rotation. [6]
b) Explain erosion and dilation with examples. [4]

- Q3)** a) Write Haar transform matrix for $n=2$ and $n=4$. List the properties of Haar transform. [4]
b) What are different color models used in color image processing? What is color transformation and how it is performed? [6]

- Q4)** a) What is the basic idea behind color slicing? Describe the method for slicing a color image. [5]
b) Explain Head and Hat algorithm and explain its use in image registration. [5]

- Q5)** a) Explain weighted geometric feature algorithm and its use in image processing. [5]
b) Explain different techniques of color image compression. [5]

P.T.O.

- Q6)** a) Explain the Iterative closest point algorithm and its use in image processing. [5]
b) Classify image registration methods. Explain any one method in detail. [5]
- Q7)** a) Write short note on volumetric surface reconstruction. [5]
b) Explain the method of reconstructing shape from shade and shape from texture. [5]
- Q8)** a) Explain in detail-Model based reconstruction. [5]
b) Compare point base representation and volumetric representations.[5]



Total No. of Questions : 8]

SEAT No. :

P9568

[Total No. of Pages : 1

[6182]-917
M.E. (DS) (Electronics)
SYSTEM ON CHIP DESIGN
(2017 Pattern) (Semester - III) (604101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Solve any 5 questions.*
- 2) *Assume suitable data, if necessary.*

- Q1)** a) Explain manchester carry chain. [6]
b) Explain overhead analysis. [4]
- Q2)** a) Compare different memories RAM, ROM. [6]
b) List technical specification of DDR-II RAM. [4]
- Q3)** a) Explain software configurable process. [5]
b) Explain SOC low power. [5]
- Q4)** a) Explain Barrel shifter. [5]
b) Compare AHB & ASB. [5]
- Q5)** a) Explain array processor. [5]
b) Explain sequential arcs. [5]
- Q6)** a) Explain IPS in SOC. [5]
b) Explain SOC flow. [5]
- Q7)** a) Explain synchronization. [5]
b) Explain metastability. [5]
- Q8)** a) Explain dynamic voltage scaling. [5]
b) Explain leakage power. [5]



Total No. of Questions : 8]

SEAT No. :

P9569

[Total No. of Pages : 2

[6182]-918

**M.E.(Electronics)(Digital Systems)
WIRELESS SENSOR NETWORK FOR IOT
(2017 Pattern) (Semester - III) (604102)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any Five questions out of eight questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*
- 4) *Use of scientific calculator is allowed.*

- Q1)** a) What are the operating systems used in WSN? Explain any one in brief.[5]
b) What are the hardware components of Wireless Sensor Network? List and explain. [5]
- Q2)** a) Write short note on energy consumption in wireless sensor node. [5]
b) Draw and Explain architecture of Wireless Sensor Network. [5]
- Q3)** a) Explain IEEE 802.15.4 in detail. [5]
b) Explain the design issues in wireless sensor network. [5]
- Q4)** a) What is ISM band? What are frequencies available with this band. [5]
b) Explain Zigbee protocol for WSN. [5]
- Q5)** a) What are the link layer technologies in IoT. [5]
b) Compare any five points between IPv4 and IPv6. [5]
- Q6)** a) List and explain the Elements of an IoT ecosystem. [5]
b) Explain the characteristics of Big data. [5]

P.T.O.

- Q7)** a) Explain how MQTT protocol works. [5]
- b) Explain IoT reference architecture. [5]
- Q8)** a) Explain the application of IoT in Energy and Environment. [5]
- b) Write a short notes on: [5]
- i) Security and privacy in IoT
 - ii) Data Aggregation Techniques



Total No. of Questions : 8]

SEAT No. :

P-9570

[Total No. of Pages : 2

[6182]-920

M.E. (E & TC) (Communication Networks)

**MODELLING AND SIMULATION OF
COMMUNICATION NETWORKS**

(2017 Pattern) (Semester - I) (504501)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Draw neat diagram whenever necessary.*
- 3) *Figures to right indicate full marks.*
- 4) *Use of electronic pocket calculator is allowed.*

- Q1)** a) What are the parameters of performance estimation in simulation? [4]
- b) Why PN sequence generators are needed in simulation? What are its Components? Also mention properties of a PN sequence? [4]
- c) What is up sampling and down sampling in communication system? [2]

- Q2)** a) What are the different methods to map uniform random variables to an arbitrary pdf? [4]
- b) Explain the graphical technique for post processing. [4]
- c) Explain the role of simulation in communication system. [2]

- Q3)** a) Giving the example state the estimation model for band pass signal. [4]
- b) What are correlated Gaussian Numbers? Why do we need them? [4]
- c) What are the various specifications to be considered while simulating a radio channel? [2]

P.T.O.

- Q4)** a) What are the aspects for selection of sampling frequency for a simulation? [4]
b) What are various techniques to generate uniformly distributed random numbers? Compare and contrast the following algorithms. [4]
i) Mixed congruence algorithm
ii) Multiplicative algorithm with prime modulus.
c) What are the categories of a communication channel? [2]
- Q5)** a) Compare and contrast pure Monte Carlo approach and semi-analytic approach to performance estimation of a communication system. [4]
b) Enlist factors to be considered while simulating nonlinearities. [4]
c) What is direct component and Quadrature component of a signal? [2]
- Q6)** a) What is semi analytic simulation technique? Hence suggest modifications in an algorithm for BER estimation for QPSK. [4]
b) Find out the power spectral density relation in a Tapped delay line model. [4]
c) Elaborate methodology used in mapping a problem into a simulation model. [2]
- Q7)** a) Describe a two state Markov model for discrete channels with memory. [4]
b) Explain Queue system related to network modeling. [4]
c) What is the significance of relative frequency in Monte-Carlo estimation? [2]
- Q8)** a) Explain Poisson's modeling. [4]
b) What are the different techniques used to reduce run time of the Monte Carlo method? Hence explain importance of sampling. [4]
c) How to estimate the error probability in a digital transmission system? [2]



Total No. of Questions : 8]

SEAT No. :

P-9571

[Total No. of Pages : 2

[6182]-921

M.E. (E & TC) (Communication Networks)
HIGH SPEED COMMUNICATION NETWORKS
(2017 Pattern) (Semester - I) (504502)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt any five questions out of eight.*
- 2) All questions carry equal marks.*
- 3) Neat labelled diagrams must be drawn wherever necessary.*
- 4) Figures to the right indicate full marks.*

Q1) a) Compare Frame Relay with ATM. [5]

b) Explain link layer addressing in High-Speed Networks. [5]

Q2) a) Discuss TCP-Window Management System. [5]

b) Explain in detail GFR traffic management. [5]

Q3) a) Explain KARN's Algorithm in detail. [5]

b) What is Traffic and Congestion control in ATM. [5]

Q4) a) Explain Fair queuing (FQ) Algorithm. [5]

b) With the help of neat diagram discuss RTP streaming protocol. [5]

Q5) a) Explain various Video coding standards. [5]

b) Describe with diagram any one video filtering method. [5]

P.T.O.

- Q6)** a) Enlist various Image compression technique. Explain any one in detail. **[5]**
b) How does Cell Delay Variation relate to performance of Congestion control in ATM networks? **[5]**
- Q7)** a) Explain architecture of 802.11. **[5]**
b) What are various characteristics of queuing system? **[5]**
- Q8)** a) Explain H.264/AVC coder. **[5]**
b) What is the need of Wireless LANs? Explain with an example. **[5]**



Total No. of Questions : 8]

SEAT No. :

P-9572

[Total No. of Pages : 2

[6182]-922

M.E. (E & TC) (Communication Networks)

NETWORK SECURITY

(2017 Pattern) (Semester - I) (504503)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt any five questions out of eight questions.*
- 2) Assume suitable data wherever necessary.*
- 3) Right hand side number indicate marks.*

Q1) a) Explain Secure software Design in detail. [5]

b) Define & Explain cryptography with example. [5]

Q2) a) Explain in detail Network Security & operations. [5]

b) Explain security architecture with neat diagram. [5]

Q3) a) Explain conventional encryption with neat flow chart. [5]

b) Explain why access control is important. [5]

Q4) a) Explain cipher block. [5]

b) Explain operating system security management. [5]

Q5) a) Explain encapsulating security payload. [5]

b) Explain Cybersecurity with suitable example. [5]

P.T.O.

- Q6)** a) Explain Network security applications. [5]
b) Explain need of cybersecurity for wired networks. [5]
- Q7)** a) Define & describe web security requirements. [5]
b) Explain Cyberattacks with example. [5]
- Q8)** a) Define SSL & describe with neat diagram. [5]
b) Explain need of cybersecurity for online shopping. [5]



Total No. of Questions : 8]

SEAT No. :

P-9573

[Total No. of Pages : 2

[6182]-924

M.E. (E & TC) (Communication Networks)

TRAFFIC ANALYSIS AND QOS

(2017 Pattern) (Semester - II) (504507)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Figures to the right side indicates full mark*
- 3) *Draw neat diagram wherever necessary.*
- 4) *Assume suitable data, if necessary.*
- 5) *Use of electronic pocket calculator is allowed.*

- Q1)** a) Describe organization model and explain two-tire & three-tire network management organization model. [5]
- b) What are the different network traffic monitoring tools? Explain its uses. [5]
- Q2)** a) Write a short note on MPLS traffic engineering. [5]
- b) Define MAN. Explain SONET-based MAN. [5]
- Q3)** a) Describe network management functional model in detail. [5]
- b) Explain SNMP in detail. [5]
- Q4)** a) Describe in detail the TMN service management architecture. [5]
- b) Write a short note on WDM optical transport network. [5]
- Q5)** a) What are the important paraments that characterize data traffic flow.[5]
- b) Explain 802.5 token ring technology in High-Speed LANs. [5]

P.T.O.

- Q6)** a) Explain functions of ISA to manage congestion and QoS transport. [5]
b) Explain stop-and-wait ARQ in detail. [5]
- Q7)** a) Explain QoS parameters defined by ATM forum? [5]
b) Discuss the fibre channel protocol architecture. [5]
- Q8)** a) What are the key performance issues in cell delay variation? [5]
b) What are the types of Network Traffic? Explain in brief. [5]



Total No. of Questions : 8]

SEAT No. :

P-9574

[Total No. of Pages : 2

[6182]-925

M.E. (E & TC) (Communication Networks)
BROADBAND WIRELESS TECHNOLOGIES
(2017 Pattern) (Semester - II) (504508)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any 5 questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*

- Q1)** a) Discuss the advantages of Block based transmissions over continuous transmission. [5]
b) What are the advantages of CP based CDMA over single carrier FDMA? [5]
- Q2)** a) Compare multi antenna system and MIMO in detail. [5]
b) Discuss in detail about SVD based Eigen beamforming technique. [5]
- Q3)** a) Discuss the difference between Time hopping, frequency hopping with Direct sequence in ultra wideband system. [5]
b) Compare ALOHA and slotted Aloha protocols in Media Access control. [5]
- Q4)** a) Explain Carrier sense multiple access with collision avoidance MAC in detail. [5]
b) Explain the IEEE 802.11 MAC protocol with suitable diagrams. [5]
- Q5)** a) Classify the Routing Protocols in Wireless Broadband Networks with suitable examples. [5]
b) Explain the significance of outage probability for video services in a multirate DS CDMA system. [5]

P.T.O.

- Q6)** a) Discuss the types of connections supported by WiMAX. [5]
b) Explain Dynamic BW allocation algorithm (DBA) in WiMAX. [5]
- Q7)** a) Explain the architecture of Ethernet Passive Optical Networks (EPONS) in detail. [5]
b) Explain the difference between passive and active optical network.[5]
- Q8)** Write a short note on (any two) :
- a) Scalable broadband access networks [5]
b) Fault tolerance & self-healing [5]
c) Hybrid WOBAN [5]



Total No. of Questions : 8]

SEAT No. :

P-9575

[Total No. of Pages : 2

[6182]-926

M.E. (E & TC) (Communication Network)

SDR AND COGNITIVE RADIO

(2017 Pattern) (Semester - II) (504509)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer Q. No. 1 or 2, Q. No.3 or 4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams to be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Discuss the basic concept of Cognitive Radio. **[7]**

b) Give any two major applications of Cognitive Radio technology. **[6]**

OR

Q2) a) Write on various low cost Cognitive Radio platforms available. **[7]**

b) What are the improvements in Software Defined Radio technology? **[6]**

Q3) a) Discuss various worldwide frequency plans available. **[6]**

b) Write the salient features in Cognitive Radio Architecture **[6]**

OR

Q4) a) Describe the layered architecture of Cognitive Radio. **[6]**

b) Write short note on end-to-end communication in Cognitive Radio. **[6]**

Q5) a) Discuss significant features of Real Time Operating Systems. **[6]**

b) Detail the requirements of CORBA technology. **[6]**

OR

P.T.O.

- Q6)** a) Write in detail om functional overview of SCA. [6]
b) Give the technological differences between SCA and JTRS. [6]

- Q7)** a) What is the role of adaptive techniques in antennas? [7]
b) Describe salient features in smart antenna architecture. [6]

OR

- Q8)** a) How is cognitive radio system principles applied to smart antenna systems? [7]
b) Write short notes on Radio Frequency design. [6]



Total No. of Questions : 8]

SEAT No. :

P9576

[Total No. of Pages : 2

[6182]-928

M.E. (E & TC) (Communication Networks)

4G LTE CELLULAR SYSTEMS

(2017 Pattern) (Semester - III) (604501)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Describe in detail LTE advanced E-UTRAN architecture. **[7]**

b) Write on the different protocol stacks in standardization of LTE. **[6]**

OR

Q2) a) Explain the characteristic features of LTE and LTE Advanced. **[7]**

b) Explain the role of Mobility management entity in LTE. **[6]**

Q3) a) Describe in detail the research challenges in spectrum sharing. **[7]**

b) Explain how carrier aggregation is accomplished in LTE. **[6]**

OR

Q4) a) Detail the RF requirements for LTE. **[7]**

b) Illustrate the methodology involved in retransmission control. **[6]**

Q5) a) Explain the operation of OFDMA. **[6]**

b) Differentiate between co-operative MIMO and single site MIMO. **[6]**

OR

P.T.O.

- Q6)** a) What are MIMO adaptive switching schemes? [6]
b) Explain the functioning of Software Defined Radio. [6]

- Q7)** a) Write in detail on centralized CoMP architecture. [6]
b) Write short notes on Wi-Max standards evolution. [6]

OR

- Q8)** a) Give the point of differences between LTE and Wi-Max. [6]
b) Discuss the relay basic schemes in CoMP. [6]



Total No. of Questions : 8]

SEAT No. :

P9577

[Total No. of Pages : 2

[6182]-929

M.E.(Electronics & Telecommunication)(Communication Network)

MODERN COMMUNICATION RECEIVER

DESIGN AND TECHNOLOGY

(2017 Credit Pattern) (Semester - III) (604502)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions carry equal marks.
- 2) Solve any FIVE questions.
- 3) Assume suitable data, wherever necessary.
- 4) Your answer as whole will be given weightage.

- Q1)** a) Enlist the various types of image reject mixers used in direct conversion receiver. [5]
- b) With the help of suitable block diagram, explain the implementation of double conversion receiver. [5]
- Q2)** a) Explain the probability of Intercept in high Dynamic Range receiver with spectrum. [5]
- b) With the help of suitable block diagram, explain the ideal receiver. [5]
- Q3)** a) Explain the design of Automatically Switched Half-Octave Filter Banks for high frequency transceiver. [5]
- b) Why the half-octave filter banks are used for ultra wideband microwave receiver? Write in detail. [5]
- Q4)** a) Explain the term MDS (minimum discernible signal) with mathematical expressions. [5]
- b) Explain the switching mechanisms of front-end filters for best dynamic range performance. [5]
- Q5)** a) Compare & Contrast various mixer technologies used in Receiver. [5]
- b) Explain the brute force, direct, coherent mixer synthesizer with suitable diagram. [5]

P.T.O.

- Q6)** a) With suitable diagram, Explain a basic double balanced commutative FET mixer. [5]
- b) Compare the direct and indirect methods of non brute force synthesizer.[5]
- Q7)** a) Enlist different Noise Blanker & state the importance of Noise Blanker in IF receiver? [5]
- b) With a suitable Block diagram explain the fast attack, slow release AGC system in the Star-10 IF receiver. [5]
- Q8)** a) What is the significance of Automatic Gain Control (AGC) in Receivers? Explain the role of the PIN diode attenuator used in AGC. [5]
- b) Write in detail I and Q Demodulation Process. [5]



Total No. of Questions : 8]

SEAT No. :

P-9667

[Total No. of Pages : 2

[6182]-931

M.E. (E & TC) (IoT & Sensor System)

SENSORS & MEASUREMENTS

(2017 Pattern) (Semester - I) (504601)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt any 5 questions out of 8.***
- 2) Neat diagrams must be drawn whenever necessary.***
- 3) Figures to the right of questions indicate full marks.***
- 4) Assume suitable data, if necessary.***

Q1) a) Give the detailed classification of sensors based on performance and types. [5]

b) Explain the working principle of piezo-resistive and capacitive pressure sensor. [5]

Q2) a) Write short note on : [5]

- i) Avalanche Photo diodes***
- ii) Load Cells***

b) Explain the working principle of LED and semiconductor Lasers. [5]

Q3) a) Explain Micro bending process and highlight its applications. [5]

b) Justify the working principle of piezoelectric accelerometer and thermal accelerometer. [5]

Q4) a) Explain the detailed working of monolithic and optical gyroscopes. [5]

b) What is Accelerometer? How it works? List its characteristics. [5]

P.T.O.

- Q5)** a) Explain the working principle of LVDT and highlight its applications. [5]
b) Explain Fabry Perot sensor in Detail & list its applications. [5]
- Q6)** a) Write short notes on : [5]
i) Eddy current
ii) Hall Effect
b) Explain the working principle of RVDT and highlight its applications. [5]
- Q7)** a) Explain the working principle of flow sensors and give its classifications and applications. [5]
b) Write short note on temperature sensors. [5]
- Q8)** a) What is Solid state electronics based microphone and how it works? [5]
b) Explain the working principle of Acoustic sensors and give its classifications and applications. [5]



Total No. of Questions : 8]

SEAT No. :

P-9668

[Total No. of Pages : 2

[6182]-932

M.E. (Electronics & Telecommunications - IoT and Sensor System)

DATA COMMUNICATION & NETWORKING

(2017 Pattern) (Semester - I) (504602)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any 5 questions out of 8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Explain the data communication techniques and throw the lights on its types. [5]
- b) Explain the layered architecture related to data communication and networking. [5]
- Q2)** a) Explain about the motivation for cross layer protocol design with neat diagram. [5]
- b) Explain network traffic behaviour under the context of network layers and topology design. [5]
- Q3)** a) Explain the fundamentals of time synchronization and how it is fruitful to network management. [5]
- b) Explain the types of sensors which are used in sensor network of data communication and networking. [5]
- Q4)** a) What is a self-organizing network? And how does it work? [5]
- b) What do you understand by multimedia? What type of data is communicated in multimedia communication? [5]

P.T.O.

- Q5)** a) Why Software defined Networking (SDN) is important in data communication. [5]
- b) Explain the term reliability and why it is important for data communications and networking. [5]
- Q6)** a) What does scheduling mean? Write the names of two networks used in data communication and networking. [5]
- b) Explain wireless sensor networks. Why is wireless networking preferred over wired networks? [5]
- Q7)** a) What are the transport and application layers that are widely used in data communication and networking? [5]
- b) Explain the term quality of service (QoS) used extensively in data communications and networking. [5]
- Q8)** a) What is encryption? Is it different from decryption or not? Explain any encryption technique. [5]
- b) Explain the term queuing as it relates to data communications and networking. [5]



Total No. of Questions : 8]

SEAT No. :

P-9681

[Total No. of Pages : 2

[6182]-933

M.E. (Electronics & Telecommunication)

(E & TC IoT and Sensor System)

WIRELESS SENSOR NETWORK FOR IoT

(2017 Course) (Semester - I) (504603)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt to any five questions out of Eight.*
- 2) Neat diagram must be drawn whenever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume the suitable data, if necessary.*

- Q1)** a) Distinguish between RS232 and RS485 protocols. [5]
b) Explain the Zigbee. [5]
- Q2)** a) With neat block diagram explain the Ultra-wide sand. [5]
b) Explain the MiWi protocol. [5]
- Q3)** a) Distinguish between WSN vs Adhoc Networks. [5]
b) Explain the transceiver design considerations in WSNs. [5]
- Q4)** a) Explain the Low duty cycle protocols and wakeup concepts. [5]
b) Write short note on IEEE 802.15.4 MAC protocol. [5]
- Q5)** a) Write a short note on Flooding and Gossiping. [5]
b) Explain Design Principle of Gateway Protocols. [5]

P.T.O.

- Q6)** a) With neat block diagram explain the WSN architecture. [5]
b) Write short note on internet communicationn WSN. [5]
- Q7)** a) Explain 6 Lo WPAN. [5]
b) Explain the IP based WSN. [5]
- Q8)** a) Explain M2M communication. [5]
b) Write short note on Alljoyn network. [5]



Total No. of Questions : 8]

SEAT No. :

P-9578

[Total No. of Pages : 2

[6182]-942

M.E. (E & TC) (Microwave)

ELECTROMAGNETICS AND ANTENNA THEORY

(2017 Pattern) (Semester - I) (504301)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) Explain the Reflector Antennas, its different types and applications. [5]
b) Determine the length of the dipole whose input resistance is 50ohms. (194). [5]
- Q2)** a) The radius of a small loop of a constant current is $\lambda/25$. Find the physical area of the loop and compare it with its maximum aperture. (243) [8]
b) Why the spiral antenna is broadband antenna? [2]
- Q3)** Describe the following antennas with illustrative structural diagrams, radiation pattern, features [10]
a) Log-Periodic Antenna
b) H-Plane pyramidal and conical horn antenna
- Q4)** It is desired to design an aperture antenna, with uniform illumination, so that the directivity is maximized at an angle of 30° from the normal to the aperture. Determine the optimum dimensions and its associated directivity when the aperture is a) Square b) Circular (696). [10]

P.T.O.

- Q5)** a) Give the comparison of various radiating elements such as infinitesimal dipole, small dipole, finite length dipole and half wave length dipole. **[6]**
- b) Explain the salient features of Rhombic Antenna and explain the working principle with relevant diagram. **[4]**
- Q6)** a) Determine the null to null beamwidth of a broadside array when array length is 15λ and number of elements are 25. **[5]**
- b) Give the structural features and radiation characteristics of following antennas: **[5]**
- i) Lens antenna
- ii) Long Wire Antenna
- Q7)** a) Design an rectangular microstrip antenna using a substrate (RT/Duroid 5880) with dielectric constant of 2.2, $h = 0.15$ so as to resonate at 10GHz (820). **[6]**
- b) Explain the following method of analysis in Microstrip patch antenna
i) Finite Difference Time Domain (FDTD). **[4]**
- Q8)** a) With reference to uniform plan wave explain the following parameters: Reflection Coefficient, VSWR. **[5]**
- b) Explain the following terms with respect to microstrip antenna: Quality factor, bandwidth, input impedance. **[5]**



Total No. of Questions : 6]

SEAT No. :

P-9579

[Total No. of Pages : 2

[6182]-943

M.E. (E & TC) (Microwave)

RF AND MICROWAVE CIRCUIT

(2017 Pattern) (Semester - I) (504302)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Use of electronics pocket calculator is allowed.*
- 4) *Assume suitable data if necessary.*
- 5) *Figures to the right indicates full marks.*

- Q1)** a) Explain with working of following microwave components using proper diagrams: **[6]**
i) Wilkinson Power divider
ii) Two hole Coupler
b) Explain working of two hole directional coupler. **[4]**
- Q2)** a) Derive the expressions for propagation constant, impedance, and power flow for the lossless coaxial line. **[6]**
b) For an arbitrary N port microwave device, derive the relationship between transmission (ABCD) matrix and impedance matrix. **[4]**
- Q3)** a) How inter symbol interference affects on wireless communication? Derive expression for roll of factor. **[5]**
b) Write short note on conversion gain and distortion. **[5]**
- Q4)** a) Derive the equation of scattering matrix for two hole directional coupler. **[5]**
b) Write a note on dynamic range and sources of noise in microwave circuits. **[5]**

P.T.O.

- Q5)** a) Describe in detail construction and operating principal of PIN diode. **[6]**
b) Explain MESFET in terms of Construction, Working & frequency response. **[4]**
- Q6)** a) How RF diode differs from conventional semiconductor diode? Explain construction and working of Schottky diode with related parametric derivation. **[6]**
b) Explain operating principle of Tunnel diode. **[4]**



Total No. of Questions : 6]

SEAT No. :

P-9580

[Total No. of Page : 1

[6182]-944

M.E. (E & TC) (Microwave)

MICROWAVE MEASUREMENTS

(2017 Course) (Semester - I) (504303)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions.*
- 2) *Assume suitable data, if necessary.*

- Q1)** a) State the basic principles of Transmission Lines. [5]
b) What are the sources of uncertainty in RF and microwave measurement? [5]
- Q2)** a) What is attenuation? Explain How to reduce Attenuation? [5]
b) Draw and Explain the set-up used for reduction of Noise in measurement. [5]
- Q3)** a) Draw and explain the practical set-up for calibration of Network Analyzer. [5]
b) Give the advantages of power sensors in Microwave Measurement. [5]
- Q4)** What are the different methods to reduce noise. Which factors are considered to calculate signal to Noise ratio. [10]
- Q5)** Explain the different parameters to be considered for attenuation measurement. Draw a practical circuit to measure attenuation, with graph. [10]
- Q6)** Write short notes on :
a) MMIC Measurement Techniques. [5]
b) Inductive Voltage divider. [5]



Total No. of Questions : 6]

SEAT No. :

P-9581

[Total No. of Pages : 2

[6182]-946

M.E. (E & TC) (Microwave)

COMPUTATIONAL ELECTROMAGNETICS

(2017 Course) (Semester - II) (504307)

Time : 3 Hours]

[Max. Marks : 50

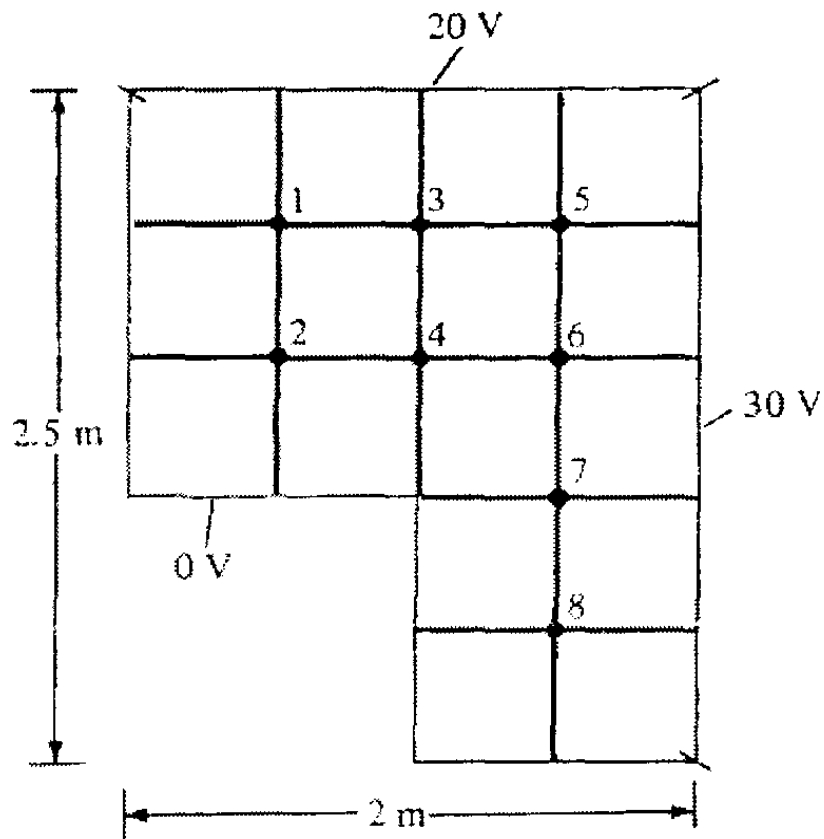
Instructions to the candidates :

- 1) Answer any five questions.*
- 2) Neat diagram must be drawn wherever necessary.*
- 3) Use of electronics pocket calculator is allowed.*
- 4) Assume suitable data if necessary.*
- 5) Figures to the right indicate full marks.*

- Q1)** a) Write short note on absorbing boundary conditions for FDTD. [5]
b) Explain strengths and weaknesses of method of moments. [5]
- Q2)** a) Explain how MOM is used for solution of integral equations with an example. [5]
b) Explain overview of various computational method. [5]
- Q3)** a) Explain Helmholtz wave equation and how do you solve it numerically. [5]
b) Explain the principle of variational method with a suitable example. [5]
- Q4)** Explain the Rayleigh Ritz method of solution of electromagnetic problem explain the procedure to solve Poisson's equation using this method. [10]

P.T.O.

Q5) Find the voltages at all the nodes using FDTD method for the problem given below. Use iterative method up to 3 steps. **[10]**



Q6) Find the functional for ordinary differential equation. **[10]**

- a) $y'' + y + x = 0$, where $0 < x < 1$.
- b) Subject to $y(0) = y(1) = 0$.



Total No. of Questions : 6]

SEAT No. :

P-9582

[Total No. of Pages : 2

[6182]-947

M.E. (E & TC) (Microwave)

RF AND MMIC TECHNOLOGY

(2017 Pattern) (Semester - II) (504308)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Use of electronics pocket calculator is allowed.*
- 4) *Assume suitable data if necessary.*
- 5) *Figures to the right indicates full marks.*

Q1) a) Explain fabrication technology used for design of HEMT device. **[6]**

b) Explain the structure of micro strip coupler. **[4]**

Q2) a) Explain performance issues of MMIC technology. **[6]**

b) Write a note on multi chip module technology. **[4]**

Q3) a) Explain multilayer technique used in MMIC. **[5]**

b) What is difference between Micro strip couplers and splitters. **[5]**

Q4) a) Explain synthesis process of n port network. **[5]**

b) Explain network matrix decomposition. **[5]**

P.T.O.

- Q5)** a) Explain harmonic balance method used in microwave circuit. **[6]**
b) What are the features and applications of CAD package. **[4]**
- Q6)** a) How digital modulator is designed by MMIC technology? Explain. **[6]**
b) Explain MMIC test system. How MMIC Measurement is carried out?**[4]**



Total No. of Questions : 8]

SEAT No. :

P-9583

[Total No. of Pages : 3

[6182]-948

M.E. (E & TC) (Microwave)

WIRELESS COMMUNICATION SYSTEM

(2017 Pattern) (Semester - II) (504309)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) a) Draw the cell broadcast message format with reference to BMC protocol in UMTS. **[6]**

b) Compare the soft handover and hard handover in UMTS. **[4]**

Q2) a) What is WCDMA, what are its technology features with respect to the following: Multiplexing, duplexing, Modulation scheme, maximum data rate? **[5]**

b) If a total of 66MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses four-cell reuse. **[5]**

Q3) a) In a digital cellular system, given a carrier frequency $f_c=1000\text{MHz}$ and the mobile velocity is 150km/hr, calculate the received carrier frequency if the mobile **[4]**

i) Directly towards the transmitted.

ii) In a direction perpendicular to the direction of arrival of the transmitted signal.

P.T.O.

- b) Define the terms co-channel, co-channel interferences. Derive the D/R ratio using the basic Hexagonal cell geometry. [6]

Q4) Write a detailed note Okumara Propagation Model for various cases,also explain its applications. [10]

Q5) a) Assume a receiver is located 10km away from a 50W transmitter. The carrier frequency is 6GHz and free space propagation is assumed, $G_t=1$ and $G_r=1$. Find the power at the receiver, the magnitude of the E-field at the receiving antenna. [5]

b) What are the main features of following standards? [5]

- i) AMPS
- ii) GSM
- iii) IS-95
- iv) DECT
- v) CDMA-2000

Q6) a) Consider a cellular system in which total available voice channels to handle the traffic are 960. The area of each cell is 6km^2 and the total coverage area of the system is 2000km^2 . Calculate the system capacity if the cluster size $N=4$. How many times would cluster of size have to be replicated to cover the entire cellular area? [6]

b) Define the following terms with reference to teletraffic engineering. [4]

- i) Erlang
- ii) CCS
- iii) Blocked call held
- iv) Blocked call held

Q7) Write a short note on the following with reference to UMTS. [10]

- i) UTRAN
- ii) GGSN
- iii) HLR

Q8) Consider a wireless network with the following data:

[10]

Total population: 400,000

Subscriber penetration: 50%

Average call holding time for mobile-to-land and land-to-mobile calls: 200 seconds

Average call holding time for mobile-to-mobile calls: 60 seconds

Average calls per hour for mobile-to-land and land to mobile 4

Average calls per hour for mobile-to-mobile:5

Traffic distribution: Mobile-to-land =30%; Land-to-mobile=50%; and mobile-to-mobile =20%

- a) Calculate the total traffic in Erlangs, and
- b) If each switch can support 3000 Erlange, how many switches are required in the network?



Total No. of Questions : 8]

SEAT No. :

P-9584

[Total No. of Pages : 2

[6182]-950
M.E. (E & TC) (Microwave)
EMI AND EMC TECHNIQUES
(2017 Pattern) (Semester - III) (604301)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer any five question*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Use of Calculator is allowed.*
- 5) Assume Suitable data if necessary.*

- Q1)** a) Define the terms Electromagnetic Interference(EMI) and Electromagnetic Compatibility(EMC), what are the various causes for EMI and the need for compatibility. **[5]**
- b) What is cross talk, how does it affect the performance in PCB? Explain various techniques for minimizing the cross talk in PCB. **[5]**
- Q2)** a) What is the role of prediction in EMI and explain any of the prediction techniques with relevant mathematical expressions and illustrative diagrams. **[8]**
- b) Give an example for radiated and conducted EMI. **[2]**
- Q3)** a) Explain amplitude culling and frequency culling with respective to transmitter and receiver circuit with relevant circuit diagrams and compare both the methods. **[8]**
- b) What are precautions to be taken while earthling? **[2]**
- Q4)** a) Explain all the various types of testing methods in detail. **[6]**
- b) Classify the interferences based on receiver and interference bandwidths. **[4]**

P.T.O.

- Q5)** a) Explain common mode and differential mode filter design for power line. **[5]**
b) Write short notes on MIL-STD and FCC regulation. **[5]**
- Q6)** a) How isolation transformers are used to suppress the interference? Explain with relevant diagrams. **[6]**
b) What is application of lumped element low pass filter? Explain with relevant circuit diagram and design considerations. **[4]**
- Q7)** Write a detailed note on various methods measurement of Radiated Emission and Radiated susceptibility. **[10]**
- Q8)** a) Draw the structure of typical anechoic chamber mentioning various equipments required. What are the considerations to be made while designing? **[6]**
b) Explain the procedure for antenna radiation pattern measurement in anechoic chamber. **[4]**



Total No. of Questions : 8]

SEAT No. :

P9585

[Total No. of Pages : 2

[6182]-951

M.E.(E & TC) (Microwave)

RADAR AND SATELLITE COMMUNICATIONS

(2017 Pattern) (Semester - III) (604302)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any Five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) a) Derive the Radar range equation? Explain the parameters that affect the Radar range. **[5]**

b) Define and explain the concept of Radar Cross Section (RCS) with a suitable equation. **[5]**

Q2) a) Explain the basic principle of pulse radar system and list its applications. **[5]**

b) Explain the radar display methods in detail. **[5]**

Q3) a) Write short note on Tracking Radar. **[5]**

b) Describe the antenna beam shapes for different types of radars. **[5]**

Q4) a) Explain the terms related to radar. **[5]**

i) Unambiguous range

ii) Pulse repetition Frequency

b) Write short note on weather prediction radar system. **[5]**

Q5) a) State and explain Kepler's three laws regarding planetary motion. **[5]**

b) Calculate the look angle of geostationary satellite if an earth station latitude and longitude are 52° N and 0° , the satellite longitude (subsatellite point) is 66° E. **[5]**

P.T.O.

- Q6) a)** Explain the methods to launch the satellite in geostationary orbit. **[5]**
- b) A satellite at a distance of 40,000Km from the surface of the earth, radiates a power of 10 Watt in the direction of earth station. The transmitting antenna gain is 17 dB. If operating frequency is 11GHz and the gain of receiving antenna is 52.3dB and effective area is 10 m². Calculate path loss, EIRP and received power. **[5]**
- Q7) a)** Explain the various losses in uplink and downlink design of satellite. **[5]**
- b) Write a note on AOCS in satellite. **[5]**
- Q8) a)** Explain the following terms of look angle calculations. **[5]**
- i) Elevation angle
 - ii) Azimuth angle
 - iii) Visibility test
- b) Explain the various Multiple Access Techniques used for satellite communication. **[5]**



Total No. of Questions : 8]

SEAT No. :

P-9664

[Total No. of Pages : 3

[6182]-953

M.E. (E & Tc) (Signal Processing)

MATHEMATICS FOR SIGNAL PROCESSING

(2017 Pattern) (Semester - I) (504401)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) All questions are compulsory.
- 2) Assume suitable data if necessary.
- 3) Use of non-programmable calculator is allowed.

- Q1)** a) A coin is tossed up four time in seccession, Determine the probability of obtaining three heads. [6]
- b) Consider a random experiment consisting of drawing of two cards in succession from a deck of cards without replacing the first card drawn. Find the probability of obtaining two red aces in two draws. [6]

OR

- Q2)** a) The Joint probability density of random variables X and Y is : [6]
- $$F_{X,Y} = 1/4 \exp(-|x|-|y|):$$
- for $-\infty < x < \infty, -\infty < y < \infty,$
- Calculate the probability that:
- $X < 1$ and $Y < 0$
- b) A binary symmetric channel has a conditional probability of error $p = 10^{-5}$ and the messages m_0 and m_1 are generated at the source with equal probability, Determine : [6]
- i) The probability of receiving a response of r_0
 - ii) The probability of receiving a response of r_1
 - iii) The probability that m_0 was sent and given that r_0 is received.
 - iv) The probability that m_1 was sent and given that r_0 is received.

P.T.O.

Q3) a) Determine the rank of the following matrices. [7]

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{pmatrix}$$

b) Use Gauss-Jordan method to find the inverse of the matrix. [6]

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{pmatrix}$$

OR

Q4) a) Reduce the matrix to its normal form and hence find its rank : [7]

$$\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

b) Verify given matrix is Orthogonal : [6]

$$A = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{pmatrix}$$

Q5) a) Prov that : [6]

$$[B \times C, C \times A, A \times B] = [ABC]^2$$

$$A \times \{B \times (C \times D)\} = B.D (A \times C) - B.C(A \times D)$$

b) Prove that formula : [6]

$$(B \times C), (A \times D) + (C \times A). (B \times D) + (A \times B). (C \times D) = 0$$

$$\text{and hence show that } \sin (C + \Phi) \sin(\theta - \Phi) = \sin^2 \theta - \sin^2 \Phi$$

OR

Q6) a) Find the moment about a line through the origin having direction $2\mathbf{i}+2\mathbf{j}+\mathbf{k}$ due to a 30 kg force acting at a point $(-4, 2, 5)$ in the direction of $12\mathbf{i}-4\mathbf{j}-3\mathbf{k}$. [6]

b) Prove that [6]

$$\sin (A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos (A+B) = \cos A \cos B + \sin A \sin B$$

Q7) a) Explain Power Spectral Density (PSD). Also explain its various properties. [7]

b) A sinusoidal signal generator of deterministic frequency f_0 and amplitude A is switched on. The starting time is random such that $x(t)=A\sin(2\pi f_0 t + \Phi)$.

Find the autocorrelation of the random process. Also sketch the same. [6]

OR

Q8) a) Consider a sinusoidal process $X(t)$ denoted by : [7]

$X(t) = A \cos (2\pi f_c t + \Phi)$; where A and f_c are constants and Φ is uniformly distributed random variable.

$$f_\Phi(\theta) = 1/2\pi, 0 \leq \theta \leq 2\pi$$

$$= 0, \text{ elsewhere}$$

Show that the random sinusoidal process is ergodic in both mean and autocorrelation function.

b) Discuss the transmission of random process through linear filters. [6]



Total No. of Questions : 8]

SEAT No. :

P-9586

[Total No. of Pages : 2

[6182]-954

M.E. (E & TC) (Signal Processing)
DIGITAL IMAGE & VIDEO PROCESSING
(2017 Pattern) (Semester - I) (504402)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any Five Question*
- 2) *Neat diagrams must be drawn wherever necessary*
- 3) *Figures to the right side indicate full Marks.*
- 4) *Assume the suitable data, if necessary.*

Q1) a) With respect to digital image explain the following. **[5]**

- i) Connectivity between pixels
 - ii) Distance measures between pixels
- b) Explain Pseudo, color image processing in detail. Write its application. **[5]**

Q2) a) Explain separability of unitary transforms & basis images. **[5]**

- b) Explain the following along with application. **[5]**
- i) Power law transformation
 - ii) Vnsharp Marking

Q3) a) Explain homo-morphic filtering approach for image enhancement. **[5]**

b) Explain JPEG baseline encoder in detail. **[5]**

Q4) a) What is Global thresholding? What is its drawback? How adaptive thresholding overcomes it? Write algorithm for adaptive thresholding. **[5]**

b) With the help of morphological operations on binary images explain how the following operations are performed. **[5]**

- i) Boundary extraction ii) Region/Hole filling iii) Thickening

P.T.O.

- Q5)** a) Describe the basic concept of Fourier descriptor for representation of boundary. List its properties and advantages. [5]
b) With block diagram, explain JPEG compression model in detail. [5]
- Q6)** a) Explain Region merging & region splitting techniques. [5]
b) What is difference between image restoration and image enhancement? Explain the image degradation model in detail. [5]
- Q7)** a) State the difference between analog and digital video? Explain basic video codes with suitable block diagram. [6]
b) Explain different color models used in Video compression. [4]
- Q8)** a) Explain basics of color video? Hence explain color conversion into YCbCr color space along with its advantages. [6]
b) What is Pseudo coloring and False coloring? Give its application. [4]



[6182]-955

M.E. (E & TC) (Signal Processing)
ADVANCED DIGITAL SIGNAL PROCESSING
(2017 Pattern) (Semester - I) (504403)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) What is FIR filter? State the advantages and Disadvantages of FIR Filters. [6]

b) What is the necessary and sufficient condition for the linear phase characteristic of an FIR filter? [2]

OR

Q2) a) What is bilinear transformation? Sketch the mapping of s-plane to z-plane in Bilinear transformation. [6]

b) What is Frequency Warping? [2]

Q3) a) What is Multirate DSP system? State its applications and advantages. [6]

b) What is LMS adaptive algorithm? [2]

OR

Q4) a) Consider the discrete time signal. [6]

$$x(n) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Determine the down-sampled version of the signals for the sampling rate reduction factors a) $D = 2$; b) $D = 3$; c) $D = 4$

b) State the Weiner Hopf Equation. [2]

P.T.O.

Q5) a) What does “characterization of Random Processes” mean? Explain the process. [9]

b) What is Ergodicity? Explain.

Are the wide sense stationary process and Ergodic Process reasonable assumptions for a video signal? [8]

OR

Q6) a) Explain what is meant by the term “ergodicity”. Is the sinusoid $X(t) = A\cos(\omega t + Q)$ with random phase- Q uniformly distributed on $[0; 2\pi]$ ergodic? (There is no justification required). [9]

b) Compare Time averages and Ensemble averages. [8]

Q7) a) Explain Super Harvard Architecture in detail. [9]

b) Write a brief note on Booth’s Multiplication algorithm. [8]

OR

Q8) a) Explain the process of implementation of FIR filter with the help of an example. [9]

b) Write the advantages and applications of FIR Filters and IIR Filters. State the applications of Decimation and Interpolation. [8]



Total No. of Questions : 8]

SEAT No. :

P-9588

[Total No. of Pages : 2

[6182]-957

M.E. (E & TC) (Signal Processing)
BIOMEDICAL SIGNAL PROCESSING
(2017 Pattern) (Semester - II) (504407)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any Five Question.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full Marks.*
- 4) *Assume the suitable data, if necessary.*

Q1) a) Explain cell junction with the help of diagram. **[5]**

b) Explain Heart Structure with diagram. **[5]**

Q2) a) How to use Eigen analysis for bio-signals processing. State its advantages and limitations. **[5]**

b) Explain Electrical Activity of heart with diagram. **[5]**

Q3) a) Write note on different Medical Images. **[4]**

b) Write a note on CT scan. **[6]**

Q4) a) What is the purpose of PET scanning? **[4]**

b) With the sequence of events and waves explain cardiac cycle and draw PQRST waveform. **[6]**

Q5) a) Write a note on MRI and Functional MRI (fMRI). **[4]**

b) Write a note on Eigen Analysis Spectral Analysis. **[4]**

c) List time frequency Methods in processing of Bio-signals. **[2]**

P.T.O.

- Q6)** a) Explain Radon Transform, Inverse Radon Transform. [4]
b) Explain in detail electromechanical models of signal generation. [4]
c) List any two characteristics of Noise. [2]
- Q7)** a) Define following : [4]
i) Edge
ii) Line
iii) Points
iv) Corners
b) Explain Spectral Analysis of Heart rate. [4]
c) Write a note on volumetric representation. [2]
- Q8)** a) Explain Noise reduction by Ensemble Averaging. [4]
b) Write a short note on RLS and Lattice Filter. [4]
c) List parametric system modelling. [2]



Total No. of Questions : 8]

SEAT No. :

P-9589

[Total No. of Pages : 2

[6182]-958

M.E. (E & TC) (Signal Processing)

SPEECH PROCESSING

(2017 Pattern) (Semester - II) (504408)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions out of eight.*
- 2) *All questions carry equal marks.*
- 3) *Neat labelled diagrams must be drawn wherever necessary*
- 4) *Figures to the right indicate full marks.*

- .
- Q1)** a) Explain vocal card & vocal tract for the speech production mechanism in details. [5]
- b) Explain how pitch frequency is measured with help of AMDF?. [5]
- Q2)** a) Explain voiced & unvoiced speech signal & explain how it is useful for decision making. [5]
- b) Mention different speech Parameters. Explain any two in details. [5]
- Q3)** a) Explain i) Pitch & ii) formats [2]
- b) Explain i) LPC ii) Melscale. [2]
- c) Establish relation between formats & linear prediction coefficients & mention its importance. [5]
- Q4)** a) Explain LTV model for speech production system? How LII system vary from LTV model [6]
- b) Explain leuison Durbin algorithm of speech processing in details. [4]

P.T.O.

- Q5)** a) What is live spectral pair frequencies. Explain the transformation from LPC to LSP. [5]
b) Explain: [5]
i) Forward linear prediction
ii) Autocorrelation Method
- Q6)** a) Explain the wavelet analysis of speech signal [5]
b) What is the Burq algorithm & What is its importance in speech processing. [5]
- Q7)** a) What is the quantization? How it is achieved in speech coding?. [5]
b) Explain Homomorphic processing. [5]
- Q8)** a) Explain uniform & non uniform quantizers in details. [5]
b) Explain the speech coding standard G 726.. [5]



Total No. of Questions : 8]

SEAT No. :

P-9590

[Total No. Of Pages : 2

[6182]-959
M.E. (E & Tc) (Signal Processing)
Computer Vision
(2017 Pattern) (Semester - II) (504409)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt any 5 questions.*
- 2) Figures to the right indicate full marks.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Make suitable assumptions wherever necessary.*

- Q1)** a) Explain with schematic diagram of digital camera. List any four specifications. [5]
- b) Write Application and challenges in Computer Vision. [5]
- Q2)** a) Explain 2D and 3D transformation in Computer Vision. [5]
- b) Explain Perspective camera model with suitable diagram. [5]
- Q3)** a) Explain eight point algorithm in stereo vision for calibrated cameras. [4]
- b) Explain Stereo parameters. [6]
- Q4)** a) Explain simple stereo system with suitable diagram. [4]
- b) Explain volumetric representations in detail. [6]
- Q5)** a) Explain how to do feature alignment using least squares. [4]
- b) Explain with neat diagram image stitching. [4]
- c) Define following image features: - i) Edge & ii) line. [2]

P. T. O.

- Q6)** a) Explain meaning of good feature and describe basic idea of Harris Corner detector with relevant mathematics. [4]
- b) Explain scale invariant feature transform. [4]
- c) Define following image features:- i) Corners & ii) Curve. [2]
- Q7)** a) Explain Kalman filter and list its applications. [4]
- b) Explain Iterative Estimation w.r.t. Optical flow. [4]
- c) Define motion field in images. [2]
- Q8)** a) Describe any one differential technique for motion estimation. [4]
- b) What is brightness consistency in optical flow estimation? [4]
- c) Define optical flow in images. [2]



Total No. of Questions : 8]

SEAT No. :

P-9675

[Total No. of Pages : 2

[6182]-962

M.E. (E & Tc) (Signal Processing)

STILL IMAGE AND VIDEO COMPRESSION

(2017 Pattern) (Semester - III) (604402)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) What is of Vector Quantization? Explain the advantages of Vector Quantization over Scalar Quantization. **[6]**

b) Explain the Linde-Buzo-Gray Algorithm. **[6]**

OR

Q2) a) Write a short note on Tree-Structured Vector Quantizers. **[6]**

b) Explain the Concept of compression using fractals. **[6]**

Q3) a) Write a short note on Image Compression. **[6]**

b) Explain the Scaling Function with respect to wavelets. **[6]**

OR

Q4) a) Explain the Multiresolution Analysis with respect to wavelets. **[6]**

b) Write a short note on JPEG 2000 compression standard. **[6]**

Q5) a) Explanation why the systems in MPEG-2 different from those in MPEG-1. **[5]**

b) Explain picture types-I, P, B and D pictures? **[8]**

OR

P.T.O.

- Q6)** a) Explain the MPEG - 1 decoder. [6]
b) Explain spatial and temporal redundancies. [7]

- Q7)** a) Explain Fast motion estimation in H.264 and Prediction selection. [6]
b) Write a short note on power estimation techniques. [7]

OR

- Q8)** a) Name two image and video compression standards & give their specific applications. [6]
b) Explain the following terms with respect to H.261. [7]
i) Intra - frame compression.
ii) Inter - frame compression.



Total No. of Questions : 08]

SEAT No. :

P-9592

[Total No. of Pages : 2

[6182]-964

M.E. (E & TC) (VLSI & EMBEDDED SYSTEMS)

DIGITAL CMOS DESIGN

(2017 Pattern) (Semester - I) (504201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Assume suitable data if necessary.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Use of nonprogrammable calculator is allowed.*

- Q1)** a) Explain MOSFET capacitance model. [5]
b) Draw stick diagram of Inverter and explain in brief. [5]
- Q2)** a) What is meant by design margin? Explain with example. [5]
b) What are the sources of cross talk? How to minimize cross talk? [5]
- Q3)** a) Explain various steps involved in CMOS fabrication process. [4]
b) Draw the cross section of CMOS Inverter and mention various materials. [4]
c) Write note on Gate capacitance. [2]
- Q4)** a) With the help of suitable diagrams, explain interconnect delays. [4]
b) Explore delays in multistage logic networks. [4]
c) Write note on delay in logic gate. [2]
- Q5)** a) Explain transmission gate, its utilities and limitations. [4]
b) What is metastability? What are solutions to it? [4]
c) Write note on compound gate. [2]

P.T.O.

- Q6)** a) Write HDL code for FSM of simple lift controller for ground plus two floor building. Assume suitable data. [4]
b) What are the techniques to minimize hazards? [4]
c) Write note on Mealy and Moore machines. [2]
- Q7)** a) Explore the low power design techniques in detail. [4]
b) With the help of diagram, explain BiCMOS circuit. [4]
c) Explain performance improvement using various materials. [2]
- Q8)** a) What is need of sense amplifier? Explain in detail. [4]
b) Explain NORA logic. [4]
c) Compare the circuit families. [2]



Total No. of Questions : 8]

SEAT No. :

P-9593

[Total No. of Pages : 2

[6182]-965

M.E. (E & TC) (VLSI and Embedded Systems)

RECONFIGURABLE COMPUTING

(2017 Pattern) (Semester - I) (504202)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions out of eight questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) a) Distinguishing between General Purpose Processor and Domain specific processor. [4]

b) Define terms : [4]

i) Run Time

ii) Reconfigurable

iii) Specialization

c) List the applications of Reconfigurable Computing. [2]

Q2) a) Give Von Neumann computer architecture and explain its flow for program execution. [4]

b) With labeled diagram, explain PAM as RC platform. [4]

c) Draw and explain multi-context LUT. [2]

Q3) a) What do you mean by Context? Explain various types of context with diagram. [5]

b) Explain instruction level parallelism and give its effects. [5]

P.T.O.

- Q4)** a) Explain Non-frequently reconfigurable systems and its application. [5]
b) Explain Relocation and Defragmentation w.r.t. RC point of view. [5]
- Q5)** a) Explain in detail the main components of system on programmable Chip. [5]
b) Discuss Run time reconfiguration and compile time reconfiguration. [5]
- Q6)** a) Explain FPGA flow design diagram. [5]
b) Draw and explain the transfer of system from PCB to system on programming chip. [5]
- Q7)** a) Write short note on: Architecture of an adaptive cryptographic system. [5]
b) What is partially reconfigurable computing, give example of any present device. [5]
- Q8)** a) Explain High performance computing w.r.t. FPGA. [5]
b) Explain Pattern matching by automation based text searching. [5]



Total No. of Questions : 8]

SEAT No. :

P-9594

[Total No. of Pages : 2

[6182]-966

M.E. (E & TC) (VLSI & Embedded System)

EMBEDDED SYSTEMS DESIGN

(2017 Pattern) (Semester - I) (504203)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) List the features of Embedded system and Explain the design metrics [5]
i) Time to market ii) NRE cost
b) List different life cycle models and Explain the spiral model in detail for embedded system design. [5]
- Q2)** a) Discuss Raspberri Pi and Arduino microcontroller with respect to [6]
i) Board details ii) IDE
iii) Applications.
b) Explain the advantages and disadvantages of system on chip. Compare it with General purpose processor. [4]
- Q3)** a) Draw and explain block diagram of LPC 1768 ARM Cortex M3 based microcontroller. [5]
b) What is system control block? Explain with suitable diagram. [5]
- Q4)** a) Draw and explain Cortex Microcontroller Software Interface Standard (CMSIS) structure. [5]
b) Explain with suitable diagram, the USB protocol with respect to LPC 1768. [5]
- Q5)** a) Explain Embedded Linux System architecture. [5]
b) Compare BIOS v/s Boot loader. [5]

P. T. O

- Q6)** a) With neat diagram, explain the structure of device driver with respect to Embedded Linux. [5]
b) Explain the file structure used in Embedded Linux. [5]
- Q7)** a) Explain design of Digital camera as Embedded system case study. [6]
b) Discuss the importance of EMI/RFI certification in Embedded system.[4]
- Q8)** a) Explain design of Automated meter reading as Embedded system case study. [6]
b) Explain types of documents to be prepared for embedded real time products. [4]



Total No. of Questions : 8]

SEAT No. :

P-9595

[Total No. of Pages : 2

[6182]-968

M.E. (E & TC) (VLSI & Embedded Systems)

ANALOG CMOS DESIGN

(2017 Course) (Semester - II) (504207)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any five questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data if necessary.*

Q1) a) Draw a circuit diagram of cascade current mirror source and derive expression for its output resistance. **[5]**

b) Explain how MOSFET can work as a Switch, Diode and Active Resistor. **[5]**

Q2) a) Explain small signal high frequency model for MOS transistor with the help of diagram. Also draw large signal model. **[5]**

b) With suitable diagram, explain two stage op-amp with slew enhancement. **[5]**

Q3) a) Compare various types of inverting CMOS amplifiers on the basis of voltage gain and output resistance. **[5]**

b) What is the use of micro power op-amp? Explain the techniques used in micro power op-amp. **[5]**

Q4) a) What are the different types of mixers. Explain square law MOSFET mixer with neat circuit diagram. **[5]**

b) Explain need of frequency compensation. State various techniques of frequency compensation. **[5]**

P.T.O.

- Q5) a)** Write a short note (Any two) : **[5]**
- i) Folded cascade amplifier
 - ii) CMOS Differential amplifiers
 - iii) Explore Neutralisation and Unilaterisation
- b) Which are dominant noises in CMOS op-amp? Explain the techniques to reduce this noise. **[5]**
-
- Q6) a)** What is need of compensation in multistage amplifiers. Explain Miller Compensation in two stage CMOS op-amp. **[5]**
- b) Write a note on stability of op-amp and its effect on slew rate. What are different way to improve stability? **[5]**
-
- Q7) a)** Explain bandwidth estimation techniques in detail. **[5]**
- b) Draw a schematic of single ended Low Noise Amplifier (LNA). What are its drawbacks and how these are overcome in differential LNA. **[5]**
-
- Q8) a)** Draw a comparator and discuss its static and dynamic characteristics. What is typical application of comparator. **[5]**
- b) Explain in brief the design considerations for RF chip design. **[5]**



Total No. of Questions : 8]

SEAT No. :

P-9596

[Total No. of Pages : 2

[6182]-969

M.E. (E&TC) (VLSI and Embedded System)

SYSTEM ON CHIP

(2017 Pattern) (Semester - II) (504208)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary*

Q1) a) What do you mean by Control Flow Modeling and enlist the Limitations of Data Flow Models? **[4]**

b) What is Metastability? Explain the Problems Due to Metastability? **[6]**

Q2) a) Write a short note on data flow model and its advantages? **[4]**

b) What is the Need for Concurrent Models? **[6]**

Q3) a) Write a short note on RISC pipeline? **[4]**

b) Explain different Timing Parameters for Digital Logic? **[6]**

Q4) a) Explain the difference between the following terms : **[6]**

i) Control hazard and data hazard

ii) Delayed branch and conditional branch

b) Explain different Timing Parameters for Digital Logic? **[4]**

P.T.O.

- Q5)** a) Write a note on Simulation-Synthesis Mismatch? [4]
b) Write a note on Bus Synchronization along with the Challenges in it and Enable Synchronization Method'? [6]
- Q6)** Explain the difference between the following terms : [10]
a) Control hazard and data hazard
b) One-Way and Two-Way Handshake
- Q7)** a) Explain different Timing Parameters for Digital Logic? [4]
b) What are the Design Issues and Techniques for image codec? [6]
- Q8)** a) Explain energy management techniques for SoC design. [4]
b) What are the Design Issues and Techniques for image codec? [6]



Total No. of Questions : 8]

SEAT No. :

P-9597

[Total No. Of Pages : 2

[6182]-970
M.E. (E & TC)
(VLSI & EMBEDDED SYSTEMS)
Embedded Automotive Systems
(2017 Pattern) (Credit System) (Semester - II) (504209)

Time : 3 Hours]

[Max. Marks : 50

- Instructions:**
- 1) *Solve any five questions.*
 - 2) *Assume suitable data if necessary.*
 - 3) *Neat diagrams must be drawn wherever necessary*

- Q1) a)** List basic systems that make up an automobile and explain their major components and functions. **[5]**
- b) Discuss the future trends in automotive embedded system. **[5]**
- Q2) a)** Explain with suitable diagram Fuel Injectors in Petrol engine. Also describe security & warning system **[5]**
- b) Discuss vehicle emission & environmental health. Hence comment about Emission control strategies. **[5]**
- Q3) a)** What is Hall effect? Explain a position sensor using principle of Hall effect? **[5]**
- b) Outline the construction of flap type & Hot wire type air flow sensor with suitable example. **[5]**
- Q4) a)** What is tyre pressure monitoring system? How does it work? **[5]**
- b) Explain working principle of throttle valve used for speed control. **[5]**

P.T.O.

- Q5)** a) Differentiate Throttle body Injection and Multi-point fuel injection system. [5]
- b) With suitable block diagram explain automatic cruise control system. Enlist sensors used in such system. [5]
- Q6)** a) Compare analog & digital cruise control system. [5]
- b) How electronic ignition control works? [5]
- Q7)** a) Write a short note on Advance Trends in Automotive Electronics. [5]
- b) With respect to CAN explain error handling and protocol extension. [5]
- Q8)** a) Compare LIN & CAN with respect to automotive applications. [5]
- b) Comment on Flexible time triggered communication on CAN [5]



Total No. of Questions : 8]

SEAT No. :

P-9598

[Total No. of Pages : 2

[6182]-972

M.E. (E & TC) (VLSI & Embedded Systems)
TESTING AND VERIFICATION OF VLSI CIRCUITS
(2017 Pattern) (Semester - III) (604201)

Time : 3 Hours]

[Max. Marks : 50

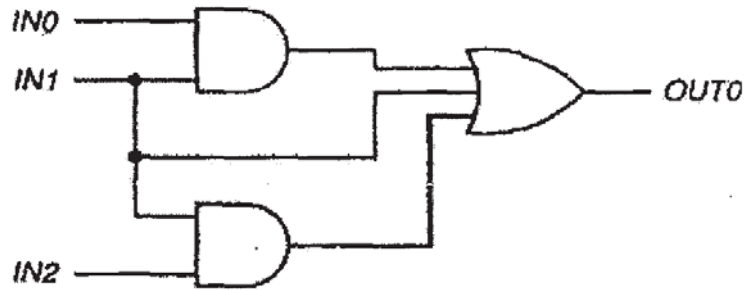
Instructions to the candidates:

- 1) Answer any five Questions.*
- 2) Figures to the right indicate full marks.*
- 3) Assume suitable data, if necessary.*

- Q1)** a) Explain Principles of testing? Explain Production testing in brief. [4]
b) Explain how cost and difficulty of chip testing have greatly affected by recent n VLSI technology trends. [4]
c) Differentiate between verification and testing in VLSI process. [2]
- Q2)** a) State any fault fault models and explain any two in brief. [4]
b) Compare Functional Vs. structural Testing. [4]
c) State the “Rule of 10”. [2]
- Q3)** a) Explain Fault Equivalence of combinational circuits with suitable examples. [5]
b) What is need of Fault Simulation? Explain parallel fault simulation algorithm. [5]
- Q4)** a) Write Short Note on techniques of Memory testing. [4]
b) Explain briefly Observability and Controllability. [4]
c) Find min. tests that together test all single stuck-at faults in a two-input AND gate. [2]

P.T.O.

Q5) a) Calculate SCOAP Measure for following circuit. [5]



b) With the help of suitable diagram explain LFSR and Signature Compaction of BIST. [5]

Q6) a) Write Short Note on “Analog Fault Models”. [5]

b) How Analog Testing is different than Digital Testing. [5]

Q7) a) Explain different test pattern generation methods for BIST. [5]

b) Explain IDDQ Current testing. [5]

Q8) a) Write a short note on “SoC Testing”. [4]

b) State need and limitation of Formal verification Techniques. [4]

c) State Importance of Hardware emulators in Hardware design. [2]



Total No. of Questions : 8]

SEAT No. :

P9599

[Total No. of Pages : 2

[6182]-973

M.E.(E & TC) (VLSI and Embedded Systems)

ASIC DESIGN

(2017 Credit Pattern) (Semester - III) (604202)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Attempt any 5 questions out of 8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data, if necessary.*

- Q1)** a) Draw design flow of ASIC Design process and explain each step in detail. [5]
b) What is test bench explain with one example. [5]
- Q2)** a) What are different types of ASIC? Classify and explain in details. [5]
b) Write VHDL code and test bench code for 8 bit counter. [5]
- Q3)** a) Differentiate floor-planning and placement in ASIC Design. [4]
b) Write short note on gate level mixed mode testing. [4]
c) List various CAD tools used in ASIC Design process. [2]
- Q4)** a) With neat labeled diagram explain clock distribution technique in ASIC.[4]
b) Compare global routing and detailed routing. [4]
c) What is placement in ASIC Design? Write its goals and objectives. [2]
- Q5)** a) How false path detection is carried out in application specific intergrated circuits. [4]
b) What is the need of time optimization? Give one example which describes time optimization in ASIC. [4]
c) Differentiate pre layout simulation and post layout simulation techniques.[2]

P.T.O.

- Q6)** a) What are different signal integrity issues? How it occurred in ASIC design. [4]
b) What is functional simulation? Where it is required? Give one example. [4]
c) What are the time related constraints? Explain with one example. [2]
- Q7)** a) Describe with one example static timing analysis. [4]
b) Describe how memory testing is carried out in ASIC design. [4]
c) What are different physical faults occurred in ASIC . [2]
- Q8)** a) Describe in brief Automatic Test Pattern Generator. [4]
b) Write short note on Joint Test Action Group. [4]
c) Write types and features of existing EDA tools. [2]



[6182]-998

M.E. (Information Technology)

MATHEMATICAL FOUNDATION OF INFORMATION TECHNOLOGY

(2017 Pattern) (Semester - I) (514401)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) Solve any two.

- a) Suppose that a pair of fair dice is tossed and let the random variable X denote the sum of the points. i) Obtain the probability distribution for X .
ii) Construct a graph for this probability distribution. [5]
- b) Explain Messy genetic algorithm with example. [5]
- c) Using your own institution and your own definition of universe of discourse, write & plot fuzzy membership functions for the designing of Washing Machine by considering parameters : {Dirt, Grease, Wash Time}. Assume Dirt & grease on a scale of 0 – 100, WashTime on a scale of 0 – 60. [5]

Q2) Solve any two.

- a) Explain graph traversals algorithm with examples. [5]
- b) Two random sample drawn from two normal populations are : [5]

Sample 1	20	16	26	27	23	22	18	24	25	19		
Sample 2	27	33	42	35	32	34	38	28	41	43	30	37

Test using variance ratio at 5% & 1% level of significance whether the two populations have the same variances. ($F_{0.05,11,9} = 3.11$, $F_{0.01,11,9} = 5.20$)

- c) A continuous random variable X has the probability density given by [5]

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Find i) $E(X)$, ii) $E(X^2)$ iii) $E(X^3)$

P.T.O.

Q3) Solve any two.

- a) Set up an analysis of variance table for the following per acre production data of three varieties of wheat, each grown on 4 plots and state if the variety differences are significant. [5]

Plot of Land	Per acre production data		
	Variety of wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

- b) Describe the concepts involved in real-coded genetic algorithm. [5]
 c) Explain simplex algorithm with example. [5]

Q4) Solve any two.

- a) Consider two fuzzy sets given by [5]

$$\tilde{E} = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.3 \\ 0.2 & 0.9 \end{bmatrix} \end{matrix} \quad \tilde{I} = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.8 & 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

Obtain fuzzy relation $\tilde{U}$ as a composition between the fuzzy relations.

- b) Let $G=(V,E)$ be an undirected graph with k components and $|V|=n$ and $|E|=m$. prove that $m \geq n-k$. [5]
 c) Explain types of problems in optimization. [5]

Q5) Solve any two.

- a) Prove Max-flow Min-cut theorem. [5]
 b) Genetic theory states that children having one parent of blood type A and the other of blood type B will always be of one of three types A, AB, B and that the proportion of three type will on an average be as 1 : 2 : 1. A report states that out of 300 children having one A parent and B parent, 30% were found to be type A, 45% type AB and remainder type B. Test the hypothesis by χ^2 test. ($\chi^2_{0.05,2} = 5.991$) [5]
 c) Differentiate between geometric and harmonic mean. State the conditions when these mean are applicable. [5]



Total No. of Questions : 7]

SEAT No. :

P-9601

[Total No. of Pages : 2

[6182]-999

M.E. (Information Technology)

**ADVANCE SOFTWARE ENGINEERING & PROJECT
MANAGEMENT**

(2017 Pattern) (Semester - I) (514402)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions from seven questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) a) Explain in detail Prescriptive Process Model. **[5]**

- b) Define the concept of Software Requirement Engineering. Give requirement elicitation in terms of Functional and nonfunctional requirements for managing web based travel planning system that helps users plan journeys. **[5]**

Q2) a) Define and Explain the term “Refinement” with suitable eg. **[5]**

- b) Explain Web Design pyramid with suitable diagram **[5]**

Q3) a) Enlist Agile methods. Explain any four in brief. **[5]**

- b) Describe the XP concepts of refactoring and pair programming in your own words. **[5]**

Q4) a) Explain with eg. PERT and CPM techniques. **[5]**

- b) Discuss various code cloning techniques with eg. **[5]**

P.T.O.

Q5) a) Explain why program inspections are effective technique for discovering errors in a program. What types of error are unlikely to be discovered through inspections? [5]

b) Explain the various challenges in software project maintenance. [5]

Q6) a) Explain Ishikawa's 7 tool model [5]

b) Explain Service-oriented Software Engineering with respect to case study. [5]

Q7) Write a short note on any two : [10]

a) User Interface Design

b) GERT

c) Petrinets

d) CMMI-IPPD



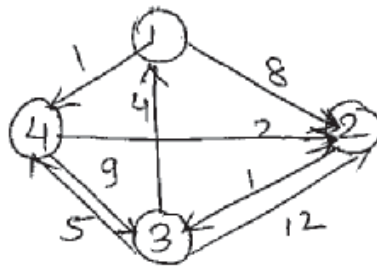
[6182]-1000

M.E. (Information Technology)
APPLIED ALGORITHMS
(2017 Pattern) (Semester - I) (514403)

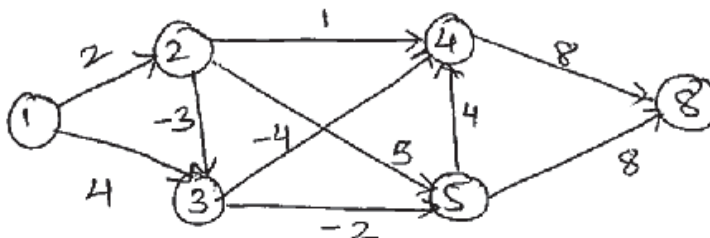
*Time : 3 Hours]**[Max. Marks : 50**Instructions to the candidates:*

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Compare Prim's and Kruskal's algorithm. [3]
 b) Write algorithm for quick sort and analyze the algorithm. [3]
 c) Apply Floyd's Warshall's Algorithm for the graph. [4]



- Q2)** a) Explain Master theorem and solve the recurrence relation by Master theorem [3]
 i) $T(n) = 8T(n/2) + n^2$
 ii) $T(n) = 2T(n/2) + n \log n$
 iii) $T(n) = 3T(n^{1/3}) + \log 3^n$
 b) Write and explain Rabin-Karp algorithm. [3]
 c) Write the Bellman Ford Algorithm and solve the following [4]



Q3) a) Take the following list of function and arrange them in ascending order justify your answer [3]

i) $n\log(n)$, $n + n^2 + n^3$, 2^n , $\text{sqrt}(n)$

i) n^2 , 2^n , $n\log(n)$, $\log(n)$, n^3

i) $n\log(n)$, n^8 , $n^2 / \log(n)$, $(n^2 - n - 1)$

b) Write Johnson's algorithm and explain its example. [3]

c) Solve the following salesman problem using approximation algorithm.[4]

0	11	10	9	6
8	0	7	3	4
8	4	0	4	8
11	10	5	0	5
6	9	5	5	0

Q4) a) Write Jarvis's march algorithm and explain suitable example. [3]

b) Explain the following terms : [3]

i) Bipartite Graph

ii) Max Flow min cut theorem

c) Explain Navie string matching algorithm with example. [4]

Q5) a) What is randomized algorithm? Explain types of randomized algorithms. [5]

b) What is the Red Black Tree? Give its properties. Insert 10,11,100,95,25 and 55 in the tree and show its stages. [5]

Q6) a) Prove that "clique problem is NP complete" [5]

b) What is online algorithm? Write and explain obstacle traversal algorithm. [5]

- Q7)** a) Show the result of inserting the keys,
F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y, D, Z, E in order into
an empty B-tree with minimum degree 2. [5]
- b) Discuss various parallel computational modules and explain how they
achieve the parallelism in algorithmic design. [5]
- Q8)** a) Write Short note on : [5]
- i) PRAM model of Computations
 - ii) K server problem
- b) Write algorithm for pointer doubling problem, solve with example. What
is its time complexity. [5]



Total No. of Questions : 6]

SEAT No. :

P-9603

[Total No. of Pages : 2

[6182]-1001
M.E. (Information Technology)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (514404)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt any five questions.*
- 2) Assume suitable data, if necessary.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Figures to the right indicate full marks.*

Q1) a) List and explain different types of Research in detail. **[5]**

b) Distinguish between research method and research methodology. **[5]**

Q2) a) Discuss the functions of research design. **[5]**

b) List tools to manage Bibliographic Data. Explain any one tool in detail. **[5]**

Q3) a) State and explain the techniques of data collection. **[5]**

b) Describe database statistical functions to summarizing Data and explain the use of principal component analysis. **[5]**

Q4) a) Mention the different types of reports, particularly pointing out the difference between technical report and descriptive report. **[5]**

b) Discuss different types of technical papers. **[5]**

P.T.O.

- Q5)** a) Discuss three pass methods to read a scientific paper efficiently. [5]
b) Elaborate different Steps in writing Report. [5]
- Q6)** a) What are Intellectual property rights? Explain the process of patenting a research finding. [5]
b) Explain the Indian patent law and cyber laws in detail. [5]



Total No. of Questions : 5]

SEAT No. :

P-9604

[Total No. of Pages : 2

[6182]-1003

M.E. (Information Technology)
CYBER SECURITY & FORENSICS
(2017 Pattern) (Semester - II) (514407)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) Solve any two :

- a) What is Risk and Threat? How risk analysis is done? [5]
- b) Write short note on [5]
 - i) Security Triad: CIA
 - ii) Phishing and Spoofing
- c) A message along with checksum travels from machine A to machine B. During the movement 3 message bits get modified. At machine B the checksum is recalculated and obviously it does not match with the checksum received along with the message. Justify with reason which security principle is violated. [5]

Q2) Solve any two :

- a) Explain The Clark-Wilson Model. [5]
- b) What is State Machine Model? Explain BLP model in detail and draw multics interpretation of BLP. [5]
- c) What is Biba Model, Explain in detail. [5]

Q3) Solve any two :

- a) Explain cross-site request forgery attack with example. [5]
- b) Explain digital signature and X.509 certification format. [5]
- c) Explain the role of third-party arbiter in successful implementation of public key infrastructure. [5]

P.T.O.

Q4) Solve any two :

- a) Explain forensic analysis and explain how to apply scientific methods to digital forensics [5]
- b) What do you mean by hypothesis formulation? [5]
- c) What criteria the digital data should satisfy so as to used as alibi in forensics law. [5]

Q5) Solve any two :

- a) Explain how bit locker works against data theft in windows forensics. [5]
- b) Which attributes of NTFS master file are useful for forensic analysis? [5]
- c) Explain the evidence preservation mechanisms for digital data in forensic law. [5]



Total No. of Questions : 8]

SEAT No. :

P-9605

[Total No. of Pages : 2

[6182]-1004

M.E. (Information Technology)
CLOUD AND DATA TECHNOLOGIES
(2017 Pattern) (Semester - II) (514408)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Discuss following Cloud-Enabling Technology: [6]
i) Web Technology
ii) Service Technology
b) Write a note on Virtualization Technology [4]
- Q2)** a) Demonstrate Multi-tier system architecture in Cloud Computing [6]
b) Explain key issues related to workload management for Cluster Computing. [4]
- Q3)** a) Explain Non-Disruptive Service Relocation Architecture. [6]
b) Explain the functionality of Failover System [4]
- Q4)** a) Discuss various parameters to improve Quality of Service (QoS) in Cloud Computing [6]
b) Explain concept of Zero Downtime Architecture. [4]
- Q5)** a) Write a short note on : [6]
i) Rapid Provisioning Architecture
ii) Hubris
b) Explain the concept of Data Visualization and Fraud Detection [4]

P.T.O.

- Q6)** a) Discuss various issues and challenges in Big data. [4]
b) Explain with suitable examples: [6]
i) Data Science and Risk
ii) Data Leakage and Model.
- Q7)** a) Explain Models and Techniques for Cloud-Based Data Analysis [5]
b) Explain concept of Big data technologies [5]
- Q8)** a) Discuss Cost Metrics and Pricing Models of Cloud Computing [5]
b) Explain relationships of Big Data and Cloud Computing. [5]



[6182]-1005
M.E. (Information Technology)
ITOOR
(2017 Pattern) (Semester - II) (514409)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, rest all questions are compulsory.*
- 2) *Assume Suitable Data, if necessary.*
- 3) *Neat Diagram must be drawn wherever necessary.*
- 4) *Figures to the right indicate full marks.*

Q1) a) Solve using Simplex Method **[6]**

Minimize $Z = 2x_1 + x_2$, subject to the constraints

$$x_1 + 2x_2 \geq 24$$

$$2x_1 + 3x_2 \geq 36$$

$$2x_1 + x_2 = 30 \text{ and } x_1, x_2 \geq 0$$

b) Write Short note on “Deriving Solution from the Model”. **[4]**

OR

Q2) a) Consider the following LP problem **[6]**

Maximize $Z = 3x_1 + x_2$

Subject to : $x_1 + 2x_2 \leq 14$

$$x_1 + x_2 \leq 10 \text{ and } x_1, x_2 \geq 0$$

- i) Convert this problem to a minimization problem.
 - ii) Solve the problem of (i) by the graphical method
- b) Discuss the types of mathematical model. **[4]**

Q3) Attempt any two questions :

- a) Explain the steps in Vogel's Approximation method in the Transportation problem. [5]
- b) Consider the following profit and requirement table of a transportation problem, find an optimal solution. [5]

16	14	20	5	3	60
4	10	12	15	19	35
8	18	5	13	15	40
22	45	20	18	30	

- c) Consider the problem of assignment 4 jobs to 4 machines. The assignment costs are given below. Find an optimal way of assignment. [5]

17	-	22	20	16	18
19	18	15	-	16	14
-	21	14	13	15	15
20	12	18	12	-	19
16	13	22	15	17	-

Q4) Attempt any two questions :

- a) Write any four limitations of Game Theory [5]
- b) i) For what value of λ , the game with following payoff matrix is strictly determinable? [5]

		Player B		
		B1	B2	B3
Player A	A1	6	2	λ
	A2	λ	-7	-1
	A3	4	λ	-2

- ii) Find the ranges of values p and q which will render the entry (2, 2) a saddle point for the game.

		Player B		
		B1	B2	B3
Player A	A1	5	2	4
	A2	q	10	7
	A3	6	4	p

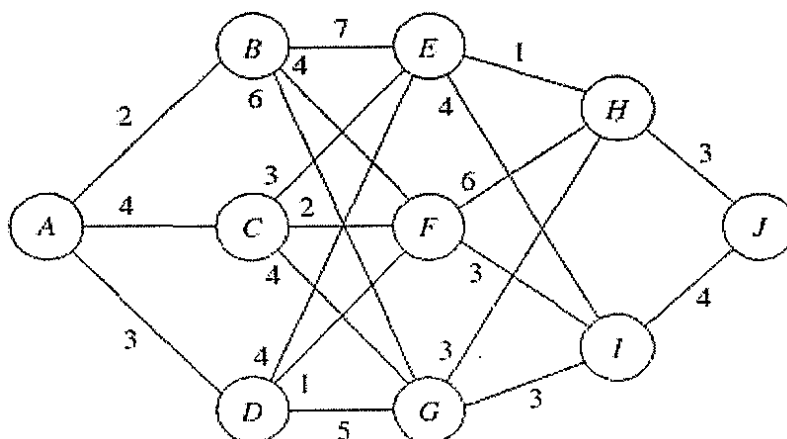
[5]

- c) Determine Value of Game. Also indicate that they are fair or strictly determinable. [5]

	B1	B2	B3	B4
A1	-5	-2	10	-3
A2	7	4	-2	-5
A3	1	9	6	0
A4	2	3	8	-1

Q5) Attempt any two questions.

- a) Write short note on dynamic programming and its application. [5]
b) Use dynamic programming for optimal path from location A to location J (Target Location) of given road network. Cost across locations are mentioned. [5]



- c) Write short note on characteristics of dynamic programming. [5]

Q6) Attempt any two questions :

- a) Explain the algorithm for finding a critical path. [5]
b) Explain the rules devised by Fulkerson for network construction. [5]
c) The utility data for a network are given below. Determine the total float and identify the critical path. [5]

Activity :	1-2	1-4	1-7	2-3	3-6	4-5	4-8	5-6	6-9	7-8	8-9	9-10
Duration :	2	6	8	5	4	4	6	3	3	2	2	5



Total No. of Questions : 7]

SEAT No. :

P-9608

[Total No. of Pages : 2

[6182]-1008

M.E. (Information Technology)
ADVANCE OPERATING SYSTEMS
(2017 Pattern) (Semester - III) (5144014)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Figures to right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Differentiate among the following types of operating systems with respect to properties and features. **[5]**

- i) Time Sharing
- ii) Parallel Processing
- iii) Network
- iv) Distributed

b) Explain design issues of Distributed Operating System. **[5]**

Q2) a) What are the features of good message passing system? **[5]**

b) Explain the Architecture of Distributed Shared Memory with suitable diagram. **[5]**

Q3) a) Explain the implementation Of RPC Mechanism along with stub generation. **[5]**

b) Enlist various consistency models in Distributed Shared Memory. Explain any two consistency models. **[5]**

P.T.O.

- Q4)** a) Discuss the Lamport's Clock Synchronization Algorithm. [5]
b) Discuss the features of good Distributed File system. [5]
- Q5)** a) Explain the probe based distributed algorithm for deadlock detection in a distributed systems. [5]
b) Explain the design principle of Distributed File System. [5]
- Q6)** a) Explain the scheduling in Mobile OS. [5]
b) Explain the Election Algorithms in brief. [5]
- Q7)** Write short notes on following : (Any Two) [10]
a) Android Architecture
b) Thrashing
c) Kernel Design in Symbian OS
d) Distributed Computing Models



[6182]-1010

**M.E. (Instrumentation & Control - Process
Instrumentation)**

**MATHEMATICAL METHODS IN INSTRUMENTATION
(2017 Pattern) (Semester - I) (506101)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Assume Suitable data if necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.

- Q1)** a) Define linear combination of vector with suitable examples. [4]
 b) Show that the vectors $(1, 1, 1)$, $(1, 1, 0)$, $(1, 0, 0)$ is linearly independent in R^3 . [4]
 c) Determine basis and dimension of vectors $(3, 8, -3, -5)$, $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ in a vector space $V = R^4$. [2]

- Q2)** a) Let $\bar{v}_1 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$ $\bar{v}_2 = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$ be an orthonormal set of vectors in R^3 , if $\bar{u} = (1, 2, 3)$ find orthogonal projection $\bar{u}$ on W and orthogonal component of $\bar{u}$ to W. [5]
 b) Use Gram-Schmidt process to transform the basis $\{(1, 1, 1), (0, 1, 1), (1, 2, 3)\}$ for R^3 into an orthogonal basis for R^3 . [5]

- Q3)** a) Explain Role of Numerical method in Process Instrumentation and Solve by Gauss - Elimination method [5]
 $10x_1 + 2x_2 + x_3 = 9,$
 $2x_1 + 20x_2 - 2x_3 = -44,$
 $-2x_1 + 3x_2 + 10x_3 = 32.$
 b) Determine the value of y when $x = 0.1$,by euler modified method
 given that $\frac{dy}{dx} = 1 + xy$, $y(0) = 1$ and $h = 0.1$. [5]

P.T.O.

- Q4)** a) If on an average one ship in every ten is wrecked, find the probability that out of five ship expected to arrive, 4 at least will arrive safely. [5]
- b) Between 2 P.M. and 3 P.M. The average number of phone call per minute coming into company are 5. Find the probability that during one particular minute, there will be 2 or less calls. [5]
- Q5)** a) Let X be a random variable define by a density function [5]
 $f(x) = e^{-x}, x \geq 0$. and $f(x) = 0$, otherwise. Find
 i) $E(X)$,
 ii) $E(X^2)$
- b) The first four central moments of a distribution are 0, 2.5, 0.7 and 18.75 . Comment on the skewness and kurtosis of the distribution. [5]
- Q6)** a) A joint PDF of two continuous random variable X and Y is [6]
 $f(x, y) = c(2x + y), 0 < x < 1, 0 < y < 2$. and $f(x, y) = 0$, otherwise.
 i) Find the value of constant C
 ii) Find $p\left[X > \frac{1}{2}, Y < \frac{3}{2}\right]$
- b) Explain the terms [4]
 i) Moments
 ii) Moments Generating Function
- Q7)** a) Find singular value decomposition of the matrix [5]

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$
- b) If on an average one ship in every ten is wrecked, find the probability that out of five ship expected to arrive, 4 at least will arrive safely. [5]



Total No. of Questions : 7]

SEAT No. :

P-9610

[Total No. of Pages : 2

[6182]-1011

M.E. (Instrumentation & Control) (Process Instrumentation)
TRANSDUCER DESIGN
(2017 Pattern) (Semester - I) (506102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables slide rule, mollies charts, electronic pocket calculator and steam table is allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Write short note on chemical sensor. [5]
b) Suggest the suitable non contact type transducer used to measure the temperature of furnace, elaborate the concept with neat sketch. [5]
- Q2)** a) Explain static and dynamic characteristic of instruments. [5]
b) Write a short notes on MEMS. [5]
- Q3)** a) Explain chemical sensor in detail with neat sketch and state its applications. [5]
b) Differentiate between torque, angular velocity and power. [5]
- Q4)** a) Suggest suitable scheme to measure torque of a rotating shaft using strain gauge. [5]
b) Specify the different level measurement approach's in the industry, suggest suitable transducer used for liquid level measurement. [5]

P.T.O.

- Q5)** a) Explain gas sensors in brief and state its applications. [5]
b) Classify and state selection criteria of transducers and illustrate performance characteristics of transducer. [5]
- Q6)** a) Specify the need of signal conditioning, and explain signal level and bias changes. [5]
b) Explain the role of ADC and DAC in interfacing of sensor with suitable example. [5]
- Q7)** a) Explain the concept of loading, divider circuits, and bridge circuits in brief. [5]
b) Comment on pH and conductivity, suggest suitable scheme for pH measurement of water in tank. [5]



Total No. of Questions : 6]

SEAT No. :

P-9611

[Total No. of Pages : 1

[6182]-1012

M.E. (Instrumentation & Control) (Process Instru.)

INDUSTRIAL AUTOMATION

(2017 Pattern) (Semester - I) (506103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Use of Calculator is allowed.*
- 5) Assume Suitable data if necessary.*

Q1) Develop programmable ladder diagram for Two lights are to flash on and off at different intervals. One is on 4 seconds and off 4 seconds. The other is on for 7 seconds and off 7 seconds. Two lights are to flash alternately, one for 4 seconds, one for 7 seconds. After five cycles both lights must go off. **[10]**

Q2) What are the components of DCS? Explain with suitable example. **[10]**

Q3) What are the OPC Technologies? **[10]**

Q4) Compare MODBUS, Fieldbus and Profibus. **[10]**

Q5) Develop point to point and multi-drop network configuration using HART protocol. **[10]**

Q6) Develop sequential function chart for motor speed control using VFD.**[10]**



Total No. of Questions : 7]

SEAT No. :

P-9612

[Total No. of Pages : 2

[6182]-1013

M.E. (Process Instrumentation & Control)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester - I) (506104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt any five questions.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks..
- 4) Use of logarithmic tables slide rule, mollier charts, electronics pocket calculator and steam table is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) What are different types of research? Give examples for each. [5]

b) Explain the errors in selecting a research problem [5]

Q2) a) List Static and dynamic characteristics of instruments used in experimental set up [5]

b) Explain criteria for good research in detail. [5]

Q3) a) Enlist and elaborate method of linear regression analysis. [5]

b) Use following data to workout linear regression analysis.. [5]

Sr.No.	Price	Sales
1	100	118
2	120	85
3	140	97
4	160	72
5	180	79

Find linear regression equation, Coefficient of correlation.

P.T.O.

- Q4)** a) Describe sensitivity theory in detail. [5]
b) Illustrate multiscale modeling in brief. [5]
- Q5)** a) Comment on common errors occur while writing research aims and objectives. [5]
b) Specify the difference between simple hypothesis and composite hypothesis. [5]
- Q6)** a) Describe the layout of research proposal. [5]
b) Enumerate the different methods of collecting data. [5]
- Q7)** a) Elaborate the different sections in a project proposal. [5]
b) Explain different steps in writing research report. [5]



Total No. of Questions : 6]

SEAT No. :

P-9613

[Total No. of Page : 1

[6182]-1015

M.E. Instrumentation & Control (Process Instrumentation)

PROCESS DYNAMICS AND CONTROL

(2017 Pattern) (Semester - II) (506107)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data if necessary.*

Q1) Comment on Interaction with suitable example. Elaborate procedure to find RGA matrix of two input two output system. **[10]**

Q2) Elaborate dynamic behaviour of second order systems with suitable example. **[10]**

Q3) Elaborate procedure to obtain an empirical model from the plant data. **[10]**

Q4) Develop feed-forward plus feedback control strategy to improve the performance of heat exchanger. **[10]**

Q5) Develop Inferential control strategy for composition control in distillation column. **[10]**

Q6) Design IMC based PID controller for FOPDT system as given below: **[10]**

$$\frac{Kpe^{-\theta s}}{(Tps + 1)}$$



Total No. of Questions : 5]

SEAT No. :

P-9614

[Total No. of Pages : 2

[6182]-1016

M.E. (Instrumentation and Control) (Process Instrumentation)

EMBEDDED SYSTEM DESIGN

(2017 Pattern) (Semester - II) (506108)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figure to the right candidates indicate full marks.*
- 4) *Use of electronic pocket calculator.*
- 5) *Assume suitable data, if necessary.*

Q1) Attempt any two of the following :

- a) Elaborate with neat diagram, AVR general purpose registers in CPU of AVR μ C. **[5]**
- b) Draw neat block diagram and elaborate watchdog timer of AVR microcontroller. **[5]**
- c) Analyze with suitable block diagram timer/counter prescalers of AVR μ C. **[5]**

Q2) Attempt any two of the following :

- a) Draw a neat block diagram and Describe ARM memory architecture. **[5]**
- b) Draw a neat block diagram and elaborate ARM Bus Technology. **[5]**
- c) Elaborate register usage in thumb. Justify how code density will be improved using Thumb. **[5]**

Q3) Attempt any two of the following :

- a) Describe 12 C Protocol. **[5]**
- b) Elaborate how multiple processors can be connected on RS-485 bus. **[5]**
- c) Describe the four primary benefits of CAN protocol provides. **[5]**

P.T.O.

Q4) Attempt any one of the following :

- a) Design embedded system using 8051 μ C for temperature monitoring and control. Draw and explain the system with suitable block diagram and circuit diagram. [10]
- b) Design & Implement Wireless Transceiver for Data Acquisition using AVR microcontroller and Wireless Sensor Network. Elaborate the scheme. [10]

Q5) Attempt any one of the following :

- a) It is decided to design home automation system. Develop the system using LPC2148 ARM-7 microcontroller to display the parameters on LCD display. Provide the facility to communicate the information through SMS. For this system.
 - i) Identify different parameters for home automation system. [2]
 - ii) Enlist selected sensors. [2]
 - iii) Develop block diagram. [2]
 - iv) Elaborate the operation of the system with respect to block diagram. [2]
 - v) Develop the algorithm for the system. [2]
- b) Identify different parameters for health monitoring system. Develop the system using LPC2148 ARM-7 microcontroller to display the parameters. Provide the facility to communicate the information through zig-bee module. For this system.
 - i) Identify different parameters for health monitoring system [2]
 - ii) Enlist selected sensors. [2]
 - iii) Develop block diagram. [2]
 - iv) Elaborate the operation of the system with respect to block diagram [2]
 - v) Develop the algorithm for the system. [2]



Total No. of Questions : 5]

SEAT No. :

P-9615

[Total No. of Pages : 2

[6182]-1017

M.E. (Instrumentation & Control)
(CONTROL SYSTEM
(2017 Pattern) (Semester - II) (506109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right candidates indicate full marks.*
- 4) *Use of electronic pocket calculator.*
- 5) *Assume suitable data, if necessary.*

Q1) Attempt any two Questions

- a) The response of system is $y(t) = x(t) + c$. Test whether system is linear or ono-linear. [5]
- b) Discuss the term Jump Resonance in brief. [5]
- c) Illustrate the following physical non linearities; Dead zone, Saturation, Friction. [5]

Q2) Attempt any two Questions

- a) Write short notes on limit cycle. [5]
- b) Brief the describing function of saturation nonlinearities. [5]
- c) Elaborate the construction of phase trajectories by isocline method.[5]

Q3) Attempt any two Questions

- a) Consider the non-linear systems described by the equation [5]

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_1^2 - x_2$$

Obtain equilibrium states.

P.T.O.

- b) Consider a non-linear system described by equations [5]

$$\dot{x}_1 = -3x_1 + x_2$$

$$\dot{x}_2 = x_1 - x_2 - x_2^3$$

Investigate the stability of equilibrium state. Find a region of stability using Krasovski method. (Consider $P = I$)

- c) Check the stability of the system described by [5]

$$\dot{x}_1 = -3x_1 + x_2$$

$$\dot{x}_2 = x_1 - x_2 - x_2^3$$

Use Krasovskii method.

Q4) Attempt any two Questions

- a) Describe the term zero dynamics with example. [5]
b) Explain the terms Stabilization and tracking. [5]
c) Discuss input Intentional and Incidental nonlinearities with example. [5]

Q5) Attempt any one Questions

- a) Elaborate the concept of sliding mode control using [10]
Sliding surface, Reachability, Reaching law, chattering effect.
b) Illustrate the term sliding mode control with example. [10]



Total No. of Questions : 5]

SEAT No. :

P-9616

[Total No. of Pages : 2

[6182]-1019

M.E. (Instrumentation & Control) (Process Instrumentation)
ADVANCED SIGNAL PROCESSINGS
(2017 Pattern) (Semester - III) (606101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *All questions are compulsory.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of electronic pocket calculator.*
- 5) *Assume suitable data, if necessary.*

Q1) Attempt any two Questions.

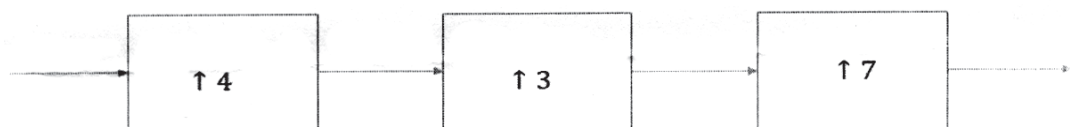
- a) Determine the response of LTI system when input $x(n) = \{1 \ 2 \ 3\}$ and impulse response $h(n) = \{3 \ 2 \ 2\}$. Determine the response of the LTI system by DFT method. [5]
- b) List different types of windows used in FIR filter design, sketch their response and also write their mathematical sequence. [5]
- c) Compute DFT using DIT FFT Algorithm. $x(n) = \{1 \ 2 \ 1 \ 2 \ 1 \ 2 \ 1 \ 2\}$. [5]

Q2) Attempt any two Questions.

- a) Discuss the benefits of STFT. Explain one application of STFT. [5]
- b) Write short note on wavelet packets. [5]
- c) Write short note on multiresolution analysis. [5]

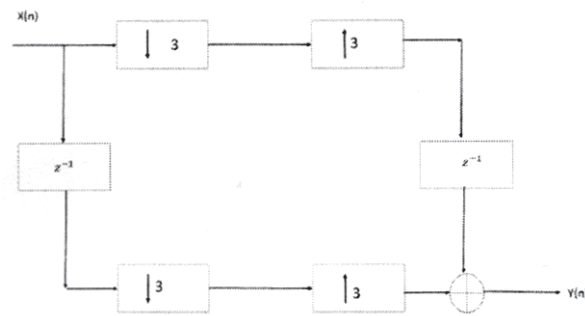
Q3) Attempt any two Questions

- a) Determine the output $y(n)$ in terms of input $x(n)$ for the multirate system shown in figure. [5]



P.T.O.

- b) Consider multirate system as shown in figure, find $y(n)$ as function of $x(n)$. [5]



- c) Distinguish between LMS and RLS filters. [5]

Q4) Attempt any two Questions.

- a) Write short note on energy spectrum of discrete time signal. [5]
b) Illustrate the Levinson-Durbin algorithm with normal equations. [5]
c) Explain ARMA modeling for power spectrum estimation. [5]

Q5) Attempt any one Question.

- a) Determine the frequency resolution, variability and figure of merit of the Bartlett, Welch (50% Overlap) and Blackman Tukey method when $x(n)$ has 1000 samples (i.e. $N = 1000$) and quality factor is 16 (i.e. $Q = 16$). [10]
b) Compute period gram of the signal vector, $x(n) = \{2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2\}$ and sketch the period gram. [10]



Total No. of Questions : 7]

SEAT No. :

P-9617

[Total No. of Pages : 2

[6182]-1020

M.E. (Instrumentation & Control) (Process Instrumentation)

BUILDING AUTOMATION

(2017 Pattern) (Semester - III) (606102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagram must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam table is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Classify fire based on the substance that is used as a fuel for the fire. [6]

b) Write a short notes on Ionization detector. [4]

Q2) a) Explain 26 & 31 bit weigand format ,explain how reader and host are connected in weigand format. [6]

b) Write a short notes on fire tube boiler. [4]

Q3) a) What is AHU? Explain AHU with neat sketch. [6]

b) Write a short notes on lighting control in BAS system. [4]

Q4) a) What is Vapour compression cycle? Explain any one type of Compressor used in vapour compression cycle. [6]

b) Write a short on Unitary equipment. [4]

Q5) a) Explain DDC architecture with neat sketch. [6]

b) Write a Short notes on Lon Works. [4]

P.T.O.

- Q6)** a) Explain Integrated building management system in data centre with neat sketch. [6]
b) Write a short notes Green Buiding. [4]
- Q7)** a) What is the difference between project life cycle & project management life cycle. [6]
b) Write a short notes on Role of Architect in Project life cycle management. [4]



Total No. of Questions : 7]

SEAT No. :

P-9618

[Total No. of Pages : 2

[6182]-1023

M.E. (Mechanical) (Automotive Engineering)
AUTOMOTIVE SAFETY AND REGULATIONS
(2017 Pattern) (Semester - I) (502303)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Use of logarithmic tables, slide rules and electronic pocket calculator is allowed.*
- 5) Assume suitable data, if necessary.*

- Q1)** a) What do you understand by vehicle safety? What are the objectives of it? List out various provisions of safety aspects incorporated in a modern car. [5]
- b) What is vehicle crashworthiness? What are various characteristics of it? [5]

- Q2)** a) Describe various causes for occupant injury. Which parts of human body are affected by injury? Explain. [5]
- b) Explain role for ergonomics in vehicle safety. [5]

- Q3)** a) Explain the types of safety glasses? What are the requirements of it? [5]
- b) Describe with neat sketch automobile seat. Explain various parts of it. [5]

- Q4)** a) What testing procedure is followed for lamps and reflectors in automotive testing laboratory? Explain beam pattern and specification followed for head lamp testing as per CMVR rules. [5]
- b) Describe luminous meter and its principle. [5]

P.T.O.

- Q5)** a) CMVR rules and standards. Describe the devices used for measuring the absolute luminous flux of lamps and luminous flux for industrial and regular lamps? [5]
- b) Draw neat sketch of specifying dimensions and color code as per motor vehicle act 1988 for following signs. [5]
- i) No entry or straight prohibited.
 - ii) Right turn prohibited.
 - iii) Cross road.
 - iv) Parking both sides.
- Q6)** a) Explain briefly what procedure to be followed for passing a permit of commercial vehicles by regulatory authority RTO. [5]
- b) Explain with neat sketch Adoptive front lighting system for vehicles. [5]
- Q7)** Write short notes on the following (Any two) : [10]
- a) Anthropometry data for automobile occupancy.
 - b) Door locks and retention system.
 - c) Testing tracks for vehicles.



Total No. of Questions : 7]

SEAT No. :

P-9619

[Total No. of Pages : 2

[6182]-1025

M.E. (Mechanical) (Automotive Engineering)

NOISE VIBRATION AND HARSHNESS

(2017 Pattern) (Semester - II) (502308)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions out of 7.*
- 2) *All the questions should be solved in one answer book and attach extra supplements if required.*
- 3) *Neat diagram must be drawn whenever necessary.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume suitable data, if necessary, but state the assumption clearly.*

Q1) a) Describe the different types of accelerometer mounting for vibration measurements. **[5]**

b) Explain the different techniques used to noise control? **[5]**

Q2) a) Show that as the distance from a point source doubles, the sound intensity level decrease by 6 dB. **[5]**

b) Explain the Pass by Noise measurement method for Vehicle. **[5]**

Q3) a) The noise signal of a certain machinery in a workshop was processed in an audio-frequency analyser which was equipped with the standard octave filters. The results obtained were as follows : **[10]**

Central frequency (Hz)	63	125	250	500	1000	2000	4000	6000
SPL (dB)	84	85	84	81	72	68	66	52

Determine the overall sound pressure level in dB

P.T.O.

- Q4)** a) Explain digital signal processing technique. [5]
b) Explain Structure borne noise and Air borne noise in car. [5]
- Q5)** a) What do you mean by sound intensity mapping? What is its significance in NVH? [5]
b) Explain in detail different sound fields and state its significance. [5]
- Q6)** The lower and upper frequencies an octave band are 11, 312 and 22, 614 Hz. Determine [10]
a) The central frequency of the 1/1 octave band.
b) The intermediate frequency of the 1/2 octave band and
c) The intermediate frequencies of the 1/3 octave band
- Q7)** a) Explain in details of Experimental Modal Analysis. [5]
b) Describe Anechoic chamber and Reverberation chamber. [5]



Total No. of Questions : 7]

SEAT No. :

P-9620

[Total No. of Pages : 2

[6182]-1026

M.E. (Mechanical) (Automotive Engineering)

AUTOMOTIVE CHASSIS DESIGN

(2017 Pattern) (Semester - II) (502309)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables, slide rules and electronic pocket calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) What do you understand by double wish bone suspension system? Give its advantages during different kinematic conditions of the vehicle.

[5]

b) Design and draw a valve spring of a petrol engine for the following operating conditions :

[5]

Spring load when the valve is open = 400 N

Spring load when the valve is closed = 250 N

Maximum inside diameter of spring = 25 mm

Length of the spring when the valve is open = 40 mm

Length of the spring when the valve is closed = 50 mm

Maximum permissible shear stress = 400 MPa

Q2) a) Compare manual steering with power, assisted steering. Describe, hydraulic power steering system with neat diagram.

[5]

b) Explain with the help of neat sketches different types of steering column.

[5]

Q3) a) Write short note on ABS giving its advantages over normal braking system.

[5]

b) Explain with the help of neat sketch Direct-acting vacuum servos.

[5]

P.T.O.

- Q4)** a) Draw cross section of a tyre explaining the role of various components in that. What are the factors improving road grip? [5]
- b) Explain the parameters on which the tread wear fatigue life and road grip depend. [5]
- Q5)** a) Explain spring stresses in rigid six wheelers. [5]
- b) With the help of neat sketch explain working of six wheeled vehicle. Discuss its application and the method of calculating wheel torque on different wheels. [5]
- Q6)** a) With the help of neat diagram explain the working and construction of disc brake. [5]
- b) Explain tyre construction with help of neat sketch. [5]
- Q7)** Write short notes on the following (**Any two**) : [10]
- a) Electrical Power Steering system.
- b) Hydraulic braking system.
- c) Tread design.
- d) Types of tracks used for vehicle evaluation.



Total No. of Questions : 07]

SEAT No. :

P-9621

[Total No. of Pages : 3

[6182]-1028

M.E. (Mechanical) (Automotive Engineering)

VEHICLE DYNAMICS

(2017 Pattern) (Semester - III) (602313)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any FIVE questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *All questions carry equal marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Draw the tire axis system as recommended by the SAE and explain various angles and forces associated with the tire. **[4]**
- b) A passenger car weighing 15 kN is equipped with the radial-ply tires. The coefficient of rolling resistance of the tire is given by $f_r = 0.0136 + 4 \times 10^{-8} V^2$, where V is the speed of vehicle in kmph. At a speed of 60 kmph, calculate-
- i) The power required to overcome rolling resistance of tire and
 - ii) The retardation rate of car due to rolling resistance of tire when brakes are applied. **[6]**
- Q2)** a) Discuss the effect of wheel lock up during braking on the directional stability and control of a road vehicle. **[2]**
- b) The mass of a passenger car including four tires is 1500 kg. Each of the tires weighs 225 N, has an effective diameter of 67 cm and a radius of gyration of 28 cm. The engine develops a power of 45 kW at 4000 rpm. The equivalent weight of the rotating parts of the driveline is 450 N with a radius of gyration of 10 cm. The transmission efficiency is 88% and the total reduction ratio of the driveline in the second gear is 7.7 to 1. The vehicle has a frontal area of 1.67 m² and the aerodynamic drag coefficient is 0.45. Assume standard atmospheric temperature and pressure condition as 15 °C and 1.013 bar and gas constant for air R=287 J/kg-K. The average coefficient of rolling resistance is 0.015. Under these conditions determine **[8]**

P.T.O.

- i) Speed of the car,
 - ii) Aerodynamic resistance to the car and
 - ii) Acceleration of the car on a level road
- Q3) a)** Define reference frame, toe-in, toe-out, wheel camber, caster and kingpin angle with respect to the suspension kinematics. **[5]**
- b) Explain roll center analysis of vehicle suspension. **[5]**
- Q4) a)** Write an expression for the steer angle required to negotiate a given curve in case of a two axle road vehicle and define the under-steer coefficient. Describe the various steady-state handling characteristics with the help of this coefficient. (No derivation). **[4]**
- b) A passenger car weighs 20 kN and has a wheelbase of 2.8 m. The center of gravity is 1.27 m behind the front axle. If a pair of radial-ply tires, each of which has a cornering stiffness of 46 kN/rad are installed in the front, and a pair of bias-ply tires, each of which has a cornering stiffness of 33 kN/rad are installed in the rear. The average steering gear ratio is 22. **[6]**
- Determine -
- i) Whether the vehicle is understeer or oversteer
 - ii) The required angle in degrees at steering wheel during a constant radius test with turn radius of 12 m at a constant forward speed of 40 kmph.
- Q5) a)** Discuss in detail the effect of natural frequency of sprung mass and damping factor on the performance of vehicle suspension. **[4]**
- b) The sprung mass of a passenger car weigh 14 kN and the un-sprung parts weigh 8% of sprung mass. When car is stationary the suspension spring and tire deforms by 175 mm and 20mm respectively. **[6]**
- Determine:
- i) Vertical stiffness of suspension spring and tire.
 - ii) The two natural frequencies of the bounce motion of the sprung and unsprung mass.

- Q6)** a) Explain the method of calculating RMS value of the vehicle response when it is excited by the random road profile with known power spectral density. [5]
- b) Write short note on the model of ISO road profile. [5]
- Q7)** a) Explain the effect of wheelbase filtering on the vibration response of vehicle suspension to the road roughness. [5]
- b) Explain various tests that can be used to evaluate handling characteristics of vehicles. [5]



Total No. of Questions : 7]

SEAT No. :

P-9622

[Total No. of Pages : 2

[6182]-1029

M.E. (Mechanical) (Automotive Engineering)

AUTOTRONICS

(2017 Pattern) (Semester - III) (602314)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat Diagram must be drawn whenever necessary.*
- 3) *Figures to the right indicate marks.*

Q1) a) Classify different types of resistors used in Automotive circuits. [5]
b) What is Clamping Diode? Explain the use of it in Vehicle Air conditioning. [5]

Q2) a) What is Reference Voltage Sensor? Explain the Throttle position sensor in computer controlled Engine. [5]
b) Explain how the misfire detection is done in Electronic Ignition System. [5]

Q3) a) Explain Digital EGR Valve in context with Emission control Systems. [5]
b) Explain the need of 42-V system. [5]

Q4) a) Explain Diagnostic Trouble Code (DTC). [5]
b) Explain the testing of temperature sensor in vehicle. [5]

Q5) a) Explain different components of automotive Air-conditioning system. [5]
b) Classify Communication Networks in vehicular communication system. [5]

P.T.O.

- Q6)** a) Explain the mechanism to reduce roll moment. [5]
- b) Explain Side slip Angle, Yaw and Roll Feedback in context with rollover control function. [5]

Q7) Write short Note (Any Two) :

- a) Air Bag System Sensor. [5]
- b) Tire pressure Sensor. [5]
- c) Gas Analyzer. [5]



Total No. of Questions : 5]

SEAT No. :

P-9623

[Total No. of Pages : 2

[6182]-1031

M.E. (Mechanical) (CAD Manufacture & Engineering)

ADVANCED MACHINE DESIGN

(2017 Pattern) (Semester - I) (502402)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Solve any two from each question.*
- 2) *Assume suitable data, if necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of electronic pocket calculator is allowed.*

Q1) Attempt any two.

- a) Derive the compatibility equations in Cartesian co-ordinate systems. [5]
- b) Derive the compatibility equations in Polar co-ordinate system. [5]
- c) Explain stress matrix and mention equations of equilibrium. [5]

Q2) Attempt any two

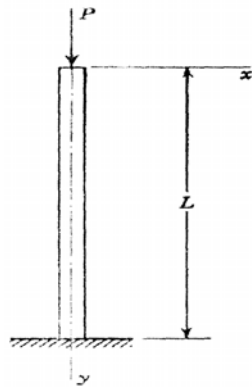
- a) Explain stress-strain relation for brittle and ductile material. [5]
- b) Describe significance of theories of failure. [5]
- c) Explain the following theories of failures with graphical representation and applications. [5]
 - i) Maximum elastic strain theory
 - ii) Distortion energy theory

Q3) Attempt any two.

- a) Explain the procedure to estimate the life of the member subjected to creep. [5]
- b) Following creep data at certain temperature is known [5]
 $S_1 = 10.5 \text{ MPa}$, $\epsilon_1 = 0.012\%$ per 1000hr.
 $S_2 = 14 \text{ MPa}$, $\epsilon_2 = 0.025\%$ per 1000hr.
Determine the constants of hyperbolic sine law and calculate the creep rates for stress values of 20 MPa and 28 MPa.

P.T.O.

- c) Determine the buckling loads of the fixed end column as shown in Fig. by using the Rayleigh-Ritz method. [5]



Q4) Attempt any two

- Describe transverse shear effect in composite laminates. [5]
- What is the hybrid material and discuss its applications? [5]
- The work cycle of a mechanical component subjected to completely reversed bending stresses consists of the following three elements:
 - $\pm 350 \text{ N/mm}^2$ for 85% of time
 - $\pm 400 \text{ N/mm}^2$ for 12% of time
 - $\pm 500 \text{ N/mm}^2$ for 3% of time

The material for the component is 50C4 ($S_{ut} = 660 \text{ N/mm}^2$) and the corrected endurance limit of the component is 280 N/mm^2 . Determine the life of the component. [5]

Q5) Attempt any two

- Explain the Maxwell and Kelvin models. [5]
- Explain high cycle and low cycle fatigue. [5]
- Explain the concept of energy balance during crack growth [5]



Total No. of Questions : 7]

SEAT No. :

P-9624

[Total No. of Pages : 2

[6182]-1032
M.E. (Mechanical) (CADME)
COMPUTER AIDED DESIGN
(2017 Pattern) (Semester - I) (502403)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer any five questions from the following.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data, if necessary.*
- 5) Use of calculator is allowed.*

Q1) a) Explain modeling and viewing features of CAD. **[5]**

b) Differentiate between Hermite cubic curve and Bezier curve. **[5]**

Q2) Explain the parametric representation of surface of revolution and tabulated surface. **[10]**

Q3) a) Differentiate between Hermite Bicubic surface and Bezier surface. **[5]**

b) Explain COONs surface. **[5]**

Q4) Explain different drafting features available in CAD. **[10]**

Q5) a) Explain the tolerances and annotation representation methods of CAD drafting. **[5]**

b) Explain the role of manual explode view in assembly. **[5]**

P.T.O.

Q6) Explain the role of analysis, synthesis and AI in Tolerance modeling. **[10]**

Q7) a) Explain the collaborative design approaches in brief. **[5]**

b) Explain STEP data exchange architecture. **[5]**



Total No. of Questions : 7]

SEAT No. :

P-9625

[Total No. of Pages : 1

[6182]-1033
M.E. (CADM&E)
COMPUTER INTEGRATED MANUFACTURING
(2017 Pattern) (Semester - II) (502407)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer any five questions from the following.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume suitable data,if necessary.*
- 5) Use of calculator is allowed.*

- Q1)** Explain the role of CAD/CAM in CIM? **[10]**
- Q2)** What is Flexible Manufacturing system? Explain with example? **[10]**
- Q3)** Explain the role Integrative manufacturing planning and control? **[10]**
- Q4)** Explain the role of Kanban, JIT, and 5S in Lean Manufacturing. **[10]**
- Q5)** Write short note on MRP-I and Explain the principle of PDM & PLM. **[10]**
- Q6)** Explain the concept of lean production? What are the benefits of lean manufacturing? **[10]**
- Q7)** Discuss in detail capacity planning & materials Requirements Planning in the context of Computer Integrated Manufacturing. **[10]**



Total No. of Questions : 7]

SEAT No. :

P-9626

[Total No. of Pages : 1

[6182]-1034

M.E. (CAD-CAM) (Mechanical)

**INDUSTRIAL PRODUCT DESIGN & PRODUCT
LIFECYCLE MANAGEMENT**

(2017 Course) (Semester - II) (502408)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer any five questions from the following.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume suitable data, if necessary.*

Q1) Explain challenges in new product development with example. **[10]**

.

Q2) Detail about the process concept generation along with technique involved in it. **[10]**

Q3) What is Benchmarking & how to do an effective competitive analysis? Explain with example. **[10]**

Q4) Explain PLM strategy to develop product-data and product workflow in PLM. **[10]**

Q5) What are the threads of PLM? Explain in details. **[10]**

Q6) With your own automobile product, show the detailed product life cycle and technology life cycle of that particular product and explain them in details. **[10]**

Q7) How you will do financial analysis is collaborative product development. **[10]**



Total No. of Questions : 7]

SEAT No. :

P-9627

[Total No. Of Pages : 2

[6182]-1035
M.E. (Mechanical) (CADME)
Automated Manufacturing System Modeling
(2017 Pattern) (Semester - II) (502409)

Time : 3 Hours]

[Max. Marks : 50

- Instructions:**
- 1) Answer any five questions.*
 - 2) Neat diagram must be drawn whenever necessary.*
 - 3) Use of non programmable calculator is allowed.*
 - 4) Assume suitable data, if necessary.*

Q1) a) Explain the principles and Strategies of Automation **[5]**

b) Compare different types of Automation **[5]**

Q2) Explain FMS and its Planning and Implementation Issues **[10]**

Q3) Explain use of pneumatic sensors and amplifiers with suitable examples **[10]**

Q4) Explain the procedure for building Mathematical Model of a manufacturing Plant. **[10]**

Q5) Describe the sensor technologies used for Automated inspection **[10]**

P. T. O.

- Q6)** a) Differentiate between Markov Chain Model and Markov Model of transfer lines used for system modeling. [5]
- b) Explain the procedure adopted for developing pneumatic circuits for automatic die casting machine [5]
- Q7)** a) Explain the Classical Petri Nets and Stochastic Petri Nets used for modeling of Manufacturing Systems [5]
- b) Explain role of Artificial neural networks in manufacturing automation with suitable example [5]



Total No. of Questions : 7]

SEAT No. :

P-9628

[Total No. of Pages : 2

[6182]-1037

M.E. (Mechanical - CADM & E)

SIMULATION MODELING

(2017 Pattern) (Semester - III) (602413)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer the five questions front following.*
- 2) Neat diagram must be drawn wherever necessary.*
- 3) Figures to the right indicates full marks.*
- 4) Assume suitable data, if necessary.*

Q1) Explain the following terms.

[10]

- a) Explain the simulation procedure of continuous system.
- b) Explain the Triangular Distribution

Q2) Explain the following terms.

[10]

- a) System & System environment
- b) Components of a system.
- c) Stochastic nature of output data.

Q3) List the areas of application related to simulation modeling. And list the limitations of simulation.

[10]

Q4) Explain different performance measures in manufacturing system modelin.[10]

Q5) Discuss in detail Weibull distribution and Empirical continuous distribution.[10]

P.T.O.

Q6) What is necessity of verification & validation of model? **[10]**

Q7) Define Modeling of System Randomness. Explain various Sources of Randomness and their effect on Machine Downtime. **[10]**



Total No. of Questions : 7]

SEAT No. :

P-9629

[Total No. of Pages : 2

[6182]-1038
M.E. (Mechanical) (CADME)
OPTIMIZATION TECHNIQUES
(2017 Course) (Semester - III) (602414)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Answer any five questions from the following.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Use of calculator is allowed.*
- 5) Assume suitable data if necessary.*

Q1) Solve by simplex method **[10]**

Minimize $f = -3x_1 - 3x_2$

Subject to $x_1 - x_2 \leq 1$

$$3x_1 - 3x_2 \leq 9$$

$$x_1 \geq 0, x_2 \geq 0$$

Q2) Explain sensitivity or post optimality analysis & Discuss classification of optimization problem. **[10]**

Q3) Solve $x^2 + 54/x$ by successive quadratic interpolation method. **[10]**

Q4) Discuss Exhaustive search method & Bounding phase method for bracketing. **[10]**

P.T.O.

Q5) Find optimum solution using variable elimination method, **[10]**

Minimize $F(x_1, x_2) = [x_1^2 + x_2 - 11]^2 + [x_1 + x_2^2 - 7]^2$

Subject to $g_1(x) = 26 - (x_1 - 5)^2 - x_2^2 \geq 0$

$$g_2(x) = 20 - 4x_1 - x_2 \geq 0$$

$$x_1, x_2 \geq 0$$

Q6) Explain the concept of particle Swarm optimization. **[10]**

Q7) Minimize the function using exhaustive search method **[10]**

$$F(x) = x^2 + (54/x)$$



Total No. of Questions : 7]

SEAT No. :

P-9630

[Total No. of Pages : 2

[6182]-1040

M.E. (Mechanical) (Energy Engineering)
ADVANCED THERMODYNAMICS
(2017 Pattern) (Semester - I) (502502)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicates full marks.*
- 4) *Use of steam tables, electronic, calculators are allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) What is the Physical significance of the two constants that appear in the Van der Waals equation of state? On what basis they are determined?**[4]**
- b) Define compressibility Factor Z. What is the Physical significance of the compressibility factor Z? **[3]**
- c) State and explain Kay rule. **[3]**
- Q2)** a) State and explain Gibbs phase rule. **[4]**
- b) A Vessel of volume 0.04m^3 contains a mixture of saturated steam at a temperature of 250°C . The mass of the liquid present is 9kg. Find the pressure, the mass, the specific volume, the enthalpy, the entropy and the internal energy. **[6]**
- Q3)** a) State and explain Third law of thermodynamics. **[4]**
- b) One Kg of water at 0°C is brought into contact with a heat reservoir at 90°C . When the water has reached 90°C Find (i) Entropy change of water (ii) Entropy change of the heat reservoir (iii) Entropy change of the universe. **[6]**

P.T.O.

Q4) a) Explain the following term **[4]**

- i) Dead state
- ii) Irreversibility

b) In an heat exchanger (parallel flow type) water enter at 50°C and leaves at 70°C while oil (specific gravity = 0.82, specific heat = 2.6 kJ/kgk) enters at 240°C and leaves at 90°C. If the surrounding temperature is 27°C. Determine the loss in availability on the basis of one kg of oil per second. **[6]**

Q5) a) What is Joule-Thomson coefficient? Describe the inversion line and the maximum inversion temperature. **[6]**

b) Derive the Maxwell relations. **[4]**

Q6) a) Methane (CH_4) is burned completely with the stoichiometric amount of air during a steady flow combustion process. If both the reactants and the products are maintained at 25°C and 1 atm and the water in the products exists in the liquid form, determine the heat transfer from the combustion chamber during this process? What would your answer be if combustion were achieved with 100% excess air? **[6]**

Substance	$\bar{h}^\circ_f$ (Kj/kmol)
CH_4	-74850
O_2	0
N_2	0
$\text{H}_2\text{O}_{(l)}$	-285830
CO_2	-393520

b) Explain the term fagacity and Activity. **[4]**

Q7) Write notes on (any two) : **[2 × 5 = 10]**

- a) Exeray Destruction
- b) Nerst heat theorem and heat death of universe
- c) Adiabatic flame temperature



Total No. of Questions : 8]

SEAT No. :

P-9631

[Total No. of Pages : 2

[6182]-1041

M.E. (Mechanical - Energy Engineering)

NON-CONVENTIONAL ENERGY SOURCES

(2017 Pattern) (Semester - I) (502503)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of non programmable calculator is allowed .*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Explain role of energy corporations in India to combat the issue of energy crisis. [5]
b) Explain in brief world energy scenario? [5]
- Q2)** a) Explain with neat diagram the working of concentric solar collector. [6]
b) Write a note on Solar module. [4]
- Q3)** a) Explain availability and limitations of solar energy. [5]
b) Enumerate the basic principle of wind energy. [5]
- Q4)** a) Explain turbine theory with respect to hydroelectric system? [5]
b) Explain the sources of biomass in brief. [5]
- Q5)** a) Explain dry steam geothermal energy with a neat sketch? [5]
b) What are the main types of OTEC power plants? Describe working of OTEC? [5]
- Q6)** a) Write a short note on wave energy conversion? [5]
b) Write down the classification and components of Tidal power Plant and enumerate advantages and disadvantages of it? [5]

P. T. O

- Q7)** a) What is the need of hybrid energy systems? [2]
b) How are Gasifiers classified? What is Pyrolysis? [4]
c) Explain Bio mass-Diesel system. [4]

- Q8)** Write a short note on any two : [10]
a) Fossil fuel reserves in India
b) Flat Plate collectors
c) Types of Wind mills



Total No. of Questions : 7]

SEAT No. :

P-9632

[Total No. of Pages : 2

[6182]-1042

M.E. (Mechanical Energy Engineering)
ADVANCED HEAT TRANSFER
(2017 Pattern) (Semester - II) (502507)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer Any Five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right of each question indicate full marks.*
- 4) *Assume suitable data wherever necessary and mention the same clearly.*
- 5) *Use of steam tables, mollier chart and calculator is allowed.*

Q1) a) What do you mean by initial and boundary conditions. Explain in brief (i) Heat flux boundary condition (ii) convection boundary condition [5]
b) A steel pipe with 50 mm OD is covered with a 6.4 mm asbestos insulation ($k = 0.166 \text{ W/m}^\circ\text{C}$) followed by 25 mm layer of fiber glass insulation ($k = 0.0485 \text{ W/m}^\circ\text{C}$). The pipe wall temperature is 393 K and outside insulation temperature is 311 K. Calculate interface temperature between asbestos and fiber glass. [5]

Q2) a) Consider the base plate of a 1200-W iron that has a thickness of $L = 0.5 \text{ cm}$, base area of $A = 300 \text{ cm}^2$, and thermal conductivity of $k = 15 \text{ W/m}^\circ\text{C}$. The inner surface of the base plate is subjected to uniform heatflux generated by the resistance heaters inside and the outer surface loses heat to the surroundings at $T = 20^\circ\text{C}$ by convection. Taking the convection heat transfer coefficient to be $h = 80 \text{ W/m}^2\text{C}$ and neglecting heat loss by radiation, obtain an expression for the variation of temperature in the base plate and evaluate the temperatures at the inner and the outer surfaces. [5]
b) What do you mean by Thermal Contact Resistance? What are methods to reduce the same? [5]

Q3) Write Short Notes on following : [10]

- a) Local and average Heat Transfer Coefficient
- b) Hydraulic diameter
- c) Thermal entrance region and length

P.T.O.

- Q4) a)** Explain physical significance of following dimensionless numbers. [5]
- Richardson number
 - Stanton Number
 - Peclet Number
- b) Experimental results for the local heat transfer coefficient h_x for flow over a flat plate with an extremely rough surface were found to fit the relation [5]
- $$h_x(x) = x^{-0.1}$$
- where x (m) is the distance from the leading edge of the plate. Develop an expression for the ratio of the average heat transfer coefficient for a plate of length x to the local heat transfer coefficient h_x at x .
- Q5) a)** Explain following terms of mixed convection with neat sketch. [5]
- Assisting flow
 - Opposing flow
 - Transverse flow
- b) A plate type 200 mm x 300 mm heater is immersed in water bath at 40 °C. The larger side of heater is maintained vertical. The heater rating is 3.5 kW. Determine the steady state temperature attended by the heater using following thermo physical properties. Consider heat Transfer from one side of the plate [5]
- $k = 0.667 \text{ W/m } ^\circ\text{C}$, $\nu = 0.415 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 977.8 \text{ Kg/m}^3$, $C_p = 4187 \text{ KJ/kg } ^\circ\text{C}$, $\beta = 4.15 \times 10^{-4} \text{ per degree Kelvin}$
- For heat transfer by convection use following correlation
- $$\text{Nu} = 0.13 (\text{Ra})^{(0.33)}$$
- Q6) a)** Write note on : [5]
- Nucleate boiling
 - Drop and film condensation
- b) Explain with neat sketch different regimes of pool boiling curve. [5]
- Q7)** Explain the following in brief : [10]
- Lambert's cosine law
 - Intensity of radiation
 - Solid Angle
 - Irradiation
 - Radiosity



Total No. of Questions : 8]

SEAT No. :

P-9633

[Total No. of Pages : 2

[6182]-1043

M.E. (Mechanical Energy Engineering)
ENERGY CONVERSION & ENVIRONMENT
(2017 Pattern) (Semester - II) (502508)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) a) Explain different energy sources. **[6]**

b) Describe conversion of chemical energy of the fuel into thermal energy. **[4]**

Q2) The following particulars refer to a stage of a Parson's steam turbine comprising one ring of fixed blades and one ring of moving blades; Mean diameter of blade ring = 75cm, R.P.M. = 3500, Steam velocity at exit of the blades 170 m/s, Blade outlet angle = 20°, Steam flow through blades = 8 kg/s.

Draw the velocity diagram and find the following **[10]**

- a) Blade inlet angle,
- b) Tangential force on the ring of moving blades,
- c) Power developed in a stage.

Q3) a) Explain any combined cycle with heat recovery boiler with sketch. **[4]**

b) Enumerate lean burn engines. **[6]**

Q4) a) Enumerate formulation of standards against the radioactive protection. **[5]**

b) Explain the environmental aspects of a nuclear power generation with remedial actions. **[5]**

P.T.O.

- Q5)** a) Explain efficient compressor operation with examples. [5]
b) Explain capacity of the compressor and various efficiencies of the compressor. [5]
- Q6)** a) Describe advantages of adopting Diesel power plants. [4]
b) Explain Energy Saving Opportunities in Cooling Towers. [6]
- Q7)** Explain in detail Reheat and intercooling methods for improvement of the efficiency of gas turbine power plant with cycle analysis. [10]
- Q8)** a) Explain design procedure for Pelton turbine. [5]
b) Explain working principle of Francis turbine. [5]



Total No. of Questions : 7]

SEAT No. :

P-9634

[Total No. of Pages : 2

[6182]-1047

M.E. (Mechanical) (Energy Engineering)
ENERGY SYSTEMS MODELING AND OPTIMIZATION
(2017 Pattern) (Semester - III) (602514)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of scientific calculator is allowed*
- 5) *Assume suitable data, if necessary.*

Q1) a) Explain Information Flow Diagram with suitable example. [5]

b) Find the root of equation $x \cdot \log_e x = 1.2$ using Modified Newton-Raphson Iteration Criteria. Take three iterations. Assume initial guess = 2. [5]

Q2) a) Explain the types of system study. [6]

b) Explain various types of simulation models for system study. [4]

Q3) a) A person wants to decide the constituents of a diet which will fulfil his daily requirements of protein, fats and carbohydrates at minimum cost. The choice is to be made from different types of foods. The yields per unit of these foods are given below in the table. [5]

Food type	Yield per unit			cost per unit
	Proteins	Fats	Carbohydrates	
1	3	2	6	45
2	4	2	4	40
3	8	7	7	85
4	6	5	4	65
Minimum Requirement	800	200	700	

Formulate the linear programming model for the problems.

P.T.O.

- b) Maximize $f(x)=6x-x^2$. Use 1-Dimensional search procedure. Take initial upper and lower bounds 4.8 and 0 respectively and tolerance error is 0.04. [5]

Q4) Attempt any two : [10]

- a) For common heat exchangers, such as the parallel and counter flow heat exchangers shown in Figure 1.5, discuss the development of a simple mathematical model to analyze the system.
- b) Explain different steps in model development.
- c) Find the polynomial for the data by using Newton's interpolation formula

x	0	1	2	3	4	5
y=f(x)	1	1	7	25	61	121

Find the value of y at x = 0.5

- Q5)** a) Explain in details search techniques Univariate method. [5]
- b) The objective function for an optimization problem is taken as the total income, which involves an income of five units on item A and seven units on item B. The former requires 2.5 hours of cutting and 1.5 hours of polishing, whereas item B requires 4 hours of cutting and 1 hour of polishing. If the total labor hours available for cutting are 4000 and for polishing 2000, formulate the optimization problem and solve it by the simplex algorithm to obtain the optimum. [5]

- Q6)** a) Solve the following differential equation using Euler's method under the boundary conditions x = 1.3 and y = 2. Find y at x = 1.8 in 10 steps.[6]

$$\frac{dy}{dx} = \frac{(x+y)}{(y^2 - \sqrt{xy})}$$

- b) Use the RK second order order methods to estimate y(0.2) of the following equation with h=0.1. $y'(x)=x+y$, $y(0)=1$. [4]

- Q7)** a) Define simple interest, compound interest, present worth factor, future worth factor. [5]
- b) Given the equation $y'(x) = (2y)/x$ with $y(1) = 2$, estimate y(2) using Euler's method. [5]
- Take h=0.25.



Total No. of Questions : 8]

SEAT No. :

P-9635

[Total No. of Pages : 2

[6182]-1049

M.E. (Mechanical-Mechatronics)
SYSTEM MODELLING, IDENTIFICATION AND
SIMULATION
(2017 Pattern) (Semester - I) (502801)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) Describe the process of parametric identification using least squares method. [5]
- b) Explain the concept of non-parametric identification and its applications. [5]

- Q2)** a) Describe the Linear Kalman filter and its applications in state estimation. [5]
- b) Discuss the use of Extended Kalman filter in parameter estimation.[5]

- Q3)** a) Draw a suitable flowchart and explain the process involved in “Black Box Identification”. [5]
- b) Write five exclusive points of comparison between supervised and unsupervised learning. [5]

- Q4)** a) In comparison to the Kalman Filter approach, discuss the advantages offered by the Extended Kalman filter approach. [5]
- b) Draw the flow chart and explain the identification of ARX model using Least Square method. [5]

P.T.O.

Q5) a) List the advantages and the disadvantages of the state space modelling approach. [5]

b) Draw the signal flow diagram for below transfer function. [5]

$$\frac{C(s)}{R(s)} = \frac{(s+3)}{(s^2+10s+24)}$$

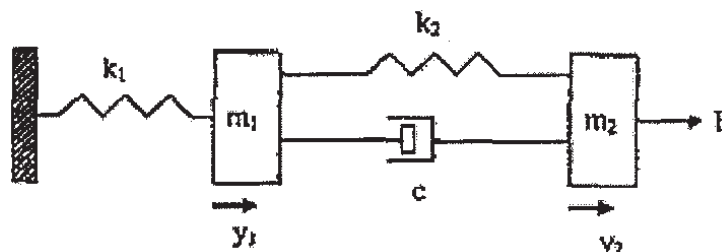
Q6) a) Draw the flow chart and explain the process of identification of a frequency domain model. [5]

b) Draw a suitable block diagram explain the process of “Model Identification”. [5]

Q7) a) For the following transfer function, draw the equivalent signal flow diagram. [5]

$$\frac{C(s)}{R(s)} = \frac{12}{(s+1)(s+2)(s+3)}$$

b) Figure shows a mechanical system with two masses and two springs. Drive an expression for the mathematical model of the system. [5]



Q8) a) Convert the below state space model into a transfer function. [5]

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t)$$

$$y = [1.5 \quad 0.625] x$$

b) List the criterion for selection of fuzzy membership functions. [5]



Total No. of Questions : 8]

SEAT No. :

P-9669

[Total No. of Pages : 2

[6182]-1050
M.E. (Mechanical Mechatronics)
CONTROL SYSTEMS - I
(2017 Pattern) (Semester - I) (502802)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions.*
- 2) *Draw neat diagrams wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of a calculator is allowed*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Explain the difference between open-loop and closed-loop control systems. Provide examples for each. [5]
b) Discuss the concept of state space modeling. How is it useful in control system analysis? [5]
- Q2)** a) Define State Transition Matrix and explain its significance in the context of control systems. [5]
b) Describe the solution to the state equation. How is it related to the behavior of a dynamic system? [5]
- Q3)** a) Provide an Overview of the Lyapunov Criterion for stability analysis in control systems. [5]
b) Explain the Routh-Hurwitz Criterion. How is it used to determine the stability of a system? [5]
- Q4)** a) Define controllability and observability. Discuss their importance in system analysis. [5]
b) Describe how to determine controllability and observability from the block diagram of a system. [5]

P.T.O.

- Q5)** a) Introduce Full State Feedback (FSF) Control. What is its significance in control system design? [5]
b) Explain the process of FSF using pole placement. How are poles strategically placed? [5]
- Q6)** a) Discuss the transformation of a state space model into control canonical form. [5]
b) Apply pole placement to a system in control canonical form. [5]
- Q7)** a) Describe the use of Ackermann's Formula in pole placement and its advantages. [5]
b) Define a linear observer (estimator). Discuss its role in control system design. [5]
- Q8)** a) Explain the design of a reduced order observer using pole placement. [5]
b) Explain the design of a command tracking system with integral control and FSF. [5]



Total No. of Questions : 7]

SEAT No. :

P-9636

[Total No. of Pages : 2

[6182]-1051

M.E. (Mechanical - Mechatronics)
SENSORS, TRANSDUCERS & INTERFACING
TECHNIQUES
(2017 Course) (502803) (Semester - I)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt any five questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat diagrams, figures wherever necessary.*
- 4) *Use of scientific calculator is allowed.*
- 5) *Assume suitable data if required.*

Q1) a) Explain following terms - **[5]**

- i) Static & dynamic error
- ii) Correlation coefficient
- iii) Standard deviation

b) Discuss the general considerations of data analysis. **[5]**

Q2) a) What are the main advantages of digital instruments over analog? Draw neat block diagram of digital storage oscilloscope & explain function of each block. **[5]**

b) State the various applications of piezoelectric transducer and explain any one in detail. **[5]**

Q3) a) Explain the working of following gauges using neat diagram - **[5]**

- i) Pirani thermal conductivity gauge
- ii) Ionization gauge

b) Draw neat diagram of Diaphragm & Bellow gauge & explain its working. **[5]**

P.T.O.

- Q4)** a) Explain the working of hot wire & hot film anemometers. [5]
b) With the help of neat diagram explain working of LDA. [5]
- Q5)** a) State the different techniques of temperature measurement & explain the most widely used technique in detail. [5]
b) Write short notes on - [5]
i) Heat flux meter
ii) Viscosity measurement
- Q6)** a) Using neat diagram explain how the o/p can be measured for resistance strain gauge. [5]
b) Discuss the need of temp compensation used in the strain gauges using neat diagram. [5]
- Q7)** a) What is the need of signal conditioning in case of DAS? Why OP-AMP's are preferred for signal conditioning? [5]
b) Enlist the important features/capabilities of A - to - D convertor. State the various types of ADC's & explain any one ADC in detail. [5]



Total No. of Questions : 8]

SEAT No. :

P-9637

[Total No. of Pages : 3

[6182]-1053

M.E. (Mechanical-Mechatronics)

PLC PROGRAMMING

(2017 Pattern) (Semester - II) (502807)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) a) Describe the different types of timer instructions in PLC programming. [4]

b) Explain the application of incremental encoder counters in PLC systems. [6]

Q2) a) Explain the concept of sequencer output instructions in PLC programming. [4]

- b) Describe the main function of each of the following main component parts of a PLC. [6]
- i) Processor module (CPU)
 - ii) I/O modules
 - iii) Programming device

Q3) a) Convert each of the following binary numbers to decimal numbers:[4]

- i) 101
- ii) 1100
- iii) 1001
- iv) 1110

b) Draw the logic symbol, construct a truth table and state the Boolean equation for each of the following: [6]

- i) Two-input AND gate
- ii) Three-input OR gate
- iii) NOT function

P.T.O.

Q4) Design a PLC ladder diagram program to control the operation of a motorized conveyor system based on the following requirements. The program should be designed using the ladder logic programming language. **[10]**

System Requirements:

- The conveyor system consists of a motor, a start button, a stop button and two limit switches (LS1 and LS2) positioned at the start and end of the conveyor.
- The motor should start when the start button is pressed and stop when the stop button is pressed.
- If LS1 is activated (closed), indicating that an object is present at the start of the conveyor, the motor should not start.
- When LS2 is activated (closed), indicating that an object has reached the end of the conveyor, the motor should stop.
- The motor should only start if both the start button is pressed and LS1 is not activated.
- The motor should stop if the stop button is pressed or LS2 is activated.

Q5) Draw the electrical symbol used to represent each of the following PLC output control devices: **[10]**

- a) Normally Open (NO) contact
- b) Normally Closed (NC) contact
- c) Coil or relay coil
- d) Timer symbols:
- e) Alarm
- f) Heater
- g) Solenoid
- h) Solenoid valve
- i) Motor
- j) Function block symbols

Q6) Write the ladder logic program needed to implement each of the following: (assume inputs A, B, and C are all normally open toggle switches): **[10]**

- a) When input A is dosed, turn ON and hold ON outputs X and Y until A opens.
- b) When input A is dosed and either input B or C is open, turn ON output Y. otherwise, it should be OFF.
- c) When input A is dosed or open, turn ON output Y.
- d) When input A is dosed, turn ON output X and turn OFF output Y

- Q7)** a) Describe the programming of counter instructions in PLC systems. [4]
- b) Write a program that will latch on a light 20 s after an input switch has been turned on. The timer will continue to cycle up to 20 s and reset itself until the input switch has been turned off. After the third time the timer has timed to 20 s, the light will be unlatched. [6]
- Q8)** a) Explain the operation of up counters and down counters in PLC programming. [6]
- b) State the method used to reset the accumulated time of each of the following: [4]
- i) TON timer
 - ii) TOF timer



Total No. of Questions : 8]

SEAT No. :

P-9638

[Total No. of Pages : 2

[6182]-1054

M.E. (Mechanical-Mechatronics)

CONTROL SYSTEMS - II

(2017 Pattern) (Semester - II) (502808)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) Define process models and explain the concept of transfer function.

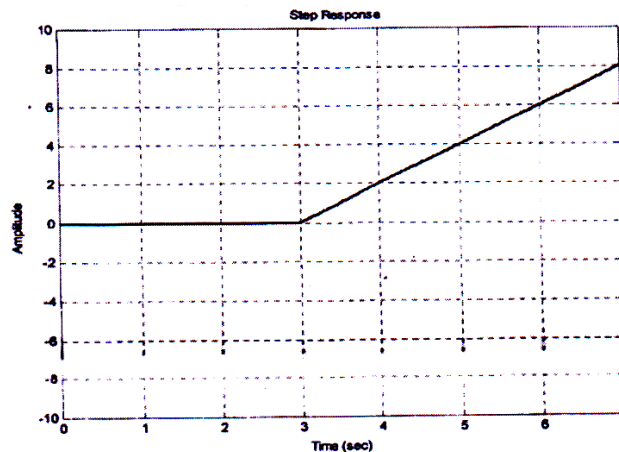
[5]

b) Describe the identification process using the time domain.

[5]

Q2) a) The step response of a model is shown below. Using suitable technique identify the model.

[5]



b) Define gain, phase, and delay margins in frequency response.

[5]

Q3) a) Derive the transfer function of a Proportional derivative control in parallel form.

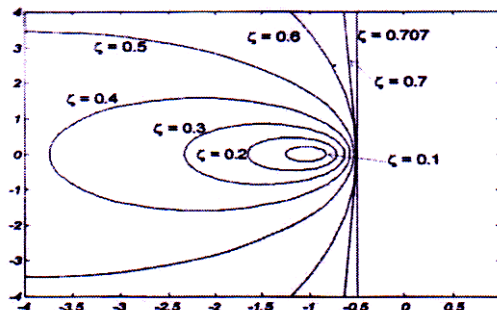
[5]

b) Explain the role of P, I, and D elements in PID control.

[5]

P.T.O.

- Q4)** a) Explain the Ziegler-Nichols method for PID tuning. [5]
 b) Describe the pole placement technique for PID tuning. [5]
- Q5)** a) Define sensitivity function and complementary sensitivity function. [5]
 b) Discuss the relationship between sensitivity and gain/phase margin. [5]
- Q6)** a) Explain the need for advanced tuning techniques. [5]
 b) Describe the Kappa Tau (KT) tuning method. [5]
- Q7)** a) In comparison to time domain analysis, discuss the advantages and the dis-advantages of the frequency domain analysis. [5]
 b) Write a short note on nyquist Stability Criterion. [5]
- Q8)** a) List the procedure for sketching the Bode Plot. [5]
 b) Determine the maximum amplitude of the closed loop for below values of damping. [5]



Total No. of Questions : 8]

SEAT No. :

P-9639

[Total No. of Pages : 2

[6182]-1055
M.E. Mechanical (Mechatronics)
INDUSTRIAL DRIVES AND ACTUATORS
(2017 Pattern) (Semester - II) (502809)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) With hydraulic circuit, explain meter in circuit with components used and applications of meter in circuit. **[5]**
- b) With respect to Construction, working and specific applications, discuss vane pump used in industrial hydraulics. **[5]**
- Q2)** Describe the classification of Electric Drive. **[10]**
- Q3)** a) Discuss hydraulic power pack with respect to components used, safety measures and symbolic representation. **[5]**
- b) Discuss in brief with hydraulic circuit symbols, types of hydraulic cylinders and their mountings. **[5]**
- Q4)** Explain how the speed control of three phase induction motor is obtained by changing frequency and voltage. **[10]**
- Q5)** a) With respect to application explain the cushioning provided in Double acting Hydraulic Cylinder. **[5]**
- b) Explain the working of Pressure Relief Valve and Sequence Valve used in industrial Hydraulics. **[5]**

P.T.O.

- Q6)** a) Explain working principle of synchronous motor. **[4]**
b) In case of separately excited DC series motor, explain constant torque constant power speed control characteristics. **[4]**
- Q7)** With hydraulic circuit and components used, discuss in brief the unloading circuits used in industrial hydraulics. **[10]**
- Q8)** Answer any two : **[10]**
- a) Active and Passive torque
 - b) Unloading Valve with applications
 - c) PLC Hydraulics and Electro-Hydraulics
 - d) AC and DC drive



Total No. of Questions : 8]

SEAT No. :

P-9640

[Total No. of Pages : 2

[6182]-1057

M.E. (Mechatronics) (Mechanical)

**MICROCONTROLLER APPLICATIONS IN EMBEDDED
SYSTEMS**

(2017 Pattern) (Semester - III) (602813)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figure to the right indicate full marks.*
- 4) *Assume suitable data, whenever necessary.*

Q1) a) Draw and explain block diagram of PIC 18F microcontroller. [6]

b) Discuss flags in STATUS register of PIC18F microcontroller in detail. [4]

Q2) a) With neat diagram explain support devices of PIC18F microcontroller. [6]

b) Explain the instructions for data transfer. [4]

Q3) a) With neat circuit diagram of LCD Interfacing tp PIC Microcontroller. [5]

b) Explain stack and instructions related to stack of PIC 18F microcontroller. [5]

Q4) a) What is Interrupt? Explain different types of interrupts. [5]

b) Draw block diagram Timer of PIC18F microcontroller. Explain use of timers. [5]

P.T.O.

- Q5)** a) With block diagram explain any one type of A/D converter. [5]
b) What is function of Capture, Compare and PWM module? [5]
- Q6)** a) Explain data transfer in Inter-Integrated circuit protocol. [5]
b) Explain data transfer in Serial Peripheral Interface protocol. [5]
- Q7)** a) Enlist features of embedded systems. Explain any two features of Embedded System. [6]
b) Explain basic concepts in serial communication. [4]
- Q8)** a) Explain hardware Aspect of the Time and Temperature Monitoring System (TTMS). [5]
b) Explain Software Aspect of the Time and Temperature Monitoring System (TTMS). [5]



Total No. of Questions : 8]

SEAT No. :

P-9641

[Total No. of Pages : 2

[6182]-1058

M.E. (Mechanical-Mechatronics)
FLEXIBLE MANUFACTURING SYSTEMS
(2017 Pattern) (Semester - III) (602814)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any Five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary*

Q1) a) Explain the components of Manufacturing system. **[6]**

b) Explain benefits of group technology. **[4]**

Q2) a) What is single station automated cells? **[4]**

b) What is automated production line? What is benefits of it? **[6]**

Q3) a) Explain the factor considered in assembly line design. **[4]**

b) What is difference between Fixed Routing and Variable Routing in Manufacturing system? **[6]**

Q4) a) Explain the difference between a dedicated FMS and a random-order FMS. **[4]**

b) Discuss the types of AGVS. **[6]**

Q5) a) Name the various elements of CNC machines. **[4]**

b) Explain the application of machine vision in robotics. **[6]**

P.T.O.

- Q6)** a) Explain Operating Characteristic Curve. [4]
b) Differentiate MRP-I and MRP-II. [6]

- Q7)** a) Write brief notes on inventory control system. [4]
b) Explain the requirement of Just In Time. [6]

- Q8)** a) The following data gives the readings for 8 samples of size 6 each in the production of a certain product. Find the control limits using mean chart.

Sample	1	2	3	4	5	6
Mean	300	342	351	319	326	333
Range	25	37	20	28	30	22

Given for $n = 6$, $A_2 = 0.483$, [6]

- b) Discuss all the goals in concurrent engineering. [4]



Total No. of Questions : 6]

SEAT No. :

P-9665

[Total No. of Pages : 2

[6182]-1060

M.E. (Printing & Graphic Communication)
PROBABILITY, STATISTICS AND REGRESSION
ANALYSIS
(2017 Pattern) (Semester - I) (508101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions compulsory.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) i) Write a short note on normal distribution.
ii) Write short note on sampling distribution.

[9]

- b) i) Write a short note on hypothesis.
ii) Write a short note on Regression.

[9]

OR

Q2) Write short notes on :

[18]

- a) Explain the term random variable. State examples of simple probability distributions.
- b) 7, 7, 9, 8, 10
19, 20, 21, 20, 22
21, 21, 17, 19, 20
8, 7, 8, 9, 10

Calculate mean, median, mode, standard deviation of the above given data.

P.T.O.

Q3) a) Explain Least square estimation. **[8]**

b) What is ANOVA and what is its significance in statistical testing. **[8]**

OR

Q4) What does Quality control mean? Explain the basis of control charts. Explain any one type of control chart. **[16]**

Q5) What do you mean by sampling? Explain Acceptance sampling. Define the OC curve. **[16]**

OR

Q6) Following are the problems observed in Offset printing. State the cause of all the problems. **[16]**

- a) Density increase
- b) Dot gain
- c) Poor contrast
- d) Low color strength



Total No. of Questions : 6]

SEAT No. :

P-9670

[Total No. of Pages : 2

[6182]-1061

M.E. (Printing Engg. & Graphic Communication)

PRINTING TECHNOLOGY MANAGEMENT

(2017 Pattern) (Semester - I) (508102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right in bracket indicate full marks.

Q1) Explain following concepts of supply chain management in printing industry. **[18]**

- a) Demand Forecasting. Long term and short term forecasts
- b) Least square forecasting
- c) Moving average forecasting

OR

Q2) Describe the concept of inventory applied to a big size printing press. Explain the model suitable to press and the Inventory systems applied in details. **[18]**

Q3) Explain following : **[16]**

- a) Quality cost
- b) Quality assurance
- c) Quality strategy development

OR

Q4) What is concept of **[16]**

- a) ISO
 - b) Six sigma,
- applied to printing industry. Explain with suitable example/Case study

P.T.O.

Q5) What are the objectives and benefits of application of statistical process control applied to printing industry in general, explain with greater details? **[16]**

Q6) Describe following with respect to SPC with suitable diagram and examples: **[16]**

- a) CL
- b) UCL
- c) LCL



Total No. of Questions : 3]

SEAT No. :

P-9642

[Total No. of Pages : 1

[6182]-1062

M.E. (Printing Engineering & Graphic Communication)

MODERN TRENDS IN PRINTING

(2017 Pattern) (Semester - I) (508103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Draw neat diagram wherever necessary.*
- 2) *Figures to right indicate full marks.*

Q1) Explain in detail sections of rotogravure. **[18]**

OR

Explain in detail sections of CI Flexo press. **[18]**

OR

Explain in detail flexo process parameters. **[18]**

OR

Explain the effect of ink rheology on gravure dot fidelity. **[18]**

Q2) Explain the working of ELS for a web press. **[16]**

OR

Explain the effect of impression roller hardness on dot reproduction. **[16]**

Q3) Mention the care and maintenance for offset plates **[16]**

OR

Explain the factors affecting ink transfer in flexography. **[16]**



Total No. of Questions : 6]

SEAT No. :

P-9682

[Total No. of Page : 1

[6182]-1063

M.E. (Printing Engg. & Graphic Communication)

RESEARCH METHODOLOGY

(2017 Course) (Semester - I) (508104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right in bracket indicate full marks.

Q1) Explain following data types of a research project with suitable example.[18]

- a) Primary Data
- b) Secondary Data

OR

Q2) Explain following : [18]

- a) Basic principles of experimental design.
- b) Literature review.

Q3) Explain different steps in preparation of a technical report. [16]

OR

Q4) What are the important aspects of making a research report presentation in digital and oral format. [16]

Q5) Explain following types of technical papers. [16]

- a) Review Paper
- b) Conference Paper
- c) Survey Paper
- d) Journal Paper

OR

Q6) What are the different steps in making research proposal. [16]



Total No. of Questions : 3]

SEAT No. :

P-9643

[Total No. of Page : 1

[6182]-1067

M.E. (Printing Engineering and Graphic Communication)

SUBSTRATE AND INK

(2017 Pattern) (Semester - II) (508109)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Assume suitable data wherever necessary.*
- 3) Figures to the right indicate full marks.*

Q1) a) Explain different polymeric substrates used in printing with suitable application. **[6]**

b) Discuss any two strength properties of paper and testing methods. **[6]**

c) Give typical formulation of offset ink. **[6]**

Q2) Calculate paper required for printing 1000 copies of 8" × 11" size 4-page brochure in 4+4 color. **[16]**

Q3) Comment on following. **[16]**

- a) QC for paste Ink
- b) Hazardous waste in Printing.



Total No. of Questions : 3]

SEAT No. :

P-9644

[Total No. of Page : 1

[6182]-1069

M.E. (Printing Engineering and Graphic Communication)

PRINTED ELECTRONICS AND RFID

(2017 Pattern) (Semester - III) (608101)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Assume suitable data wherever necessary.*
- 3) Figures to the right indicate marks.*

Q1) a) What is Printed electronics and Discuss its scope? [6]

b) Explain different functional layers used in electronics manufacturing. [6]

c) What are the traditional methods for electronics manufacturing? Explain one of them in detail. [6]

Q2) Discuss Identifying and Tagging Everything with upcoming changes in technology. [16]

Q3) Discuss use of printed electronics in Smart Packaging. [16]



Total No. of Questions : 3]

SEAT No. :

P-9645

[Total No. of Page : 1

[6182]-1070

M.E. (Printing Engineering & Graphic Communication)

ADVANCES IN CONVERTING AND PACKAGING

(2017 Pattern) (Semester - III) (608102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Draw neat diagram wherever necessary.*
- 2) Figures to right indicate full marks.*

Q1) Explain the security features used in packaging.

[18]

OR

Explain in detail varnishing techniques used for packaging.

OR

Explain in detail blown film lamination for converting.

OR

Explain in detail wet lamination technique.

Q2) Explain in Retort packaging for liquid products.

[16]

OR

PET plays an important role in packaging. Explain

Q3) Explain in detail aseptic packaging.

[16]

OR

Explain in detail FFS techniques used for packaging.



Total No. of Questions : 6]

SEAT No. :

P-9646

[Total No. of Pages : 2

[6182]-1073

M.E. (Production) (CAD/CAM)
(Manufacturing and Automation)
RESEARCH METHODOLOGY
(2017 Pattern) (Semester - I) (511102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q1 OR Q2, Q3 OR Q4, and Q5 & Q6 are compulsory.*
- 2) Assume suitable data if necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Neat diagrams must be drawn wherever necessary.*

Q1) a) What is mean by experimental research? Explain in short. [5]

b) Comment on: Ethical research. [5]

OR

Q2) a) Explain various problem solving approaches in short. [5]

b) Write short note on Group Problem Solving Techniques. [5]

Q3) a) Explain process of formulation of model based on simulation. [5]

b) Explain various Stages in Model building. [5]

OR

Q4) a) Explain various steps in design of experiments. [5]

b) Explain Methods of manipulating Variables. [5]

Q5) a) Explain Taguchi Approach to Parameter Design. [8]

b) Write short note on Robust Design. [7]

P.T.O.

Q6) Write short notes on :

[15]

- a) Format for report writing.
- b) Bi-variate analysis.
- c) Parametric Tests.



Total No. of Questions : 8]

SEAT No. :

P-9647

[Total No. of Pages : 2

[6182]-1074

M.E. (Production) (CAD/CAM)
COMPUTER AIDED DESIGN
(2017 Pattern) (Semester - I) (511301)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4 and Q5 or Q6.*
- 2) *Q7 and Q8 are compulsory.*
- 3) *Figures to the right indicate full marks.*
- 4) *Draw neat self-explanatory sketches wherever necessary.*
- 5) *Use of calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

Q1) What is parametric representation of geometrical entities used in CAD software? Explain your answer with non-parametric and parametric equations of appropriate geometric entity. **[5]**

OR

Q2) Discuss the stages in product life cycle and the importance of each stage. **[5]**

Q3) What is Homogeneous Coordinate System? Explain the necessity of Homogeneous Coordinate System for transformation of geometric entities using suitable examples. **[5]**

OR

Q4) Differentiate between STEP and IGES. **[5]**

Q5) The coordinates of four points are given by $P_0 = [2 \ 2 \ 0]^T$, $P_1 = [2 \ 3 \ 0]^T$, $P_2 = [3 \ 3 \ 0]^T$, $P_3 = [3 \ 2 \ 0]^T$. Find the equation of Bezier curve. Also find the points on the curve for $u = 0, 0.25, 0.5, 0.75$ and 1, also plot the curve for the given data. **[7]**

P.T.O.

Q6) Write the mathematical representation, application & limitations of the following surfaces. [7]

- a) Spherical surfaces
- b) Composite surface

Q7) a) Discuss in brief the basic elements of CSG model. Explain the main building operation of CSG scheme with examples. [7]

- b) What is Z-buffer algorithm for B-REP and CSG model? [8]

Q8) Write short note on any three : [18]

- a) Wire frame modeling
- b) Hidden line removal algorithm
- c) CAD/CAM integration
- d) Parametric programming
- e) Types of animations



Total No. of Questions : 6]

SEAT No. :

P-9648

[Total No. of Pages : 2

[6182]-1075

M.E. (Production) (CAD/CAM)
COMPUTER AIDED MANUFACTURING
(2017 Pattern) (Semester - I) (511302)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q1 OR Q2, Q3 OR Q4, and Q5 & Q6 are compulsory.*
- 2) Assume suitable data if necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Neat diagrams must be drawn wherever necessary.*

Q1) a) Explain working principle of rolling guide ways in CNC machines. **[5]**

b) Write a short note on drives and controllers used in CNC machines. **[5]**

OR

Q2) Write a short note on: **[10]**

- a) Facing canned cycle.
- b) CNC Programming.

Q3) a) Explain working principle of automatic welding machine with block diagram. **[5]**

b) Explain CNC-Press in detail. **[5]**

OR

Q4) a) What aspects should be considered while designing the material handling system? **[5]**

b) Write short note on Automated Guided Vehicle (AGV) system. **[5]**

P.T.O.

- Q5)** a) What is mean by automatic inspection? Explain in short contact and non-contact method of inspection. [8]
- b) Write short note on machine vision. [7]
- Q6)** a) Write note on future automated factory and its impact on society. [7]
- b) Explain use of bar code system for shop floor data collection. [8]



Total No. of Questions : 8]

SEAT No. :

P-9649

[Total No. of Pages : 2

[6182]-1077

**M.E. (Production) (CAD/CAM)/M.E. (Production)
(Manufacturing and Automation)
COMPUTER INTEGRATED MANUFACTURING
(2017 Course) (Semester - II) (511305)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4 and Q5 or Q6.*
- 2) *Q7 and Q8 are compulsory.*
- 3) *Figures to the right indicate full marks.*
- 4) *Draw neat self-explanatory sketches wherever necessary.*
- 5) *Use of calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

Q1) What is a FMC? How does FMC ensure flexibility in manufacturing? [5]

OR

Q2) What are the benefits of GT to the manufacturing industry? [5]

Q3) What are the potential benefits of CIM? How can you apply the CIM in the case of (i) Job shop production, (ii) Mass Production? [5]

OR

Q4) What is a relational database? Describe the architecture of database management system. [5]

Q5) What are the components of PLM software? Explain. [7]

OR

Q6) Which are the typical sensors that are normally used in robot? Explain. [7]

Q7) a) What are the communication interfaces used in computer-to-computer communication? Explain in brief. What are the commonly used interface cards? [8]

b) What are the components of small local area network in CIM setup? Explain. [7]

P.T.O.

Q8) Write short notes on any three :

[18]

- a) The FMC software
- b) ASRS
- c) GKS implementation in a CAD workstation
- d) ESPRIT - CIM OSA model



Total No. of Questions : 8]

SEAT No. :

P-9650

[Total No. of Pages : 3

[6182]-1078

**M.E. (Production) (CAD/CAM)
(Manufacturing and Automation)**

**ARTIFICIAL INTELLIGENCE AND ROBOTICS
(2017 Pattern) (Semester - II) (511306)**

Time : 3 Hours]

[Max. Marks : 50

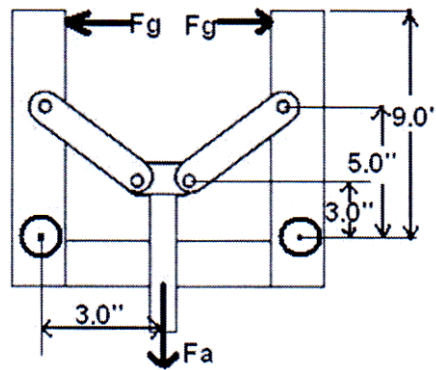
Instructions to the candidates:

- 1) *Answer any five questions*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) Explain in brief working principle of Vacuum gripper. [6]
The mechanism connecting wrist assembly is a twisting joint which can be rotated through 7 full rotations from start to end position. It is desired to have control resolution of rotation of ± 0.37 . What is the number of bit storage capacity to achieve this resolution?
- b) What is workspace? Give the functional diagram with workspace for 2PR-robot and 2RP robot. [4]
- Q2)** a) End effector of a cylindrical robot initially at (9, 0, 3) is to be moved to (4.80, 30.21, 7). [4]
Determine the rotation about X, translation along X and translation along Z.
- b) With neat sketches, explain different configurations of robot joints. [6]
- Q3)** a) Explain the geometric solution of inverse kinematics with an example of 2RP robot arm. [6]
- b) It is desired to have the first joint of a six axis robot to go from initial angle of 60° to a final angle of 95° in 6 seconds. Using a third order polynomial, calculate the joint angle at 1, 2, 3 and 4 seconds respectively. [4]

P.T.O.

- Q4) a)** For the information given in the mechanical gripper design of following figure, determine the required actuating force if the gripper force is to be 97 lb. [6]



- b) A 5cm diameter electromagnetic gripper consists of a 210 turn coil of average turns length 7 cm, mounted on a high permeability core ($\mu_r=2000$) of magnetic path length 14 cm (including 2 cm through the object). If, operated from a 24V DC supply a current of 1 Amp is drawn, calculate the maximum retention force. [4]
- Q5) a)** A robot arm with revolute joint is stationary at $\theta = 35^\circ$. It is required to move it to $\theta = 80^\circ$ in 6 seconds. Find the coefficient of cubic polynomial that accomplishes this motion and brings the manipulator to rest at goal point. Hence determine the angular position, velocity and acceleration at $t = 4$ seconds. [4]
- b) Write and explain a robot program for robot tending application. [6]

- Q6) a)** The 8×7 array of pixels indicating each element as the gray level of pixel is given below. i) Construct the histogram and find threshold value. ii) Convert it into black and white image. iii) perform shape analysis (first and second order moments, centroids, run length encoding, principal angle) [6]

7	11	13	13	13	11	13	13
10	13	10	15	15	15	14	11
9	13	14	16	16	15	15	8
8	13	14	16	15	15	15	12
9	11	14	15	15	15	15	14
7	14	14	14	14	14	14	9
8	7	6	10	10	14	8	17

- b) Write a short note on Proximity sensors. [4]

- Q7)** a) Develop a model for the transfer function of a dc motor driving a robot joint. [6]
- b) Write short note on in A.I. in robotics. [4]
- Q8)** a) What is function of encoder? Explain the principle of operation of Absolute encoder. [5]
- b) Demonstrate with suitable example the use of any two artificial intelligence techniques. [5]



Total No. of Questions : 6]

SEAT No. :

P-9651

[Total No. of Pages : 2

[6182]-1079

M.E. (Production) (CAD/CAM)
COMPUTATION TECHNIQUES IN CAD/CAM
(2017 Pattern) (Semester - II) (511307)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer Q1 OR Q2, Q3 OR Q4 and Q5 & Q6 are compulsory.*
- 2) *Assume suitable data if necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Neat diagrams must be drawn wherever necessary.*

Q1) a) Explain in short the steps in static analysis. **[4]**

b) Write a short note on. **[6]**

i) FEA tool for design.

ii) FEA element types and characteristics

OR

Q2) Defined engineering stress-strain and true stress-strain and also derive the relation between them. **[10]**

Q3) a) Explain in short numerical method for convection. **[5]**

b) Explain the numerical method for incompressible flow. **[5]**

OR

Q4) a) Solve the following differential equation using Euler's method **[5]**

$$\frac{dy}{dx} = 1 + xy$$

Given : $y(0) = 1$ and

Also find y for $x = 0 (0.1) 0.5$

P.T.O.

- b) Solve the following equation? [5]

$$\frac{dy}{dx} = x - 2y$$

Using Range-Kutta 4th order method.

Given: $y = 1$ when $x = 0$ and

Find y at $x = 0.1$ taking $h = 0.1$

- Q5)** Write short note on (any three) [15]

- a) Direct search method
- b) Gradient base method
- c) Geometric programming
- d) Point elimination methods

- Q6) a)** Solve the following unconstraint non-linear optimization problem using Simulated Annealing (SA) (only one Iteration) [7]

$$\begin{aligned} \min(z) = & (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2 \\ & + 0.1[(x_1 - 3)^2 + (x_2 - 2)^2] \end{aligned}$$

Subjected to:

$$-6 \leq x_1, x_2 \leq 6$$

- b) What is a function value corresponding to coded substring value 1101 in Genetic Algorithm (GA) to minimize the function [8]

$$f(x) = x_1^2 + 3x_2 + 10$$

With in

$$0 < x_1, x_2 \leq 3$$



Total No. of Questions : 8]

SEAT No. :

P-9652

[Total No. of Pages : 2

[6182]-1081

**M.E. (Production (CAD/CAM)/Manufacturing & Automation)
COMPUTATIONAL INTELLIGENCE IN TOOL DESIGN
(2017 Pattern) (Semester - III) (611101)**

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume Suitable data if necessary.*

Q1) a) Explain 3-2-1 principle with neat sketch. [6]

b) Explain applications of Computer Aided Jig Design. [4]

Q2) a) Explain Compound die used in sheet metal working with neat sketch.[6]

b) A hole of diameter 5mm is to be pierced centrally in a square blank 40mm × 40mm shape. Calculate Cutting force and Press tonnage. [4]

(Assume Thickness of strip = 1 mm, shear strength of material 200N/mm²)

Q3) a) Explain important Variables in Upset forging. [6]

b) Explain design procedure to design flash and flash gutter in forging.[4]

Q4) Write short notes on : [10]

a) Methods to overcome spring back in bending.

b) Combination die for sheet metal working operations.

P.T.O.

- Q5)** a) Explain the procedure to calculate number of impression while designing injection mold. [6]
b) Explain types of ejector plate assembly of Injection molding. [4]
- Q6)** a) Explain die casting process suitable for high melting point alloys. [6]
b) Explain elements of Cycle time required for die casting process. [4]
- Q7)** a) Describe optimal design of cooling system of sand casting. [6]
b) Describe functions of risers. [4]
- Q8)** Write short notes on : [10]
a) Ejection system design of Die casting.
b) Gooseneck type die casting.



Total No. of Questions : 6]

SEAT No. :

P-9653

[Total No. of Pages : 2

[6182]-1082

M.E. (Production) (CAD/CAM)

COMPUTER AIDED PRODUCTION PLANNING

(2017 Pattern) (Semester - III) (511311)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Q. No. 5 and Q. No.6 are compulsory.*
- 2) *Answers to the two sections should be written in separate answer books.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right indicate full marks.*
- 5) *Use of non.programmable calculator is allowed.*
- 6) *Assume suitable data, if necessary.*

- Q1)** a) Explain with suitable sketches patterns of demand patterns. [5]
b) Pros and cons of group technology. [5]

OR

- Q2)** a) Explain bond energy algorithm used for group technology. [5]
b) Explain the computerized method for design of facility layout. [5]

- Q3)** a) Justify the need of computer aided process planning with structure of process plan. [5]
b) What are the objectives of MRP? What is the input for MRP? [5]

OR

- Q4)** a) Briefly describe various measures of performance in single machine scheduling problem with independent jobs. [6]
b) Explain with suitable example types of simulations. [4]

- Q5)** a) Describe the steps involved in simulating manufacturing system. And also state its purpose. [8]
b) What is analytic hierarchy approach in ERP, explain with suitable application. [7]

P.T.O.

Q6) Write short notes on the following :

[15]

- a) Shop floor control - data collection.
- b) Design and manufacturing attributes in GT
- c) ERP



Total No. of Questions : 8]

SEAT No. :

P-9654

[Total No. of Pages : 2

[6182]-1084

M.E. (Production Engineering) (Manufacturing & Automation)

ADVANCED MANUFACTURING PROCESSES

(2017 Pattern) (Semester - I) (511103)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Attempt any five questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume Suitable data if necessary.*
- 5) Use of electronic pocket calculator and logarithmic tables is allowed.*

- Q1)** a) Explain the important steps followed during modern sand casting. [4]
b) How to improve casting quality by mold filling and solidification? [4]
c) Which are the limitations of conventional machining processes to fulfill demands of modern industries? [2]
- Q2)** a) Which are the modern machining processes used for advanced engineering materials? .[4]
b) Describe with sketch material removal mechanism of EDM process.[6]
- Q3)** a) Explain with neat sketch working principle Ultrasonic machining. [4]
b) How to identify welding defects and remove them? [4]
c) Differentiate between destructive and Non-destructive testing's. [2]
- Q4)** a) Describe principle of material removal, advantages and limitations of ECM. [5]
b) Explain various are welding defects with their causes and remedies. [5]

P.T.O.

- Q5)** a) A wire of 12 mm diameter is to be reduced to 7.5 mm diameter. The die angle is 15° and the coefficient of friction at dies and wire interface is 0.5. The flow stress of wire material is 340 N/mm^2 . Determine drawing stress and drawing load if the drawing speed is 0.7 m/s. [4]
- b) Explain metal high speed hot forging with neat sketch. [4]
- c) List out various the non-conventional forming processes. [2]
- Q6)** a) Explain metal stretch forming with neat sketch. [4]
- b) Explain high energy rate forming with neat sketch. [4]
- c) Differentiate between hot forming and cold forming. [2]
- Q7)** a) Explain important process parameters of electro-hydraulic forming. [4]
- b) A tube of 12mm external diameter and 1mm thickness is to be reduced to 16mm external diameter and 0.5 mm thickness. The die angle is 24° and plug angle is 16° . The coefficients of friction at die and tube interface and tube and plug (mandrel) interface is 0.5. The flow stress of tube material is 340 N/mm^2 . The tube drawing is carried at a speed of 0.4m/s. Calculate the fixed plug. [4]
- c) What is high speed blanking? [2]
- Q8)** a) Explain various rolling problems. [5]
- b) Explain forming limit diagram. [5]



Total No. of Questions : 8]

SEAT No. :

P-9655

[Total No. of Pages : 2

[6182]-1085

M.E. (Production-Manufacturing & Automation Engg.)

INDUSTRIAL AUTOMATION

(2017 Pattern) (Semester - I) (511104)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data if necessary*

Q1) a) A hydraulic system is operating at 200 bar with pump flow 20 lpm. The input power to the pump drive is 12 kW. The pump is loaded for 70% of the operating time. The overall efficiency when it is on load is 75%. If the ambient temperature is 25°C and maximum permissible fluid temperature in the reservoir is 60°C. Calculate the suitable size of the reservoir if it is of square section of size 'a' with length '1.4a'. **[6]**

b) Explain with neat sketch principle of working and application of pressure compensated flow control valve. **[4]**

Q2) a) The door of a sand feeding hopper is actuated by a double acting pneumatic cylinder. When hopper is full, the door is opened and sand fall into a wagon underneath. The door should be closed after a delay of 5 seconds to ensure that all the sand has emptied out. Draw suitable pneumatic circuit. **[6]**

b) Explain hydro-pneumatic system. **[4]**

Q3) a) What is parity flag? What is its significance? Explain with suitable example. **[4]**

b) Explain clearly the difference between microprocessor and microcontroller. **[6]**

P.T.O.

Q4) Write short notes on: [10]

- a) Position and pressure sensing in pneumatic systems.
- b) Fringe condition modules and their integration.

Q5) a) Explain with neat sketch revolving feeder. [6]

- b) What factors should be considered while deciding the type of transfer device to be used in automated systems. [4]

Q6) a) Define and explain following co-ordinate systems for industrial robots: [6]

- Cartesian Coordinate System,
- Polar co-ordinate system
- Spherical co-ordinate system

b) For a 2R manipulator, determine the joint angular displacement if: [4]

- length of each arm is 1.5 meter
- Initial co-ordinates of end effector are (1.4 m, 1.6 m)
- Final co-ordinates of end effector are (-1.4 m, 1.6 m)
- Simple cubic fit is followed.

Q7) a) What are key challenges in implantation of AI in manufacturing applications. [5]

- b) Explain applications of neural networks in manufacturing automation. [5]

Q8) Write short notes on:

- a) Need for system Modeling [5]
- b) Robot programming for press working [5]



Total No. of Questions : 8]

SEAT No. :

P-9656

[Total No. of Pages : 2

[6182]-1087

M.E. (Production) (Manufacturing & Automation)

ADDITIVE MANUFACTURING

(2017 Credit Pattern) (Semester - II)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q. 6 and Q. 7 or Q. 8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of Calculator is allowed.*
- 5) *Assume Suitable data jf necessary.*

Q1) a) Write short note on application of additive manufacturing in current scenario. [4]

b) Enlist various powder based rapid prototyping system explain any one of them in short. [5]

OR

Q2) a) Explain the role of additive manufacturing in product development. [4]

b) Enlist various solid based rapid prototyping system explain any one of them in short. [5]

Q3) a) Write short note on model slicing. [4]

b) Explain the different materials used in additive manufacturing. [5]

OR

Q4) a) Differentiate between fused deposition modeling and laminated object manufacturing. [4]

b) Explain with neat sketch different steps in additive manufacturing [5]

Q5) a) Differentiate between solid ground curing process and selective laser sintering process. [8]

b) Explain the medical and Bio-Additive manufacturing with its applications, advantages and disadvantages [8]

OR

P.T.O.

- Q6)** a) Write short note on digitization techniques [8]
b) Write short note on Laser Engineered Net Shaping (LENS) [8]
- Q7)** a) Explain the Stereo-lithography with reference to its working process, advantages and disadvantages. [8]
b) Write short note on Fused deposition modeling [8]
- OR
- Q8)** a) Explain the bio additive manufacturing with neat sketch. [8]
b) Explain with neat sketch the Electron Beam Melting process [8]



Total No. of Questions : 6]

SEAT No. :

P-9657

[Total No. of Pages : 2

[6182]-1089

M.E. (Production) (Manufacturing & Automation)

MECHATRONICS

(2017 Pattern) (Semester - III) (611102)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Q.No.5 and Q.No.6 are compulsory.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of non-programmable calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

Q1) a) Discuss the significance of mechatronics in aviation industry. [5]

b) Describe the construction and working of optical encoder. [5]

OR

Q2) a) State and explain type of sensor used in touch screen. [5]

b) A 360° wire-wound potentiometer has 300 turns and a total resistance of 1 kΩ. What is the resolution in ohms? In radians? [5]

Q3) a) State the detail steps to do subtraction of two numbers in a microprocessor. [4]

b) Develop a conceptual design of a sensors based control system for an electronic scooter. Assume suitable data if necessary. [6]

OR

Q4) a) Write the instructions to load two hexadecimal numbers 69D and 70C in the register A and B respectively. Find the greatest number and store the result in memory location 607B. [6]

b) State the characteristics of ADC (Analog to Digital Conversation). [4]

P.T.O.

- Q5)** a) Construct the ladder logic diagrams for (i) the NAND gate and (ii) the OR gate. Also develop a truth table for both logics. **[6]**
- b) Explain application of PLC system for pneumatic clamping application. **[9]**

Q6) Write short notes on the following : **[15]**

- a) Microprocessor 8086 pin diagram
- b) PID controllers
- c) Microcontroller 8051 pin diagram



[6182]-1100

M.E. (E & TC) (VLSI & Embedded System)

FAULT TOLERANT SYSTEMS

(2013 Pattern) (Semester - III) (604201)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve any Five questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume Suitable data, if necessary.

Q1) a) Construct a binary decision diagram for $f = \bar{a}bc + a\bar{b}c + abc$ considering "a" as root node. [5]

b) Explain structural model of RTL. Write on RTL model for a positive edge triggered D-Flip-Flop. [5]

Q2) a) Construct a primitive cube table for function. [5]

$$F = \bar{X}_1\bar{X}_2 + X_1\bar{X}_2 + \bar{X}_1X_2X_3$$

b) Define & explain in detail with timing diagram, different delay models w.r.t. 2 i/p And gate. [5]

Q3) a) Find the set of all tests that detects the multiple stuck fault for fault 'a' s-a-0 and fault 'b' s-a-1 for the circuit shown in fig. 1. [5]

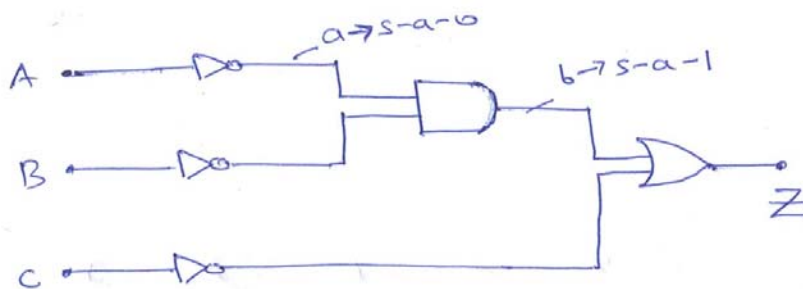


fig. 1

- b) Write short note on : [5]
- Fault sampling
 - Statistical fault analysis

- Q4)** a) Explain in detail the general fault simulation technique. [5]
- b) Define detectability. Find test vectors that determine the OR bridging fault between input A & B in fig 2 shown below : [5]

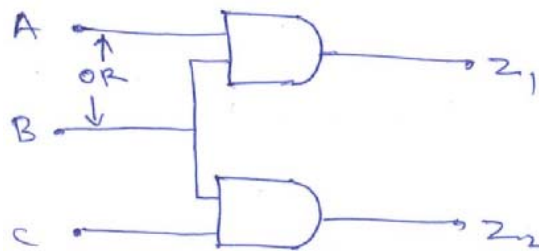


Fig. 2

- Q5)** a) Explain the necessity of compression techniques. List out different compression techniques. [5]
- b) Explain single i/p signature analyzer with suitable example. [5]
- Q6)** a) What is boundary scan? Show a design for the boundary scan cell circuit by for bidirectional I/O pin in IEEE1149.1 methodology. [5]
- b) Explain self checking combinational circuit. [5]
- Q7)** a) Explain any two techniques of DFT to increase controllability, observability and predictability. [5]
- b) Differentiate between online and offline BIST. Explain in detail the generic offline BIST architecture at the board level. [5]
- Q8)** a) Write short note on : PLA Testing. [5]
- b) List & define important factors that determine the complexity of deriving a test for a circuit. [5]



Total No. of Questions : 7]

SEAT No. :

P-9659

[Total No. of Pages : 3

[6182]-1101

M.E. (Mechanical Engineering) (Robotics & Automation)
(MERA501) : ADVANCED MATHEMATICS AND
NUMERICAL METHODS
(2021 Course) (2017 Pattern) (Semester - I)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions from the following.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Use of electronic pocket calculator is allowed.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Solve the following system of linear algebraic equations by using Gauss Elimination method. $X + 3y + 2z = 6$, $3x - 2y + 5z = 5$, $2x - 3y + 6z = 7$.
- b) Compute the solution of $x^2 - 2 = 0$ by using bisection method in $[1, 2]$.
- [10]**

- Q2)** a) Find the regression line of y on x for the data $(x, y) = (0, 4), (3, 0), (4, 2), (6, -8), (8, -7)$.
- b) Use Newton divided difference formula to derive interpolating polynomial for the data $(0, -1), (1, 1), (2, 9), (3, 29), (5, 129)$. Hence compute $y(4)$.
- [10]**

- Q3)** a) Evaluate the integral $I = \int_{-1}^1 2x^2 + 3xy + 4y^2 dx dy$ by using gauss quadrature.
- b) By means of newton divided difference formula find the value of $f(8)$, $f(9)$ from the following table
- [10]**

X	4	5	7
F(x)	48	100	294

P.T.O.

Q4) a) Find the characteristic equation and eigenvalues of $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ by

using Faddeev Laeverries method.

b) Find the numerically larger eigenvalue of the given matrix by power

method $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ where $x_0 = (1, 1, 1)$.

[10]

Q5) a) If $\frac{dy}{dx} = x + y - 1$ with $y(0) = -2$ then estimate $y(2)$ by using Euler's Method using step size 0.5.

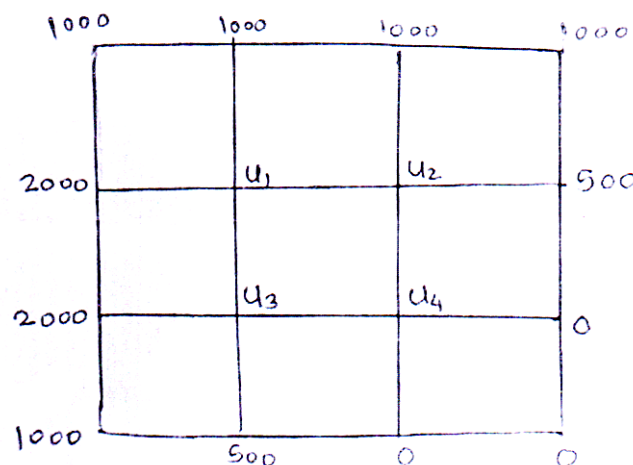
b) Use the shooting method to approximate the solution of the boundary value problem $\frac{d^2y}{dx^2} - 2y = 0, y(0) = 1.2, y(1) = 0.9, h = 0.25$.

[10]

Q6) a) Give the classification of partial differential equations. Explain the type of Laplace Equation.

b) Given the values of $u(x, y)$ on the boundary of the square in the figure evaluate the function $u(x, y)$ satisfying the laplace equation $\nabla^2 u = 0$ at the pivotal Points of this figure by Jacobi's

[10]



Method.

- Q7)** a) Find a real root of equation $3x = \cos x + 1$ by using Newton Raphson Method.
- b) If $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} - 2xy = 1, y(0) = 1, y'(0) = 0$, Evaluate $y(0.1)$ using Runge Kutta method of order 4.

[10]



Total No. of Questions : 7]

SEAT No. :

P-9660

[Total No. of Page : 1

[6182]-1102

M.E. (Mechanical Engineering) (Robotics & Automation)

MERA-502 : MACHINE VISION SYSTEM

(2021 Course) (2017 Pattern) (Semester - I)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Answer any five questions from the following.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) What are Imaging Sensors? Explain CCD and CMOS sensors considering Specifications and Interface Architectures. **[10]**

Q2) What is Image processing? Explain in detail Point Operation and Thresholding. **[10]**

Q3) Write short notes on : **[10]**

- a) Grayscale Stretching
- b) Neighborhood Operations
- c) Image Smoothing and Sharpening
- d) Edge Detection

Q4) Write Machine vision applications in manufacturing, electronics, printing, pharmaceutical and textile. **[10]**

Q5) Explain how Industrial machine vision is used in production and services. Explain with an example. **[10]**

Q6) Define Multi-Object Tracking. Differentiate between Model-based Tracking and Hybrid Tracking. **[10]**

Q7) Explain the terms : **[10]**

- a) Template based activity recognition
- b) Human Recognition Using Gait



Total No. of Questions : 7]

SEAT No. :

P-9661

[Total No. of Page : 1

[6182]-1103

M.E. (Mechanical) (Robotics & Automation)

TOTAL INTEGRATED AUTOMATION INDUSTRY4.0

(2021 Pattern) (Semester - I) (MERA503)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Answer any five questions from the following.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*

Q1) Discuss Principles and Strategies of Automation in Industry 4.0. **[10]**

Q2) Explain following terms : **[10]**

- a) Analysis of Transfer Lines Without Storage
- b) Automated Flow Lines with Storage Buffers
- c) Computer Simulation of Automated Flow Lines

Q3) Explain following terms : **[10]**

- a) Analysis for Material Handling Systems
- b) Conveyor Systems
- c) Automated Guided Vehicle Systems

Q4) Write difference between Process Industries Verses Discrete Manufacturing Industries. **[10]**

Q5) How to use Sensor Technologies for Automated Inspection. **[10]**

Q6) Write a PLC Program for any industrial application. **[10]**

Q7) Explain the sequential steps of working PLC system. **[10]**



Total No. of Questions : 7]

SEAT No. :

P-9662

[Total No. of Page : 1

[6182]-1104

M.E. (Mechanical Engineering) (Robotics & Automation)

**MERA-504 : ENGINEERING RESEARCH
METHODOLOGY**

(2021 Course) (2017 Pattern) (Semester - I)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer any five questions from the following.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*

Q1) Explain the criteria's of good research in connection with the society benefits. **[10]**

Q2) Write a note on Assessment of Quality of Journals and Articles. **[10]**

Q3) Explain the use of Standards and Codes in research Design. **[10]**

Q4) Discuss the role of Data organization and data grouping in data collection. **[10]**

Q5) Discuss the role of Deterministic and random data selection in data analysis. **[10]**

Q6) Explain the steps involve in writing a research proposal. **[10]**

Q7) Differentiate the Parametric V/s Non Parametric methods used in data collection. **[10]**

